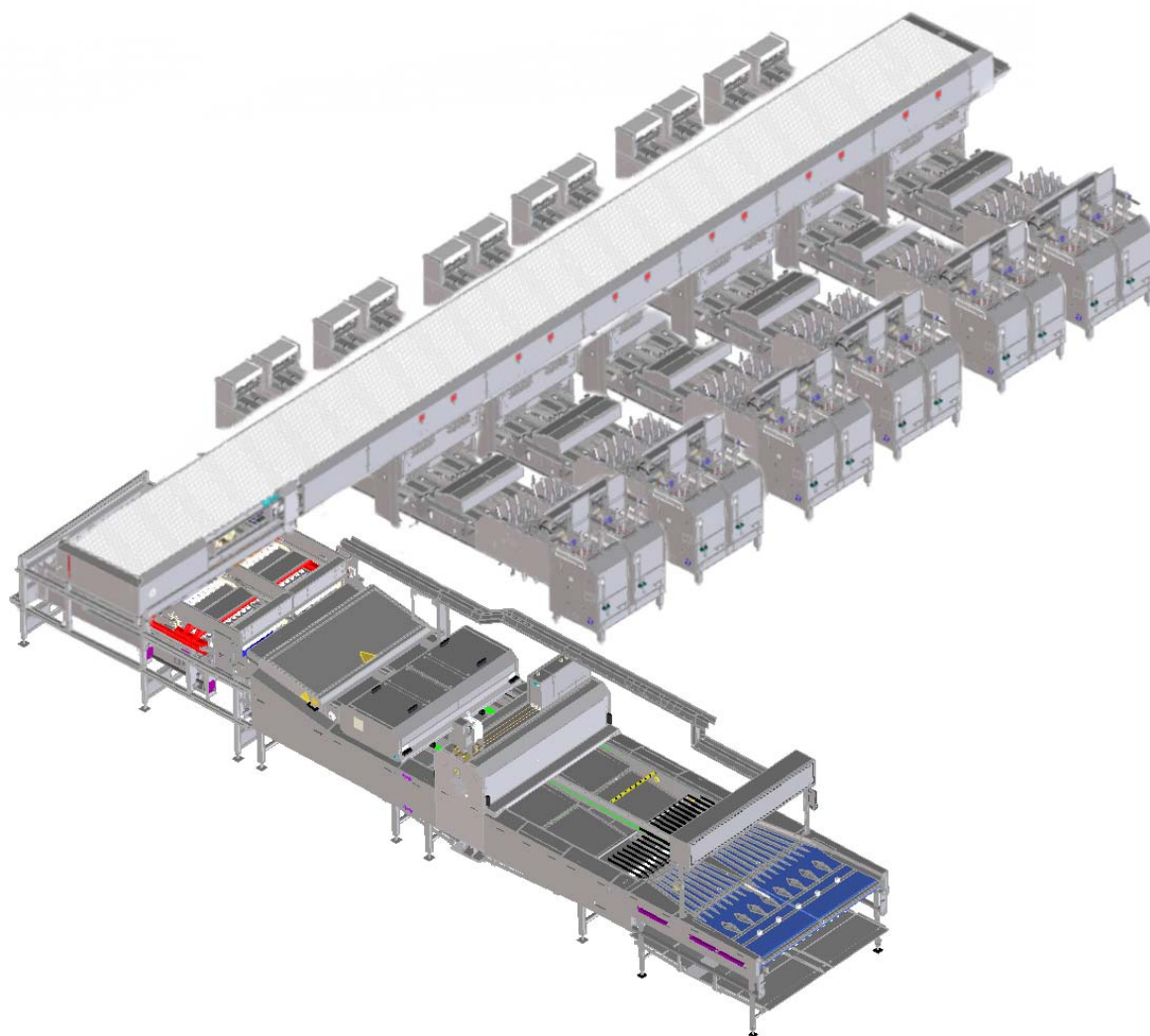


# **MOBA**



## **Service Manual**



## **Omnia FT/XF**

## Service Manual - Omnia FT/XF

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| <b>A</b> | <b>Comparative table of lubricants</b> | <b>213</b> |
| <b>B</b> | <b>TIP box OMNIA</b>                   | <b>215</b> |
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|          | <b>List of illustrations</b>           | <b>219</b> |
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## Preface

---

The documentation supplied with the Omnia FT/XF consists of four documents:

1. The user manual.

The User Manual explains how to start and stop the machine and contains safety and maintenance instructions, the adjustments and error messages. This User Manual is intended for **persons operating the machine**.

2. The Service Manual.

This contains an explanation of the procedure used to switch off the machine, the inspection of the settings and the disposal procedure. This manual is to be used by **your maintenance personnel and our service personnel**.

3. The Exploded Views and/or the 3D model.

This contains the information you need to order parts. This information is for **your maintenance staff and our service staff**.

4. Electrical and electronic diagrams.

These contain the electrical engineering and electronic diagrams, adjustments of the frequency controllers and adjustments of all other electrical and electronic components. This manual is to be used by **your maintenance personnel and our service personnel**.



---

**NOTE**

***Keep these manuals with the machine. The service technician may need information about the machine to carry out the maintenance work on the machine.***

---



# **1 Introduction**

## **1.1 Purpose**

The Omnia FT/XF is a machine, with which you can transport, weigh, grade and pack eggs. Your Omnia FT/XF is a precision machine. So you should carry out all adjustment to and maintenance on the machine with great care. This machine may not be used for work other than the work mentioned above. Have everyone read this Manual before they use the machine. This manual describes the adjustment procedures for your Omnia FT/XF.

## **1.2 Using this Manual**

This manual is intended for the service technician of the machine. Service tasks may only be carried out by authorised, trained and skilled personnel. The service technician must know and understand the contents of the manual. The version numbers of documents are indicated in the page headers of the User Manual and Service Manual. Refer to this number when you contact our service helpdesk.

## 1.3 Type designation

This manual has been written for the following machines:

| Type of machine |           |           |         |
|-----------------|-----------|-----------|---------|
| Type            | FT        | XF        |         |
| Capacity (e/h): | -         | OMNIA 85  | 30.000  |
|                 | -         | OMNIA 125 | 45.000  |
|                 | OMNIA 170 | OMNIA 170 | 60.000  |
|                 | OMNIA 250 | OMNIA 250 | 90.000  |
|                 | OMNIA 330 | OMNIA 330 | 120.000 |
|                 | OMNIA 500 | -         | 180.000 |



### NOTE

*In this manual all versions of the OMNIA grader of the above-mentioned types are described. Options or units may be installed depending upon your version of the machine.*



### NOTE

*The illustrations in this manual may be different from the actual machine before you! However, the settings and adjustments are the same procedure as described in this manual.*

Your type of machine can be read from the type plate mounted on your machine. State this type when you contact our service or sales department.

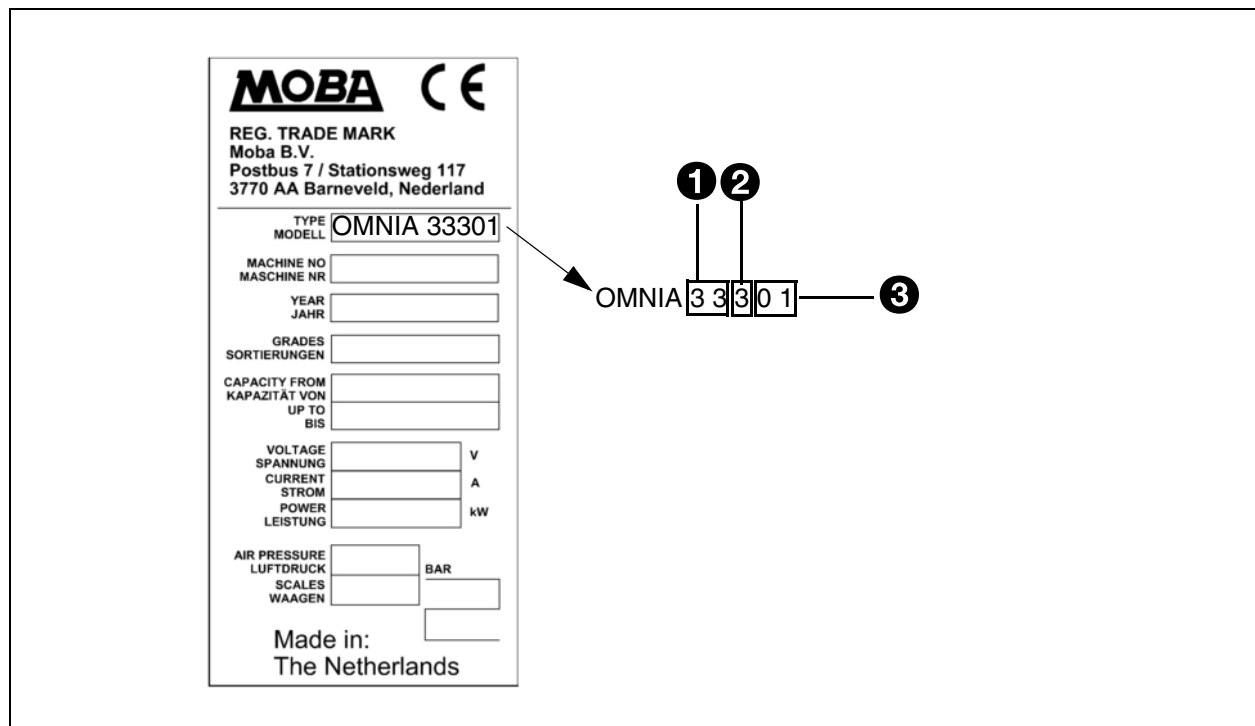


Fig. 1 - CE and type designation

- 1 - Capacity
- 2 - Type
- 3 - Run number

### 1.3.1 Capacity code

The serial number indicates the capacity of the machine:

- 85 - 1 Tracks, 6 Rows
- 125 - 2 Tracks, 6 Rows
- 17 - 2 Tracks, 6 Rows
- 25 - 4 Tracks, 12 Rows
- 33 - 4 Tracks, 12 Rows
- 500 - 6 Tracks, 18 Rows

### 1.3.2 Omnia type

The type relates to the hygiene related set up.

- 2 - XF type
- 3 - FT type

### 1.3.3 Run number

The run number is a sequential run number.

## 1.4 CE designation

The EEC conformity-declaration and the CE mark on the machine confirm that this machine meets the strict EEC guidelines 98/37/EC, 89/336/EEC and 73/23/EEC. The small plate is attached to the side of the machine electricity cabinet.

## 1.5 The Installation instructions



---

**NOTE**

*The machine must be installed and dismantled in accordance with the procedures described in the service manual.*

---



---

**NOTE**

*After installing the machine store the documents delivered with the order confirmation together with this manual.*

---

In the layout drawing you will find the location of each component and the space needed to install the system, as well as the space required to open the doors of the control panels. The following information is provided in the documentation that came with your order:

1. The workplaces
2. The applicable CE directives
3. The electrical specifications
4. The quality of the power supply
5. The supply of compressed air
6. The operating conditions
7. The storage conditions

## 8. The Installation instructions

### 1.6 Guarantee

For the guarantee provisions we refer you to the description in the general conditions of sale. These conditions apply if no other conditions have been explicitly agreed.

You **lose the right** to a guarantee on this product in the following cases:

- The machine has not been operated correctly.
- Changes have been made to the machine without permission from MOBA.
- The general safety instructions have not been observed.
- The machine has not been serviced and maintained in accordance with the instructions.



#### **NOTE**

***The parts subject to wear are not covered under this guarantee. See the attachments in the User Manual!***

### 1.7 How to order parts

You must always provide the following information when you order parts:

- Machine number
- main group of the machine
- part number and description of the part

### 1.8 Contact information

The contact information depends on the region.

Table 1 - Contact information

| Region  | Service                       | Parts |
|---------|-------------------------------|-------|
| Europe  | Service in the Netherlands    |       |
| • UK    | Service in the UK             |       |
| America | Service in the USA            |       |
| Asia    | Service in Asia & the Pacific |       |

Table 1 - Contact information

| Region        | Service                       | Parts                      |
|---------------|-------------------------------|----------------------------|
| • China       | Service in China              |                            |
| • Japan       | Service in Japan              |                            |
| • Middle East | Service in Asia & the Pacific | Service in the Netherlands |
| Australia     | Service in Asia & the Pacific |                            |
| Africa        | Service in the Netherlands    |                            |

### 1.8.1 Service in the Netherlands

**Moba b.v.**

Stationsweg 117  
3771 VE Barneveld,  
the Netherlands  
The Netherlands

PO Box 7  
3770 AA Barneveld  
The Netherlands

**General**

Telephone: +31 (0) 342 455 655  
Fax: +31 (0) 342 455 755  
E-mail: info@moba.nl  
Internet: www.moba.nl

**'Service' Department**

Telephone: +31 (0) 342 455 655  
Fax: +31 (0) 342 455 789  
E-mail: service@moba.nl

**'Parts Sales' Department**

Telephone: +31 (0) 342 455 635  
+31 (0) 342 455 637  
Fax: +31 (0) 342 455 644  
E-mail: spareparts@moba.nl

### 1.8.2 Service in the USA

**Diamond Moba Americas**

1821 Colonial Village Lane,  
Lancaster, PA 17601  
USA

Telephone: +1 717 - 397 33 37  
Fax: +1 717 - 397 33 45  
E-mail: info@dma-group.com

**Diamond Moba Americas**

23400 Haggerty Rd  
Farmington Hills, MI 48335  
USA

Telephone: +1 248 - 476 - 7100  
Fax: +1 248 - 476 - 0849  
Phone service: +1 248 - 476 - 7247  
Fax service: +1 248 - 476 - 3617  
E-mail: service&parts@dma-group.com

### 1.8.3 Service in the UK

#### **Moba UK Ltd.**

Red Lees, Ketley  
Telford Shropshire TF1 5YA  
Great Britain  
Telephone: +44 1952 - 615 911  
Fax: +44 1952 - 620 457  
E-mail: info@mobauk.co.uk

### 1.8.4 Service in Asia & the Pacific

#### **Moba Asia Sdn. Bhd.**

Unit 703, Block G  
Phileo Damansara 1,  
Jalan 16/11, Off jalan Damansara.  
46350 Petaling Jaya  
Malaysia  
Telephone: +60 3 7956 9336  
Service Hotline +60 3 7954 4133  
Fax: +60 3 7958 6233  
E-mail: info@asia.moba.com.my

### 1.8.5 Service in Japan

#### **Moba Japan Co., Ltd.**

Sannomiya Grand Building 4F  
2-2-21, Isogami-Dori  
Chuo-Ku, Kobe, Hyogo  
651-0086 Japan.  
Telephone: +81 78 262 5661  
Fax: +81 78 262 5662  
E-mail: info@mobajapan.co.jp

### 1.8.6 Service in China

## FPS Beijing Co., Ltd.

艾弗爱食品加工设备（北京）有限公司 / FPS Beijing Co  
中国北京经济技术开发区东区双羊路 15 号迪兴工业园区 邮编：100023  
FPS Beijing Co., Ltd.  
No.15 Shuangyang Road  
Dixing Industrial Zone, East District  
Beijing E&T Development Area  
100023  
P.R.China  
Telephone: (+86) 10 6789 2399  
Fax: (+86) 10 6789 2655  
E-mail: fpschina@fps-bj-moba.com.cn

## 1.9 How to read this manual

In the manual the term 'machine' is used that also includes the relative module, installation, unit or the system with the associated equipment.

The icons in the manual and on the machine's screen indicate a function or a comment. The icons are explained below.



### **NOTE**

*The illustrations in this manual can differ from those on the machine. Remember this when reading this manual and carrying out work on the machine.*

### 1.9.1 Position numbers in the text

The numbers between round brackets, e.g. (5) refer to the relevant part in the figure. Text in square brackets, such as [DETAIL], refers to a button on the keyboard or touchscreen of the main control panel.

### 1.9.2 Icons in the documentation

Icons are shown in the left-hand column of the pages in this documentation. The icons have the following meanings:



Point for attention or remark



Consult the adjustment lists supplied with the machine



Condition to be met before the adjustment can be carried out



Use of a special tool



Mechanical adjustment



Electrical adjustment



Pneumatic adjustment

-  Settings
-  Configuration of the machine's software
-  Control procedure
-  Instructions for lubrication of the machine
-  Item for periodical maintenance
-  Tip

### 1.9.3 Icons used to indicate machine parts

Instructions that concern a specific machine unit are indicated with the icons that are shown below. In the documentation, these icons appear in the header text of the relevant page.

-  Loader
-  Infeed unit
-  Transfer unit™
-  Frame
-  Packing lane
-  Take-away conveyor
-  Contiflow



## 2 Safety

Read the manual carefully before you start to use the machine/module. Read and understand the information in this manual, and observe the instructions. Carry out all work in accordance with the current work, safety and environmental regulations.

Table 2 - General safety instructions

| GENERAL SAFETY INSTRUCTIONS                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The machine must not be in operation when you remove guards, drip trays or other parts of the machine during maintenance or service.                                            |
| Stop the machine with a normal stop. Use the emergency-stop system only in emergencies.                                                                                         |
| Only personnel authorised and instructed by the machine's owner may use the machine.                                                                                            |
| Tell other employees that you are going to work with or on the machine.                                                                                                         |
| Clear away any tools and fit, any protective covers removed from the machine, where relevant, before you carry out a test run.                                                  |
| Always carry out a test run with the machine after a modification or after replacing parts.                                                                                     |
| Never remove, bridge or block any safeguards.                                                                                                                                   |
| Switch off the main switch before replacing any electrical parts. Lock the main switch with the padlock and keep the key with you.                                              |
| Only specialised persons who are authorised to do so may carry out work on the machine's electrical circuit.                                                                    |
| Take ESD preventive safety-steps before you touch electronic components. (ESD = ElectroStatic Discharge.) A discharge of electrostatic energy that can damage electronic parts. |
| Wait at least five minutes after switching off the main switch to the machine before you start work on any electronic systems.                                                  |

### 2.1 Safety instructions indicated in the documents



#### **NOTE**

*Additional information will be provided in this frame.*



#### **WARNING**

*A warning will be displayed in this frame. Users can injure themselves or cause serious damage to the product.*



#### **DANGER**

*A danger is indicated in this frame. Failure to heed this warning places the user's life in danger.*

### 2.2 Warning symbols on the machine/module

Warning symbols are attached to the machine/module. Warning symbols that become detached or are unreadable, they must be replaced with new symbols. Warning symbols are available from MOBA b.v..



This symbol is shown in places where there is a risk of injury through general causes or through different causes occurring at the same time.



This warning symbol is shown in locations with a risk of an electric shock.



This warning symbol is shown locations with a risk of injury by turning parts.



This warning symbol is shown in locations with a risk of injury by sharp parts.



This warning symbol is shown in locations with a risk of injury caused by a slippery floor.



This warning symbol is shown in locations with a risk of eye damage due to UV-C<sup>(\*)</sup> light.



This warning symbol is shown in locations with a risk of pinching parts of the body.



This warning symbol shown in locations where spraying with water or the use of a high-pressure cleaner is prohibited.



This warning symbol is displayed at locations where hearing protection must be worn.



This symbol is displayed at locations where it is necessary to read the manual in connection with safety and the safe condition of the machine.

(\*) See the glossary.

## 3 Commissioning

**WARNING**

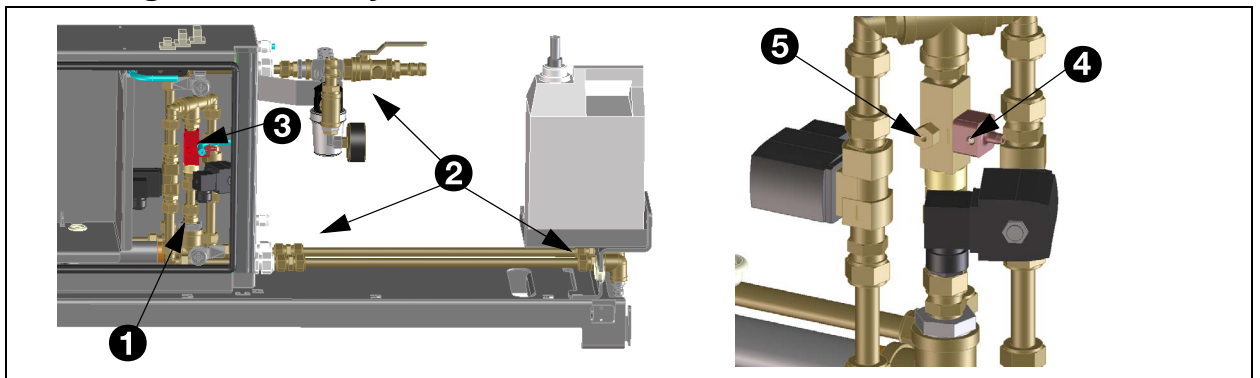
*Before using the machine/unit read this chapter carefully.*

The following units/parts of the Omnia FT/XF must be checked before commissioning.

### 3.1 Cleaning unit

- The cleaning unit can be performed with an injector. See paragraph 3.2.
- The cleaning unit can be performed with dosing pump. See paragraph 3.3.

### 3.2 Cleaning unit with injector



*Fig. 2 - Injector and clutches*

Before the unit can be brought into use, the following points must be carried out:

- 1 Connection to the water supply.

Check the level of the cleaning-agent in the jerry can. Sumabac D10 Article number: 01602005.

- 2 Check for leaks in the clutches, both inside (1) and outside (2) the control-cabinet.

**Adjustment of injectors (3):**

- 1 Open the setscrew (4)  $\pm 5$  turns.
- 2 Close the setscrew (5) for the bypass until there is just some water sucked in via the injector.
- 3 Entirely close the setscrew (4).
- 4 Open the setscrew (4) by one turn.

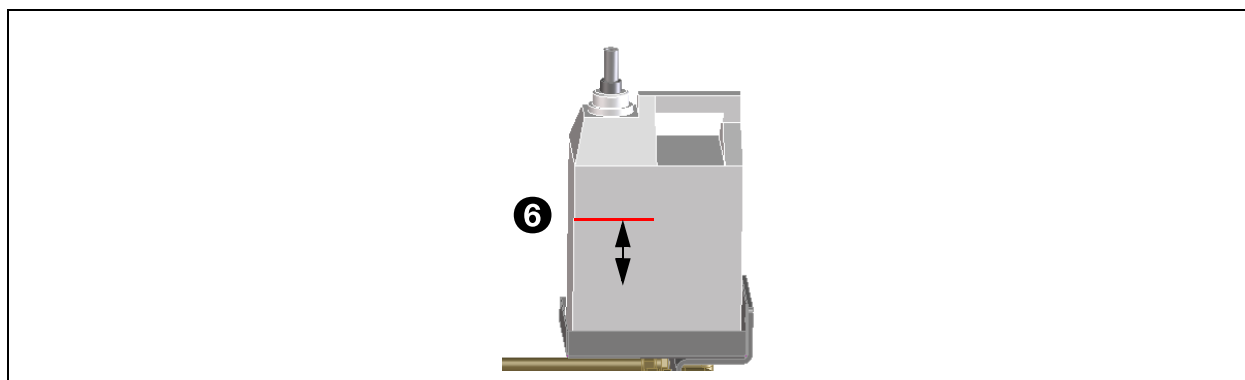


Fig. 3 - Marking jerry can

**Check the use of the cleaning-agent as follows**

- 1 Make a mark (6) at the current level in the jerry can.
- 2 Start the cleaning unit, see the cleaning instructions in the user manual.



**NOTE**

***Set the grader at the lowest speed.***

- 3 After the cleaning programme has been entirely completed, measure the level difference in the jerry can.
- 4 The level difference for:
  - Omnia 85, 125, 170: = 16 mm.
  - Omnia 250, 330: = 32 mm.
  - Omnia 500: = 48 mm.
- 5 If the case of a deviating level difference, open or close the setscrew (4) slightly more.

### 3.3 Cleaning unit with Metering pump

#### 3.3.1 Metering pump for CIP

Certain parts of the machine are cleaned with water and a cleaning agent / disinfectant. A metering pump is used for the correct dosage.

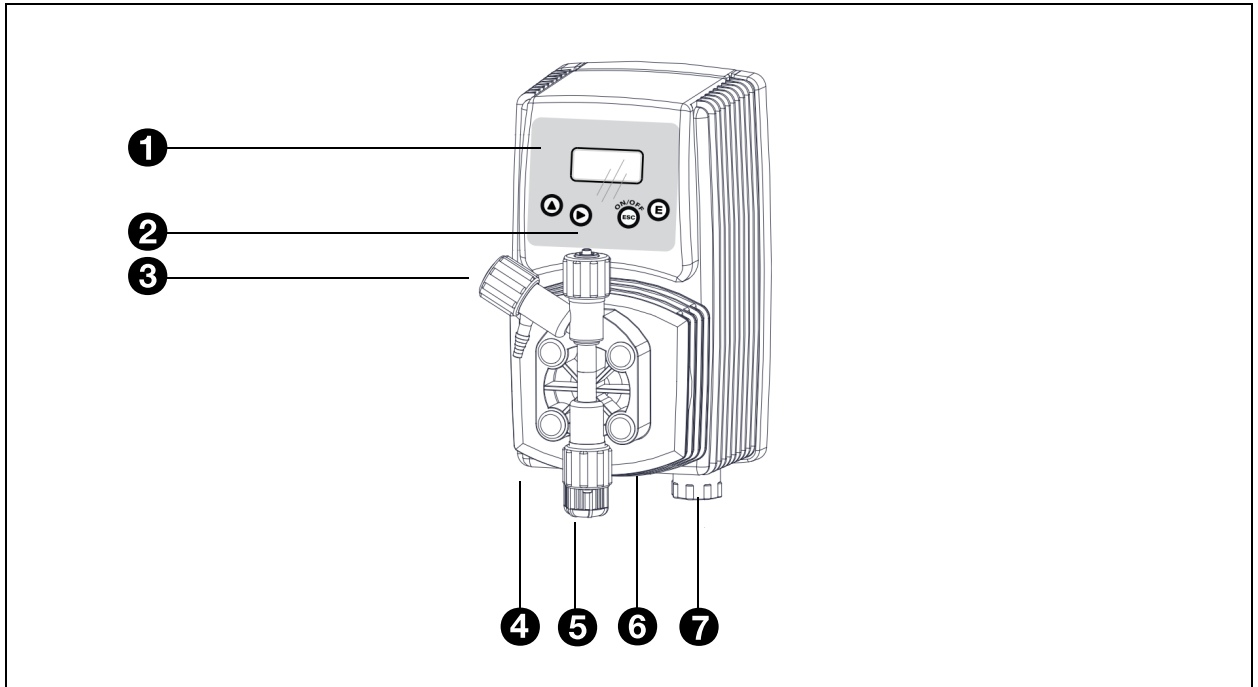
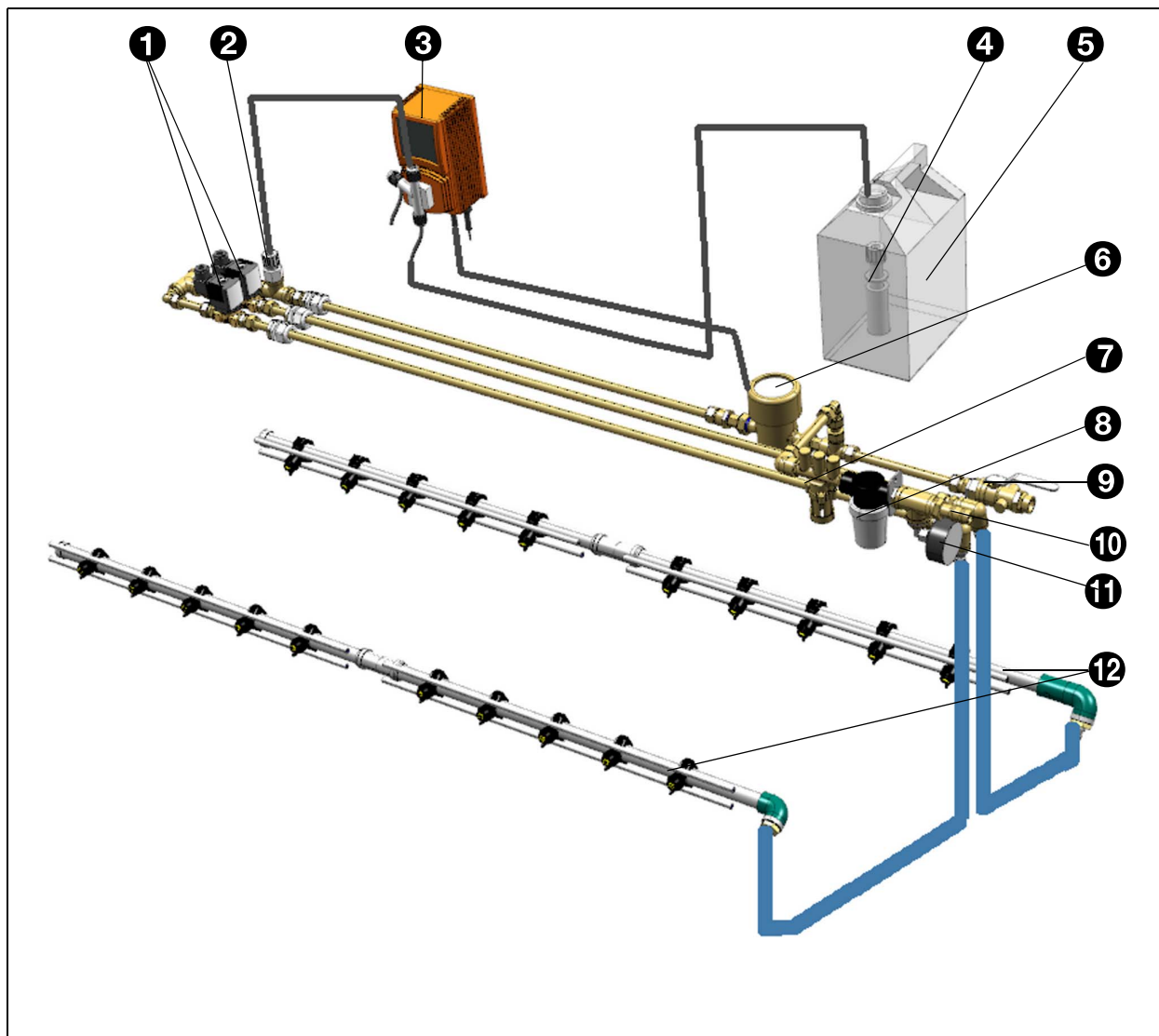


Fig. 4 -

- 1 - LCD display
- 2 - Inlet valve for outflowing cleaning agent
- 3 - Air-relief valve
- 4 - Power supply
- 5 - Suction valve for cleaning agent
- 6 - External input signal (from flow meter)
- 7 - Cleaning agent level measurement (Not in use)

### 3.3.2 Autoclean Infeed unit



*Fig. 5 - Autoclean Infeed unit*

- 1 - Valve for foam / flushing water to spray tube
- 2 - Injector
- 3 - Metering pump
- 4 - Suction hose
- 5 - Cleaning agent
- 6 - Flow meter
- 7 - Non-return valve
- 8 - Filter
- 9 - Tap
- 10 - Connection to water supply
- 11 - Water pressure gauge
- 12 - Spray tube

### Description autoclean Infeed unit

- 1 The water is connected to the water supply.
- 2 The water passes through a filter and a non-return valve to the flow meter.
- 3 The flow meter sends a number of pulses per litre to the metering pump, depending on the flow.

- 4 A suction hose is used to feed the cleaning agent to the metering pump.
- 5 Depending on the number of pulses and the settings in the metering pump, the correct amount of cleaning agent is supplied to the injector.
- 6 The injector mixes the cleaning agent and the water.
- 7 Once the relevant valve is activated, the foam is fed to the spray tubes.

**NOTE**

*Please refer to the circuit diagrams for the metering pump settings.*

### 3.3.3 Check consumption cleaning agent

Check for the correct cleaning agent dosage. The metering pump is set to a usage of 1% of the SumaBac D10 cleaning agent.

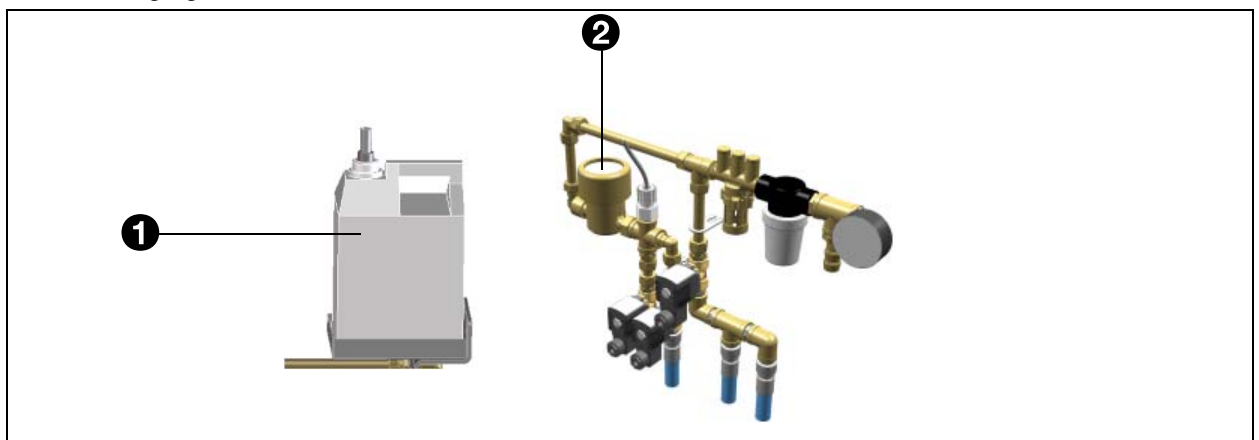


Fig. 6 - Jerrycan with water meter

- 1 - Jerrycan with cleaning agent
- 2 - Watermeter

#### Check consumption cleaning agent

- 1 Determine the contents of the jerry can with cleaning agent.
- 2 Make a note of the water meter reading.
- 3 Start the cleaning program. See the cleaning instructions in the User Manual.

**NOTE**

*Set the grader at the lowest speed.*

- 4 Record the water meter reading after the cleaning program has completely finished.
- 5 Determine the water usage and divide the result by 100.
  - The result is the number of litres of cleaning agent that is allowed to be used.
- 6 After the cleaning program has finished completely, measure the amount of cleaning agent used.
  - Compare this with the result of step 5.

### 3.4 Checking MultiDrum™

Before the unit can be adjusted, the MultiDrum™ must be checked to ensure that the outer drum is not mounted spirally. This can be done in the following manner

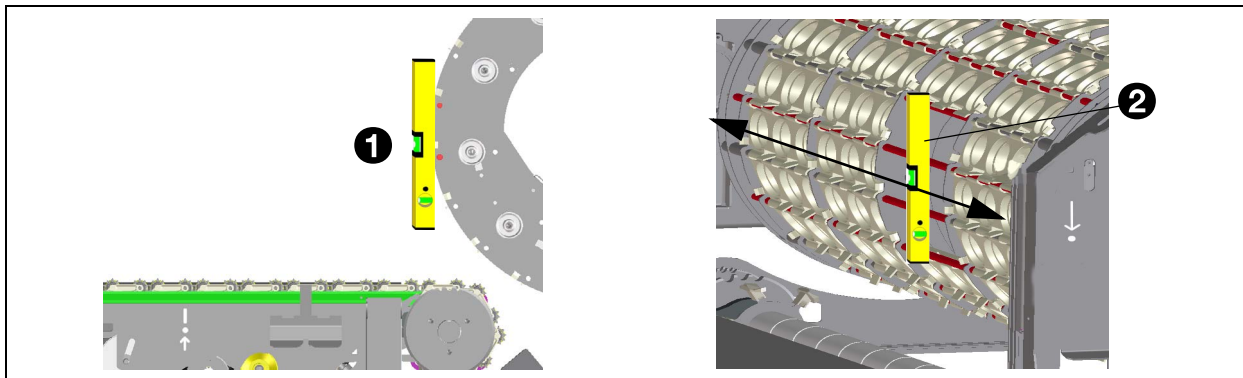


Fig. 7 - Check with level

- 1 Remove two rows of carrier sets. See paragraph 7.3.1, 'Removal of carrier sets' on page 73
- 2 Put the MultiDrum™ level (1).
- 3 Check on the lefthand side whether the shafts of the MultiDrum (2) are level.
- 4 Adjust the shafts if they are twisted, see Chapter 7.3 for the adjustment of the MultiDrum™.

### 3.5 Synchronisation Loader™ - Transfer™

The synchronisation between the Transfer™ and the Loader™ ensures that the eggs are placed correctly **between** the rolls on the laying-unit of the infeed-unit. If the eggs are laid **on** the rollers then the synchronisation is not adjusted correctly.



**NOTE**

***The synchronisation between the infeed unit and the Loader™ must be readjusted two weeks after the infeed unit is brought into use.***

For the adjustment of the synchronization of the Loader™ and Transfer™ see paragraph 9.9, 'Synchronisation Loader™ - Transfer-unit™' on page 112.



### 3.6 Loadcell-unit

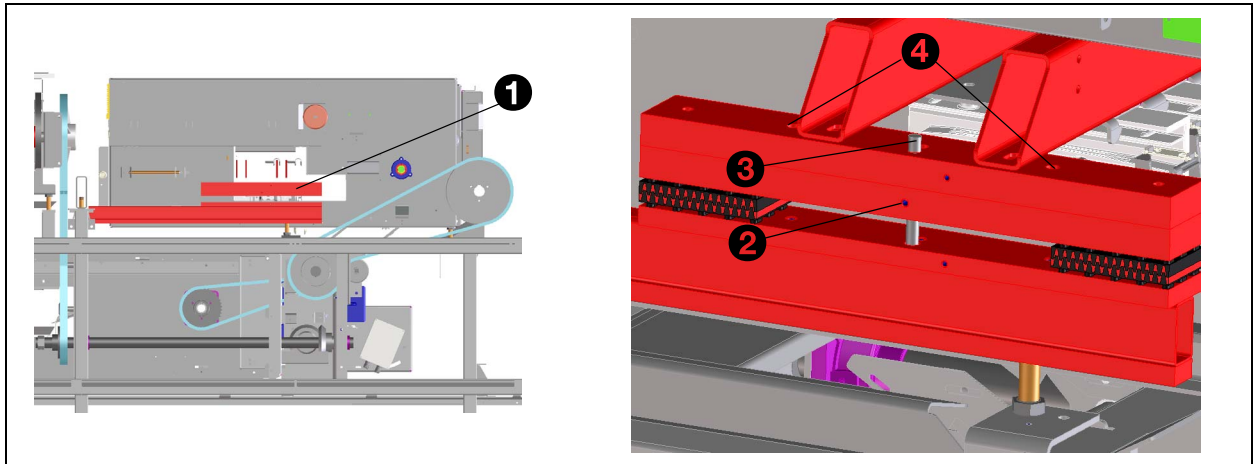


Fig. 8 - Loadcell-bar



**NOTE**

**Before you can use the loadcell unit the transport bolts (2x) (4) on both sides of the loadcell unit must be removed from the lowest beam. As a result, the loadcell bar (1) will be released from the frame.**

- 1 After the installation, loosen the Allen bolt (2).
- 2 Pull the centring/safety pin (3)  $\pm 10$  mm upwards to release the weighing unit. The pin must remain in about 4 mm of the upper part of the hole to make sure that the weighing unit can never be carried into the machine.
- 3 Tighten the Allen bolt securely.
- 4 Repeat this procedure for the other side of the loadcell-unit.



**NOTE**

**The centring/safety pin may never come out of the top of the hole, since the complete weighing unit could be carried into the machine.**



## 4 Adjustment, general

### 4.1 Drive chain tensioning without spring tensioner

- 1 Check the tension at periodic intervals in accordance with the maintenance instructions.
- 2 Determine the drive-chain' tension as follows:
- 3 Turn the drive section by hand, and locate the point where the chain is the most taut. Tension the chain so that the chain at this point is just free of tension.
- 4 When doing so, take note of the function and length of the drive chain.

### 4.2 Distance between the sensors and indicators

**NOTE**

*Electronic sensors are fitted to the machine at various locations.*

**NOTE**

*The sensors (1) are equipped with an LED which illuminates when the sensor is activated.*

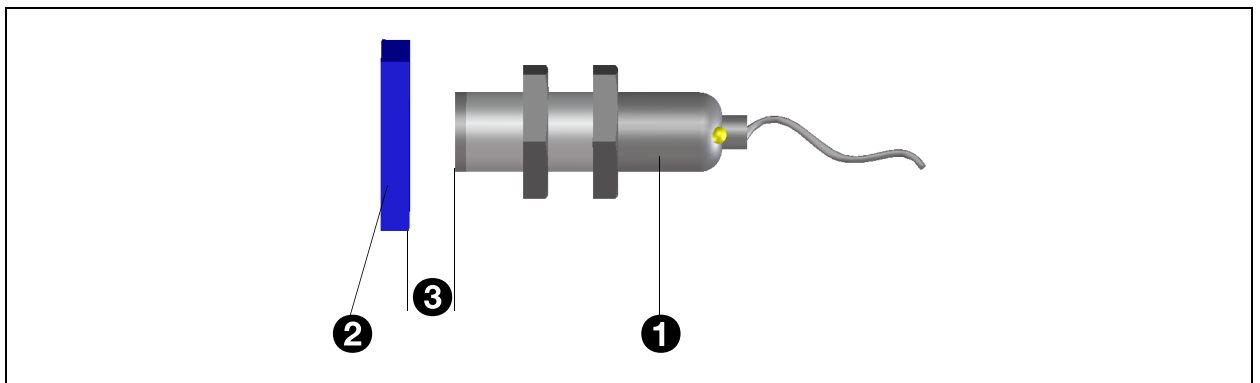


Fig. 9 - Distance indicator

- 1 - Sensor
- 2 - Indicator
- 3 - Adjustment = 2 mm

- 1 Adjust the distance between a sensor and the corresponding indicator to 2 mm for a steel indicator and 1 mm for a stainless steel indicator.

## 4.3 Locations of the circuit boards

fig. 10 lists the positions and names of the circuit boards. See manual 4 for an overview of the positions of the DIP switches.

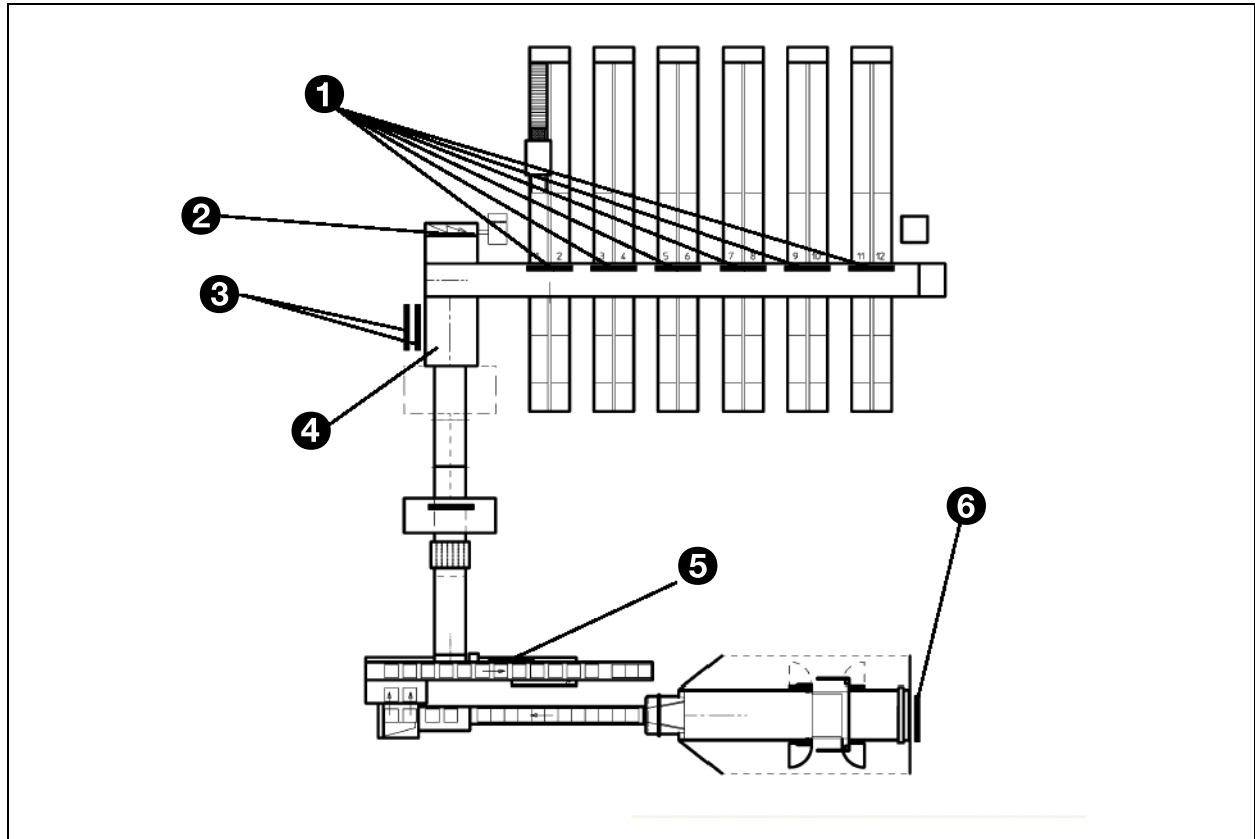


Fig. 10 - Location of the circuit boards

- 1 - Lane (CAN 1)
- 2 - Inkjet (CAN 0)
- 3 - Weigher (CAN 0)
- 4 - Synchro (CAN 0)
- 5 - Loader (CAN 0)
- 6 - CDS (CAN 0)

## 4.4 Exchanging frequency controllers and motors

The frequency controllers for the main and buffer conveyors motor are installed in the main electrical cabinet. The other controllers are installed in the electrical cabinet for the relevant part of the machine.

### 4.4.1 Exchanging frequency controllers.

- 1 Connect the new frequency controllers in accordance with the circuit diagrams.
- 2 Then set the machine speed to the maximum setting.
- 3 Start the machine and check the direction of the relevant motor's rotation.
- 4 Adjust the frequency controller.
- 5 See electrical diagram for correct settings.

### 4.4.2 Exchanging motors.



---

***DANGER***

***ALWAYS CARRY OUT THE WORK IN ACCORDANCE WITH THESE INSTRUCTIONS. AN INCORRECTLY-INSTALLED MOTOR CAN DAMAGE THE MACHINE.***

---

- 1 Make a note of the direction of rotation of the old motor.
- 2 Connect the new motor's wiring, but do not mount the motor yet.
- 3 Start the motor and check the direction of rotation.
- 4 Exchange to wires if the motor rotates in the wrong direction.
- 5 Check the direction of rotation again.
- 6 Mount the motor **ONLY ONCE THESE CHECKS HAVE BEEN COMPLETED.**

## **4.5 Service pages in the ServerPC**

The 'WinMonitor' program, will need to be started on the ServerPC to carry out specific tests on the machine. Most tests are intended solely for MOBA service technicians. The tests of value to you are indicated in this chapter and included in this manual.

### **4.5.1 Starting the 'WinMonitor' program**

- 1 Stop the machine with the normal stop procedure.
- 2 Minimize the MMI program on the ServerPC.
- 3 Start the WinMonitor program.
- 4 Program is starting in a DOS-box.

### **4.5.2 Stopping the 'WinMonitor' program**

- 1 In the DOS-box, select "Exit to DOS".
- 2 The DOS-box disappears.
- 3 Maximize the MMI program.

## 4.5.3 Description of the main menu

Table 3 - Main menu options

| The main menu offers the following options: |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File menu                                   | Only for MOBA technicians                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Clientshift data                            | Only for MOBA technicians                                                                                                                                                                                                                                                                                                                                                                                                                  |
| I/O control                                 | <p>The I/O control menu appears.</p> <p>REMEMBER THE SAFETY INSTRUCTIONS, and select the following options.</p> <ul style="list-style-type: none"> <li>- Set a speed</li> <li>- Start the machine</li> <li>- Start the I/O timing</li> </ul> <p>A help menu will be displayed on the screen.</p> <p>Enter "alt S", and press return.</p> <p>The I/O timing will now be displayed.</p> <p>Use "Esc" to go back to the previous screens.</p> |
| Scan other PCs                              | Only for MOBA technicians                                                                                                                                                                                                                                                                                                                                                                                                                  |
| View screendump                             | Only for MOBA technicians                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Moba link                                   | Only for MOBA technicians                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Canproc monitor                             | Only for MOBA technicians                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Enhanced mode                               | Only for MOBA technicians                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Message file                                | A list of error messages is displayed.                                                                                                                                                                                                                                                                                                                                                                                                     |
| Weighing                                    | <p>A check screen with an "OK" bar is displayed.</p> <p>Press return to display the weighing menu.</p>                                                                                                                                                                                                                                                                                                                                     |
| Exit to DOS                                 | Go back to the DOS prompt                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                            |



## 5 Adjustment of OMNIA XF infeed unit

### 5.1 Turning the infeed unit by hand

**NOTE**

*The machine must be turned by hand to check a number of settings.*

**NOTE**

*All drive roller chains must be tensioned so that the return section is not tensioned unless otherwise stated.*

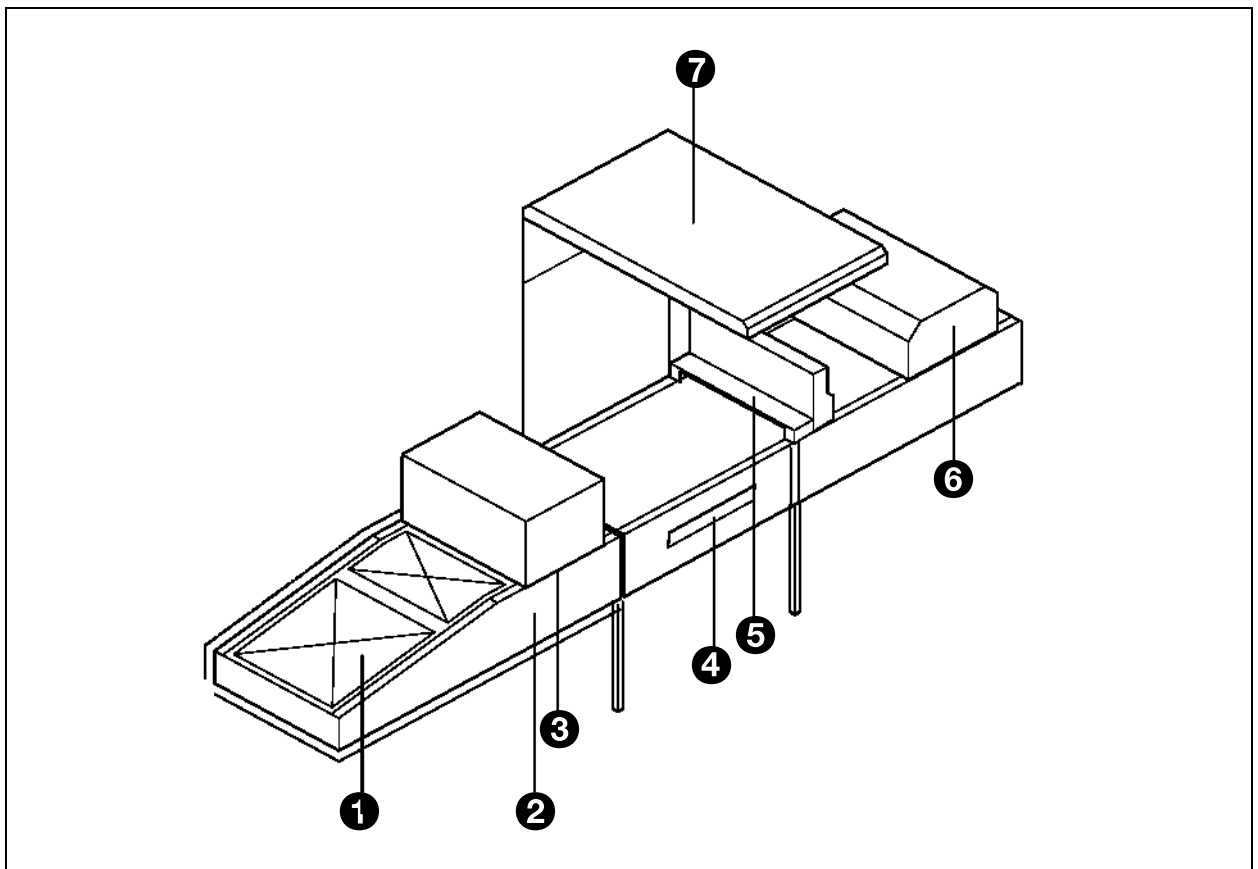


Fig. 11 - Overview of infeed unit

- 1 - Placement unit
- 2 - Roller cleaner unit
- 3 - Turning unit
- 4 - Candling lights
- 5 - Semi-automatic candling
- 6 - Crack detector
- 7 - Candling booth

## 5.2 Turning section (Grippers)

This chapter discusses the adjustment of the gripper turning section.

### 5.2.1 Adjustment of the turning unit

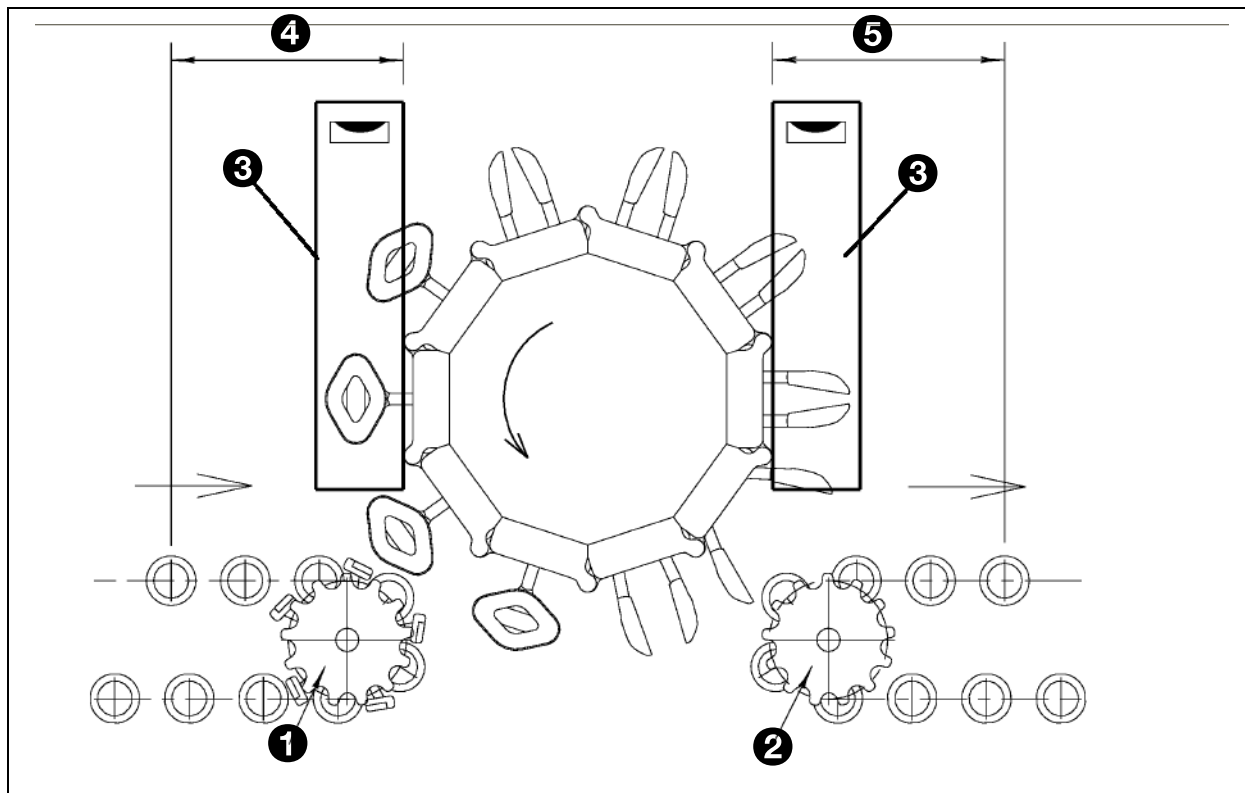


Fig. 12 - Turning section grippers

- 1 - Chain wheel
- 2 - Chain wheel
- 3 - Level
- 4 - Adjustment broad roller - turning unit = 152 mm
- 5 - Adjustment small roller - turning unit = 150-152 mm

#### Adjustment of the turning unit:

- 1 Turn the machine by hand until the turning unit is level (3) as shown in fig. 12.
- 2 Move the adjustable chain-wheels (1) on the return shaft of the broad rollers to set dimension (4), see fig. 12.
- 3 Move the adjustable chain-wheels (2) on the return shaft of the small rollers to set dimension (5), see fig. 12. (coarse adjustment)
- 4 Move the same chain-wheels (2) to adjust the timing of the small rollers such that the container just contacts the large adjustment eggs as it turns away, see fig. 13. (fine adjustment)



#### **NOTE**

***Make sure that the position of the turning unit does not change during this adjustment.***

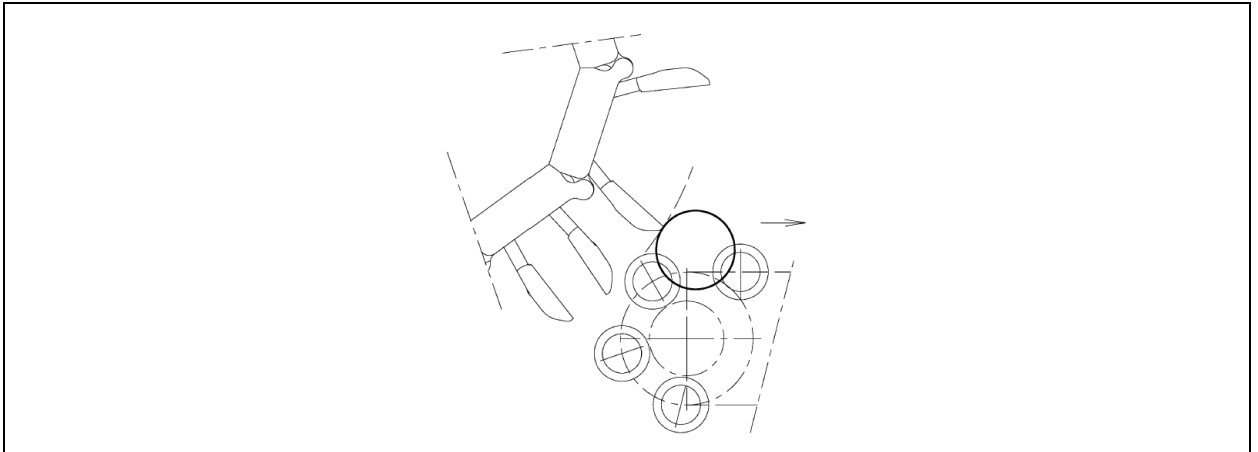


Fig. 13 - Fine adjustment of turning section grippers

## 5.2.2 Adjustment of the closing time

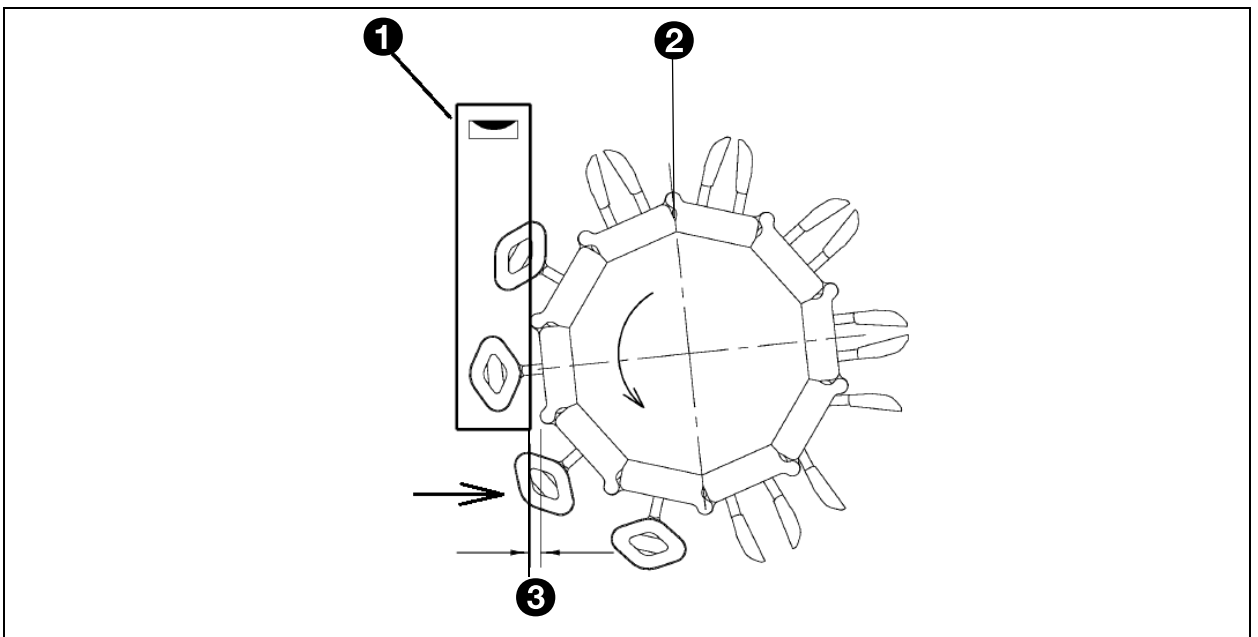


Fig. 14 - Adjustment of the closing time

- 1 - Level
- 2 - Turning unit
- 3 - Distance level - turning unit = 7 mm

### Adjustment of the closing time:

- 1 Turn the machine by hand until the turning unit is located at the position shown in fig. 14.  
At this point the grippers indicated in the figure must just start to close.
- 2 Adjust this setting by moving the inner tube relative to the drive ring.



## 5.3 Turning section, vanes with chutes

This chapter discusses the setting of the turning section, vanes with chutes.

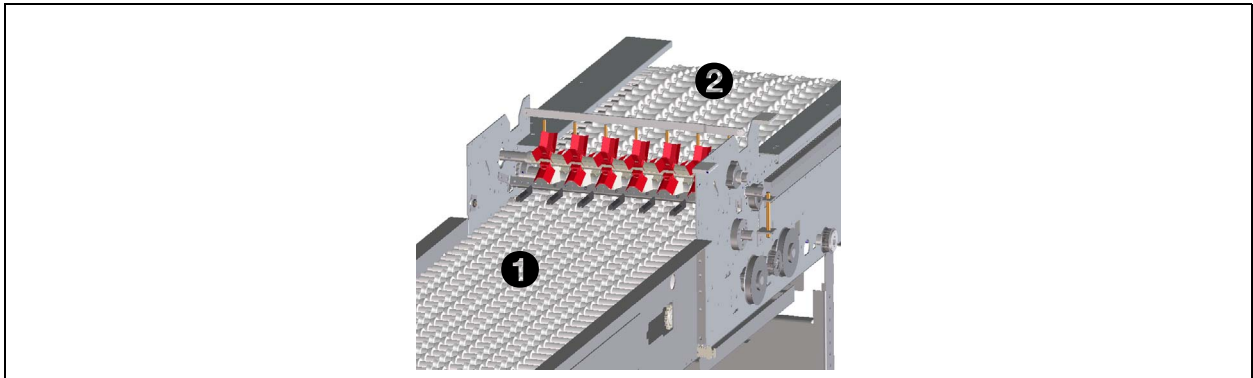


Fig. 15 - Turning unit

### 5.3.1 Adjustment of the turning pins

The turning pins guide the eggs from the upper to the lower rollers.

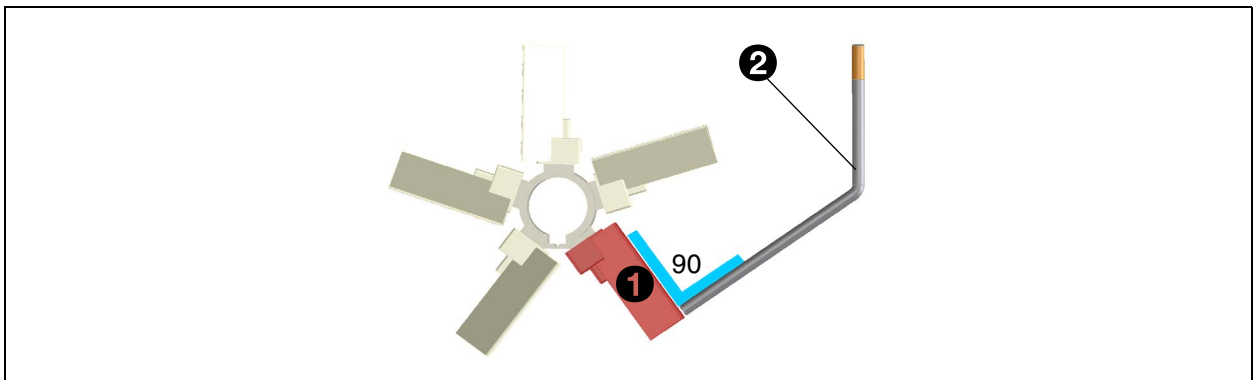


Fig. 16 - Scoop at right angles to turning pin

Turn the machine by hand until a scoop (1) of the turning unit is positioned at a right angle to the turning pin (2). See fig. 16.

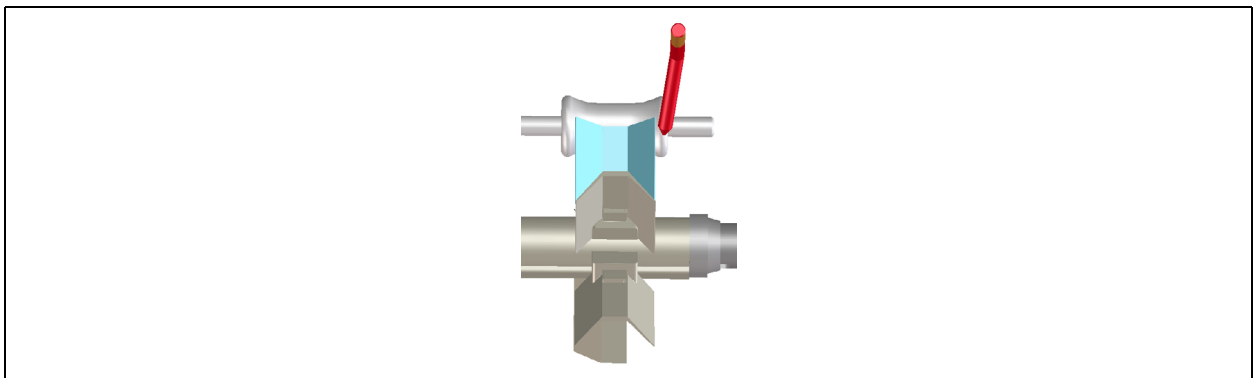
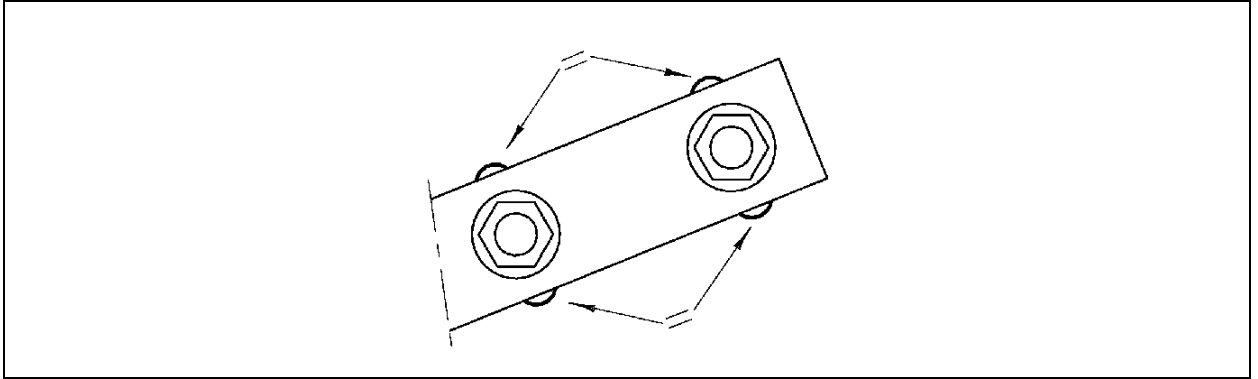


Fig. 17 - Tolerance between scoop and turning pin

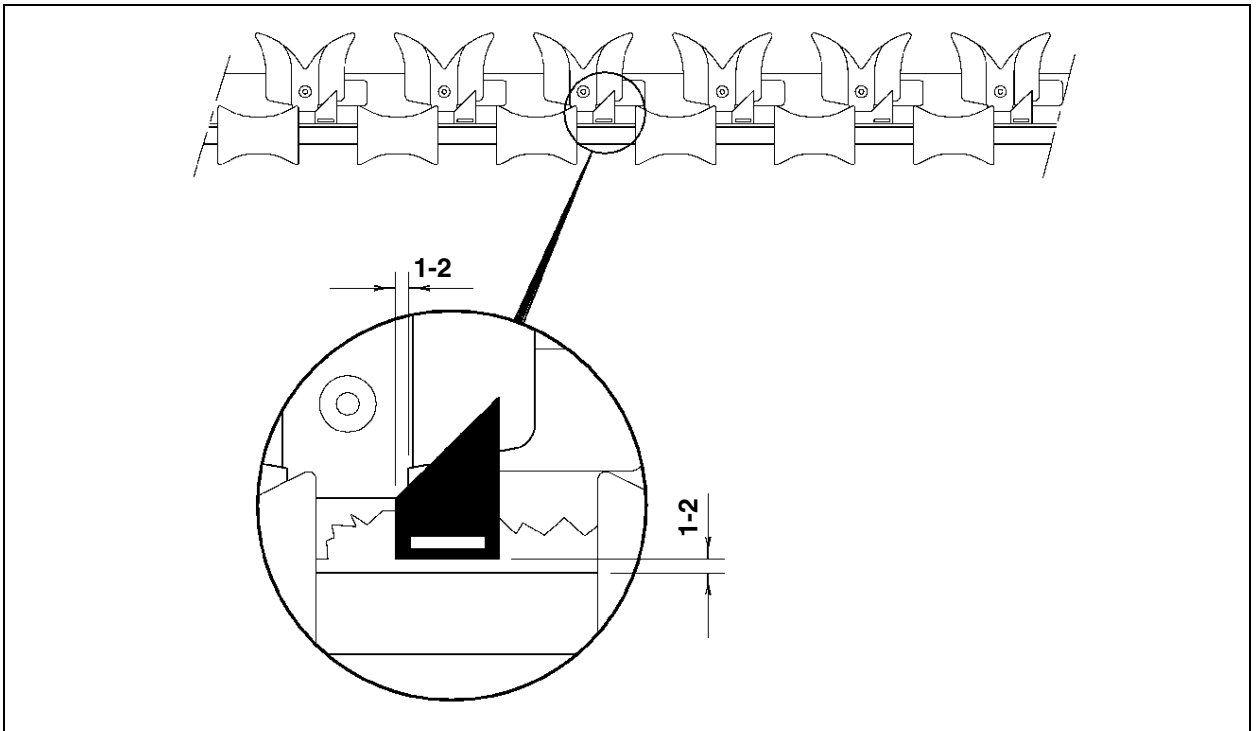
1. Mount the turning pins so that they are all positioned inside the turning unit's turning circle. See fig. 17

### 5.3.2 Adjustment of the cups and the triangular rubbers

1. Remove the turning unit's cover
2. The cup strip must be mounted at a specific angle. Check this position using fig. 18, which shows the correct position.
3. Mount the cups and the rubbers as shown in fig. 19. make sure that the cups do not touch the rollers.

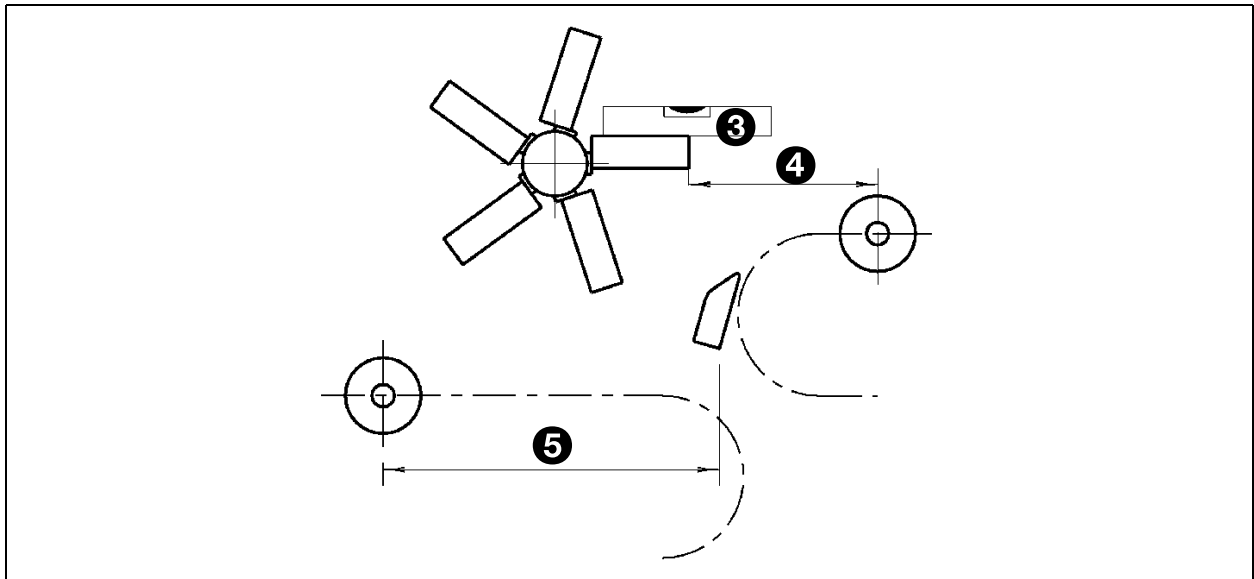


*Fig. 18 - Angle of cups strip*

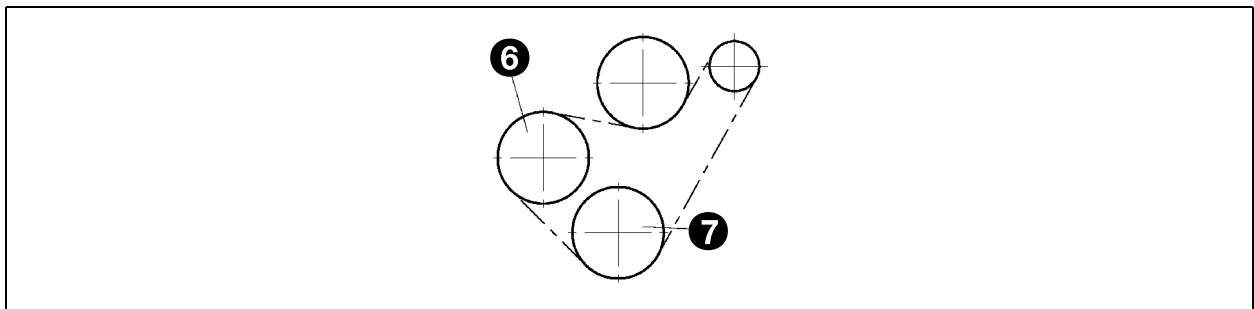


*Fig. 19 - Fitting the cups*

### 5.3.3 Adjustment of the turning unit



*Fig. 20 - Levelling the turning unit*



*Fig. 21 - Adjustment of the turning unit*

1. Remove the turning unit's cover
2. Set 1 of the vanes of the turning unit level (3) and in the direction of the first section of the infeed unit (with the broad rollers). See fig. 20.
3. Check the distance (4) from the tip of the vane to the centre of the first broad roller. This must be 80 mm. Adjust this dimension by moving chain-wheel (6) (see fig. 21) relative to the shaft.
4. Check the distance (5) from the bottom of the chute to the centre of the first small roller of the infeed unit. This must be 130 mm. Adjust this dimension by moving chain-wheel (7) (see fig. 21) relative to the shaft.



## 5.4 Semi-automatic candling

This section discusses the adjustment of the semi-automatic candling. The correct operation of this system is dependent on the adjustment of the switching disk on the Transfer unit in the frame.

### 5.4.1 Adjustment of the position of the detection unit

- 1 Start the machine and press all the inspection-buttons.
- 2 Position the unit horizontally so that the bolt is located at the middle of the slot. Fasten the unit.
- 3 Position the unit vertically such that the last button is just touched by the detection arm.
- 4 Stop the machine.
- 5 Turn the machine by hand until the "HIRESZERO" red LED on the main timing circuit board (MPS97) lights. See fig. 140
- 6 Position the unit horizontally such that point (10) of the detection arm is located on the middle of the button. See fig. 22.
- 7 Check that the unit is in line with the buttons.
- 8 Then turn the machine further by hand until the detection arms just touch the white buttons that have not been pushed in, see fig. 23.
- 9 Now position the sensor (8) so that it is just activated, see fig. 24. (Take note of the direction of rotation).
- 10 Press the black inspection buttons in the row contacted by the detection arms.
- 11 Carefully turn the machine further by hand until the last detection arm just falls free of the white button, see fig. 25.
- 12 Now position the sensor (9) so that it is just activated, see fig. 24. (Take note of the direction of rotation).

**NOTE**

**CAUTION:** It is not important how the two sensors are mounted relative to each other

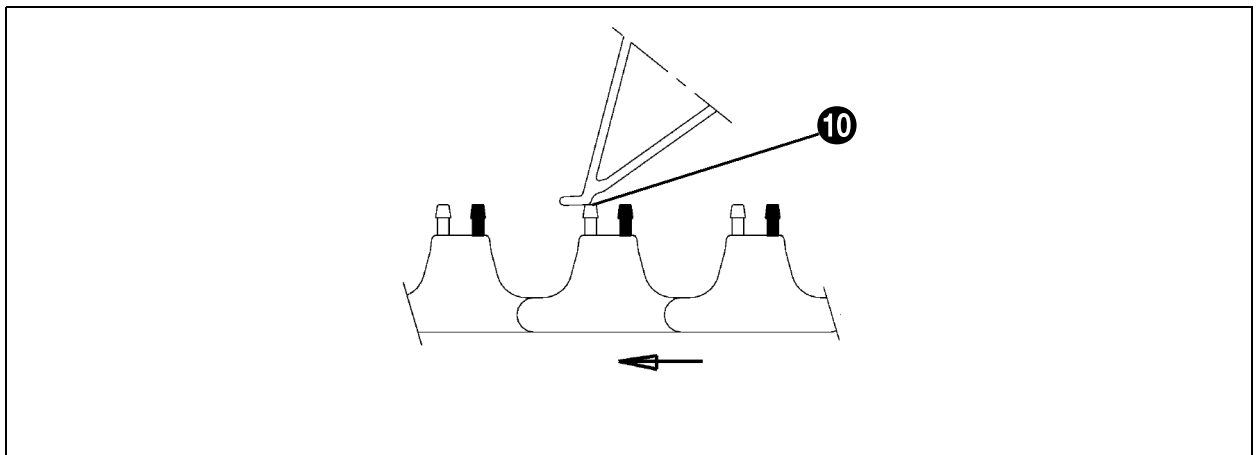
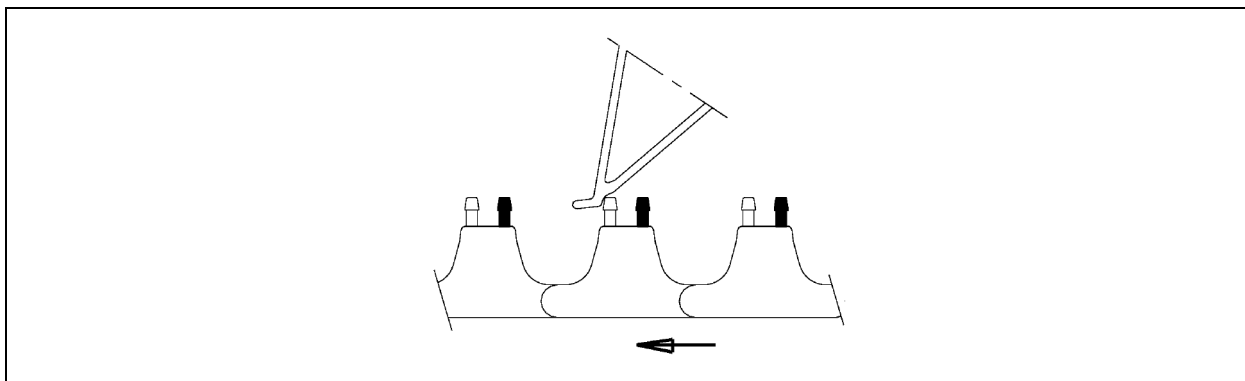
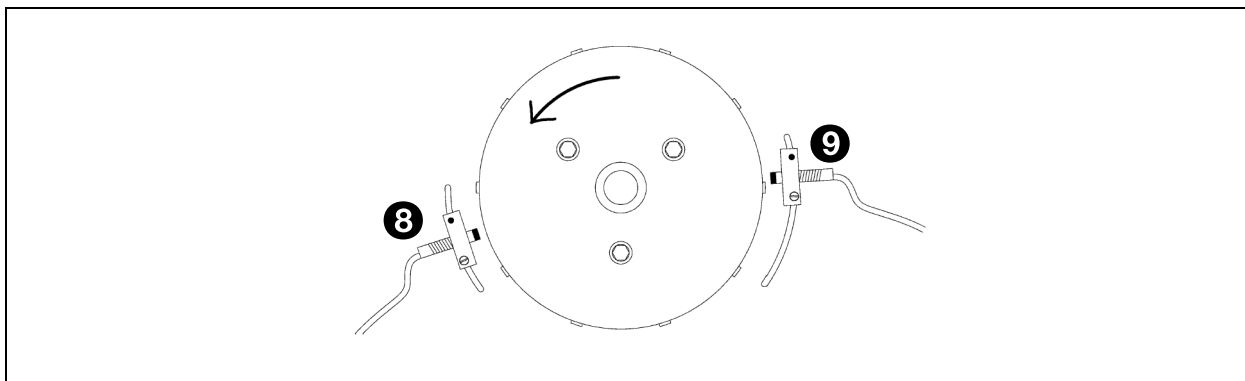


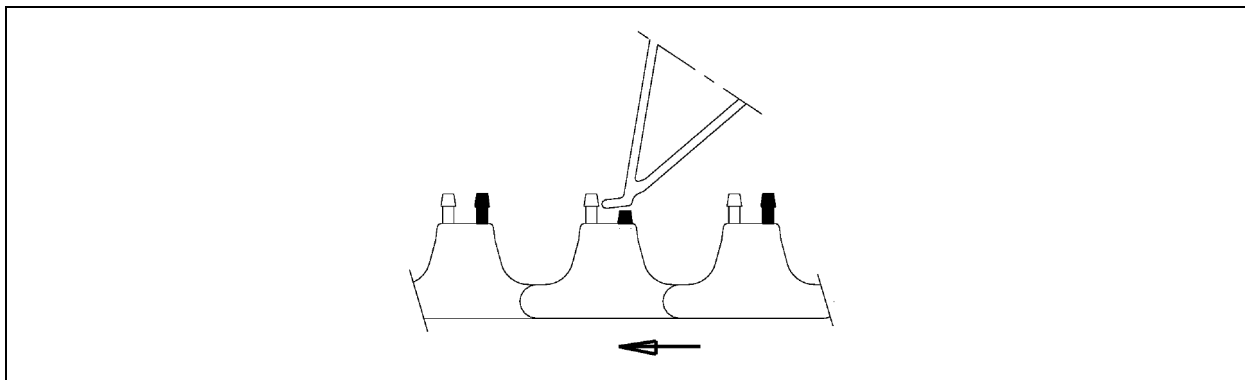
Fig. 22 - Adjustment of the position of the detection unit



*Fig. 23 - Turning by hand 1*



*Fig. 24 - Adjustment of the sensor*



*Fig. 25 - Turning by hand 2*

## 5.5 Roller cleaner unit

### 5.5.1 Adjustment of the distance between the rollers and the brushes

1. Move the guide above the broad rollers and the loop wheel above the small rollers to adjust the distance between the rollers and the brushes, see fig. 26. The brushes must just touch the middle of the rollers.

**NOTE**

*The arrow (1) indicates the direction of the eggs' travel.*

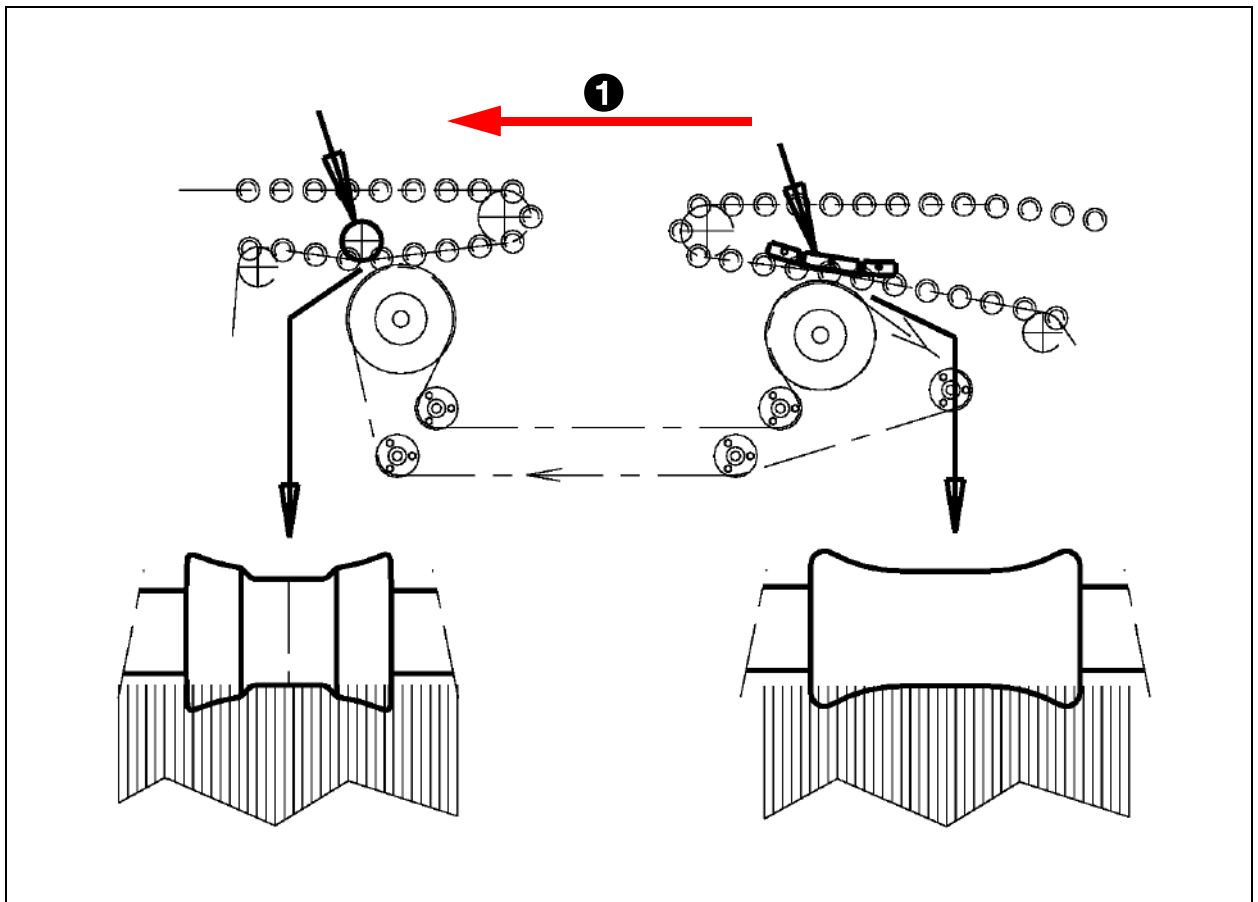


Fig. 26 - Roller cleaner unit



## 6 Adjustment sequence for OMNIA FT infeed unit

Adjustments of the machine modules have an influence on one another. The machine must be adjusted in the order that is shown below, so that its operations are properly synchronised.



### **NOTE**

*The illustrations may be different from your version of the machine; this has no influence on the settings and adjustments of your machine.*

The left and right side are determined on the basis of the direction of the movement of eggs. In fig. 27 the direction of movement of the eggs is indicated with the arrow (7).

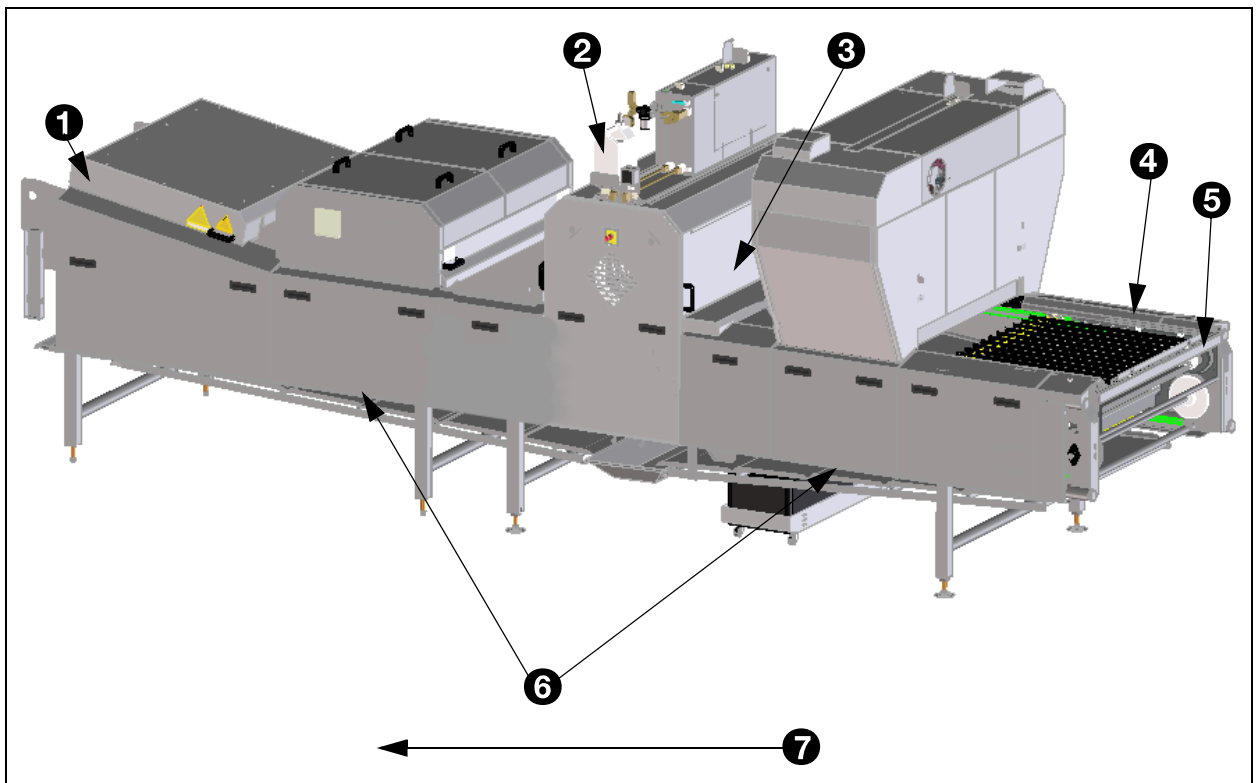


Fig. 27 - Overview of infeed unit



### **NOTE**

*Before you adjust the machine/module always check the tension of the chains first*

1. Adjustment of the transfer point from the infeed unit to the loadcell unit
2. Adjustment of the MultiDrum take-away side™
3. Adjustment of the MultiDrum infeed side™
4. Adjustment of the placement unit.
5. Adjustment of transfer-point from Loader™ to infeed unit, see manual for OMNIA grader
6. Adjustment of chain guides.
7. Direction of egg travel



## 6.1 OMNIA FT Transfer™ accessory tools

The following accessory tools are supplied with the Transfer™:

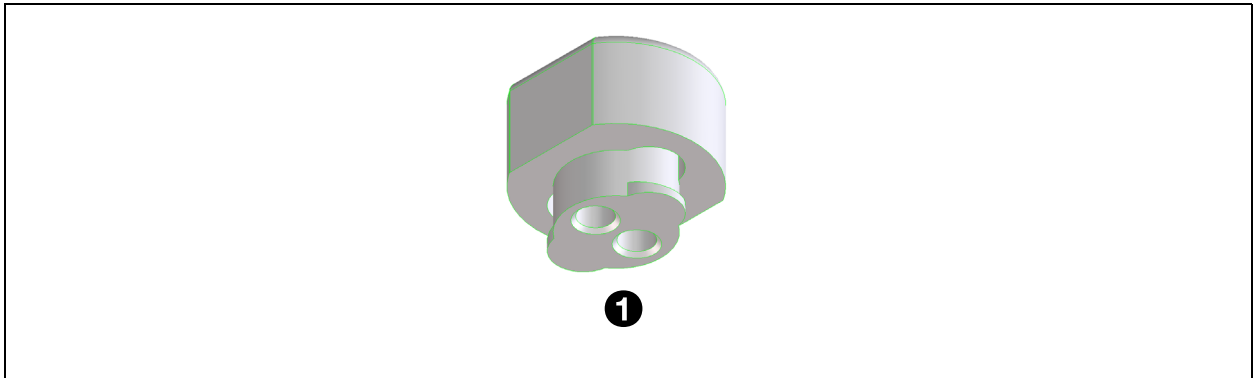


Fig. 28 - Loadcell calibration

**Loadcell carrier calibration, see fig. 28.**

1 MOBA calibration weight, plastic, 63 gram. Article number: 80206980.

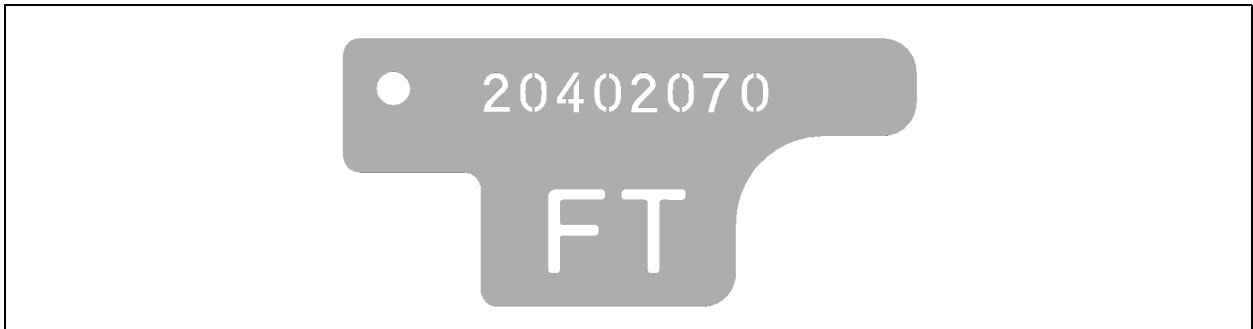


Fig. 29 - Adjusting jig

**Adjustment gauge, see fig. 29.**



Adjusting jig for connecting infeed unit to the transfer unit. Article number: 20402070.

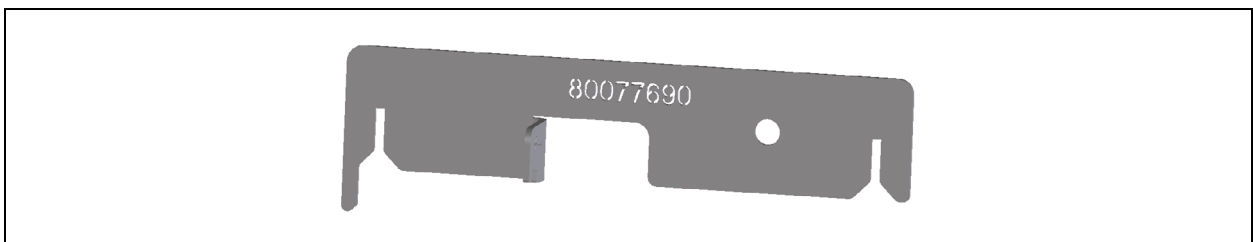


Fig. 30 - Adjustment template for the MultiDrum™

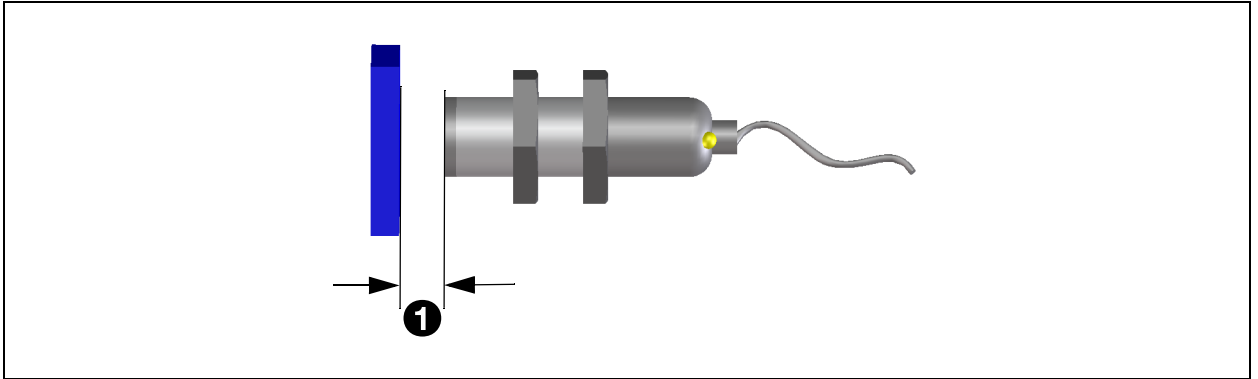
**Adjustment template for the MultiDrum™, see fig. 30.**



Adjusting jig for the unlocking time in the MultiDrum™. Article number: 80077690.

## 6.2 Sensor adjustment

The sensors that are mounted in this module are adjusted to the standard values. The standard distance between the sensor and the adjustment-disk, strip or small plate amounts to 1 mm.



*Fig. 31 - Distance of the sensor*

1. After a sensor has been replaced, use a feeler gauge to set the distance (1) to 1 mm. See fig. 31.

## 6.3 Adjustment of chain-tension

Chain-tensioners are found at various locations in the infeed unit. Follow the procedures below to apply tension to the chains.

**NOTE**

*Check the tension of the chains one month after the infeed unit has been brought into use. Subsequently in accordance with the maintenance plan in Chapter 14.*

### 6.3.1 Chain tension double roll standard infeed unit

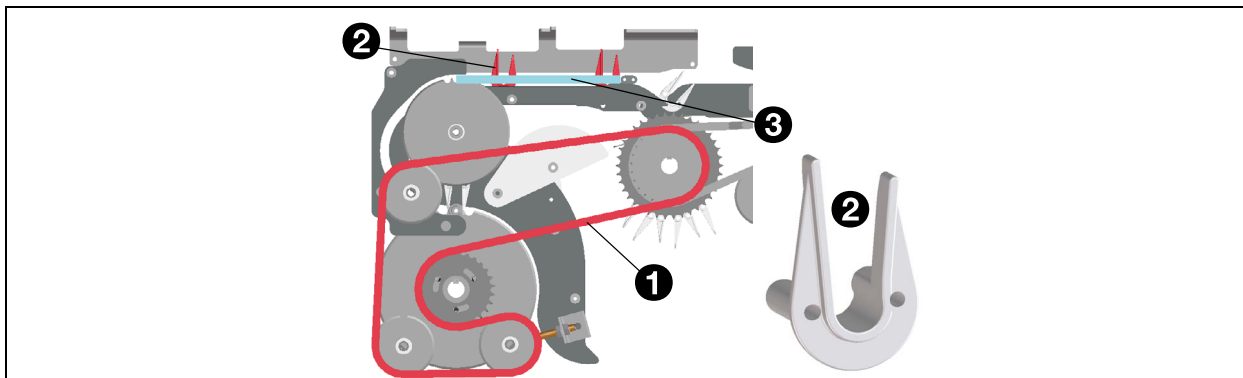


Fig. 32 - Adjustment of the chain tension

There are two chains in the placement unit of the infeed unit, see, fig. 32:

- The drive chain (1) for the system with the U-holders (2).
- The transport-chain (3) on which the U-holders are mounted.

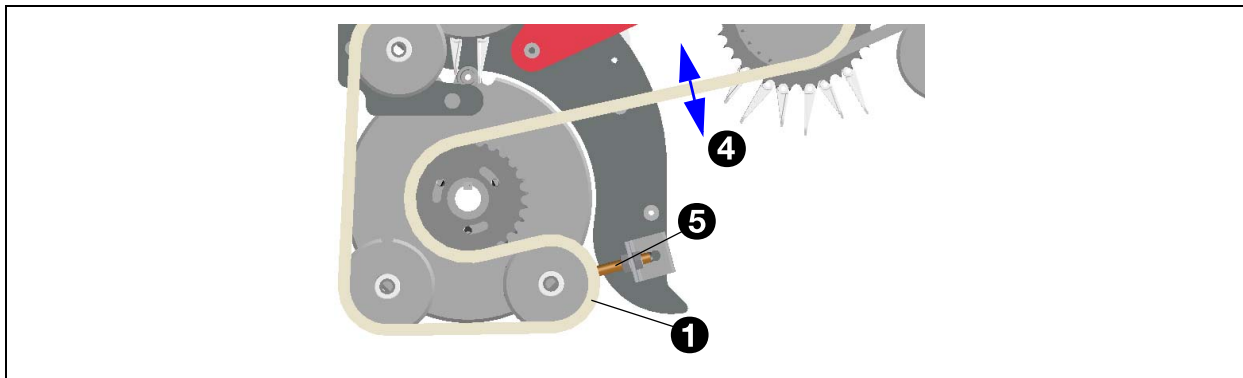
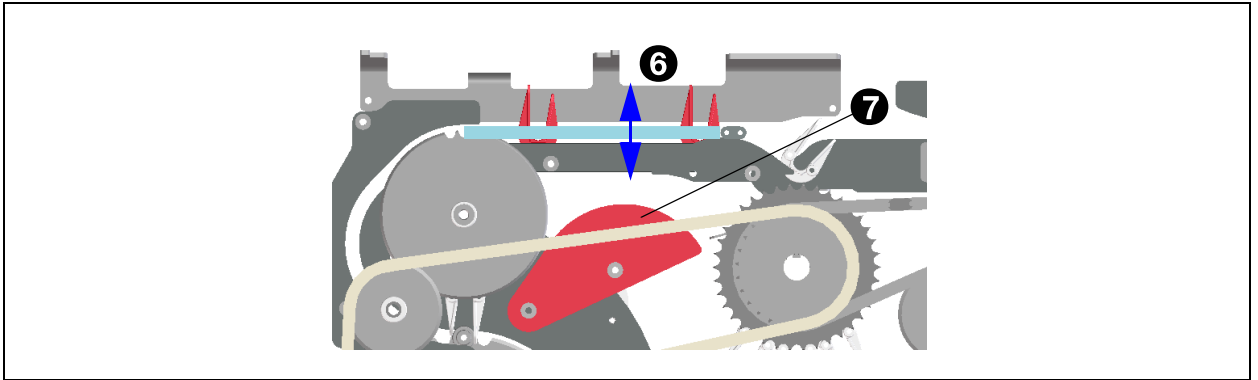


Fig. 33 - Drive-chain

**For adjustment, follow the procedure below.**

1. The tolerance (4) of the drive-chain (1) can amount to a maximum of 10 mm. See fig. 33.
2. Adjust the play with the chain-tensioner (5).

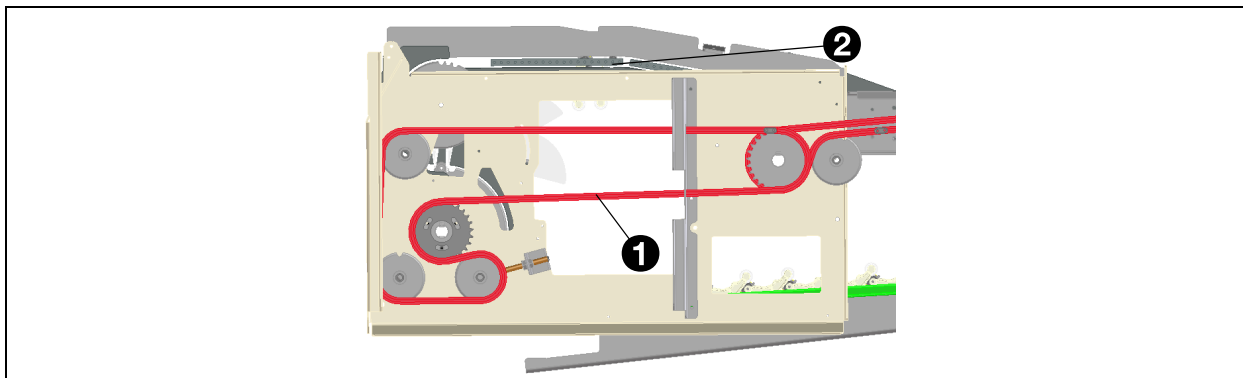


*Fig. 34 - Transport-chain*

3. The play (6) in the transport-chain can amount to a maximum of 5 mm. See fig. 34
4. Adjust the play with the chain-tensioner (7).

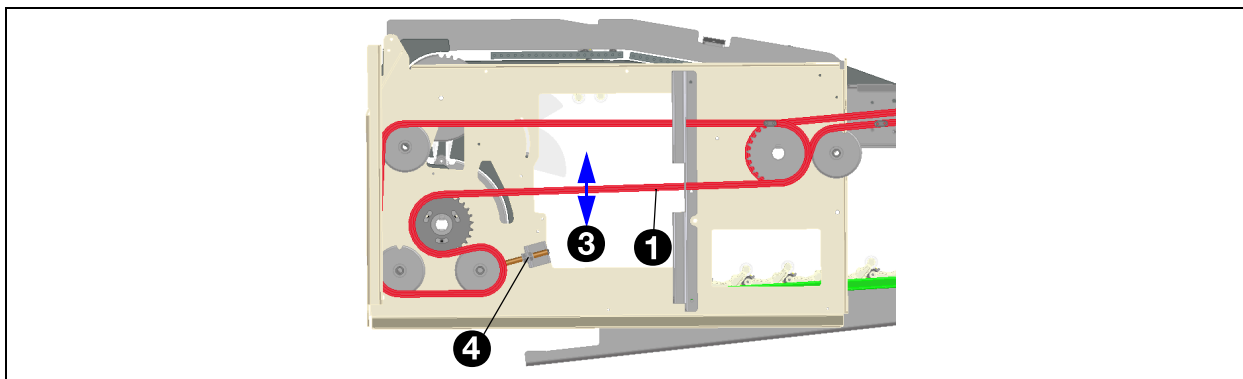
### 6.3.2 Chain tension double roll Combi infeed unit OMNIA 330.

There are two chains in the placement unit of the Combi infeed unit, see, fig. 35:



*Fig. 35 - Adjustment of the chain tension*

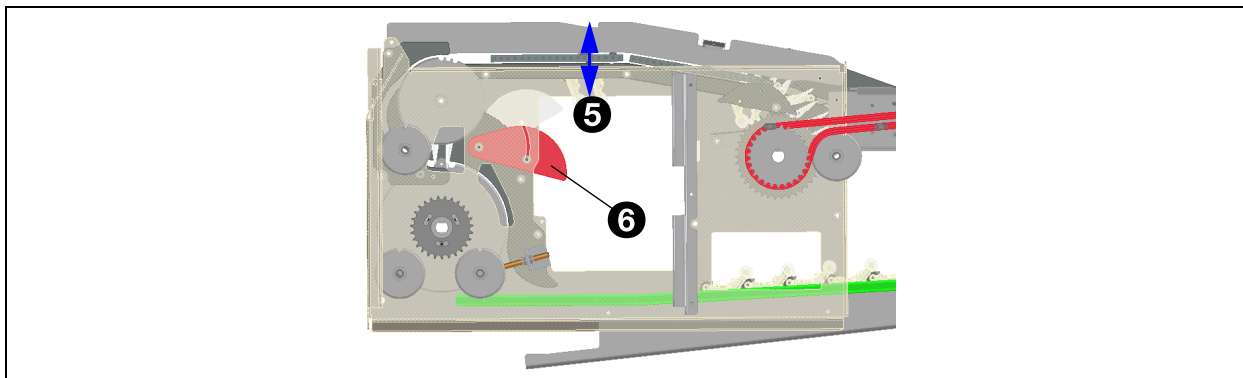
- The drive chain (1) for the system with the U-holders.
- The transport-chain (2) on which the U-holders are mounted.



*Fig. 36 - Drive chain OMNIA 330*

**For adjustment, follow the procedure below.**

1. The play (3) in the drive-chain (1) can amount to a maximum of 10 mm. See fig. 36.
2. Adjust the play with the chain-tensioner (4).



*Fig. 37 - Transport-chain*

3. The play (5) in the transport-chain can amount to a maximum of 5 mm. See fig. 37.
4. Adjust the play with the chain-tensioner (6).

### 6.3.3 Chain tension double roll Combi infeed unit OMNIA 170.

There is one chain in the placement unit of the Combi infeed unit, see, fig. 38:

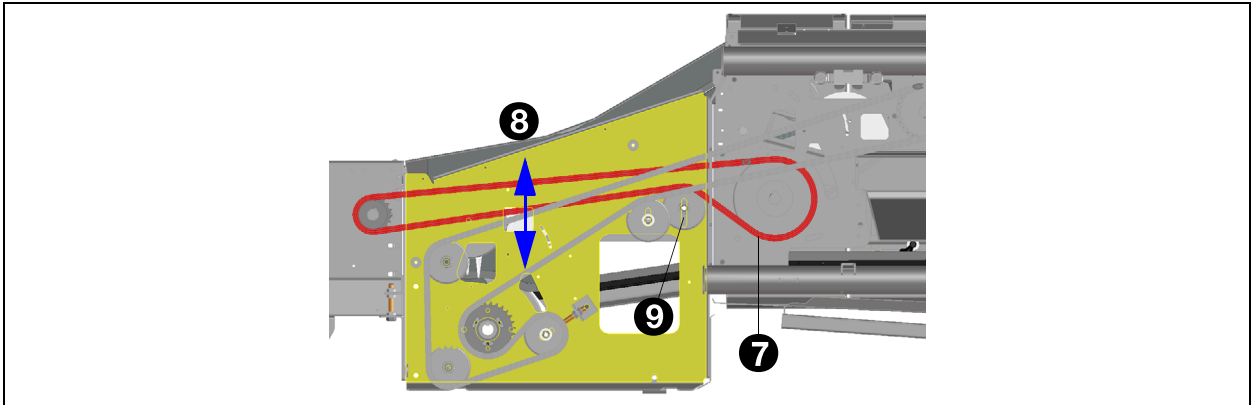


Fig. 38 - Drive chain OMNIA 170

**For adjustment, follow the procedure below:**

1. The play (8) in the drive-chain (7) can amount to a maximum of 10 mm. See fig. 38.
2. Adjust the play with the chain-tensioner (9).

### 6.3.4 Chain-tension MultiDrum™

To prevent damage to the MultiDrum™ as a result of a crash, a protector is mounted on the MultiDrum™. The tension of the drive chains is checked by two microswitches. The microswitches (2) are mounted on both sides of the MultiDrum™. See fig. 39. The microswitches (2) are included in the emergency-stop circuit. See fig. 39 and fig. 40.

When the machine is running the chain will judder. The MultiDrum™ protection is controlled by a time relay which needs to be setup for x seconds. When the delay time is expired the machine generates an emergency stop.

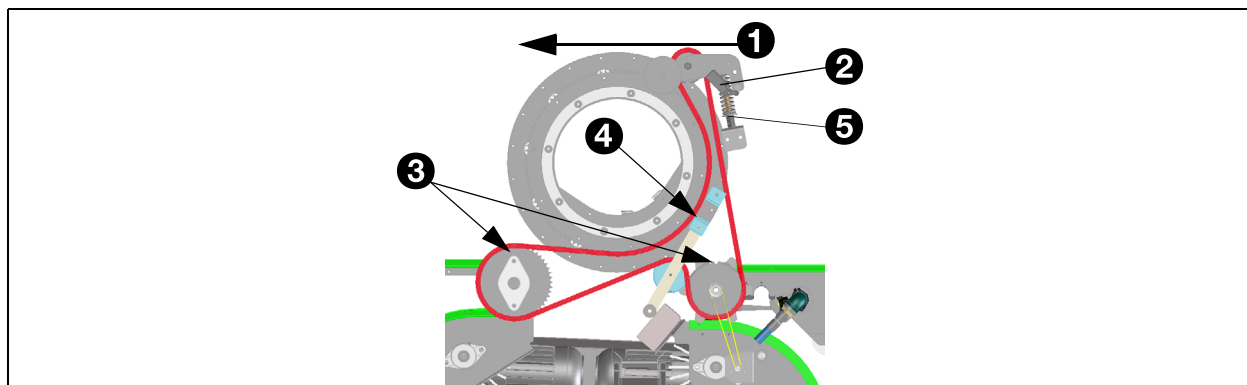


Fig. 39 - Drive chain MultiDrum™

The arrow (1) in the illustration shows the direction of the egg travel.

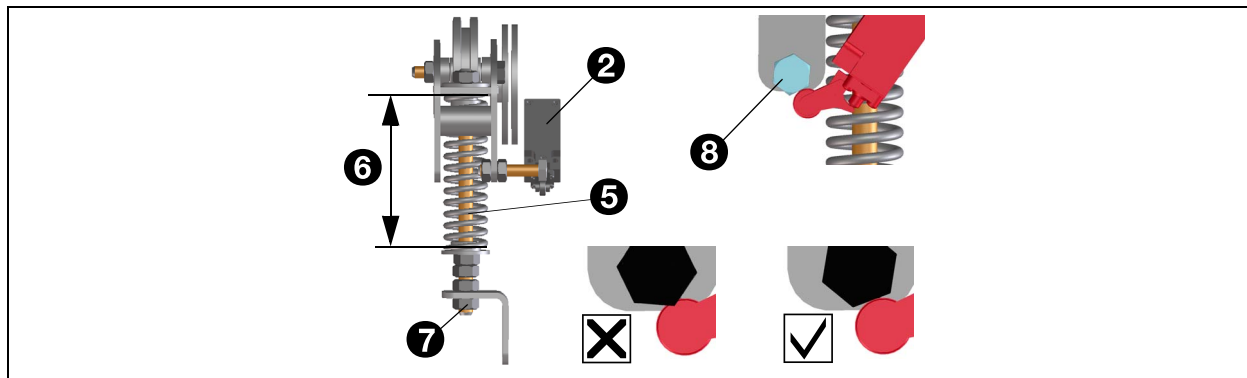


Fig. 40 - Safeguards

#### The security is adjusted as follows:

1. Loosen the bolts of the two chain-wheels (3). See fig. 39 and fig. 41.
2. Adjust the length of the spring (6): See fig. 40.
  - a. Default adjustment:
    - Omnia 85,125,170 = 135 mm.
    - Omnia 250, 330, 500 = 110 mm.
  - b. When the default adjustment is not correct and the machine stops too often with emergency stop you can set it to the minimum length:
    - Omnia 85,125,170 = 125 mm.
    - Omnia 250, 330, 500 = 100 mm.



#### **NOTE**

**Without load, the length of the spring is 160 mm.**

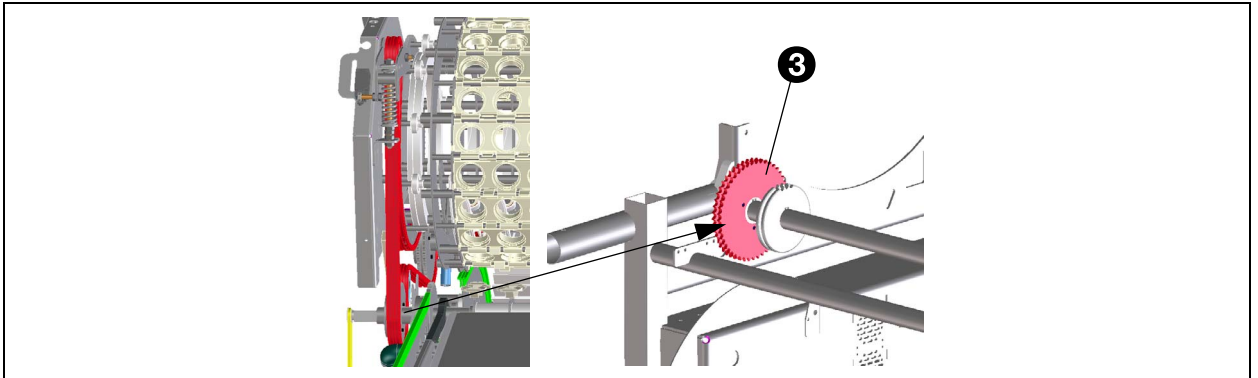


Fig. 41 - Disconnecting chain-wheels

3. Tighten the bolts of the two chain-wheels (3). See fig. 41
4. Tension the chain (4) using the chain tensioner (5) until the safeguard is pulled to the microswitch. See fig. 39 and fig. 40.

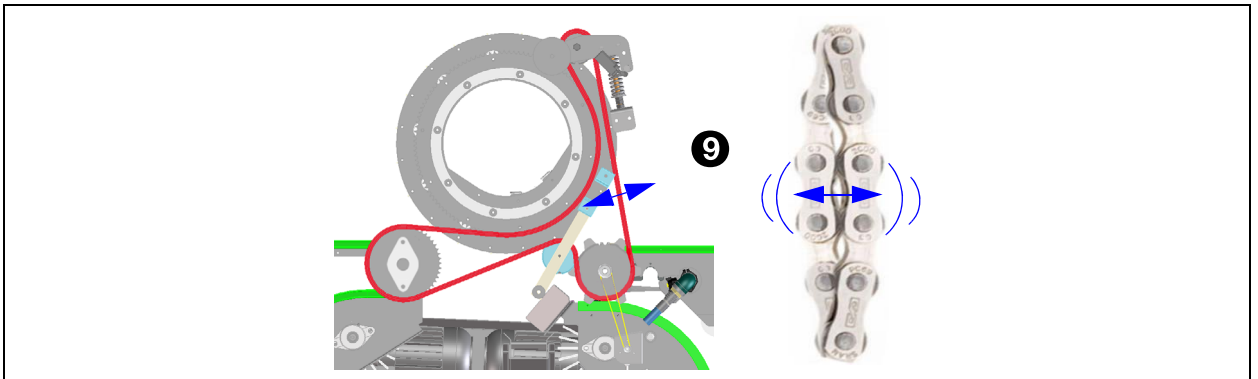


Fig. 42 - Chain deflection

5. Using the chain tensioner, set the deflection (9) of the chain to 10 mm See fig. 42.
6. Use the nuts (7) to adjust the safeguard so that bolt (8) (mounted on the arm) just touches the microswitch's roller. See fig. 40.



**NOTE**

**Check whether the bolt (8) touches the flat part of the microswitch's roller.**

7. Carry out the same adjustments on the other side of the MultiDrum™.



After tensioning the chain readjust the infeed and take-way sides of the MultiDrum™. See paragraph 7.3.

## 6.3.5 Chain tension of Spinner chain drive

The spinner chain's drive is located on the righthand side of the infeed unit. The OMNIA 500 has a spinner chain on the left and right.

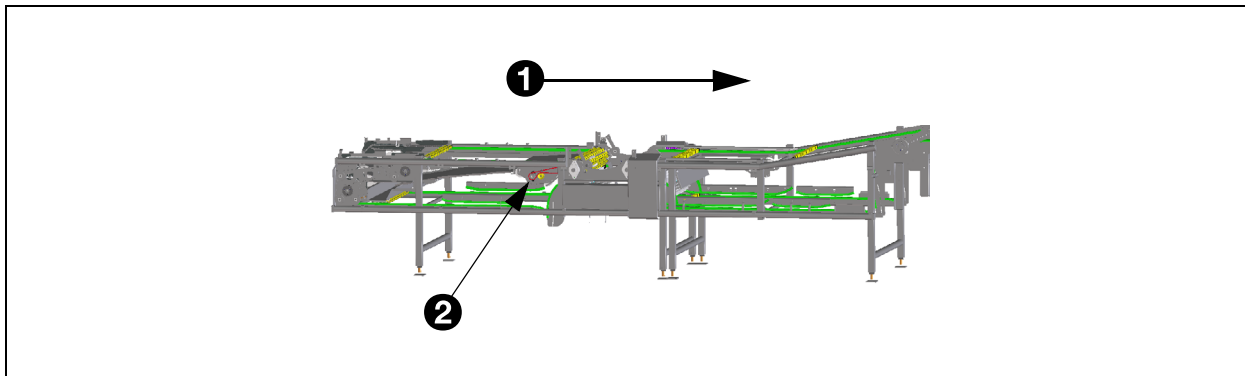


Fig. 43 - Right-hand side of the infeed unit

The arrow (1) in the illustration shows the direction of the egg travel. See fig. 43.

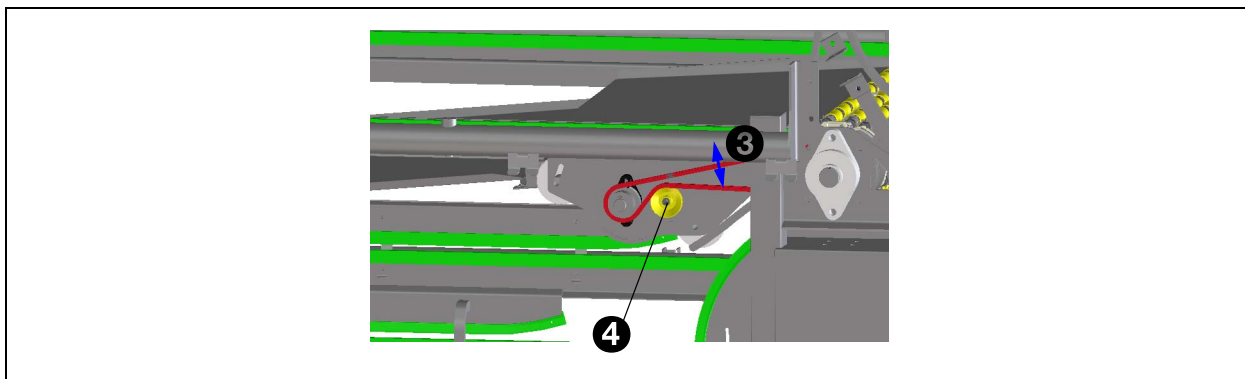


Fig. 44 - Detail of chain-tensioner

1. The play (3) in the drive-chain can amount to a maximum of 10 mm. See fig. 44.
2. Adjust the play with the chain-tensioner (4).

## 6.3.6 Chain tension of Spinner chain

The spinner chain is located on the right and left-hand side of the infeed unit.

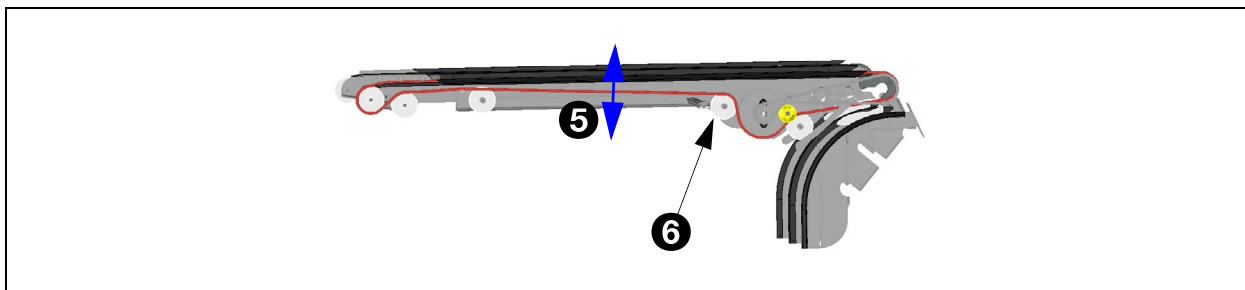


Fig. 45 - Chain tension of Spinner chain

1. The play (5) in the spinner chain can amount to a maximum of 15 mm. See fig. 45.
2. Adjust the play with the chain-tensioner (6).

### 6.3.7 How to act with shaking infeed

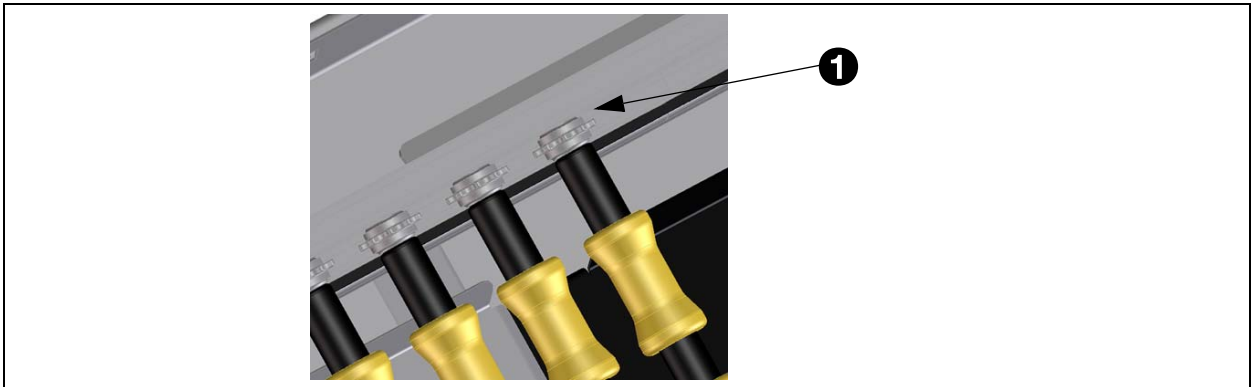
When the infeed is shaking, the following items need to be checked / adjusted:

1. Check the safety from the MultiDrum™, is it moving / vibrating then: (See paragraph 1.1.4)
  - Verify the setting of the tension spring.
  - Verify differences in drive settings from one side to the other.



*Fig. 46 - Chain rests on support as much as possible*

2. The returning part of the chain at the Double roll infeed section down section has to be adjusted in such way that the chain rests on as much support as possible. Supports as high as possible without risking jumping chain.



*Fig. 47 - Adjustment = 0,5 - 1 mm.*

3. Verify if all the rollers in the Double roll infeed section and crack section have sufficient play, 0,5 - 1 mm.
4. Check the setting of the U-chain, entering and exiting of the little rollers. (See paragraph 7.2.3)
5. Top supports from the returning part of the chain at both lay down section and crack section should not be too tight on top of the chain.

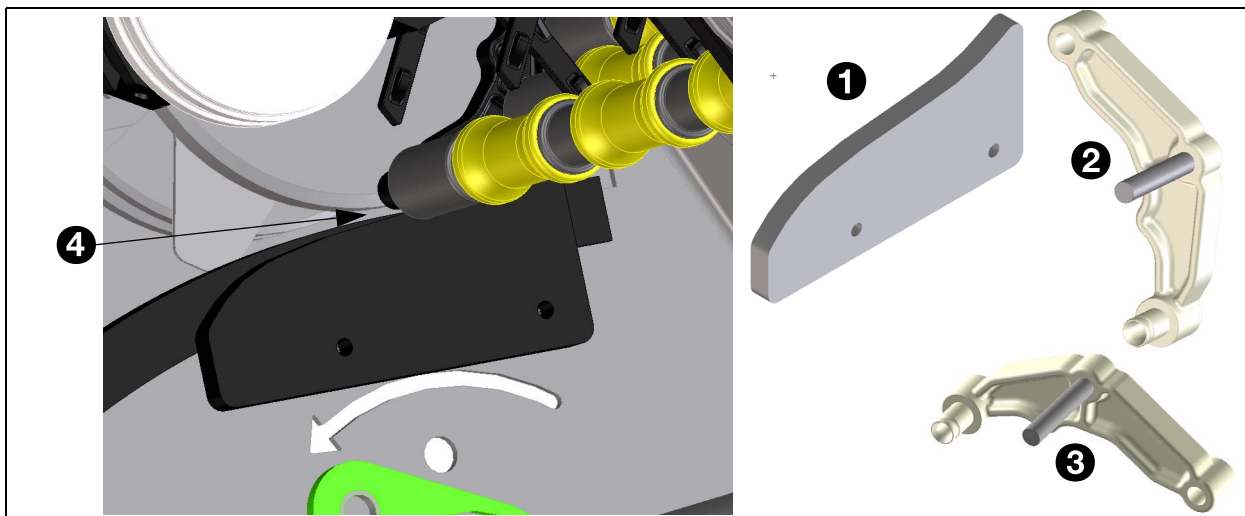


Fig. 48 - Support and holder double roller

- 1 - 80035620
- 2 - 80033270 - right
- 3 - 80033260 - left
- 4 - Play between holder and double rollers

6. Support 80035620, above brushes double rollers, check if it has enough play to the holder of the double rollers (80033260 en 80033270).
7. Check the wear on the MultiDrum™, when the curves or carriers have too much wear, the carriers won't open enough and this can disturb the balance between pick up side and lay down side.
8. Check if the carriers of the MultiDrum™ rotate freely.
9. Are the chains properly lubricated?

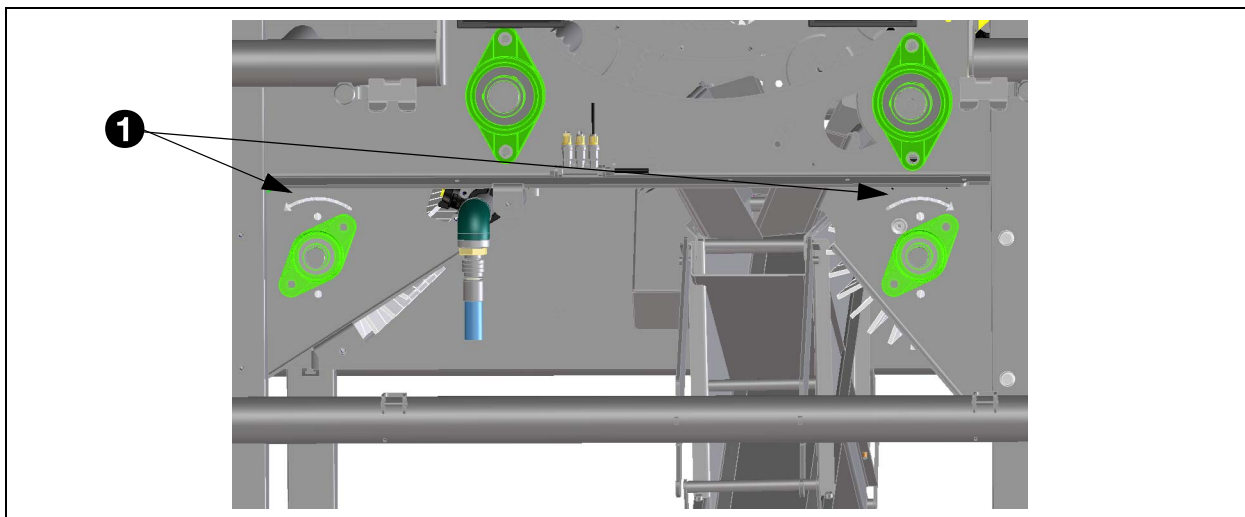


Fig. 49 - Rotation of the brushes

10. Are the cleaning brushes rotating in the correct way?
11. Spring tensioner at washer or extended section needs to have enough stretch in the tensioner (verify correct spring setting).
12. Is the extra motor (for the extended length of the infeed) strong enough?
13. Synchronisation is not working correctly.



14. Transition before / after the washer should line up properly (no hanging) and should be as short / straight/ smooth as possible.
15. Check if the shaft for the opening curve of the MultiDrum™ carriers rotates freely.
16. Mount extra wheels in the Crackroller returning section, so close to the weighing - transfer. This prevents the slip stick effect and ensures a smooth returning chain.



## 7 Adjustment of OMNIA FT infeed unit

The Infeed unit transports the eggs that are supplied by an accumulator-unit, Loader™ or a combination of both to the Transfer-unit™. A number of operations are carried out while this transport is in progress.

The adjustments of the following units are explained in this chapter.

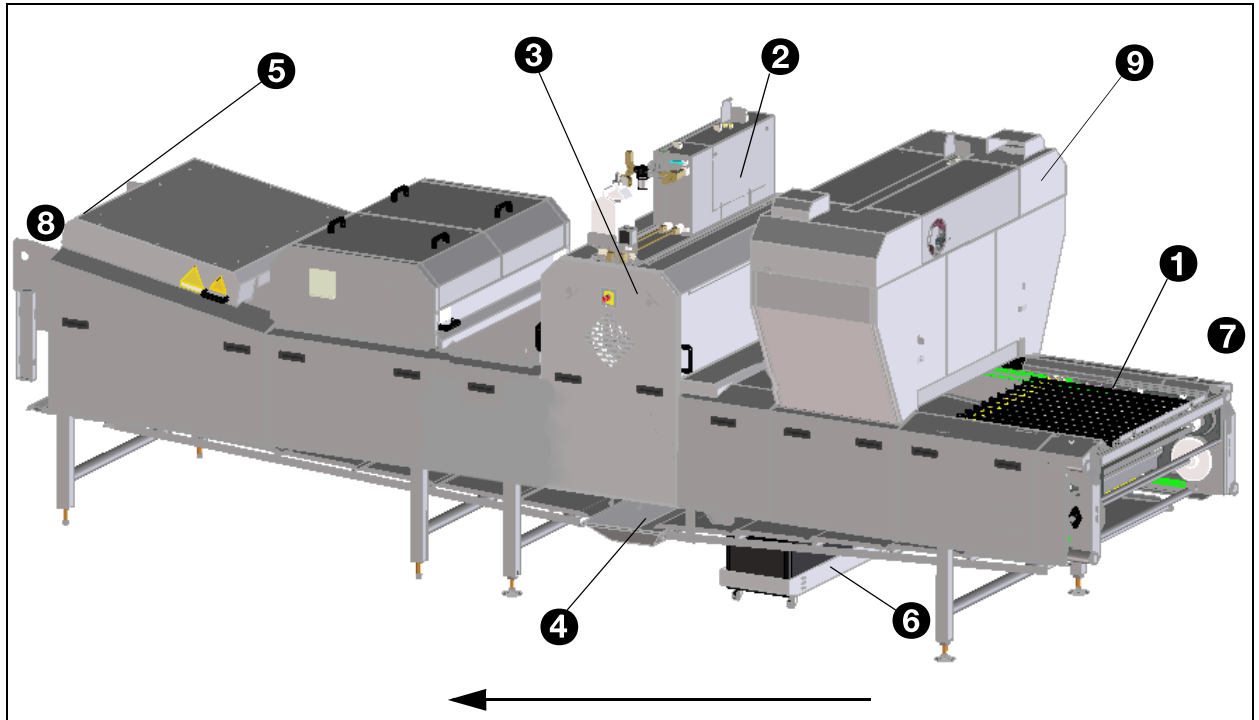
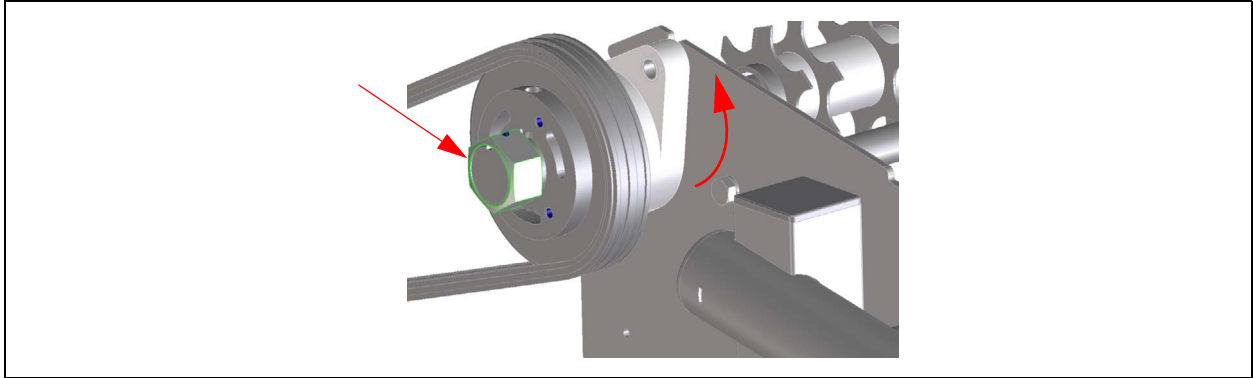


Fig. 50 - Overview of infeed unit

- 1 - Placement unit
- 2 - Cleaning unit
- 3 - MultiDrum™
- 4 - Ejection unit
- 5 - Delivery unit
- 6 - Waste water container
- 7 - Infeed-side
- 8 - Take-away side
- 9 - Egg inspector

The arrow in the illustration shows the direction of the movement of eggs.

## 7.1 Turning the transport installation by hand.



*Fig. 51 - Manual operation*

Turn the installation by hand using the central drive shaft. A hexagonal nut is mounted on the shaft for this purpose, see fig. 51. The direction of rotation is indicated by an arrow on the shield.



## 7.2 Adjustment of the laying unit

The placement unit can be fitted with:

- single rollers,
- double rollers or hygiene rollers.

The double roll or hygiene roller ensures that the dirt on an egg is not passed on to the egg on the roller in front or behind.

### 7.2.1 Adjustment of single rollers

The adjustment of the single rollers does not apply in this version!

### 7.2.2 Operation of double rolls

The Loader™ places the eggs with a pass-through of 2". The placement unit increases this pass-through distance between the eggs from 2" to 3" in order to be processed by the infeed unit.

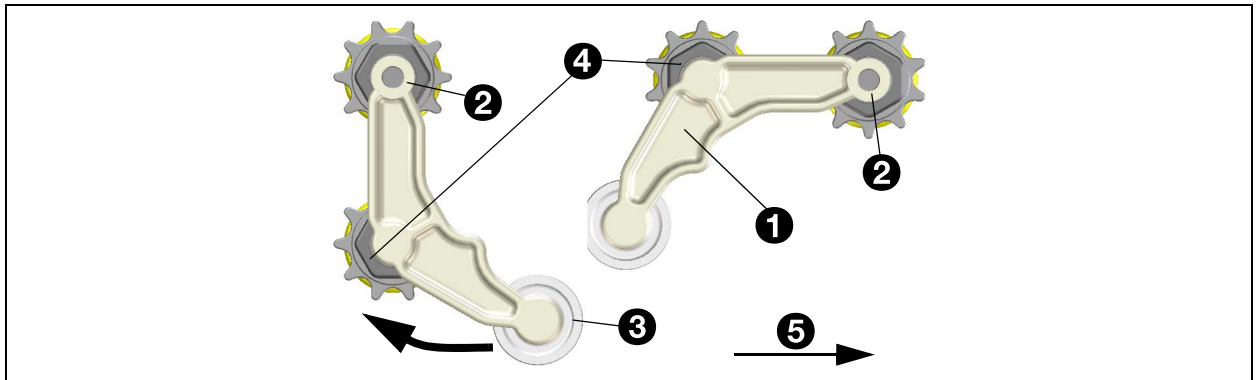


Fig. 52 - Roll set

The roller set (1) is used with the placement unit as a single roller. The first roller (2) of the double roller set is carried along by a U-holder<sup>(\*)</sup> (6). See fig. 52 and fig. 53.

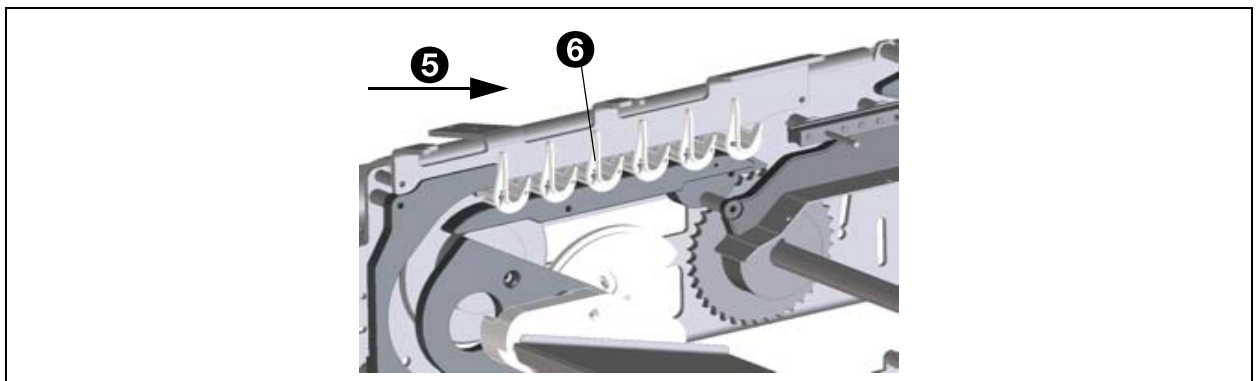
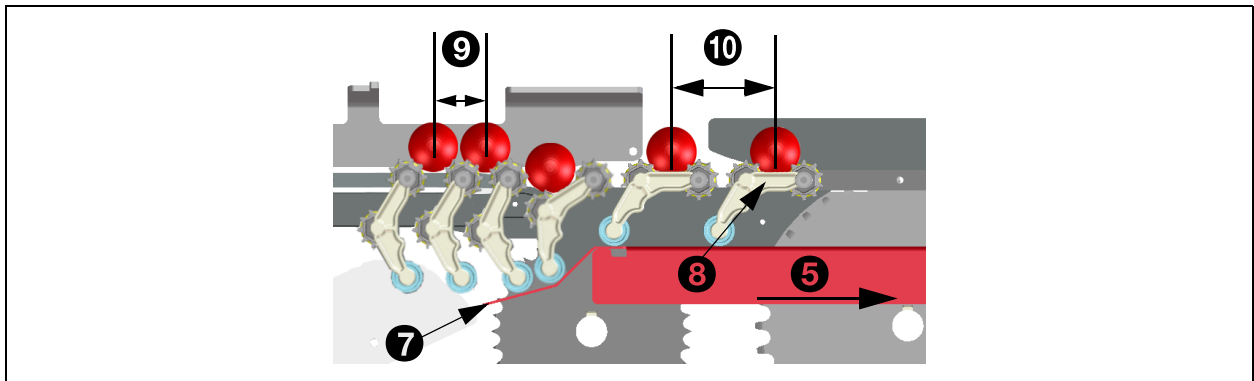


Fig. 53 - U-holder

The first roller (2) hangs in the U-holder at the start of the placement unit. After placement the wheel (3) is pushed upwards by a guide (7), fig. 54, during the transport. See fig. 52 and fig. 54.

The second roller (4) therefore rises, the egg lands on the two rollers (8) of the roller set. The pass-through distance is now increased from 2" (9) to 3" (10). See fig. 54.

(\*) See the glossary.



*Fig. 54 - Guide of double rolls*

After the pass-through distance has been increased the eggs are spun. The spinning of the eggs ensures that the eggs are properly placed on the rollers.

The arrow (5) in the illustration shows the direction of the egg travel.



### 7.2.3 Adjustment of standard infeed unit + Combi 6R

The first roller of the roller set is pulled by the pin chain out of the U-holder. The U-holder must be adjusted in such a way that the pin chain experiences no resistance.



TIP box Article number:03241800.

Check the tension of the drive chain and the chain with the U-holders. See paragraph 6.3.

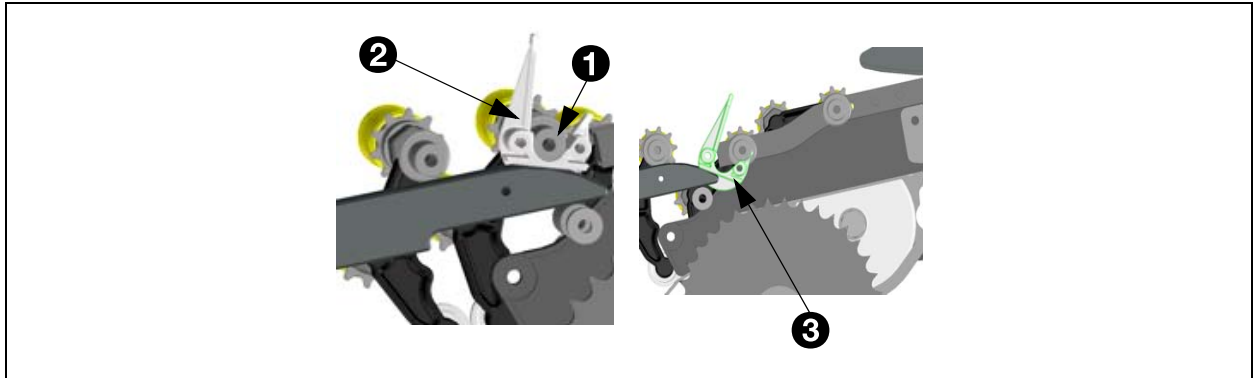


Fig. 55 - Run-out position of U-holder

Remove the protective cap from the placement unit and between the Loader™ and infeed unit.

1. The first roller (1) of the roller set is carried along by a U-holder (2). See fig. 55.
2. Just before the return path<sup>(\*)</sup> of the U-holder the transport chain pulls the roller out from the U-holder and the roller is placed on the guide in the frame.
3. The roller must not be obstructed by the lowest leg (3) of the U-holder during the transfer from the U-chain to the guide in the frame.

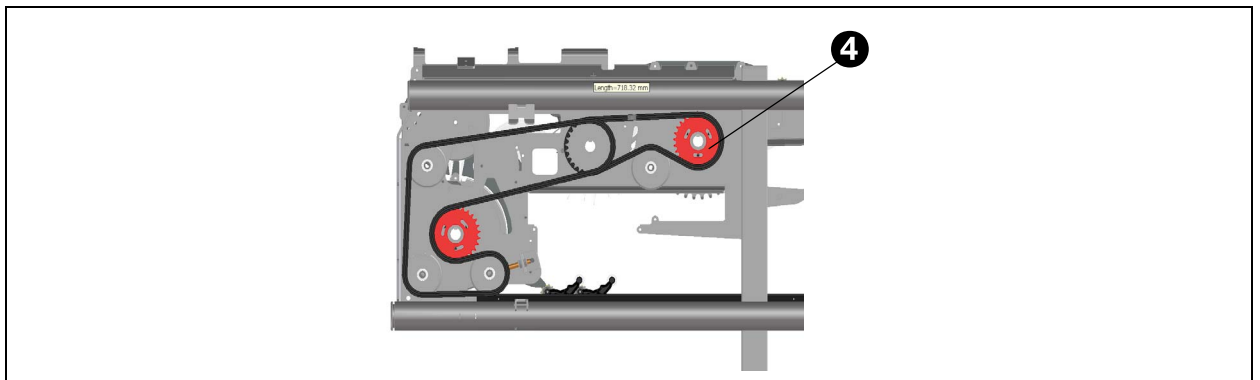


Fig. 56 - Drive of double rolls

4. Loosen the bolts of the chain-wheel (4). See fig. 56.
5. Turn the chain-wheel until the roller is smoothly passed on by the U-holder.
6. Tighten the bolts of the chain-wheel.



#### **NOTE**

**If the chain is turned so far that the U-holder presses against the roller (7), see fig. 58, first adjust the U-holder in order to continue with the above-mentioned adjustment.**

(\*) See the glossary.

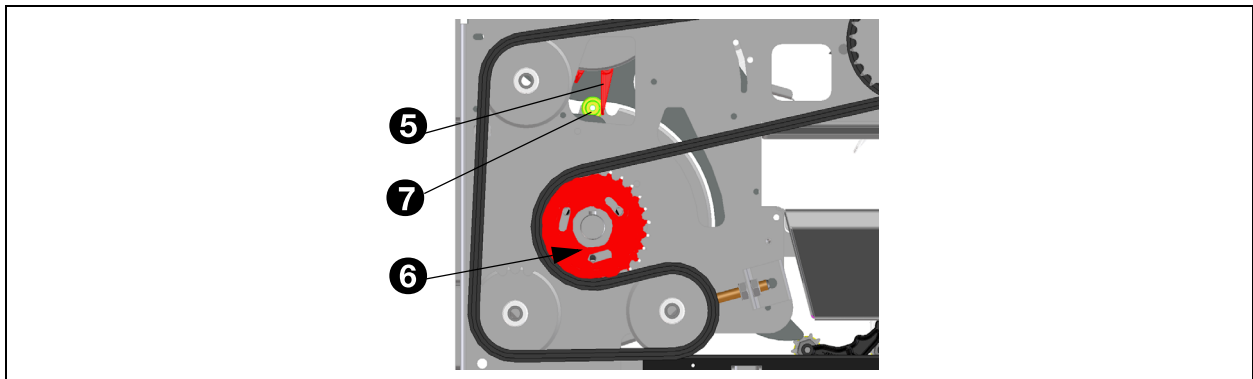


Fig. 57 - Run-in position of U-holder

After adjusting the U-holder, check whether:

1. The roller (7) during change-over by the U-holder (5), shown in fig. 57, runs clear of the leg of the U-holder. See also detail in fig. 58
2. Loosen the bolts of the chain-wheel (6).
3. Turn the chain-wheel until the roller runs clear of the leg of the U-holder.

Tighten the bolts of the chain-wheel.

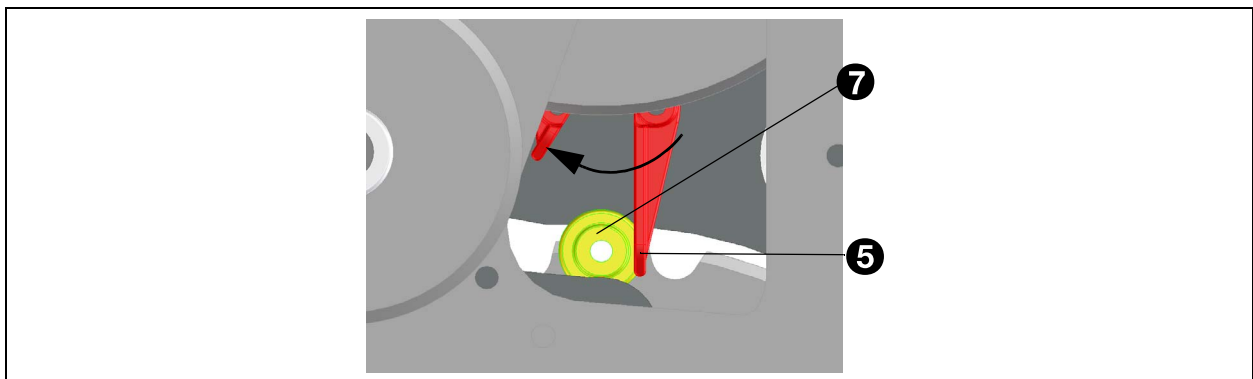


Fig. 58 - Detail of the run-in position of U-holder

Turn the machine manually using the TIP box<sup>(\*)</sup>. Check whether the rollers are pulled smoothly out of the U-holder by the pin chain. See fig. 55.



**NOTE**

**Check this adjustment on the left and right side of the infeed unit.**



**NOTE**

**Replace all protective caps on the infeed unit and Loader™.**

## 7.2.4 Adjustment of the combi infeed unit 12 R

The first roller of the double roll sets is pulled by the pin chain out of the U-holder. The U-holder must be adjusted in such a way that the pin chain experiences no resistance.



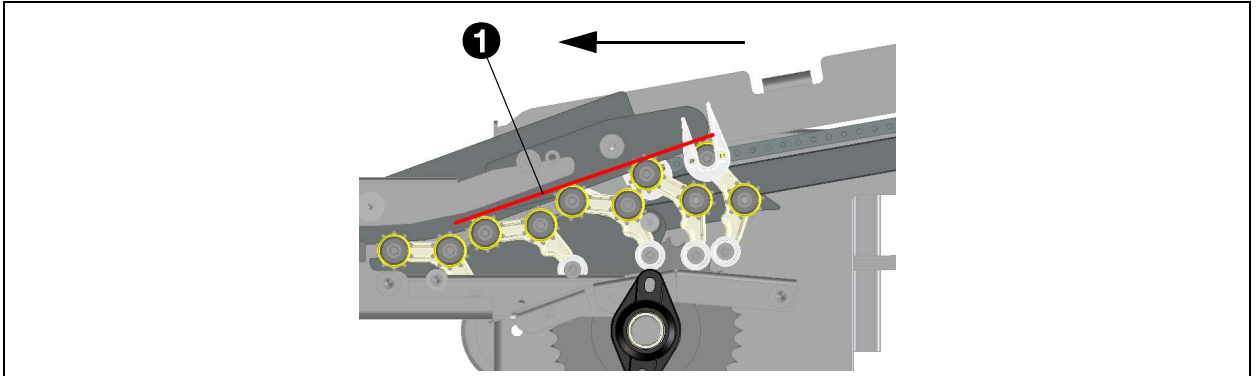
TIP box Article number: 03241800.

Check the tension of the drive chain and the chain with the U-holders. See paragraph 6.3.

<sup>(\*)</sup> See the glossary.

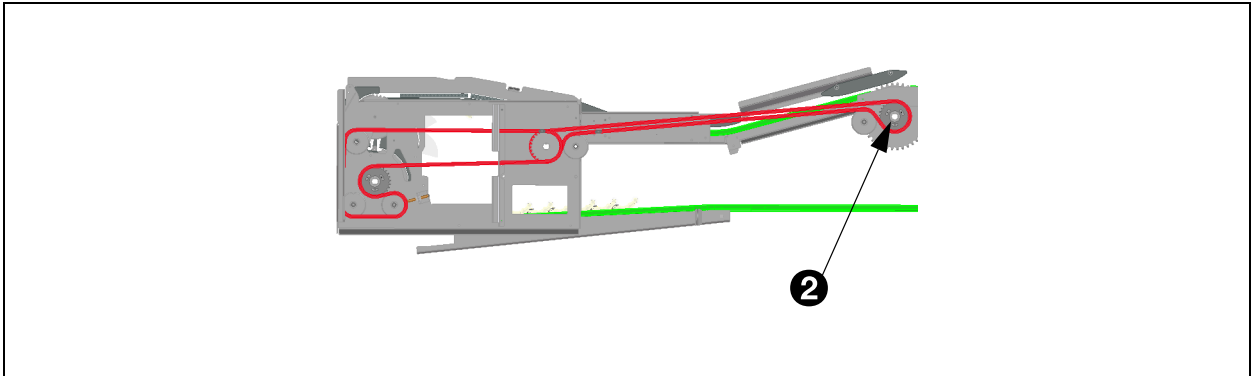


Remove the protective cap at the transfer-point of the accumulator-unit to the infeed unit and of the Loader™ and infeed unit.



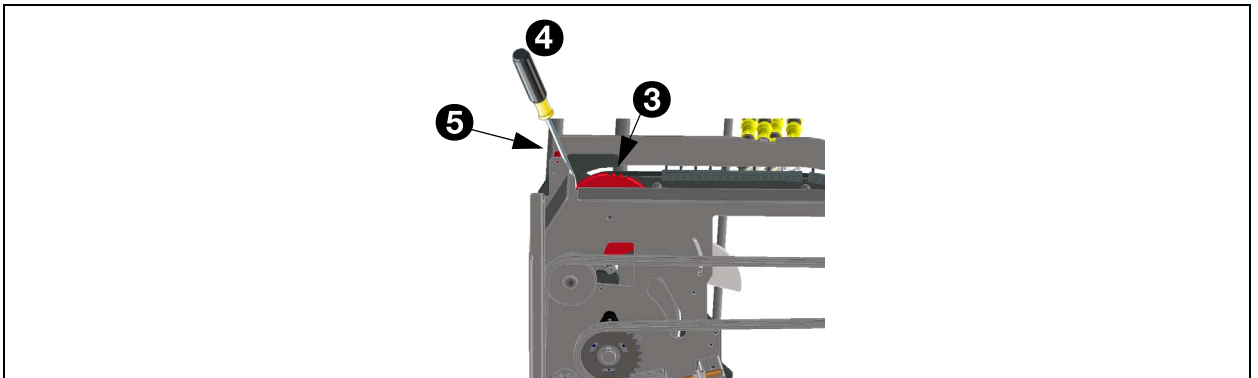
*Fig. 59 - Straight line double roll sets*

The double roller sets must be transported in a straight line (1) from the placement unit to the unit under the suction heads of the Loader™.



*Fig. 60 - Drive of U-chain*

1. Loosen the bolts (2) of the drive chain-wheel on the infeed unit. The chain-wheel (3) of the U-chain can now be turned.



*Fig. 61 - Adjustment of U-holders*

2. Turn the chain with the U-holders using a flat screwdriver (4).  
Place the screwdriver between the distance-bush (5) and the chain on the chain-wheel (3).
3. Tighten the bolts of the drive chain-wheel.

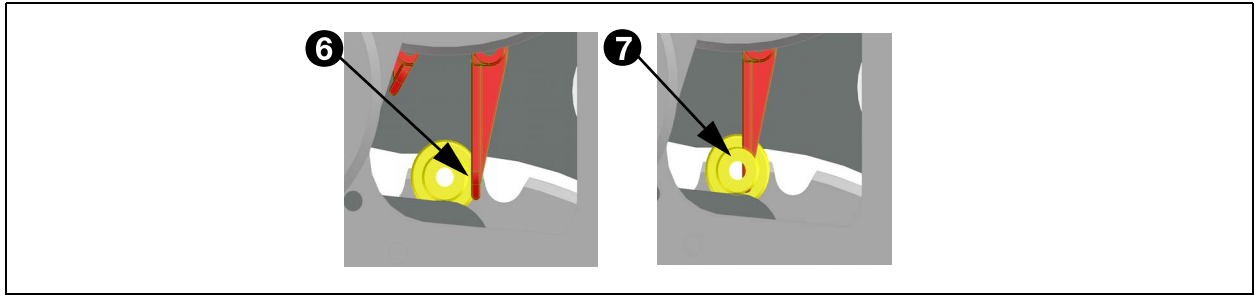


Fig. 62 - Detail of roller/U-holder



**NOTE**

*If the chain is turned so far that the U-holder presses against the roller (6), see fig. 62, first adjust the U-holder in order to continue with the above-mentioned adjustment.*

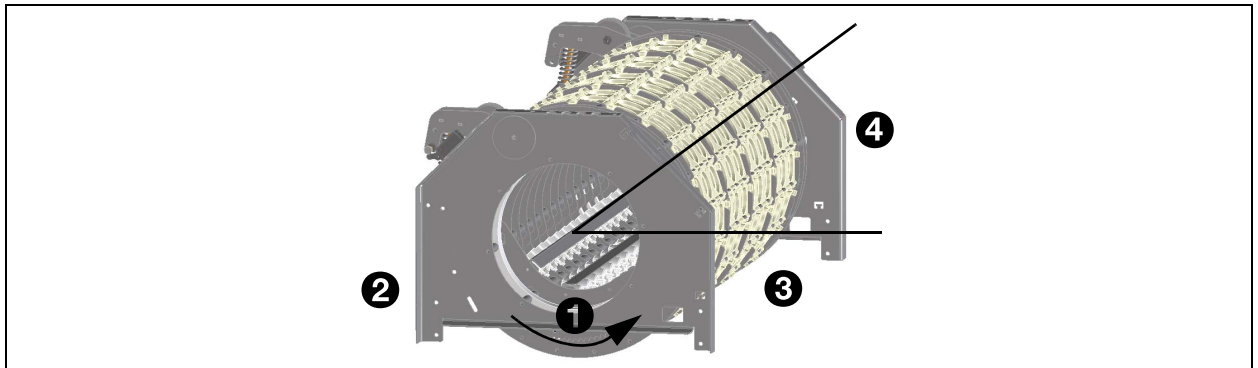


## 7.3 MultiDrum™ adjustment

The MultiDrum™ turns the eggs in the right direction to carry out a number of processing-steps and to position them correctly in the packaging.

### 7.3.1 Removal of carrier sets

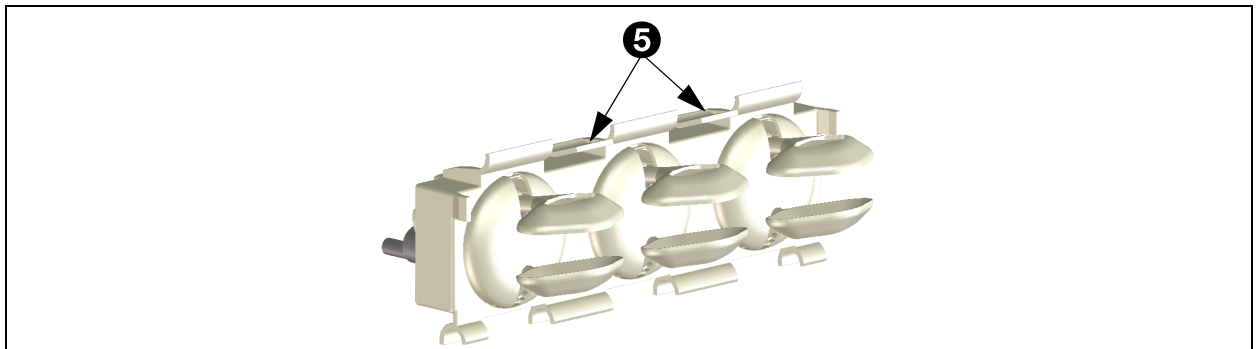
The removal of the carrier sets takes place on the outfeed-side (3) of the MultiDrum™.



*Fig. 63 - Removal angle of carrier sets*

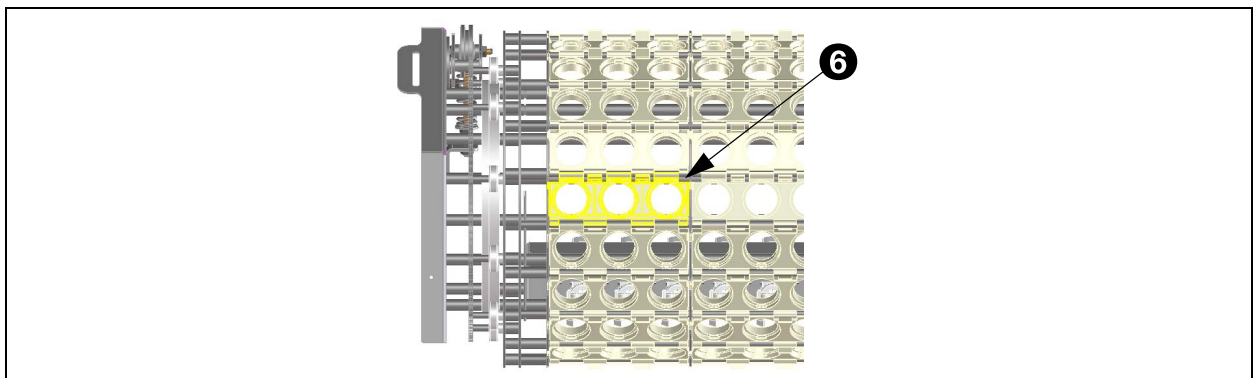
The direction of rotation of the MultiDrum™ is indicated with the arrow (1). The infed-side is indicated by (2) and the outfeed side by (3). See fig. 63.

1. The can be indicated within the angle (4) in fig. 63. Outside this angle the carriers are guided by a plastic curve.



*Fig. 64 - Detail of carrier set*

2. Press the lips (5) on the upper side of the carrier set downwards. See fig. 64.



*Fig. 65 - Carrier set detached from shaft*

3. Now pull the upper side of the carrier set loose from the shaft (6). See fig. 65.

## 7.3.2 Adjustment of infeed-side

The eggs are taken up from the rollers by the MultiDrum™.

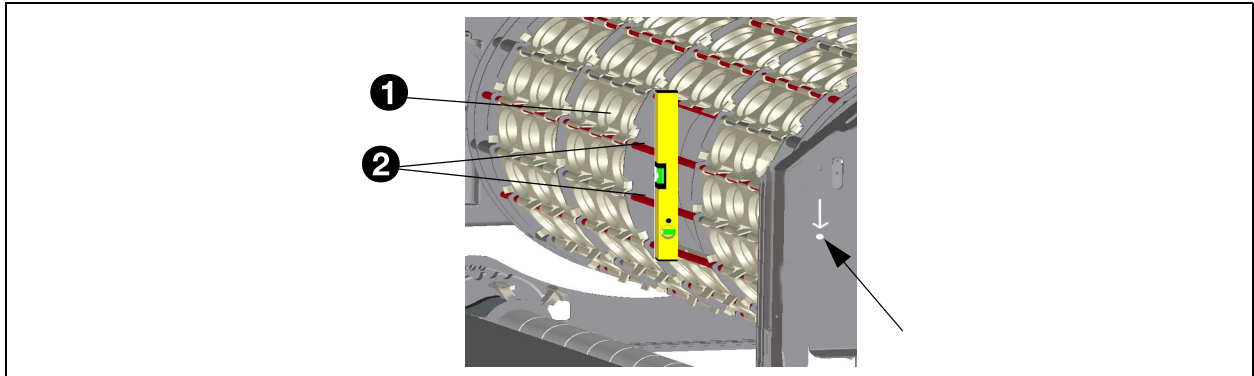


Fig. 66 - Carrier set

The time the eggs are taken up can be adjusted in the following manner:



TIP box Article number: 03241800.

Lift up the protective caps.

1. Turn the MultiDrum™ using the TIP<sup>(\*)</sup> box so that the hole in the side shield of the MultiDrum™ is in line with the hole in the chain-wheel. The shafts are now level.

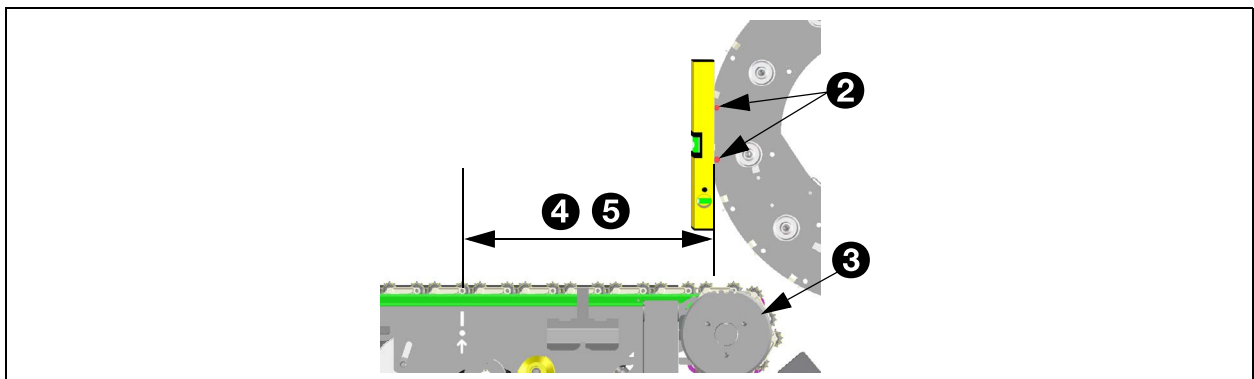


Fig. 67 - Distance between roller and shafts

2. Loosen the bolts of the chain-wheels (3). See fig. 67.
3. Do this both on the left and right side of the infeed unit.
4. Adjust the first roller of a double roller so that it is in line with the mark on the side shield. See fig. 67 and fig. 68.
5. Tighten the bolts of the two chain-wheels.
6. Close the protective caps.

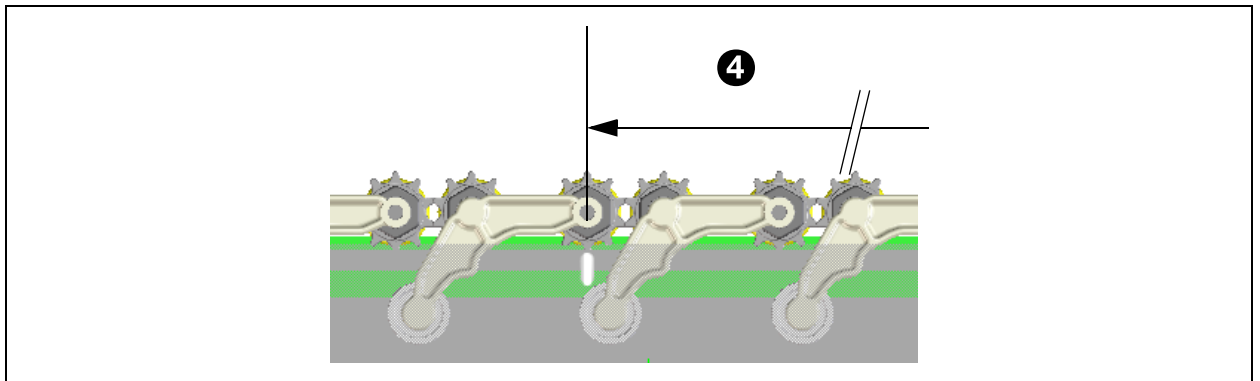


### NOTE

The above adjustment can be checked by carrying out the following steps.

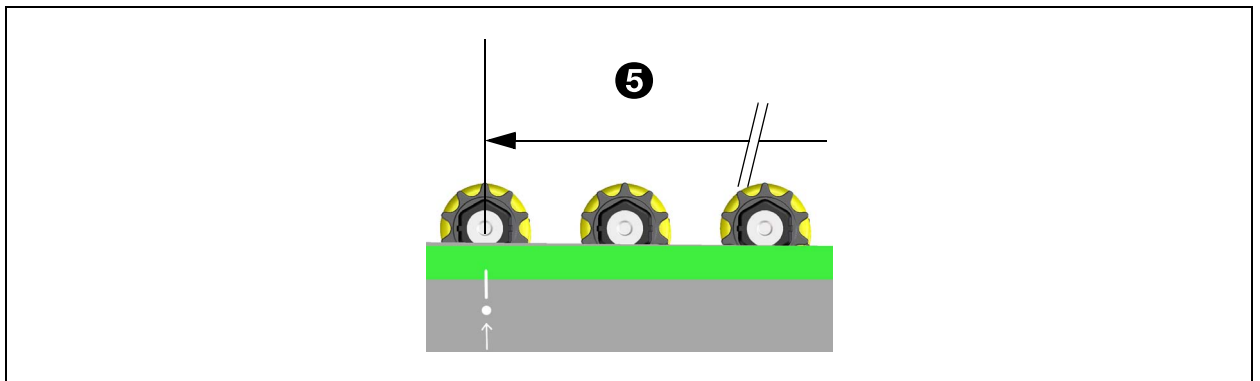
1. Remove three carrier sets<sup>(\*)</sup> (1) from the MultiDrum™. See paragraph 7.3.1.
  - Two shafts (2) are released.
2. Position a water level against the shafts. See fig. 66 and fig. 67. This must be level.

(\*) See the glossary.



*Fig. 68 - Detail of double roller*

3. Check the first roller of a single roller set (ER) at 364 mm (5) from the shafts, see fig. 67 and fig. 69. With a DR the first roller of a double roller set must be at 360 mm (4). See fig. 67, fig. 68.



*Fig. 69 - Detail of single roller*

4. Tighten the bolts of the two chain-wheels.
5. Replace the carrier sets in the MultiDrum™.
6. Close the protective caps.

## 7.3.3 Adjustment of outfeed-side

After being turned by the MultiDrum™ the eggs are transferred to the rollers; the moment of transfer can be adjusted in the following manner:

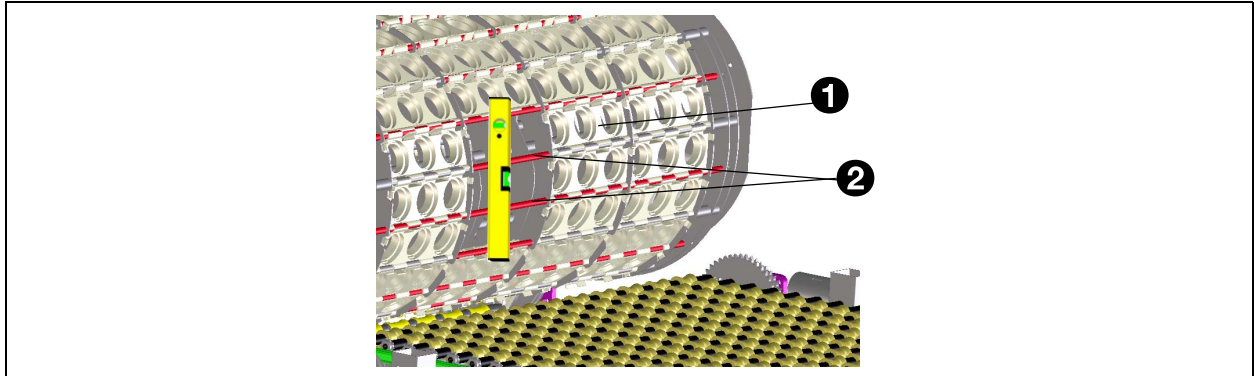


Fig. 70 - Removal of carrier sets

Lift up the protective caps.



TIP box Article number:03241800.

Small MOBA adjustment-egg. Article number: 14000990.

Large MOBA adjustment-egg. Article number: 14001000.

1. Remove three carrier sets (1) from the MultiDrum™. See paragraph 7.3.1.
  - Two shafts (2) are released.
2. Position a water level against the shafts.
3. Turn the MultiDrum™ using the TIP<sup>(\*)</sup> box so that the shafts are level.

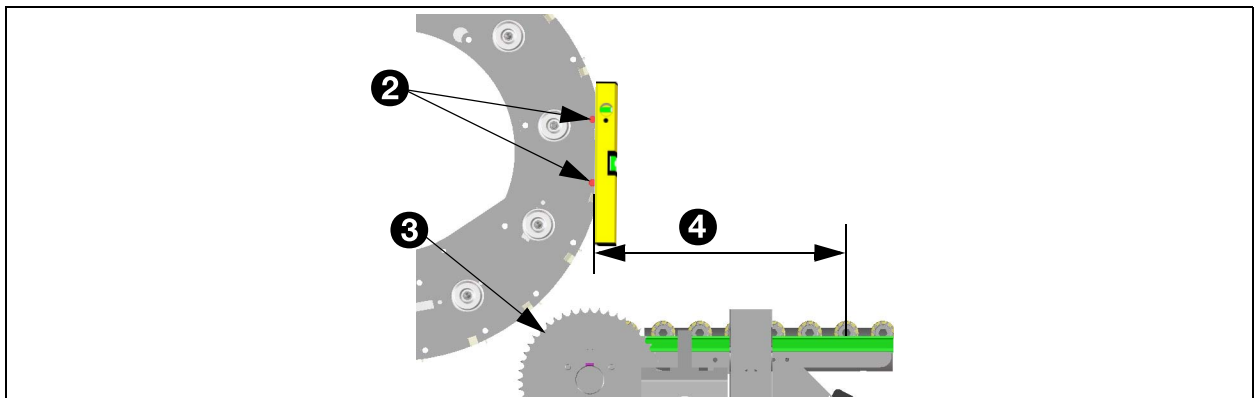


Fig. 71 - Distance between roller and shafts

4. Loosen the bolts of the chain-wheels (3). See fig. 71.
5. Do this both on the left and right side of the infeed unit.
6. Adjust a roll at 319 mm (4) from the axes.
7. Tighten the bolts of the two chain-wheels.
8. Replace the carrier sets in the MultiDrum™.

(\*) See the glossary.

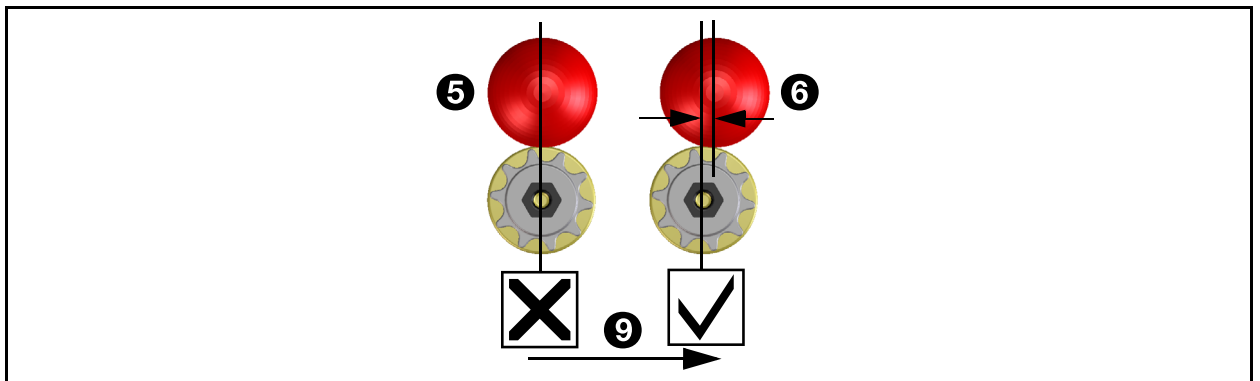


Fig. 72 - Transfer of small egg

**Place a small adjustment-egg in a carrier.**

- 1 Turn the MultiDrum™ using the TIP box.

Transport direction is indicated by the arrow (9).

When the egg is transferred by the carrier it must not be placed precisely on (5) a roller but slightly (6) above it. See fig. 72.

- 2 Now place a large adjustment-egg in a carrier.

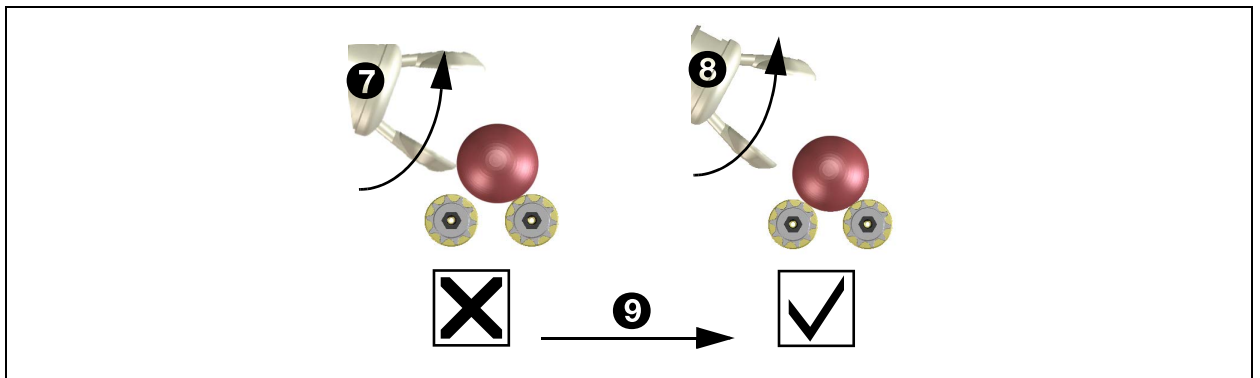


Fig. 73 - Transfer of large egg

- 3 Turn the MultiDrum™ using the TIP box.

When the egg is transferred by the carrier it must not be pushed away (7) by the carrier but be guided (8). See fig. 73.

- 4 Close the protective caps.

### 7.3.4 Infeed protection MultDrum™

If there is an object on the infeed like a tray the infeed protection before the MultiDrum™ will be activated and stop the machine.

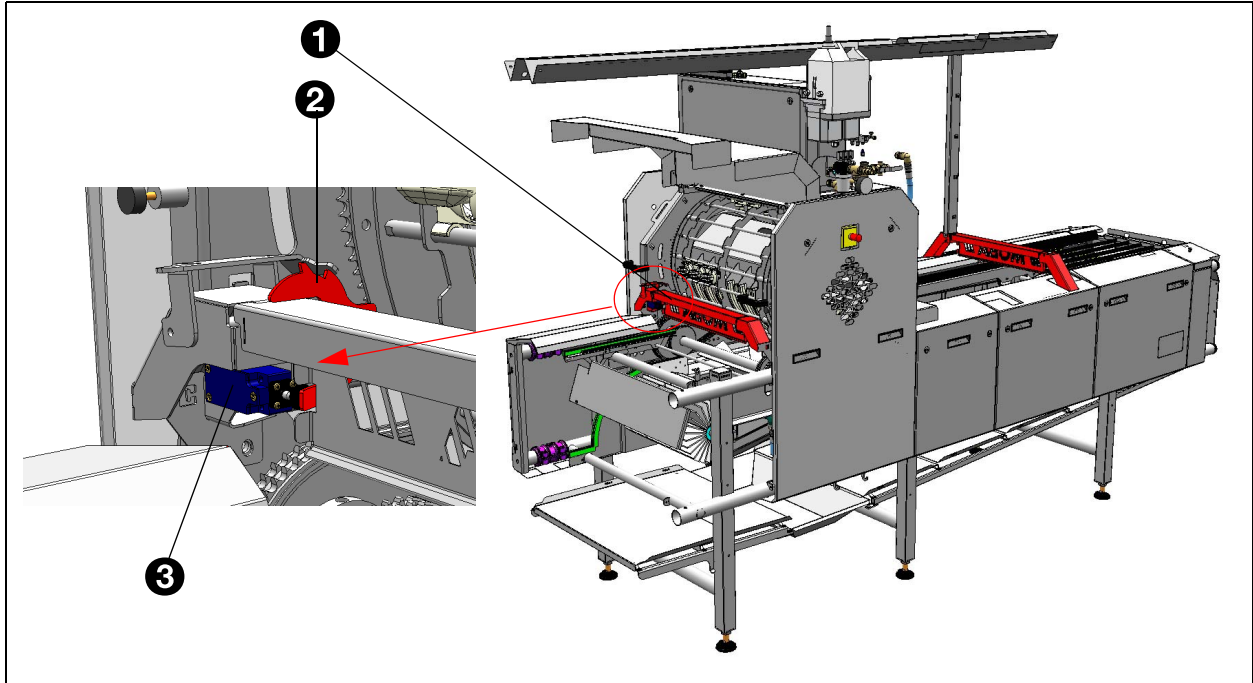


Fig. 74 - Infeed protection

- 1 - Infeed protection
- 2 - Guide plate
- 3 - Micro switch Infeed protection

#### Adjustment of the infeed protection:

- 1 The infeed protection is in the rest position.
- 2 Move the guide plate to left or right until the micro switch is activated.



### 7.3.5 Adjusting the Egg/Leaker inspector

Full details of the adjustment of the Leaker Inspector's Egg Inspector are given in the Egg Inspector's Service Manual.

Check the following settings in the Egg Inspector program:

1. Data communication with the Grader, "Distance to Data Output (mm)"
2. Unlocking time of the carriers in the Multi Drum™ "Distance to Leaker Unlock (mm)"

### 7.3.6 Alignment of carrier sets OMNIA FT 170, 250 and 330

If the carriers of the MultiDrum™ are not aligned (twisted) then carry out the following procedure:

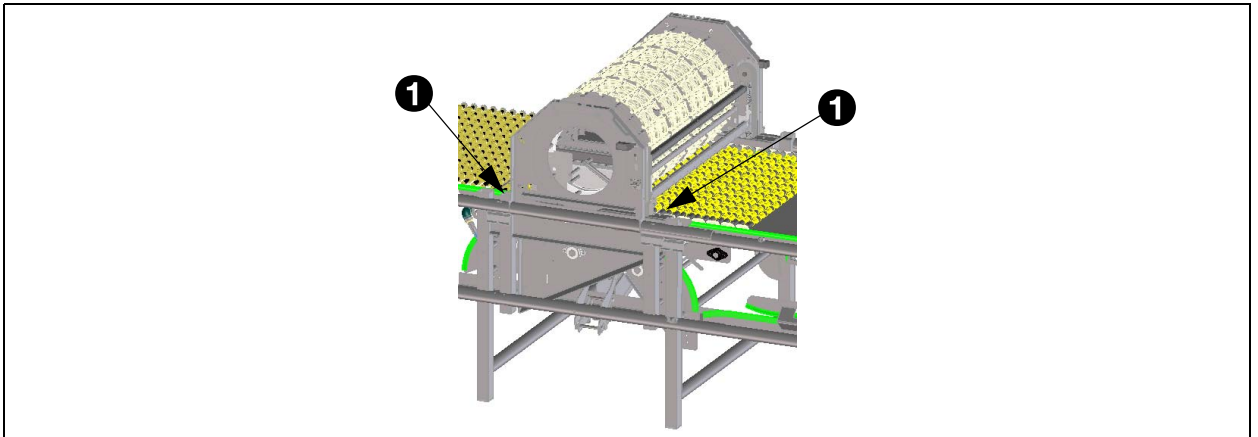


Fig. 75 - Location of the chain-wheels

The adjustment points (1) are accessible on the inside of the MultiDrum™. See fig. 75.

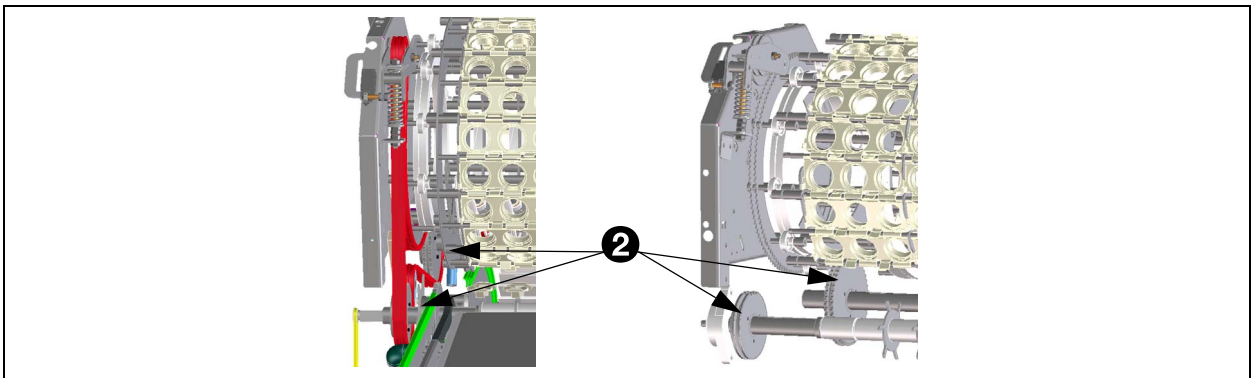


Fig. 76 - Chain-wheels MultiDrum™

1. There are three bolts on each chain-wheel (2). See fig. 76.
2. Loosen the bolts of the two chain-wheels.
3. Turn the chain-wheels until the carriers are aligned.
4. Tighten the bolts of the two chain-wheels.

Check the adjustment as described in paragraph 7.3, 'MultiDrum™ adjustment' on page 73.

## 7.4 Adjustment of chain-guides

Chain-guides (1) are mounted on the return path<sup>(\*)</sup> of the double roll transport chain (2) and of the crack roll transport chain (3). The chain guides can be adjusted.

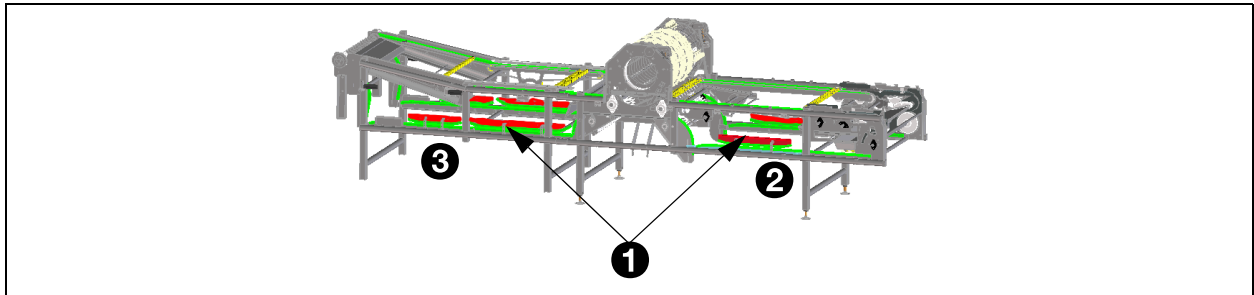


Fig. 77 - Locations of chain guides



TIP box Article number: 03241800.

- See the attachments for the use of the TIP box.

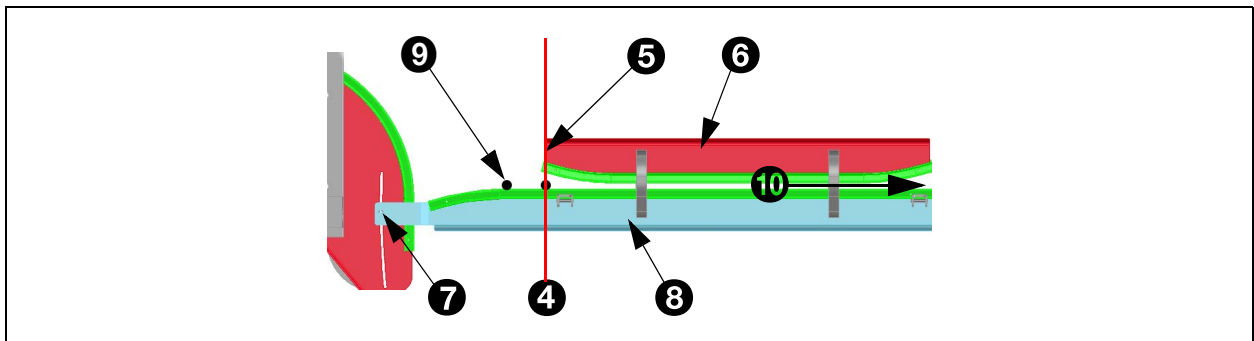


Fig. 78 - Chain guide

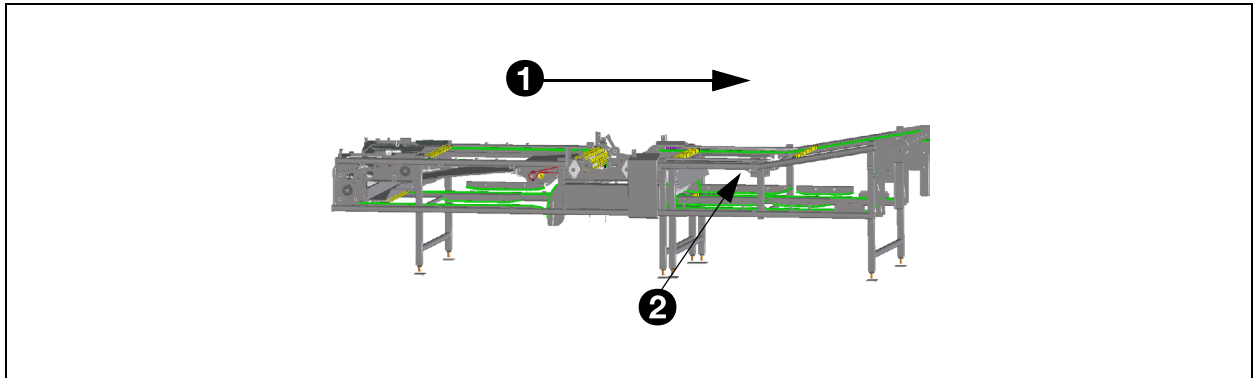
1. Turn the machine manually using the 'TIP box' until a white bush (4) mounted on the transport chain is even with the end (5) of the upper guide (6).  
See the attachments for the use of the 'TIP box'.
  2. With the height-adjustment (7) adjust the lower guide (8) in such a way that the following white bush (9) on the transport chain just rests on the lower guide.
  3. Adjust the upper guide (6) at 0.5 to 1 mm above the chain.
  4. Adjust the chain-guides to the left and right side of the infeed unit. (OMNIA FT 500: also in the middle).
- The transport direction of the chain is indicated with the arrow (10) in fig. 78.

(\*) See the glossary.



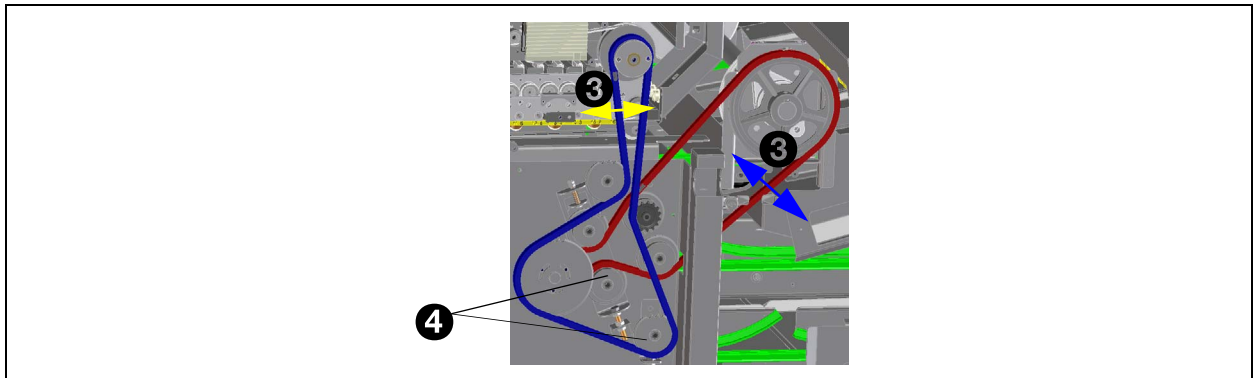
## 7.5 Chain tension drive of Crack Unit

The Crack Unit drive is located on the right-hand side of the infeed unit.



*Fig. 79 - Right-hand side of the infeed unit*

The arrow (1) in the illustration shows the direction of the egg travel. See fig. 79.

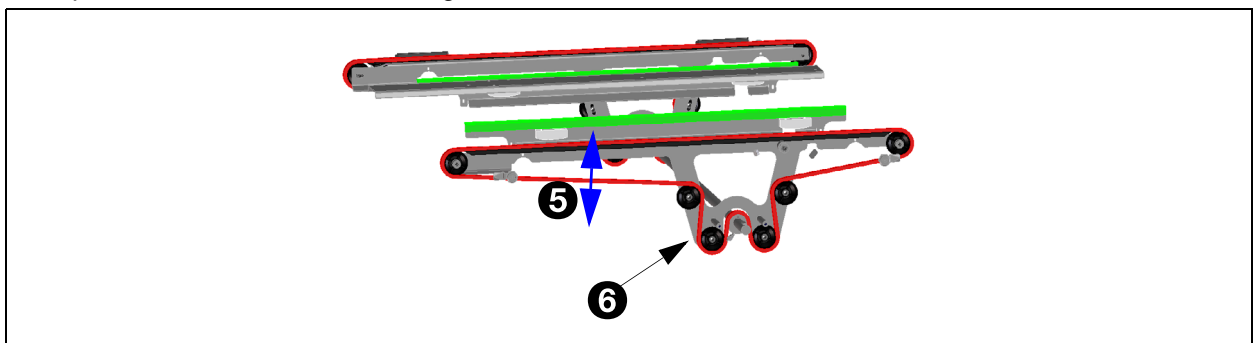


*Fig. 80 - Detail of chain-tensioner*

1. The total play (3) in the drive-chain may amount to a maximum of 2 mm. See fig. 80.
2. Adjust the play with the chain-tensioners (4).

### 7.5.1 Chain tension Spinner chain Crack unit

The spinner chain is located on the right and left-hand side of the infeed unit.



*Fig. 81 - Chain tension of Spinner chain*

1. The play (5) in the spinner chain can amount to a maximum of 15 mm. See fig. 81.
2. Adjust the play with the chain-tensioner (6).

## 7.6 Adjustment of the Crack unit spring-tensioner

A spring-tensioner has been mounted in the Crack unit to absorb the differences in chain length in relation to temperature differences when using, for example, a washer.

### 7.6.1 Mounting the spring

When mounted, the spring must have a length of 165 mm (1). See fig. 82.

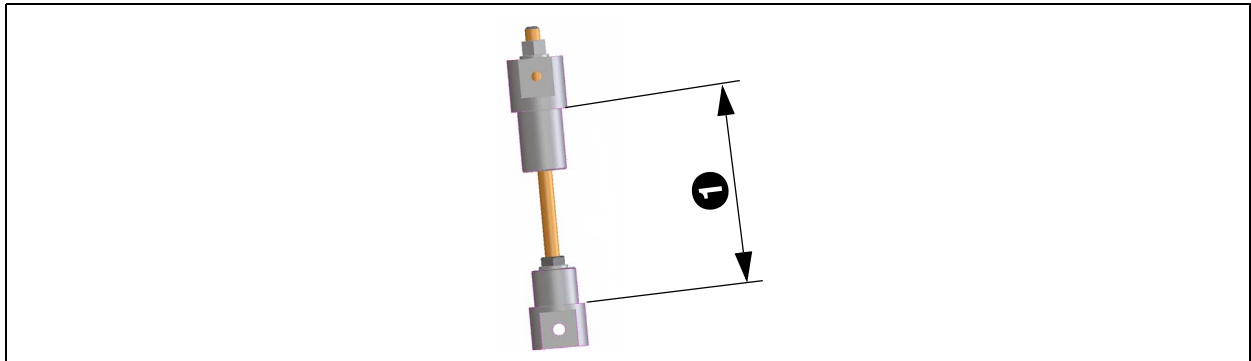


Fig. 82 - Setting the spring-tensioner length

### 7.6.2 Adjusting the compression spring

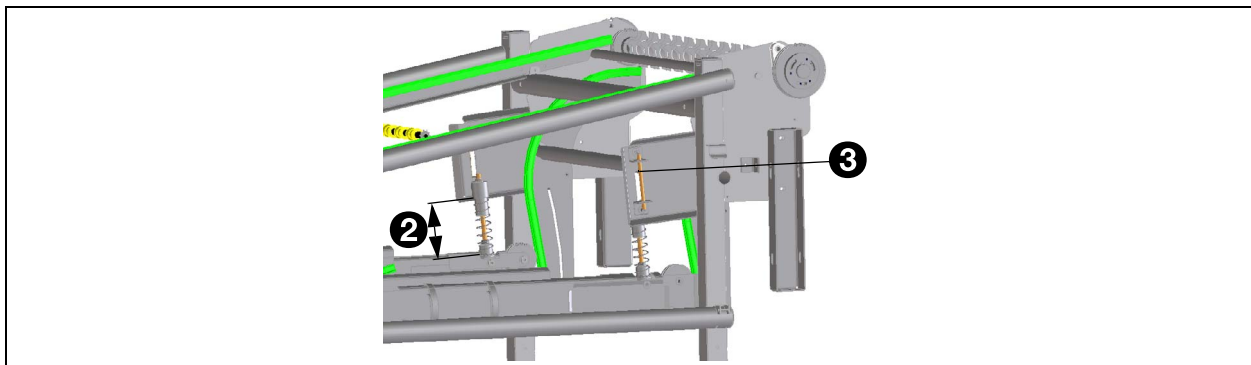


Fig. 83 - Mounted Crack unit spring-tensioner

1. Set the spring length with the tensioning direction (3) to 117 mm (2) with a cold chain. Size (2) will be approximately 160 mm when the chain has been heated. Adjust this adjustment at both sides of the infeed unit.



## 7.7 Adjustment of the transfer-point infeed unit - loadcell-unit

After the transport by the infeed unit (1), fig. 84, the eggs are transferred to the loadcell-unit (2). This transfer-point can be adjusted in two ways:

- With the aid of the centre-to-centre distance.
- With the aid of the adjustment gauge.

### 7.7.1 The centre-to-centre distance

Remove the chain (3) from the drive shaft of the infeed unit.

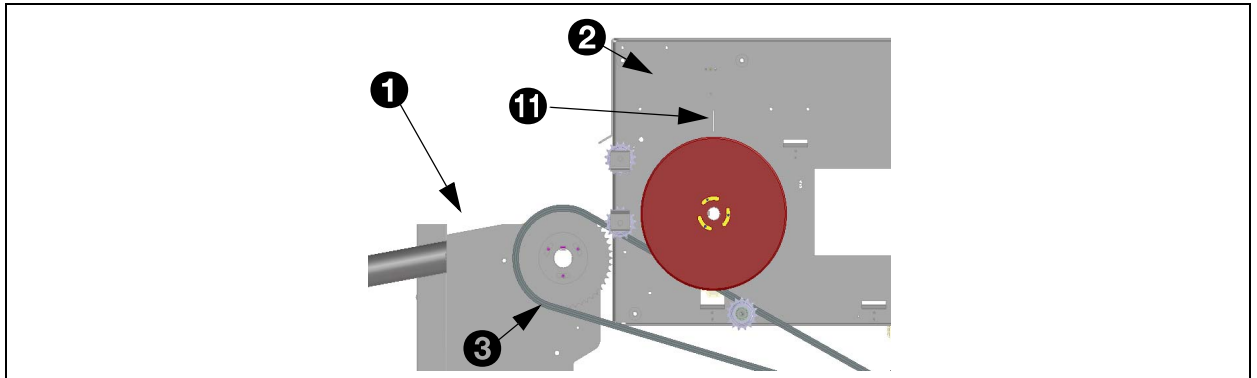


Fig. 84 - Transfer-point infeed unit - loadcell-unit

1. Adjust the distance and the height of the shafts of the infeed unit (1) and loadcell-unit (2) with the dimensions (4) 278.8 - 279 mm and (5) 87 mm. See fig. 85.

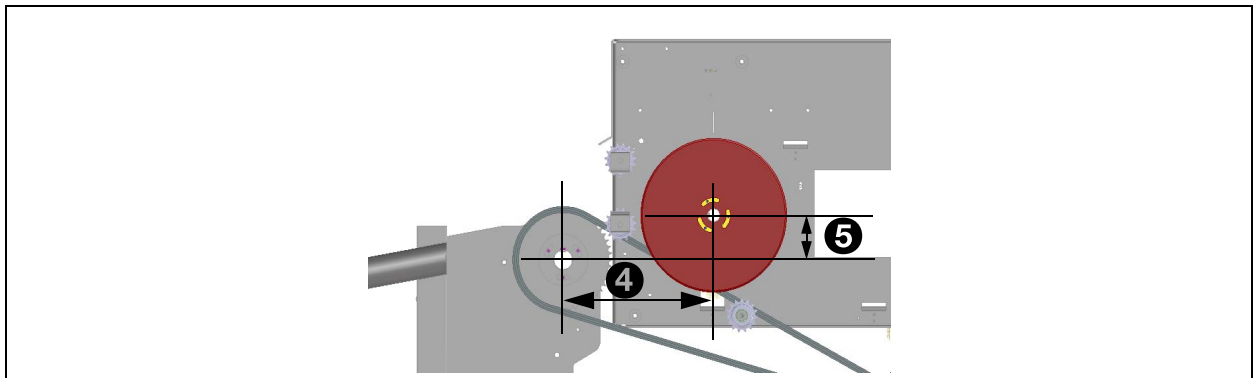


Fig. 85 - Adjustment dimensions shaft



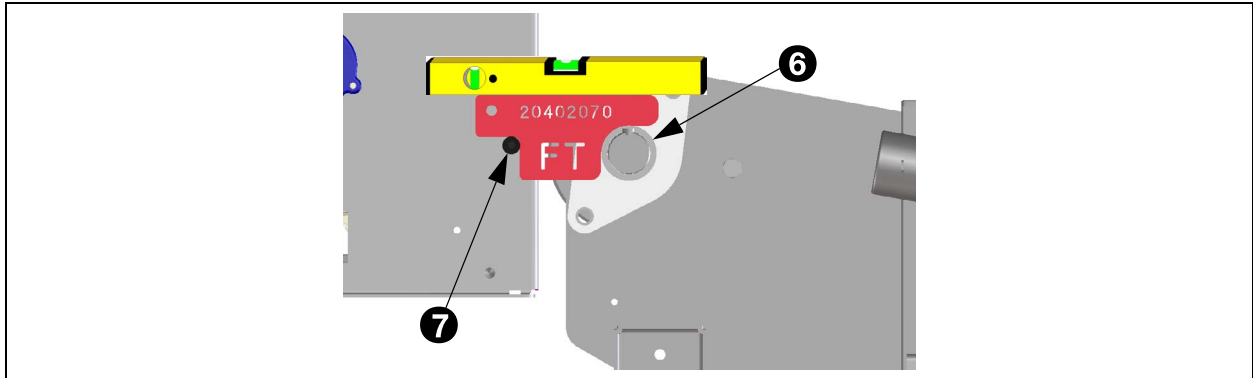
#### **NOTE**

**The stated dimensions are exact sizes, measured centre-to-centre.**

### 7.7.2 The adjustment gauge

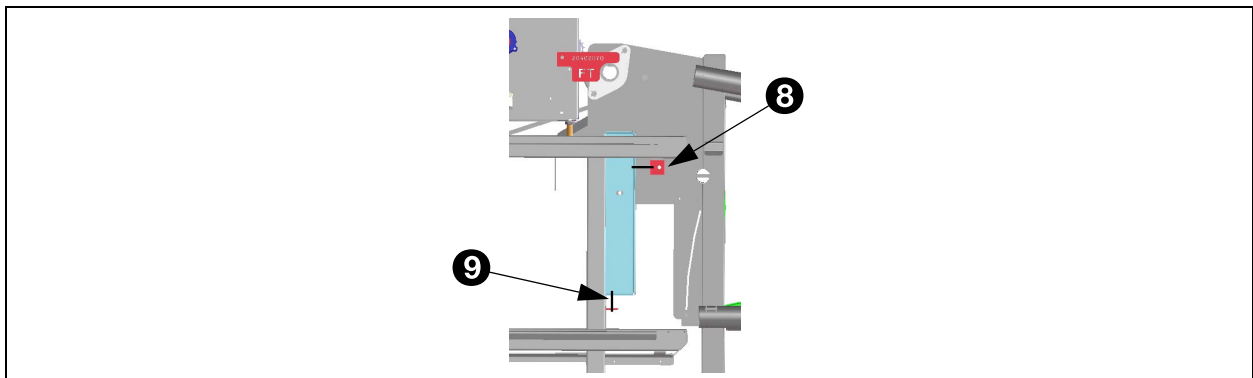
An adjustment gauge to set the distance between the two shafts is supplied with the machine. Adjusting jig for connecting infeed unit to the transfer unit. Article number: 20402070.

Remove the chain (3) from the drive shaft of the infeed unit.



*Fig. 86 - Adjustment with jig*

1. Place the jig with the radius (6) on the washer of the bearing house.  
When placing the jig, bear in mind the washer's protruding Allen bolts.
2. The other notch rests on a little shaft (7) mounted on the loadcell-unit.



*Fig. 87 - Adjusting distance and height*

3. Adjust the distance (8) and height (9) of the drive shaft of the infeed unit in relation to the loadcell-unit.  
Place a level on the jig for a correct height adjustment.



**NOTE**

**Check both sides of the infeed unit.**



### 7.7.3 Positioning rollers

To transfer the eggs at the right moment, the rollers should be positioned in relation to the carriers of the Transfer™.

Remove the chain (3) from the drive shaft of the infeed unit.

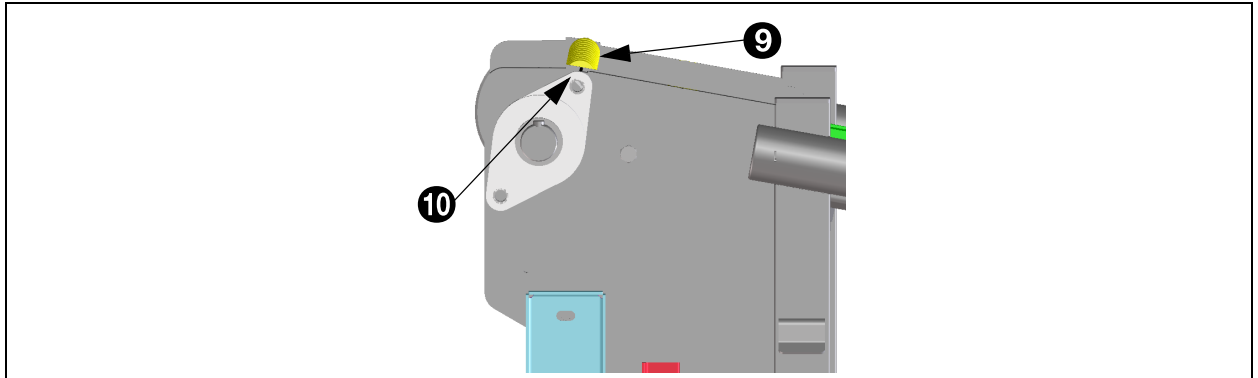


Fig. 88 - Position of the roller shaft in relation to the drive shaft

1. Align the centre of a roller shaft (9) to the centre of the bolt or notch in the shield (10) of the infeed-unit, see fig. 88.

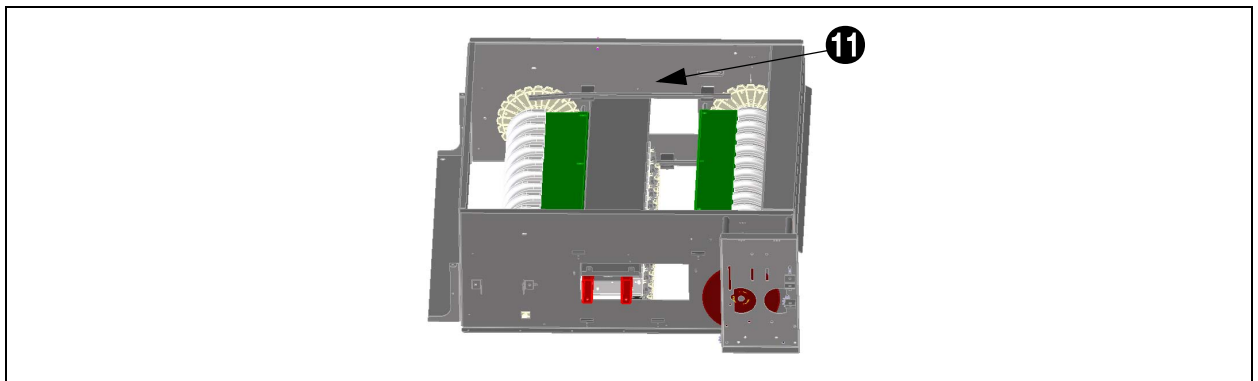


Fig. 89 - Mark in loadcell unit

2. Set the centre of a transport wheel (12) even with the mark (11) made in the strip of the loadcell-unit , see fig. 89 and fig. 90.

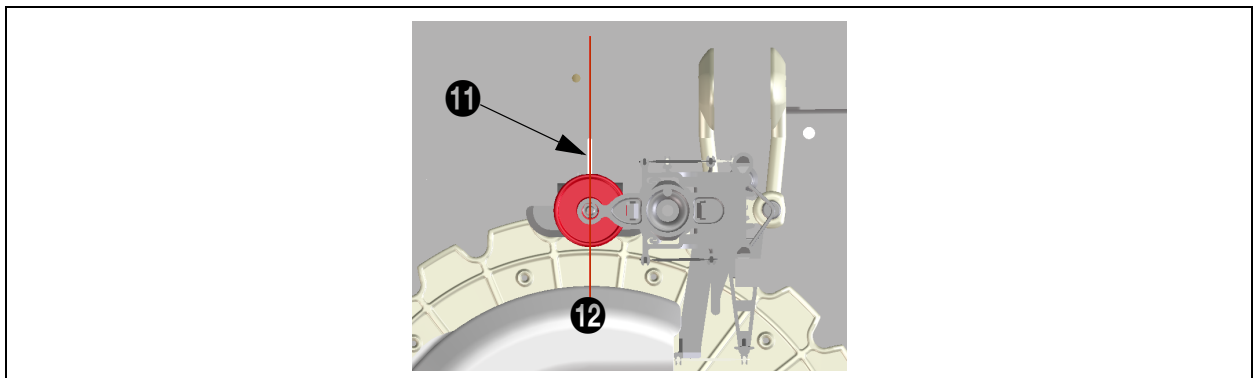


Fig. 90 - Transport wheel in relation to marks

3. Mount the chain (3) of the drive. See fig. 84



**NOTE**

**Be careful that the shafts do not turn.**





## 8 Adjustment sequence, Transfer™

### 8.1 Adjustment sequence of Transfer™

Adjustments of the machine modules have an influence on one another. The machine must be adjusted in the order that is shown below, so that its operations are properly synchronised.

**NOTE**

*The illustrations may be different from your version of the machine; this has no influence on the settings and adjustments of your machine.*

The left and right side are determined on the basis of the direction of the movement of eggs. In fig. 91 the direction of movement of the eggs is indicated with the arrow.

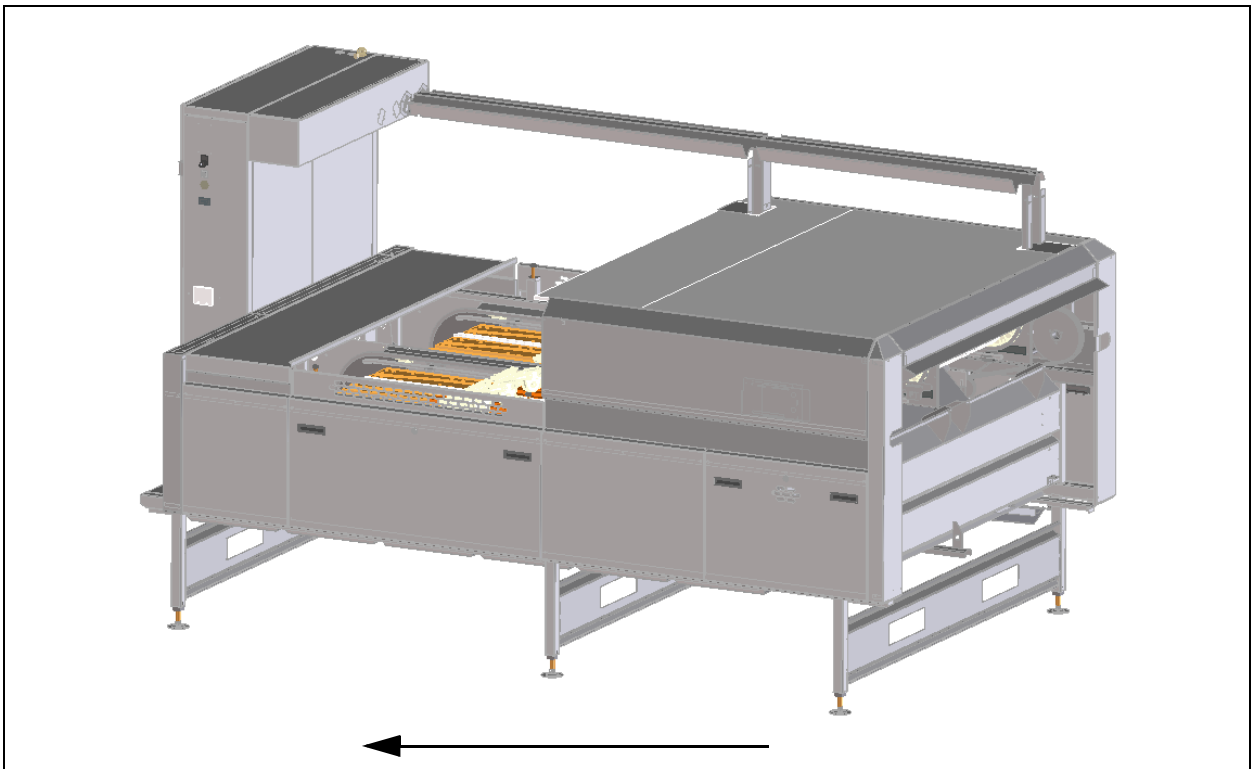


Fig. 91 - Overview of Transfer™

**NOTE**

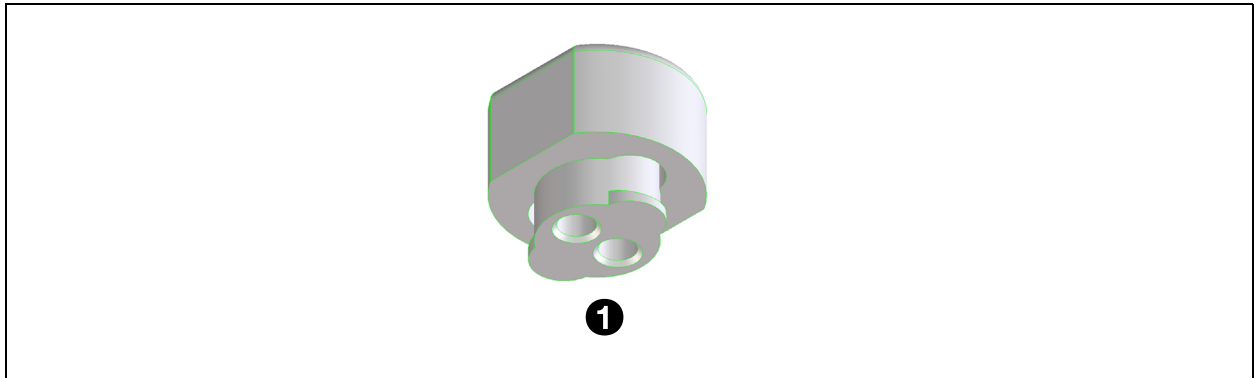
*Before you adjust the machine/module always check the tension of the chains first*

1. Adjusting sensor 'loadcell-carrier 0'.
2. Adjustment of the transfer from loadcell-carrier to transfer bridge.
3. Adjustment of the transfer from transfer bridge to frame-carrier.
4. Adjustment of the weighing moment.
5. Adjustment of placing on the Loader™.



## 8.2 OMNIA Transfer™ accessory tools

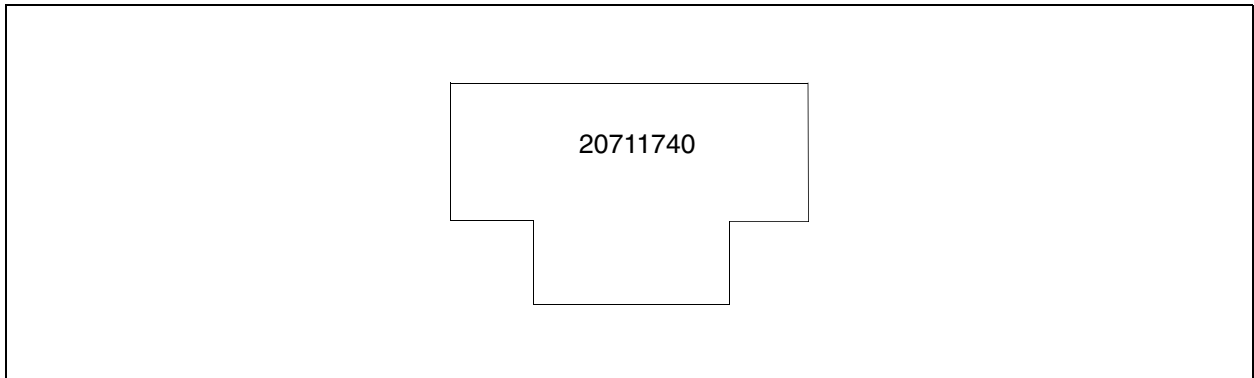
The following accessory tools are supplied with the Transfer™:



*Fig. 92 - Loadcell carrier calibration*

**Loadcell carrier calibration, see fig. 92.**

1 MOBA calibration weight, plastic 63 gram. Article number:80206980.



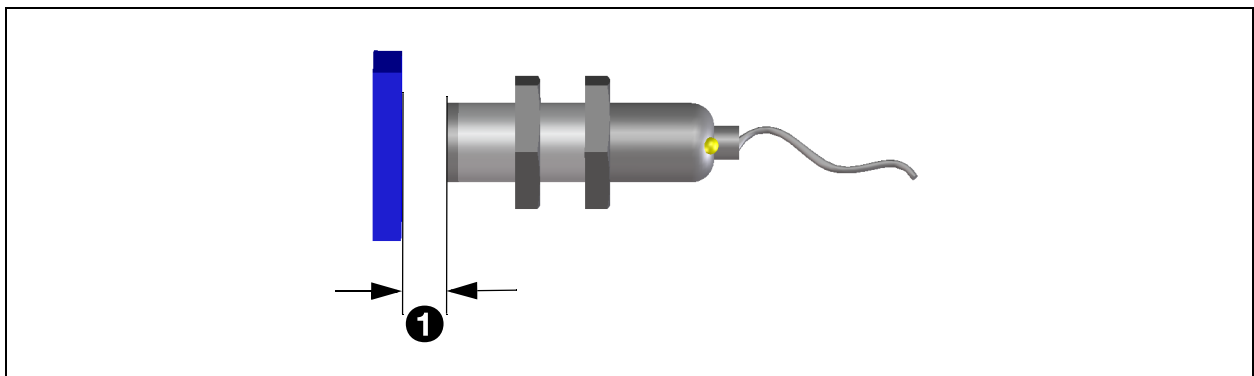
*Fig. 93 - Adjusting jig*

**Adjustment gauge, see fig. 93.**

Adjusting jig for connecting infeed unit to the transfer unit. Article number;20711740.

## 8.3 Sensor adjustment

The sensors that are mounted in this module are adjusted to the standard values. The standard distance between the sensor and the adjustment-disk, strip or small plate amounts to 1 mm.



*Fig. 94 - Distance of the sensor*



1. After a sensor has been replaced, use a feeler gauge to set the distance (1) to 1 mm. See fig. 94.

## 8.4 Adjustment of chain-tension

Chain-tensioners are fitted at various locations in the Transfer™. Follow the procedures below to apply tension to the chains.

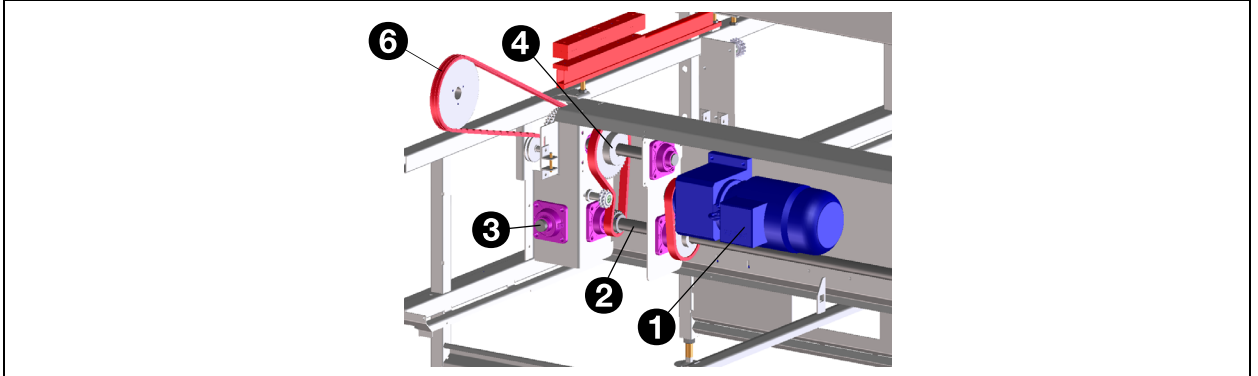


Fig. 95 - Main drive



### NOTE

**Check the tension of the chains one month after the Transfer™ has been brought into use. After the initial check, use the maintenance schedules.**

The primary motor (1) drives the main shaft (2), see fig. 95. The main shaft drives the Transfer™ (4), loadcell unit (5) and infeed unit (6), see fig. 95 and fig. 97. A right-angled transmission (3) is mounted on the main shaft to drive the frame. See fig. 97.

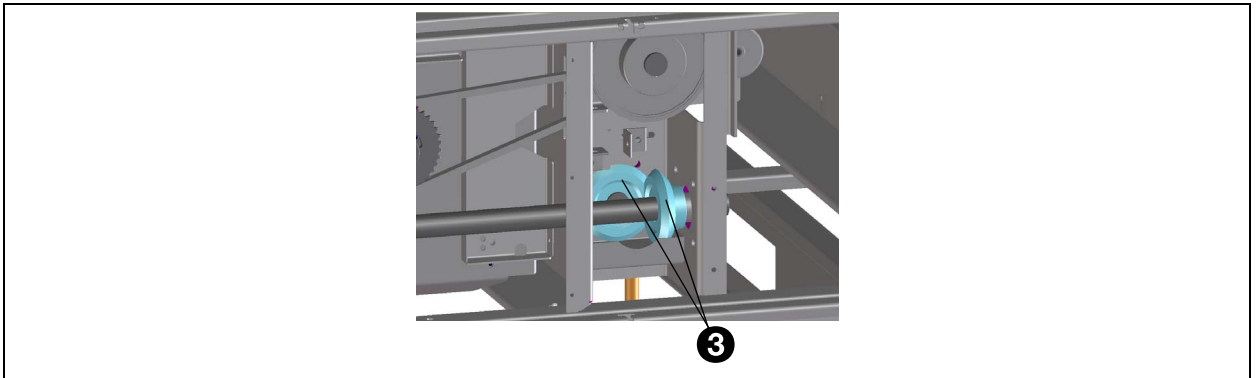


Fig. 96 - Right-angled transmission

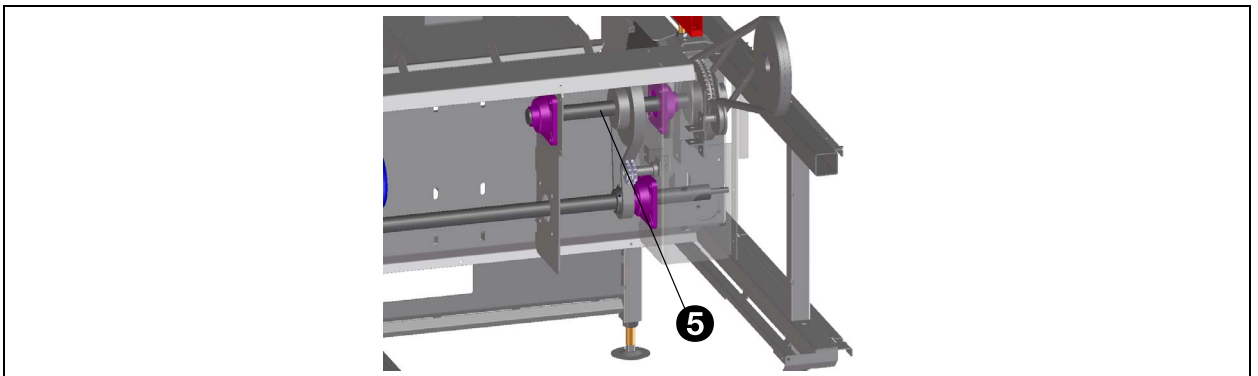
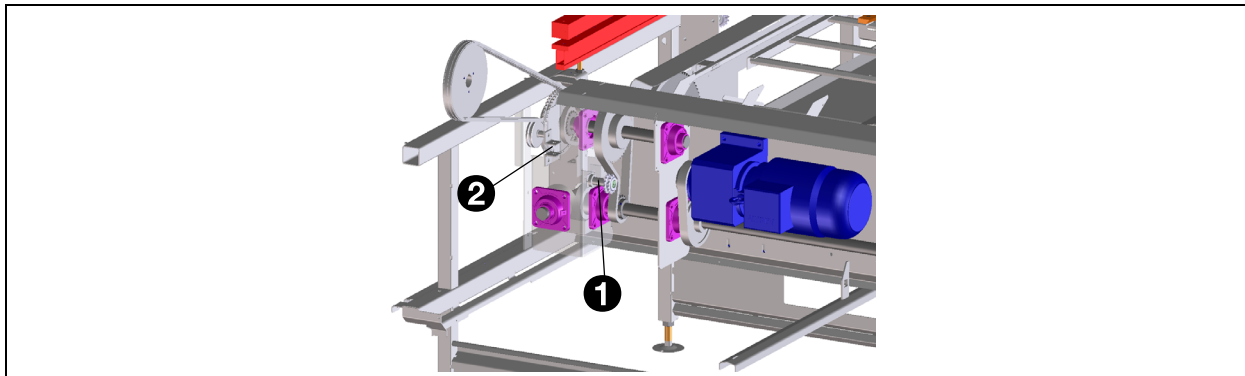


Fig. 97 - Drive of infeed-unit and loadcell-unit



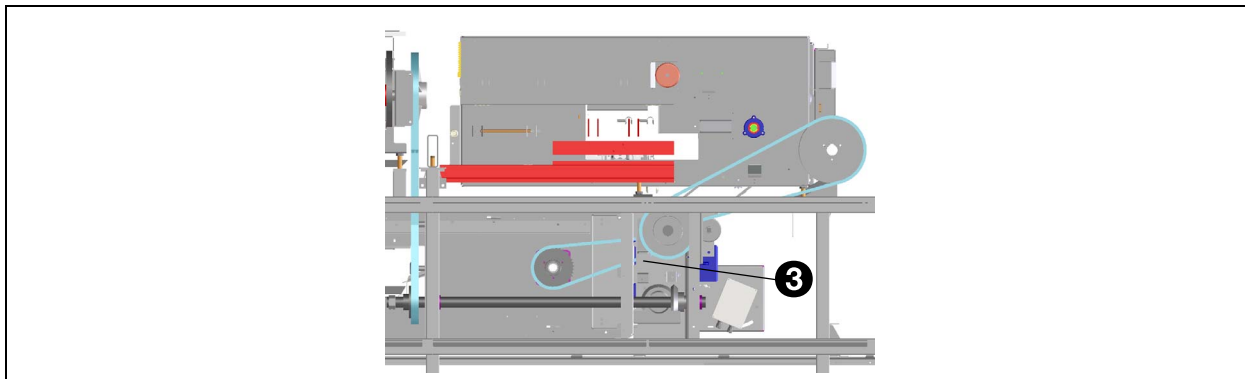
### 8.4.1 Chain tension of infeed unit and Transfer™ drive

The infeed-unit and Transfer™ are driven by the same shaft.



*Fig. 98 - Drive shaft of infeed-unit and Transfer™*

1. Adjust the chain tension of the combined drive with the chain tensioner (1). See fig. 98.
2. Adjust the chain tension of the drive of the infeed-unit with the chain tensioner (2).

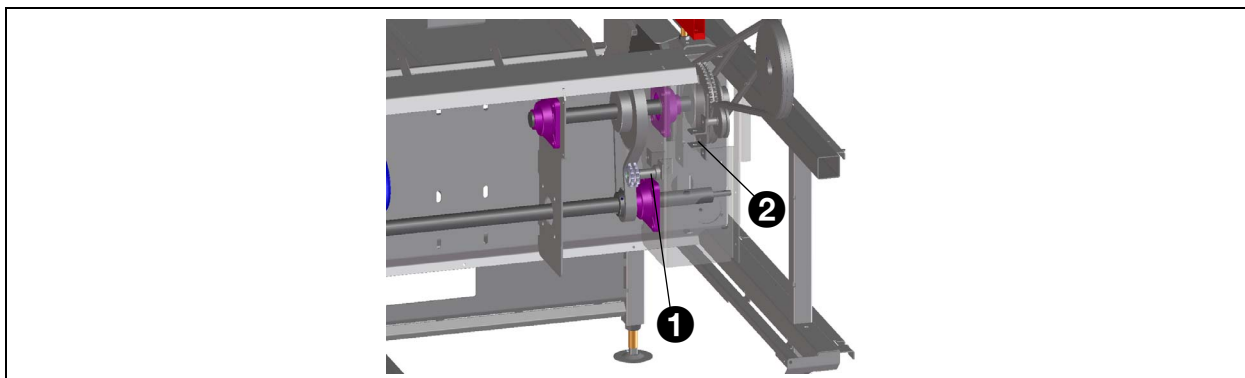


*Fig. 99 - Chain tensioner of Transfer drive shaft™*

3. Adjust the chain tension of the drive of the Transfer™ with the chain tensioner (3). See fig. 99.

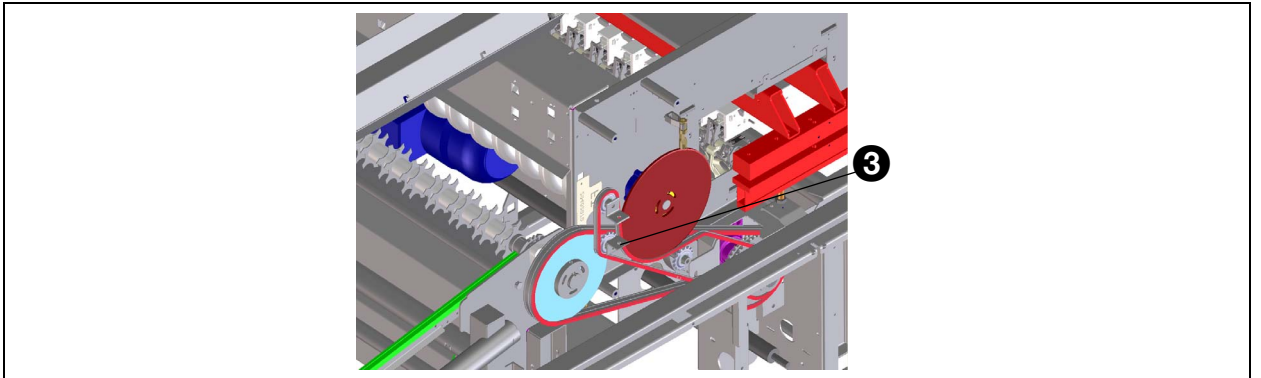
### 8.4.2 Chain tension of infeed unit and loadcell unit

The infeed-unit and loadcell-unit are driven by the same shaft.



*Fig. 100 - Drive shaft of infeed-unit and loadcell-unit*

1. Adjust the chain tension of the combined drive with the chain tensioner (1). See fig. 100.
2. Adjust the chain tension of the drive of the infeed-unit with the chain tensioner (2).



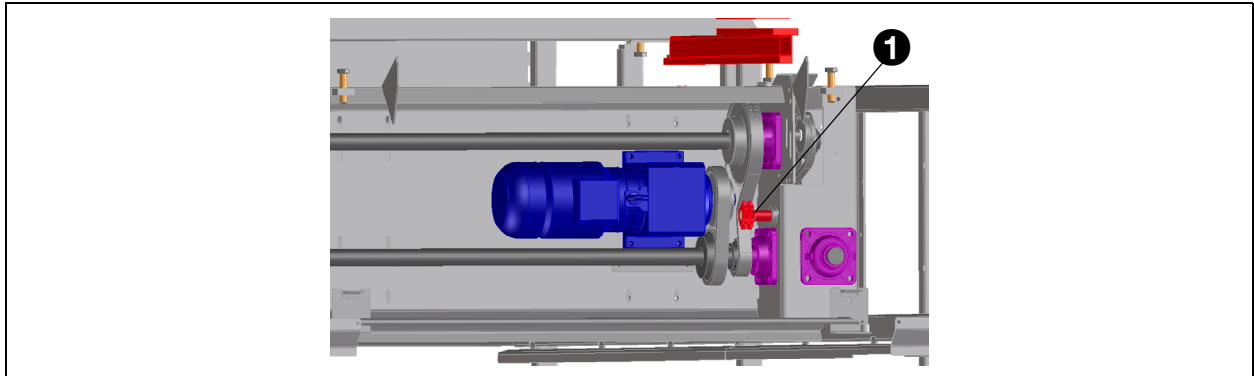
*Fig. 101 - Chain tensioner of loadcell-unit drive shaft*

3. Adjust the chain tension of the drive of the loadcell-unit with the chain tensioner (3). See fig. 101.



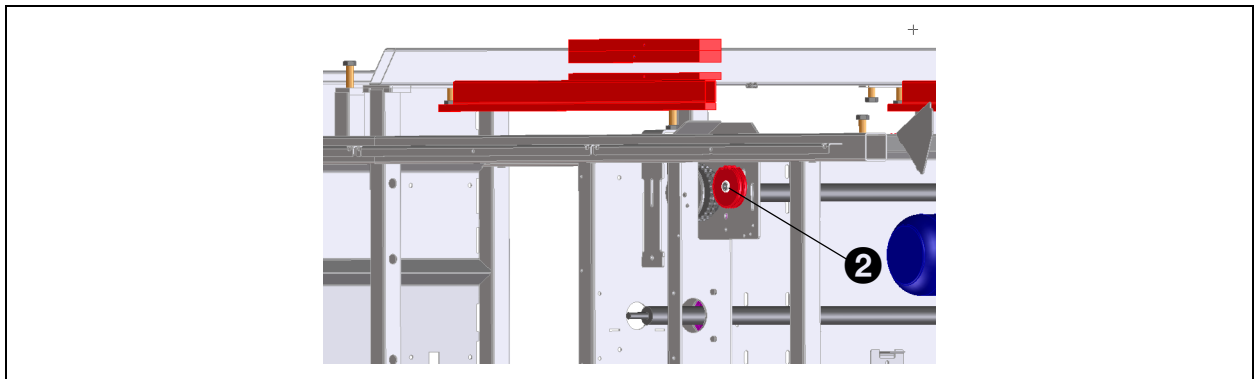
## 8.4.3 Chain tension of infeed unit, Transfer™, and loadcell unit

The infeed-unit Transfer™ and loadcell-unit are driven by the same shaft.



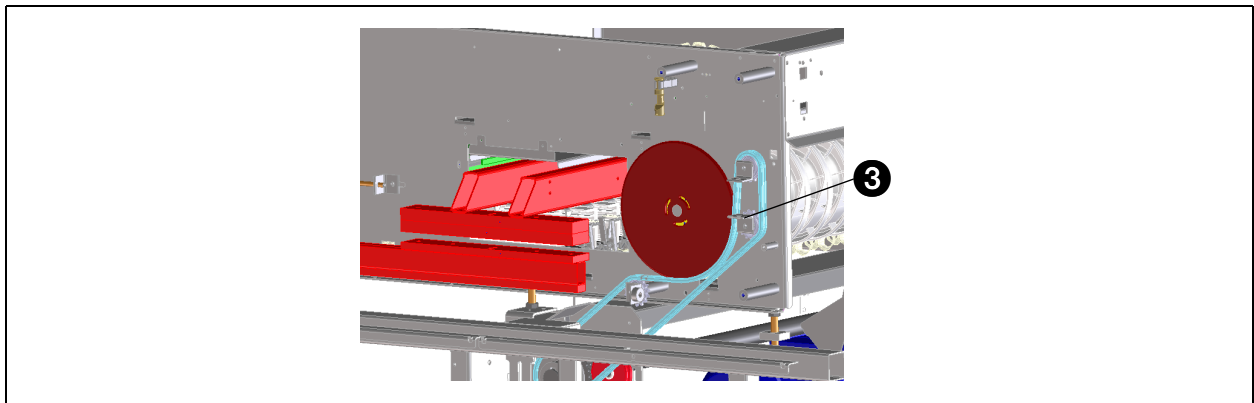
*Fig. 102 - Drive shaft of infeed-unit, Transfer™ and loadcell-unit*

1 Adjust the chain tension of the combined drive with the chain tensioner (1). See fig. 102.



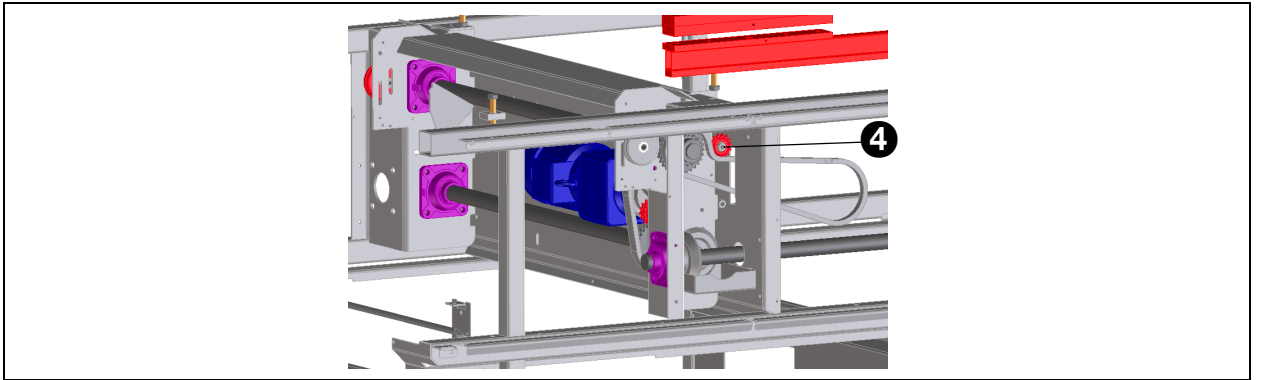
*Fig. 103 - Chain tensioner of infeed-unit drive shaft*

2 Adjust the chain tension of the drive of the infeed-unit with the chain tensioner (2). See fig. 103.



*Fig. 104 - Chain tensioner of loadcell-unit drive shaft*

3 Adjust the chain tension of the drive of the loadcell-unit with the chain tensioner (3). See fig. 104.



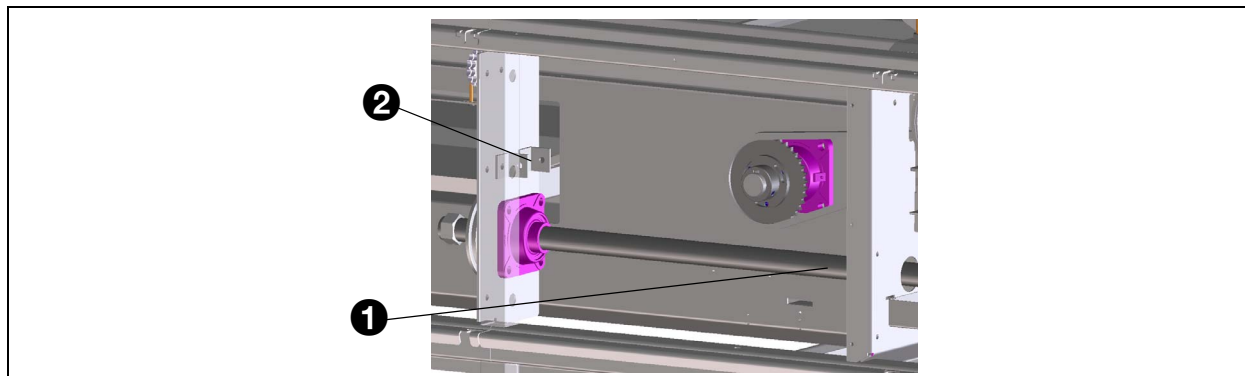
*Fig. 105 - Chain tensioner of Transfer drive shaft™*

1. Adjust the chain tension of the drive of the Transfer™ with the chain tensioner (4). See fig. 105.



#### 8.4.4 Chain tension drive frame

The main shaft in the frame is driven by the right-angled transmission (1) of the Transfer™.

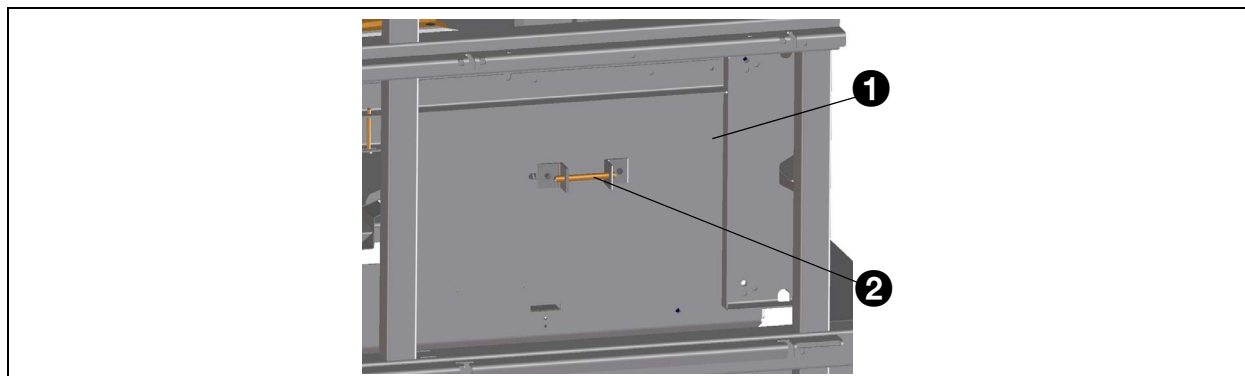


*Fig. 106 - Drive of the frame*

1. Adjust the chain tension of the drive of the frame with the chain tensioner (2). See fig. 106.

#### 8.4.5 Chain-tension of Transfer™

The chains that transport the transfer bridges can be adjusted with the chain-tensioners that are mounted on the side-strips (1) of the Transfer™.



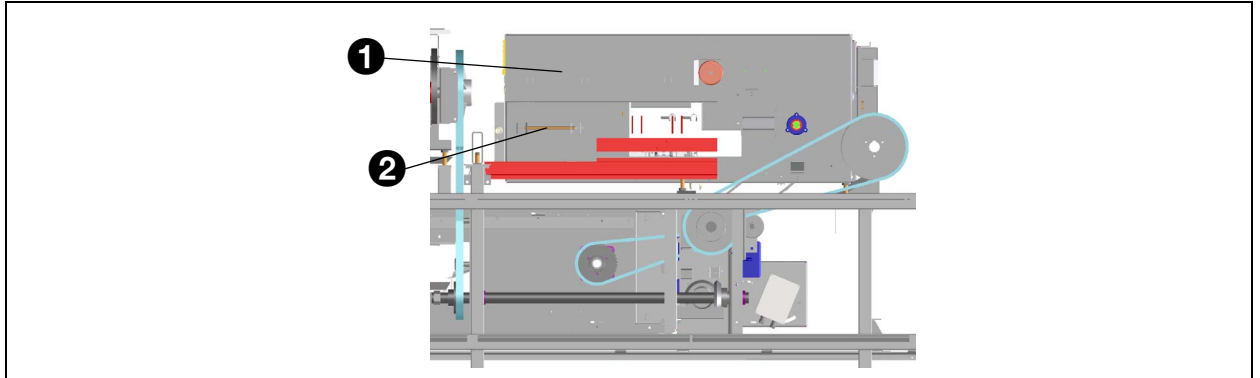
*Fig. 107 - Chain-tensioners of Transfer™*

1. Adjust the chain tension of the transport chains with the chain tensioner (2). See fig. 107.
2. Adjust the chain tensioners on both sides of the Transfer™.
3. In the case of the OMNIA FT 500, a chain tensioner is mounted within the Transfer™.



#### 8.4.6 Chain tension of loadcell-unit

The chains that transport the loadcell-carriers can be adjusted with the chain-tensioners that are mounted on the side-strips (1) of the loadcell-unit.



*Fig. 108 - Chain tensioners of loadcell-unit*

1. Adjust the chain tension of the transport chains with the chain tensioner (2). See fig. 108.
2. Adjust the chain tensioners on both sides of the loadcell-unit.
3. In the case of the OMNIA FT 500, a chain tensioner is also mounted inside the loadcell-unit.





## 9 Adjustment of the Transfer,<sup>TM</sup>

### 9.1 Overview of the adjustment of the Transfer<sup>TM</sup>

The Transfer-unit<sup>TM</sup> transfers the eggs from the infeed-unit to the carriers in the frame. During the processing of the eggs by the Transfer<sup>TM</sup> the eggs are weighed by the loadcell-unit that is mounted on the Transfer-unit<sup>TM</sup>. During the installation of the machine the loadcell-unit and part of the frame are already mounted on the Transfer-unit<sup>TM</sup>. The Transfer-unit<sup>TM</sup> is fully adjusted. The adjustments in this chapter should be used after the replacement of defect parts or for checking after installation.



#### **NOTE**

*The illustrations may be different from your version of the machine; this has no influence on the settings and adjustments of your machine.*

The left and right side are determined on the basis of the direction of the movement of eggs. In fig. 109 the direction of movement of the eggs is indicated with the arrow.

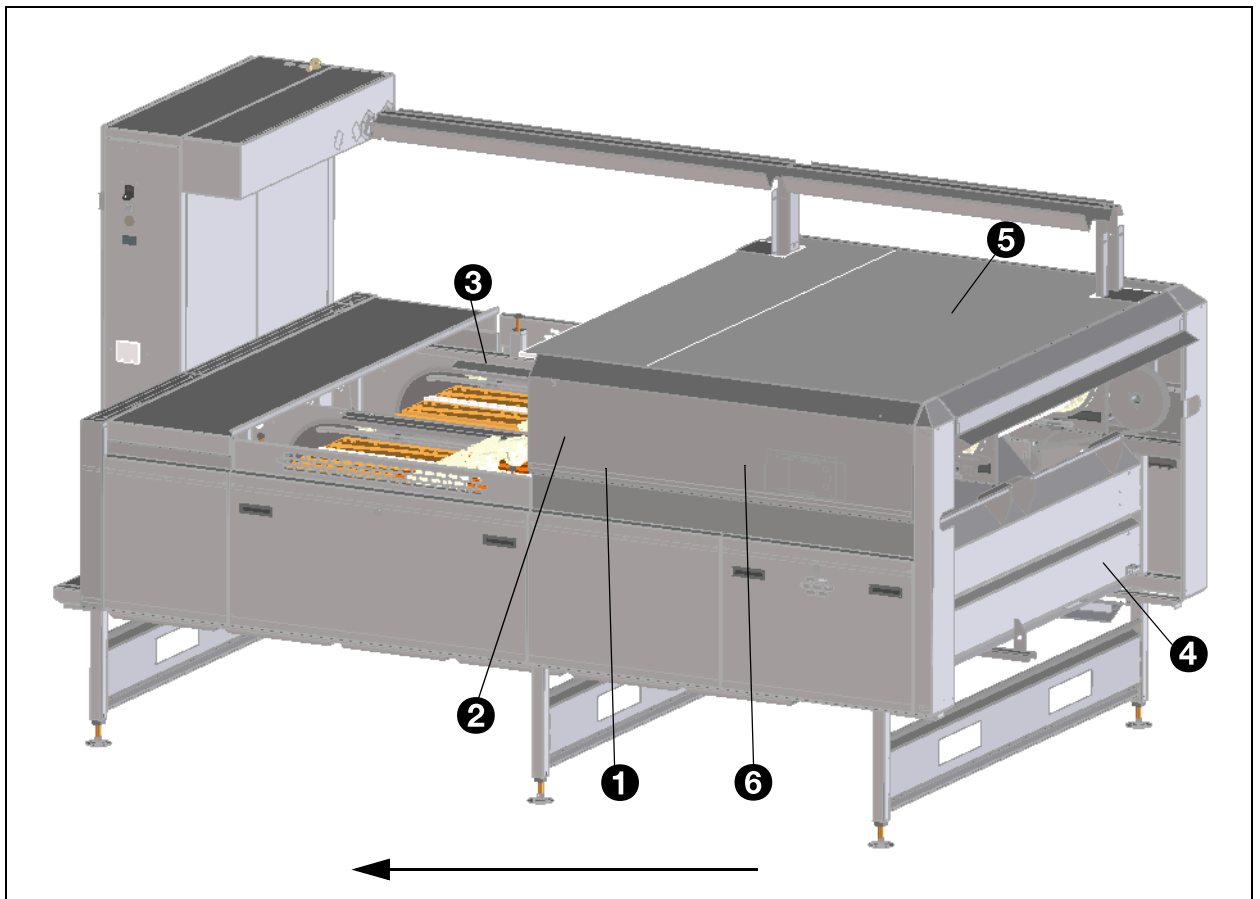


Fig. 109 - Overview of Transfer<sup>TM</sup>

- 1 - Adjusting loadcell-unit - Transfer-unit<sup>TM</sup>
- 2 - Adjusting carrier - transfer bridge
- 3 - Adjusting transfer bridge - frame carrier
- 4 - Fine-tuning synchronisation of Loader<sup>TM</sup>
- 5 - Adjusting loadcell-unit - Main timing disk frame
- 6 - Calibrating the loadcells



## 9.2 Adjusting loadcell-unit - Transfer-unit™

The height between the loadcell-unit and the Transfer-unit™ is important for the correct transfer of the eggs.

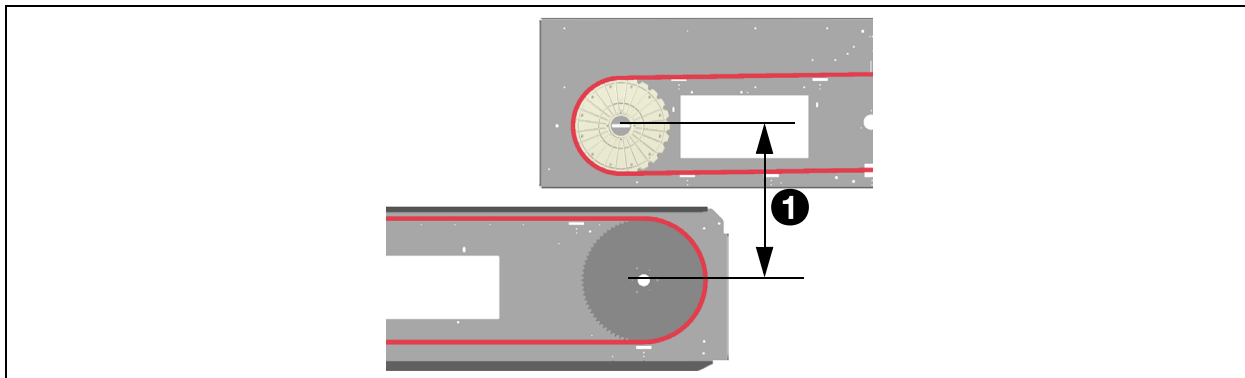


Fig. 110 - Centre-to-centre distance loadcell-unit - transfer-unit™

1. The distance (1) between loadcell-unit and Transfer-unit™ should be 506 to 507 mm.

## 9.3 Adjusting loadcell carrier in relation to the transfer bridge

The distance between the loadcell carrier and the transfer bridge is adjusted as follows.

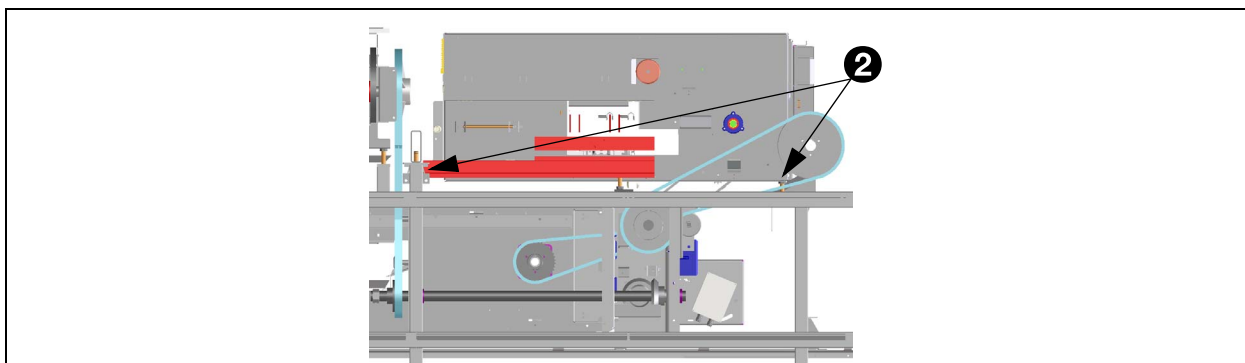


Fig. 111 - Adjustment bolts loadcell-unit - Transfer-unit™

1. Using the bolts (2) between the loadcell-unit and the frame of the Transfer-unit™ adjust the distance (3) between the carrier and the transfer bridge to 1 to 1.5 mm.

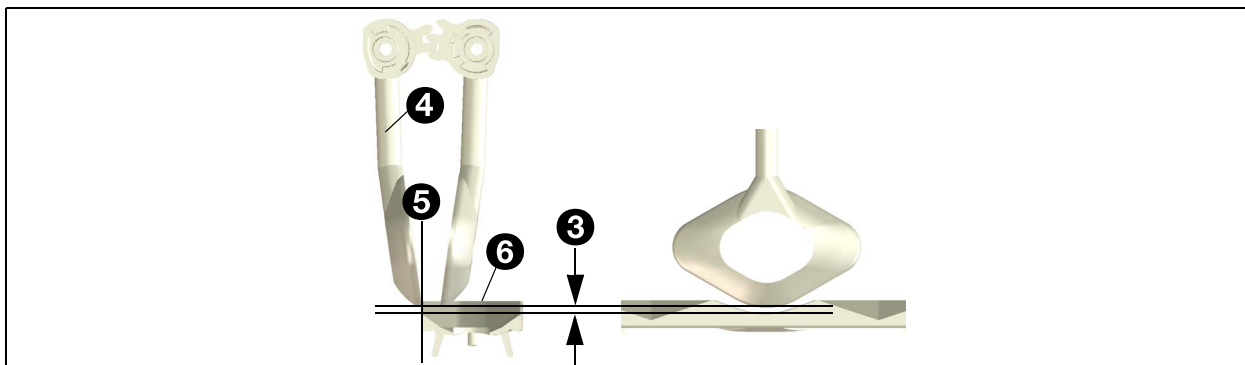


Fig. 112 - Position of the carrier in relation to the transfer bridge.



2. Just as the carrier opens, the inside of the carrier arm (4) must be in line (5) with the front of the transfer bridge (6).

**NOTE**

*The loadcell unit must be level.*

## 9.4 Disassembly transferbridge curveplate (conversion)

The curveplate ensures the raising of the transferbridge. The curveplate is changed. An existing machine can be performed with this new curveplate.

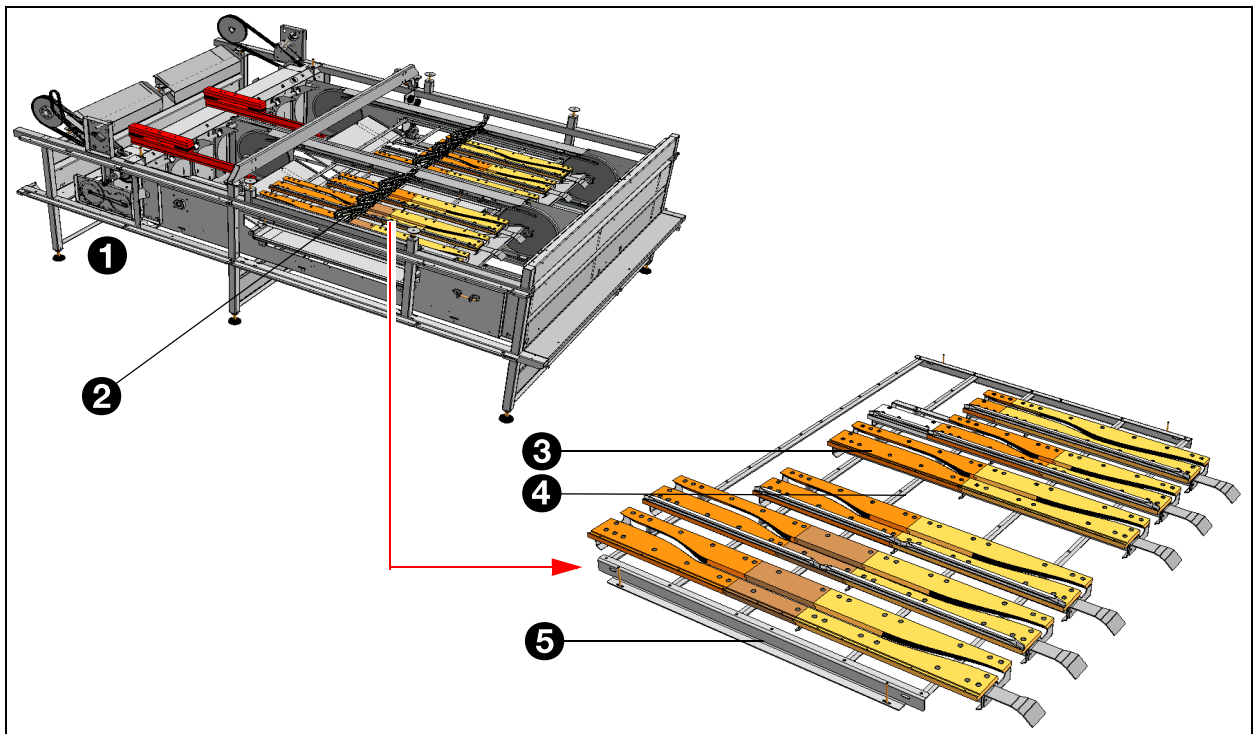


Fig. 113 - Omnia transfer

- 1 - Transfer
- 2 - Transferbridge
- 3 - Transferbridge curveplates
- 4 - Shaft
- 5 - Adjusting plate

### Disassembly transferbridge curveplates:

- 1 Disassemble all transferbridges.
- 2 Disassemble the transferbridge curveplates.
- 3 Disassemble the shafts.
- 4 Disassemble the adjusting plate.



### 9.4.1 Assembly transferbridge curveplates

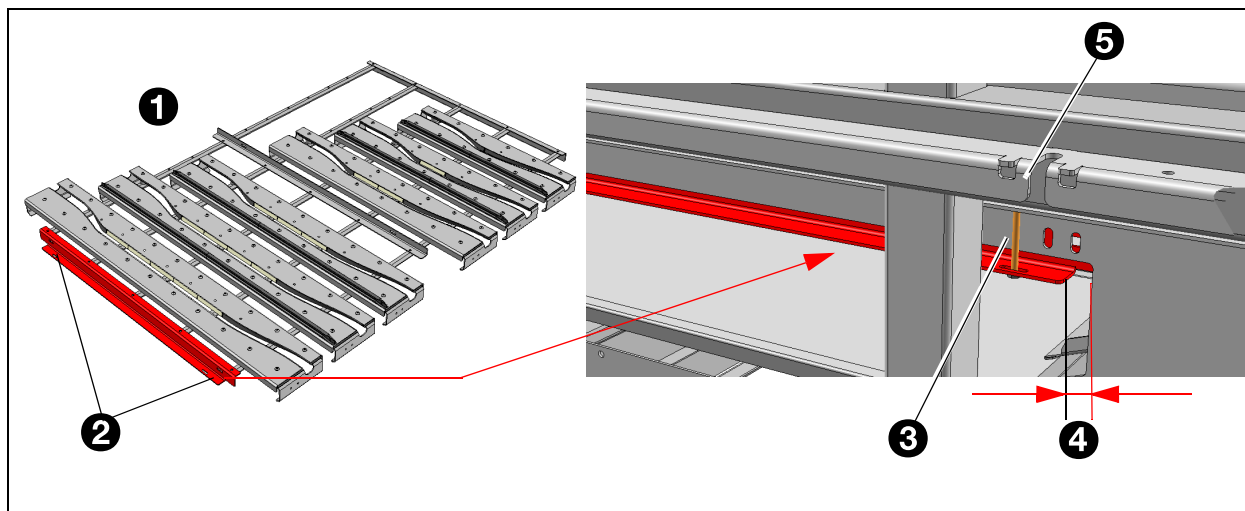


Fig. 114 - Assemble transferbridge, curveplates, shafts and adjusting plate

- 1 - Composition transferbridge curveplates
- 2 - Slot holes adjusting plate
- 3 - Set bolt (4x)
- 4 - Preset distance, adjusting plate - frame = 18 mm.
- 5 - New holes drilled for bolts of the adjusting plate

**NOTE**

*The adjustments apply to new machines and in case of conversion of the curveplates.*

**Assembly transferbridge curveplates:**

- 1 Drill new holes (4X) in the frame for the set bolts of the adjusting plate. (Only for conversion)
- 2 Install the new adjusting plate.
- 3 Install the new shafts.
- 4 Move the adjusting plate in the slot holes until the distance: Adjusting plate - frame = 18 mm.
- 5 Install the transferbridge curveplates.
- 6 Adjust the adjusting bolts (4x) until the distance transferbridge - curveplates is as shown in paragraph 9.4.2.
- 7 Install all transfer bridges.

**NOTE**

*Take care during assembly of the transferbridges the chain snapper is not mounted on the chain link.*

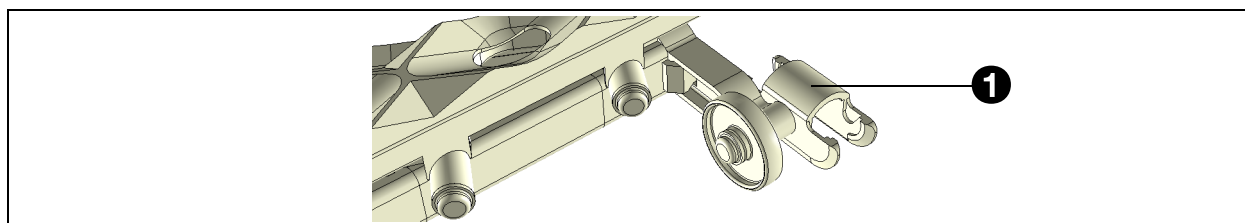


Fig. 115 - Transferbridge

- 1 - Chain snapper



### 9.4.2 Distance transferbridge - curveplate

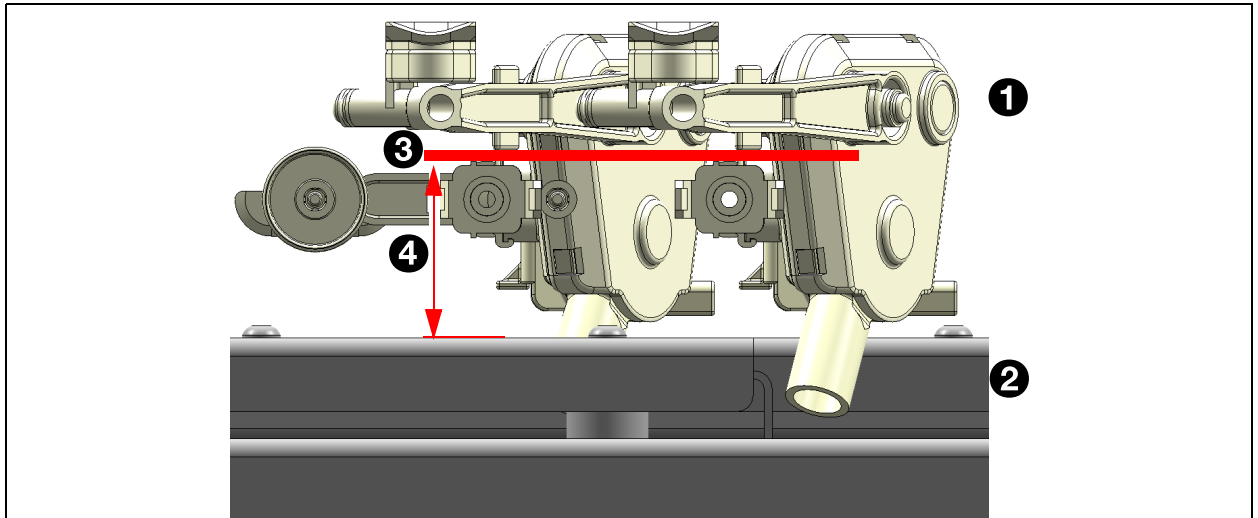


Fig. 116 - Distance transferbridge - curveplate

- 1 - Transferbridge
- 2 - Curveplate
- 3 - Ruler
- 4 - Distance transferbridge - curveplate = 68 mm.



#### **NOTE**

Adjust with the set bolt of the adjusting plate (zie fig. 114) the distance transferbridge - curveplate to 68 mm.

### 9.4.3 Adjustment curveplate

The curveplates has to be adjusted for a good transport of the eggs from the transferbridges to the frame carriers.

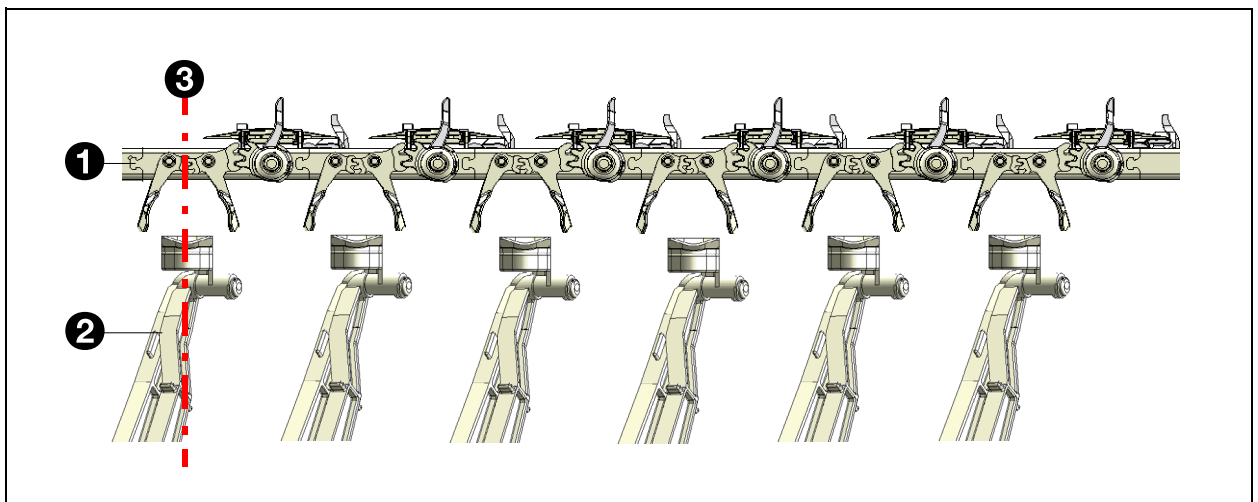


Fig. 117 - Adjustment transferbridge curveplate

- 1 - Frame carrier
- 2 - Transferbridge
- 3 - Transferbridge in the middle of the frame carrier

#### **Adjustment curveplate:**

- 1 Set the transferbridge at its highest position by using the TIP box.



- 2 Move the complete assembly of the curveplates until the transfer bridge is in the middle of the carrier.
- 3 Check this for all the other tracks, any deviation must be divided by moving the complete assembly of the curveplates.

## 9.5 Adjusting the transfer bridge in relation to the frame carrier

The transfer bridge transfers the eggs to the carriers in the frame.

### 9.5.1 Adjust Transfer in the middle of the carrier

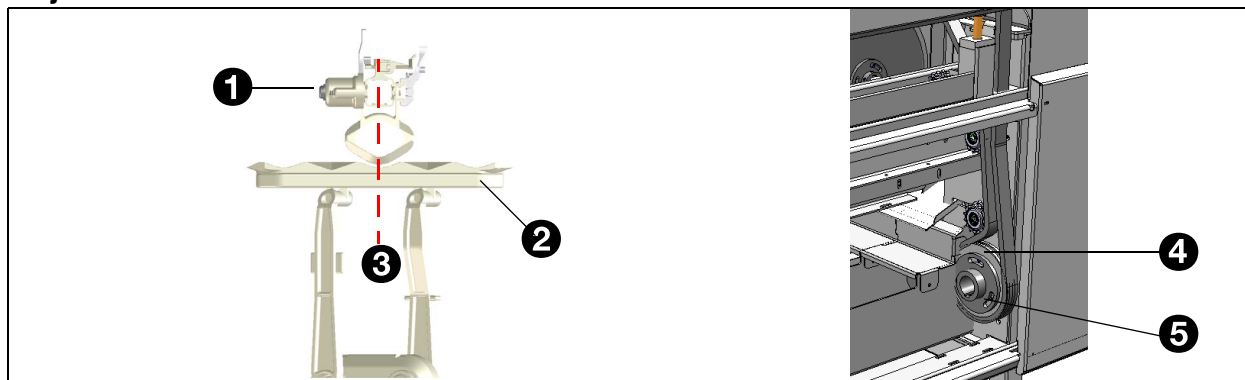


Fig. 118 - Transfer bridge in the middle of the carrier

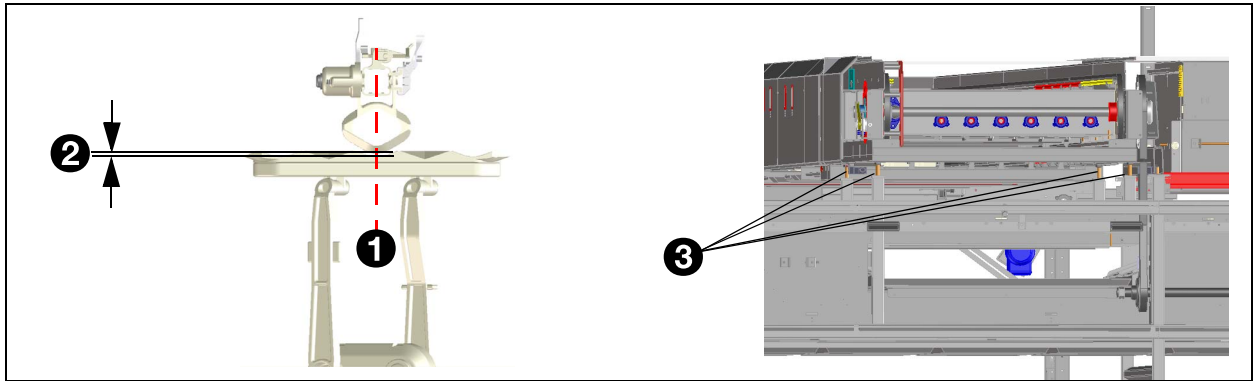
- 1 - Frame carrier
- 2 - Transfer bridge
- 3 - Transfer bridge in middle of the carrier
- 4 - Main drive chain wheel
- 5 - 3 bolts main drive chain wheel

#### Adjust transfer bridge in middle of carrier:

- 1 Set the transfer bridge at its highest position by using the TIP box.
- 2 Loosen the 3 bolts of the main drive chain wheel.
- 3 Turn the main chain with the carriers by hand until a frame carrier is in the centre of the transfer bridge.
- 4 Tighten the 3 bolts of the main drive chain wheel.



### 9.5.2 Adjust the distance transfer bridge and frame carrier



*Fig. 119 - Adjusting the height*

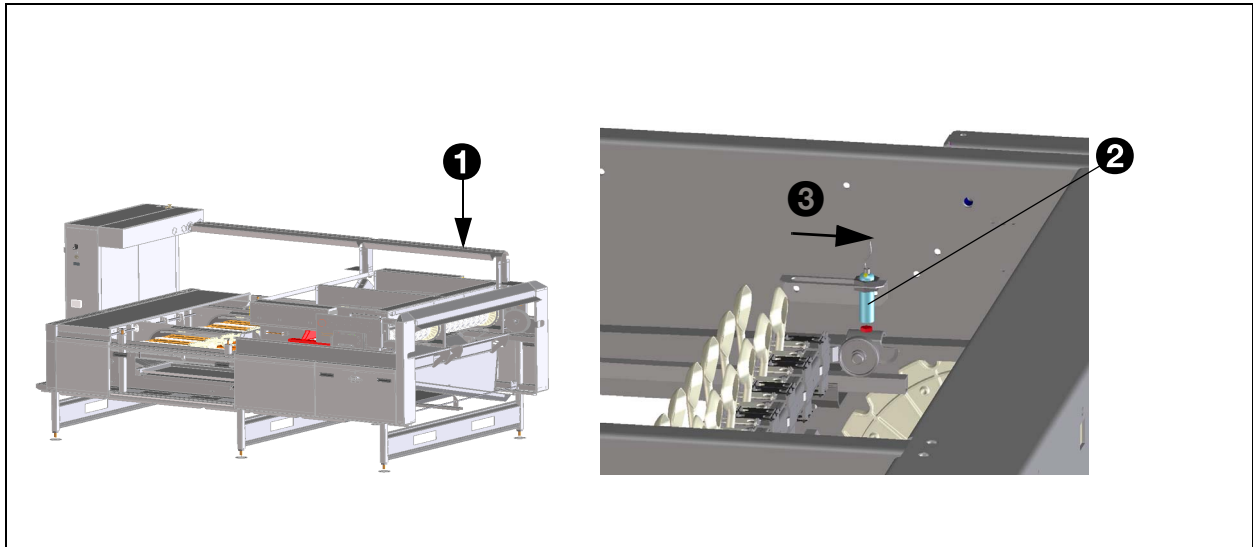
- 1 - Transfer bridge in middle of the carrier
- 2 - Distance transfer bridge - frame carrier = 2 mm
- 3 - Bolts in transfer frame

#### **Adjust the distance transfer bridge - frame carrier:**

- 1 Check the distance between the transfer bridge and the frame carrier.
- 2 Set the height to 2 mm on top of the transfer bridge using the bolts in the frame of the transfer.

## 9.6 Adjusting sensor 'loadcell-carrier 0'

A sensor is mounted at the top (1) of the loadcell-unit for detecting the 'loadcell-carrier 0'.



*Fig. 120 - Location of sensor 'loadcell-carrier 0'*

- 1 - Top of the Loadcell unit
- 2 - Sensor 'loadcell-carrier 0'
- 3 - Slotted hole

### Adjustment of the 'Loadcell-carrier 0'

- 1 Mount the sensor (2) in the outermost position of the slotted hole towards (3) infeed-unit. The loadcell-carriers move at the top of the loadcell-unit towards the infeed-unit.



## 9.7 Control of loadcell

The loadcell-unit consists of a loadcell-interface, loadcells and loadcell-carriers. If one of these parts is replaced then the entire system must be calibrated. The calibration must be performed in the following order.

- Calibration of the loadcells.
- Calibration of the loadcell carriers.

To perform the calibration procedure we use the 'carwgpu.exe' program installed on the 'Main PC'.



### **NOTE**

***Make sure that the machine is operational before you start with the calibration procedure.***

### 9.7.1 Loadcell calibration

Loadcell calibration is a separate application within the Omnia Master Control program. The loadcell calibration application should only be activated when the machine is stopped and when loadcell calibration is to be performed.

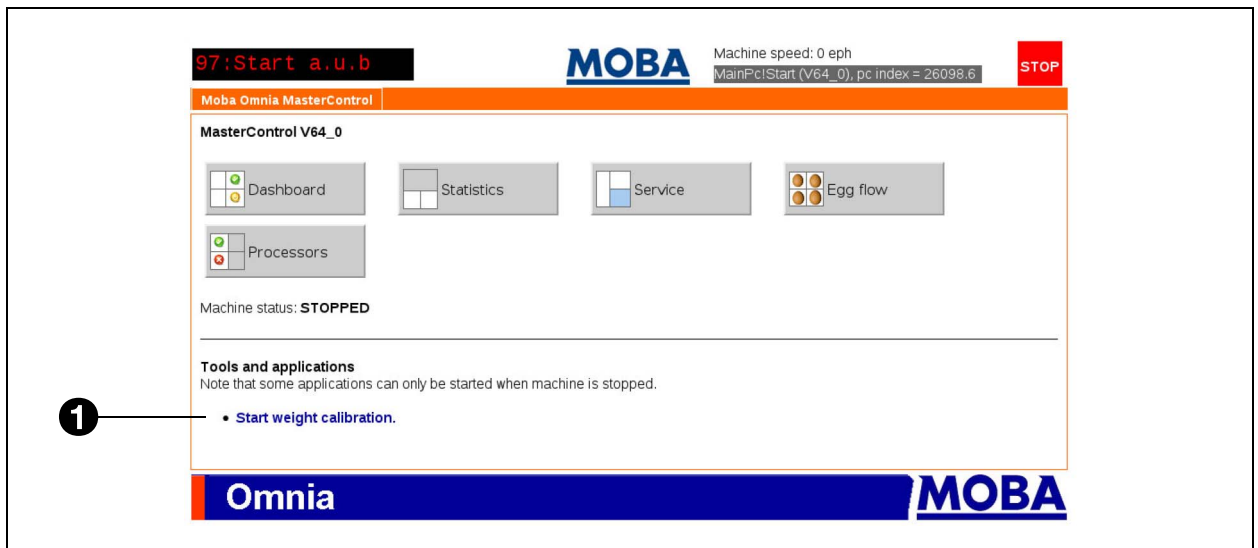


Fig. 121 - Omnia Master Control  
1 - Start weight calibration



### 9.7.2 Loadcell calibration

The loadcells are calibrated with a running machine.

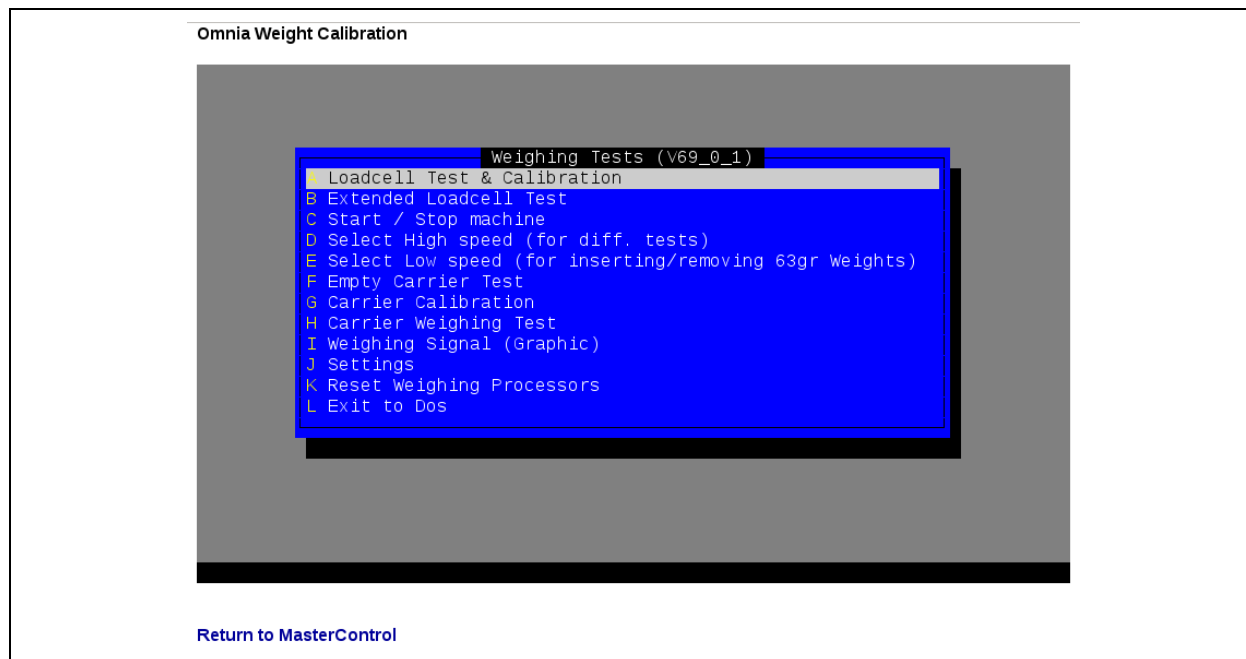


Fig. 122 - Loadcell calibration

#### Calibration of the loadcell:

- 1 Go to 'Start weight calibration' on the Omnia Master Control PC. See fig. 121.
- 2 Press [ENTER].  
The loadcell calibration program is now started.
- 3 Set the machine to the highest speed.
- 4 Go to 'Start / Stop machine' and press [ENTER].
- 5 Go to 'Start machine' and press [ENTER].
- 6 Press [ESC].
- 7 Go to 'Empty Carrier Calibration' and press [ENTER].  
The program carries out the loadcell compensation and calibrates the unit.  
Allow the machine to run until the 'Empty Carrier Calibration Ready' message is displayed after 10/10 cycles (press 'ESC').
- 8 The calibration value is compared with the old value for each loadcell.  
When these values differ then the 'Accept new Calibration value for row xx' message is displayed.
- 9 When all carriers are 'OK' [ENTER] can be pressed to accept the new values.
- 10 When 1 or more carriers in each row generate an error 'E' then the message 'One or more carriers not OK. Are you sure to accept new value' is displayed on the screen.
- 11 Press 'ENTER' for the old value, and use the right arrow and 'ENTER' to accept the new value. This is for each loadcell.
- 12 Repeat steps 7 - 9 until all rows have been accepted.
- 13 The empty loadcells have now been calibrated.



#### **NOTE**

**When all new calibration values are equal to the old values the program will skip steps 7 - 10.**



### 9.7.3 Loadcell carrier calibration

The loadcell carriers are calibrated with a running machine.

**Special tool:**

MOBA calibration weight, plastic, 63 gram. Article number:80206980.

**Calibrate the loadcell carriers:**

- 1 Go to 'Start weight calibration' on the Omnia Master Control PC. See fig. 121.
- 2 Press [ENTER].  
The loadcell calibration program is now started. See fig. 122.
- 3 Set the machine to the highest speed.
- 4 Go to 'Start / Stop machine' and press [ENTER].
- 5 Go to 'Start machine' and press [ENTER].
- 6 Press [ESC].
- 7 Go to 'Empty carrier test' and press [ENTER].  
Let the machine run until the loadcell carriers have been weighed ten times.
- 8 Go to 'Select low speed (for inserting/removing 63gr Weights)' and press [ENTER].
- 9 Place the calibration-weight in the loadcell carriers.
- 10 Go to 'Select High speed (for diff tests)' and press [ENTER].
- 11 Go to 'Carrier Calibration' and press [ENTER].  
The message 'Run at high speed' appears.
- 12 Press [ENTER].
- 13 Wait until the message appears: 'Calibration ready'.  
The message 'Save the changes' appears.
- 14 Press [ENTER].  
The calibration procedure for the loadcell-carriers has now been completed. (1 row is now ready).
- 15 Go to 'Select low speed (for inserting/removing 63gr Weights)' and press [ENTER].
- 16 Remove the calibration eggs.
- 17 Go to 'Start / Stop machine'.
- 18 Go to 'Stop machine' and press [ENTER].
- 19 Go to 'Exit to DOS' and press [ENTER].
- 20 Press [ENTER] again to confirm.
- 21 Enter 'Main PC' and press [ENTER].
- 22 Check whether all processors start up. See paragraph 9.7.4.
- 23 The machine is now ready for use.



## 9.7.4 Check of processors

After performing the calibration procedure the processors are checked by the 'mainpc' program. The message 'PrOk' (processor OK) or PrEr' (processor error) appears at the top-right of the screen.

1. In the case of the message 'PrEr', press [F8].  
The processor screen appears.
2. With the arrow keys go to the processor with the error message.
3. Press [S] (start) to restart the processor.
4. Press [R] (reset) if the processor **does not** restart.

If the processor does not restart after resetting, contact the service helpdesk of MOBA b.v.



### 9.7.5 Illuminated ticker-tape messages during the calibration.

During the calibration messages are displayed on the machine's banner. These messages relate to the calibration procedure.

Table 4 - Illuminated banner messages during the calibration.

| Message                     | Description                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Weighing test & calibration | CarWgPc program active.                                                                                                                                                                            |
| Loadcell test               | Loadcell test and calibration active.                                                                                                                                                              |
| Place Calib.Wgt             | Place the 130-gram calibration weight on a loadcell.                                                                                                                                               |
| Ic xx Calib.busy            | Loadcell xx is calibrated with the 130-gram weight (xx is the number of the loadcell).                                                                                                             |
| Ic xx Calib. OK             | Loadcell xx is calibrated and approved. The 130-gram calibration weight can be placed on the following loadcell.                                                                                   |
| Ic xx Calib.Adj.            | Calibration ready, the old calibration factor must be changed for loadcell xx. See the information on the CARWGPC screen. The 130-gram calibration weight can be placed on the following loadcell. |
| Extended Ic-test            | Extended loadcell test active.                                                                                                                                                                     |
| EmptyCarrierTest            | 'Empty Carrier Test' active.                                                                                                                                                                       |
| CarrierCalib.               | Loadcell carrier calibration active.                                                                                                                                                               |
| CarrierWeighTest            | Loadcell-carrier test active.                                                                                                                                                                      |
| WeighingSignal              | Loadcell-signal active.                                                                                                                                                                            |
| CarWgPc Ready               | The CARWGPC-program will be terminated.                                                                                                                                                            |



#### **NOTE**

*During the use of the CarWgPc program, the message for an emergency stop is not shown on the banner. The machine can only be started/stopped with the keyboard. Any messages appear on the screen of the CarWgPc program.*



## 9.8 Adjusting weighing moment

The Main timing disk mounted on the drive shaft of the frame must be adjusted in relation to the position of the 'loadcell carrier 0' of the loadcell-unit.

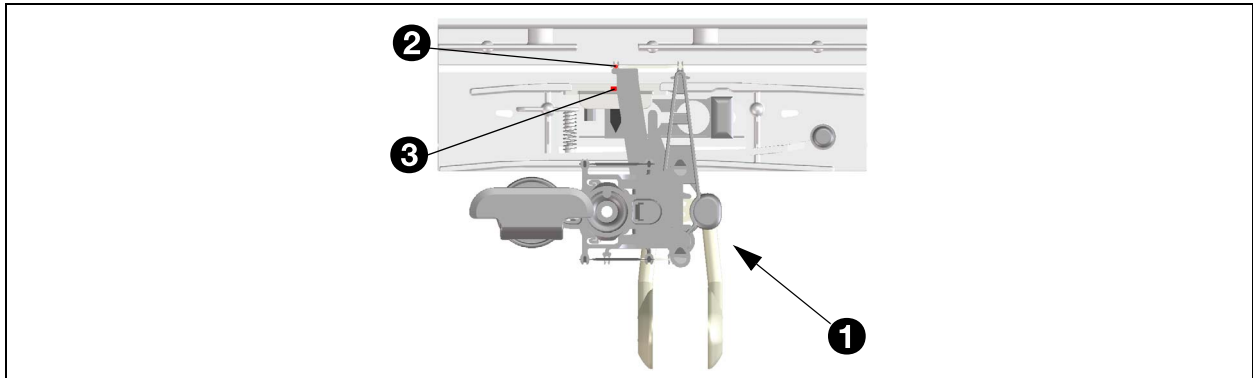


Fig. 123 - 'Loadcell carrier'

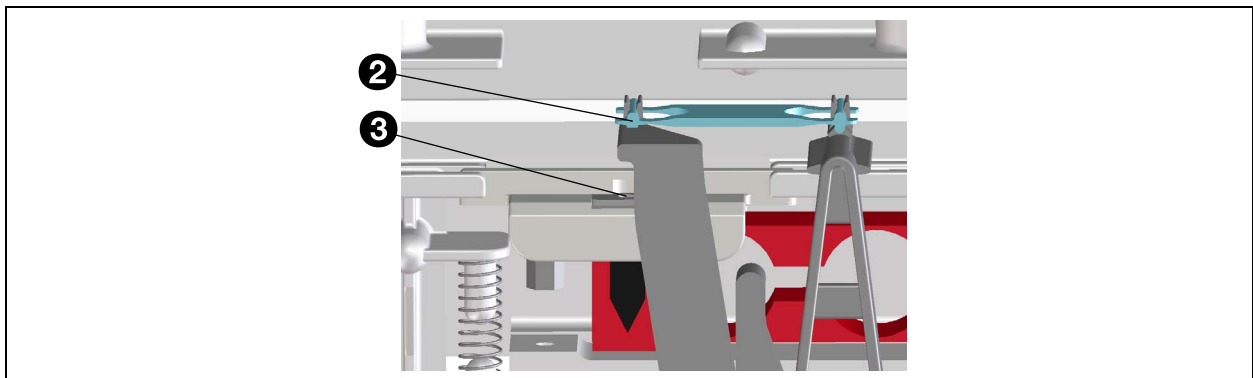


Fig. 124 - Detail of position of loadcell carrier

1. Place the loadcell carrier (1) with the first confirmation point (2) of the white line above the loadcell (3). See fig. 123 and the detailed illustration in fig. 124.
2. Loosen bolts of the Main timing disk (4). See fig. 125.
3. Turn the Main timing disk until the sensor (5) just lights up. The sensor detects the innermost two holes of the Main timing disk.



### **NOTE**

**When adjusting the Main timing disk note the rotation direction of the Main timing disk. The rotation direction is indicated with the arrow (6) on the Main timing disk. See fig. 125.**

Now finely tune the Main timing disk with the following procedure.



### 9.8.1 Fine-tuning of weighing moment

To finely tune the weighing moment, we use the graphical illustration in the CarWgPc program.

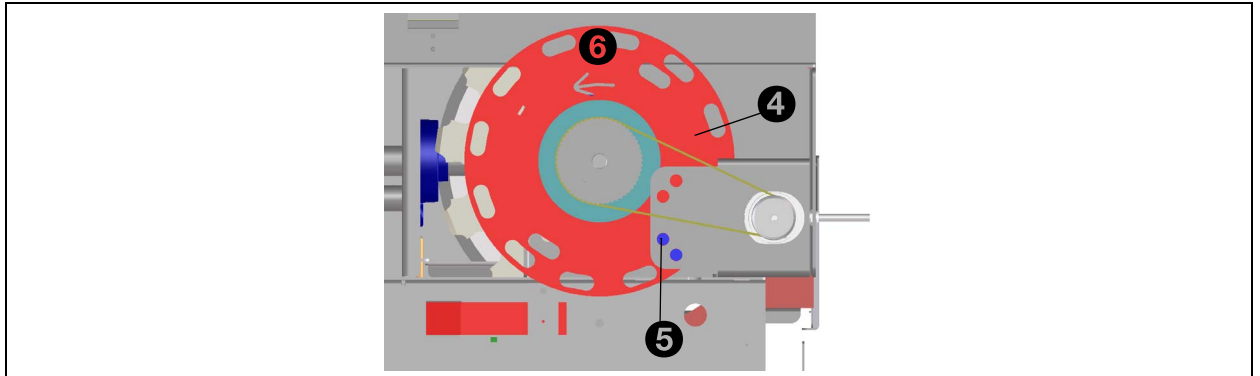


Fig. 125 - Main timing disk

1. Start the machine.
2. Go to 'Weighing Signal (Graphic)'.
3. Press [ENTER].
4. Select a loadcell.
5. Press [G].
6. The dotted line (6) on the value '100' is the switching moment of the sensor (5). See fig. 125 and fig. 126.

This moment must be between two loadcell carriers, one-third before the upward flank (7) and two-thirds after the downward flank (8).

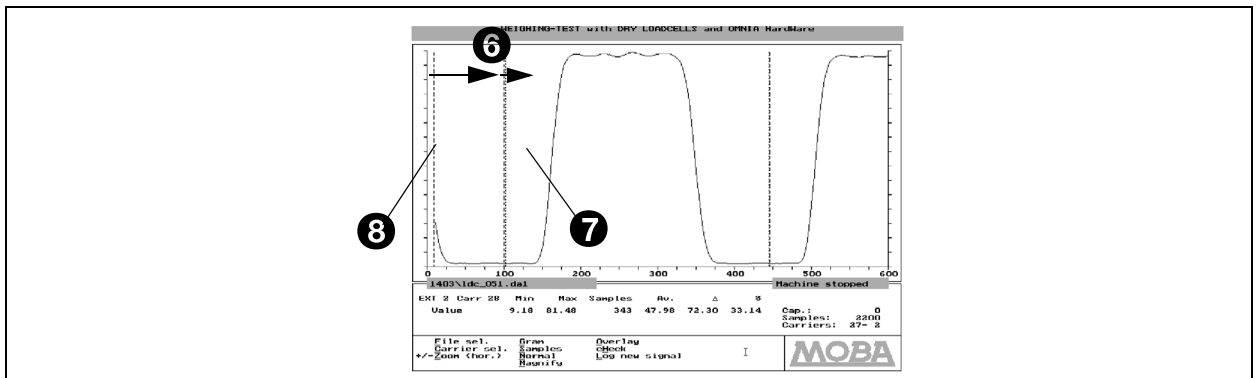


Fig. 126 - Weighing signal

7. Adjust the Main timing disk in such a way that the graphical illustration on the screen corresponds with the illustration shown in fig. 126.

Examine the graphical illustrations of all loadcells, it is possible that one loadcell differs.

Adjust the deviating loadcell by moving it a little in its suspension.

## 9.9 Synchronisation Loader™ - Transfer-unit™

A synchrobox (3) is mounted on the drive shaft of the Loader™ (1) and the Transfer™ (2). The signals from these synchroboxes are sent to the print Synchro Control MA19 (4) in the power-supply cabinet of the Loader™. The phase shifts of the signals are compared to the set machine-speed (5) that is sent to the frequency controller (6) of the Transfer-unit™. When there is a phase shift, a signal is sent to the frequency controller (7) of the Loader™ to increase or decrease the speed. In this way the speeds of both motors are tuned to one another.

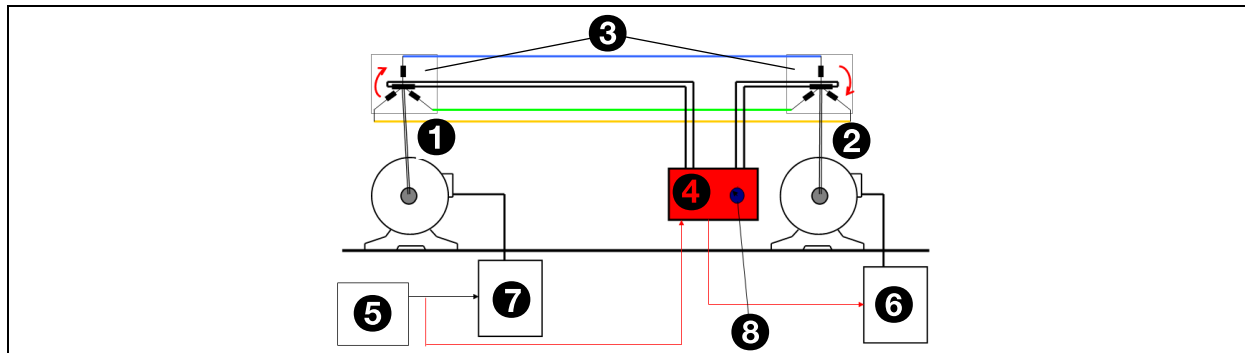


Fig. 127 - Synchro principle

A potentiometer (8) is mounted on the synchronisation print for the fine-tuning of the Loader™.

### 9.9.1 Adjustment of placement on the Loader™

The egg placement on the infeed-rolls by the Loader™ can be adjusted using the synchro adjustment (1) mounted on the Transfer-unit™.

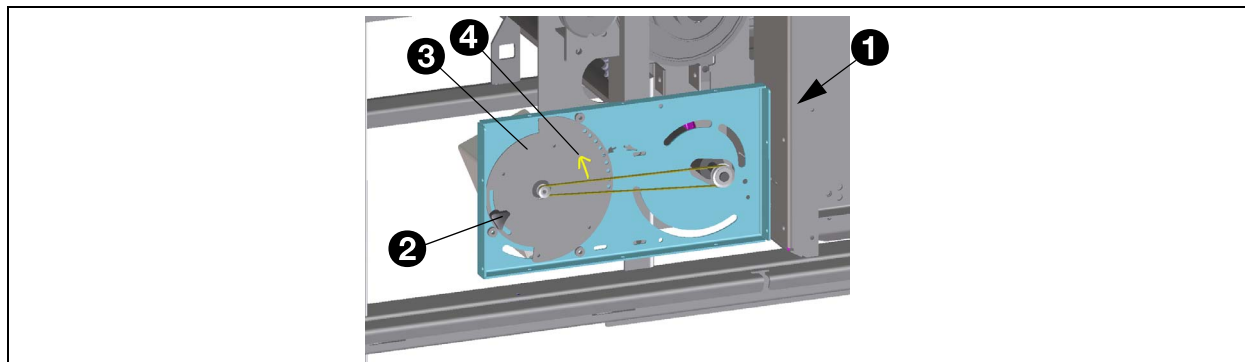


Fig. 128 - Synchro system

1. Loosen the button (2). See fig. 128.

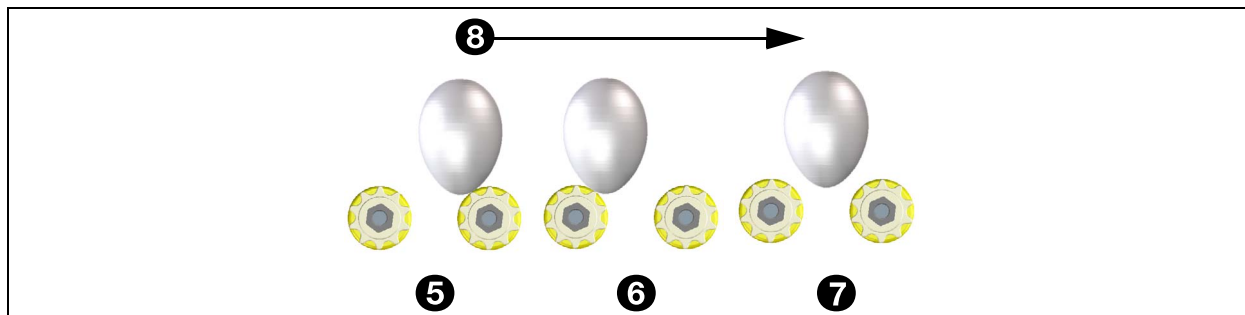


Fig. 129 - Placement by the Loader™

2. Turn the disk (3) to correctly adjust the placement of the eggs.  
 - Too soon (5), turn the disk towards the arrow. See fig. 129.



- Too late (6), turn the disk in the opposite direction to the arrow.
- Right position (7), tighten the button (2).

**NOTE**

***The arrow (4) shows the direction of the movement on the infeed-unit. See fig. 128***

There are a number of holes in the disk, moving by one hole means a 1 mm move in the placement of the eggs on the infeed-unit by the Loader™.

**NOTE**

***The arrow (8) in the illustration shows the direction of the egg flow. See fig. 129***

The toothed-belt of the synchro system must be tight enough to ensure that no teeth jump off the drive-wheel.





## 10 Overview of Frame

### 10.1 Turning by hand and overview of parts of machine

**NOTE**

*The overview lists all the options for this type of machine. The configuration of your machine can differ from this overview.*

#### 10.1.1 Turning by hand

**NOTE**

*The machine must be turned by hand to check a number of settings. To this end a nut is mounted on the Transfer's drive shaft. Use a 46 mm spanner (see fig. 130) to turn the machine by hand in the direction in which the eggs travel.*

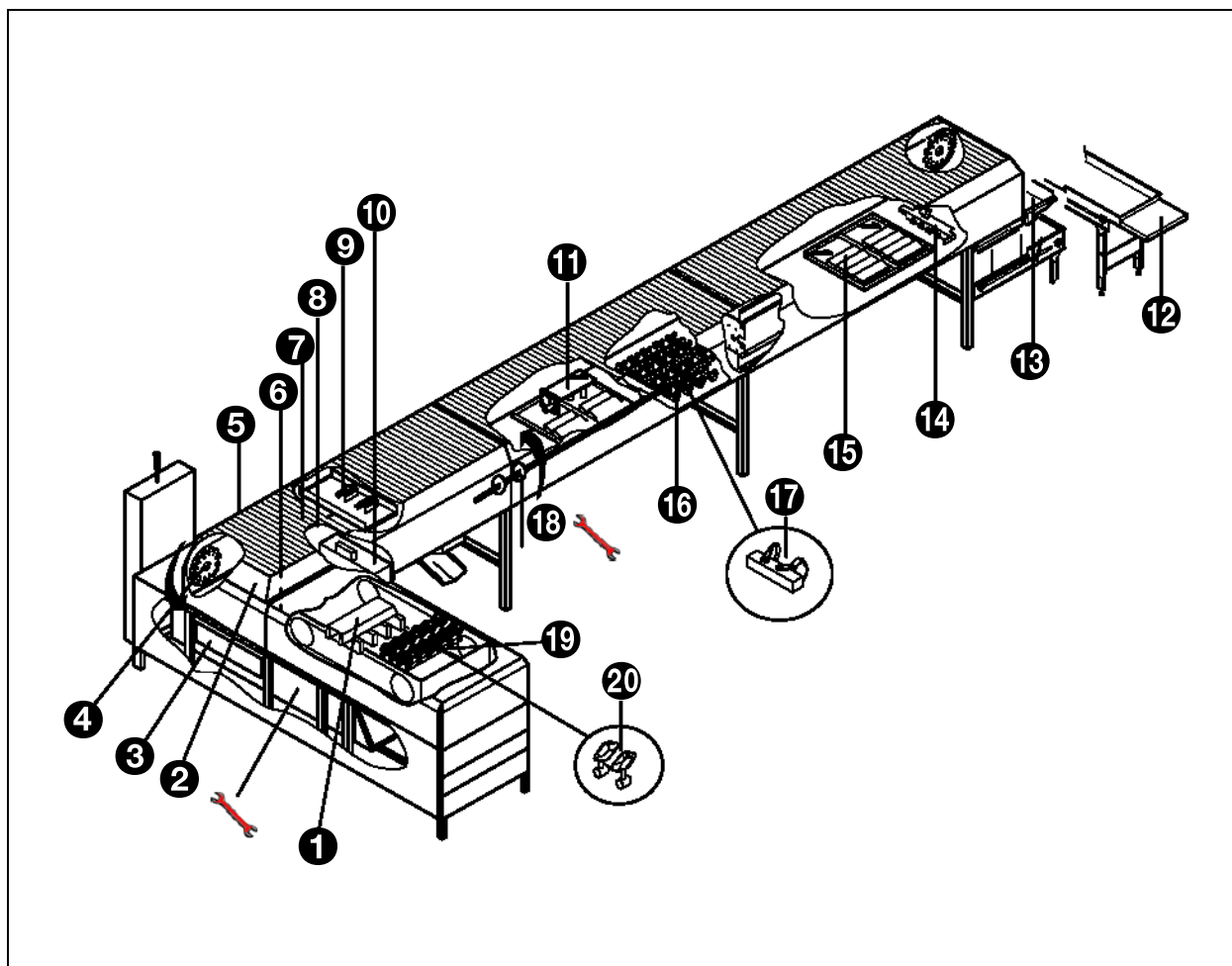


Fig. 130 - Overview of parts of machine

- 1 - Loadcell-unit
- 2 - Tracks
- 3 - Transfer
- 4 - Drive of the frame
- 5 - Main Timing circuit board
- 6 - Main timing disk
- 7 - O-carrier
- 8 - Begin check
- 9 - Inkjet
- 10 - Blood detection unit
- 11 - Linear actuator
- 12 - Egg discharge container take-away unit
- 13 - Egg discharge container frame
- 14 - Final check
- 15 - Release
- 16 - Track carriers
- 17 - Carrier
- 18 - Drive, buffer conveyor
- 19 - Loadcell carrier
- 20 - Carrier



## 10.2 Timing adjustments

This section discusses all timing adjustments from the loadcell unit to the transfer to the carriers in the frame inclusive. Check these settings if there are problems with the change-over of the eggs.

### 10.2.1 Tensioning and alignment of the main chain

1. Tension the chain by moving the 2 loop wheels at the back of the frame by the same distance. Tension the chain by feel. An over-tensioned chain will not turn smoothly. A chain that is too loose will flap just after the drive wheel.
2. Check that the track carriers move perpendicular to the frame. Determine this by checking that the 2 track carriers on top of each other are in line just after the transfer unit. Correct a misaligned main chain with the loop wheel opposite the drive wheel. Loosen the loop wheel's clamping bush and position the loop wheel such that the units are perpendicular to the frame. Tighten the clamping bush and check the setting after the unit has run for some time.

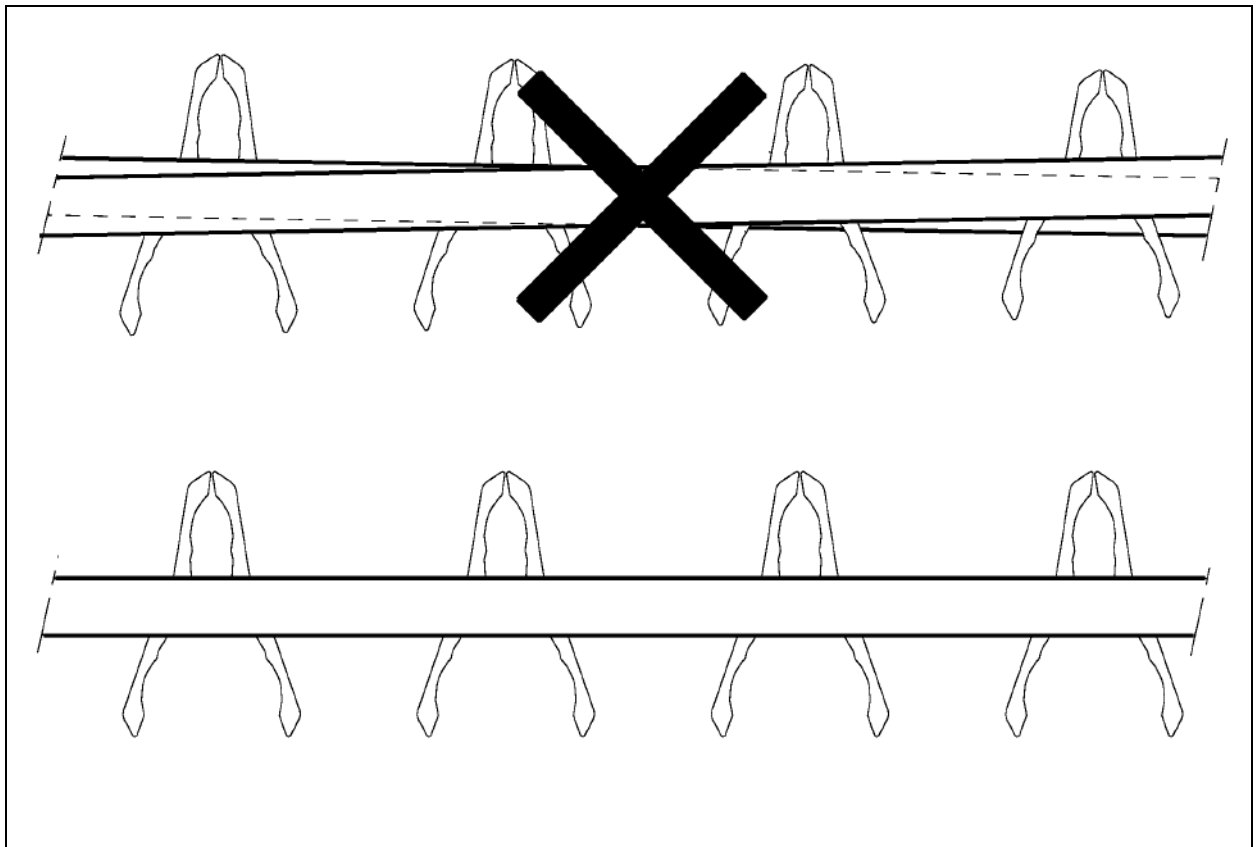


Fig. 131 - Tensioning and alignment of main chain



## 10.2.2 Carriers closer

1. Adjust the length of the arm (2x) on the closer's main shaft in accordance with fig. 132.

**NOTE**

*Note: The dimension is indicative.*

2. Turn the transfer unit by hand to the position in which the closer is in the middle between the track's scoops.
3. Push the carrier bridge as far as possible to the side of the frame, against the guide strip.
4. Turn the machine so that the closers are locked to the maximum extent.
5. Slide the locking flap that is now released towards the closing arm and check that the remaining play is 1 mm.
6. Loosen the chain-wheel fastening bolt (5) on the closer shaft and adjust the closers (2x) in accordance with fig. 133.

**NOTE**

*The closer must be at the point of closing the carriers*

**NOTE**

*Once the individual closers have been adjusted the closing time of all the closers can be adjusted in 1 go with the adjustable chain-wheel (6), see fig. 133. The narrowest carrier must then be set to 1 mm.*

**NOTE**

*Never make the adjustments wide than are necessary, since otherwise excessive vibrations will arise which reduce the accuracy of the loadcell system.*

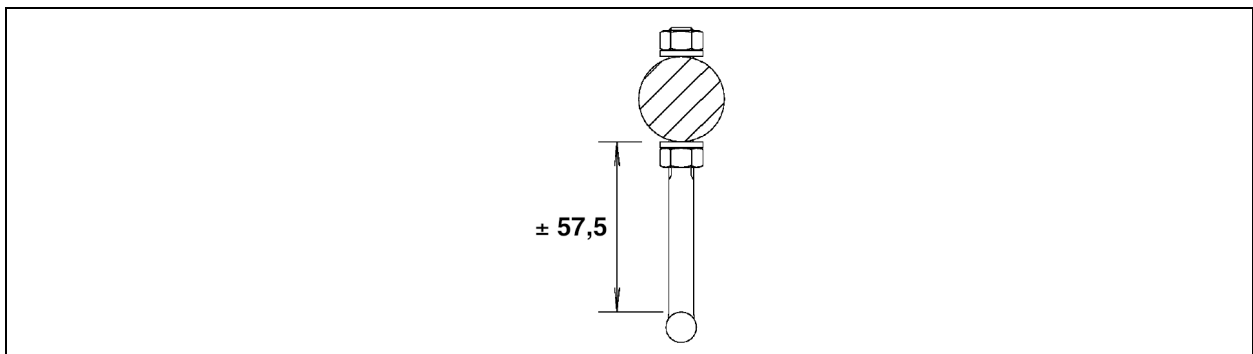


Fig. 132 - Length of arm

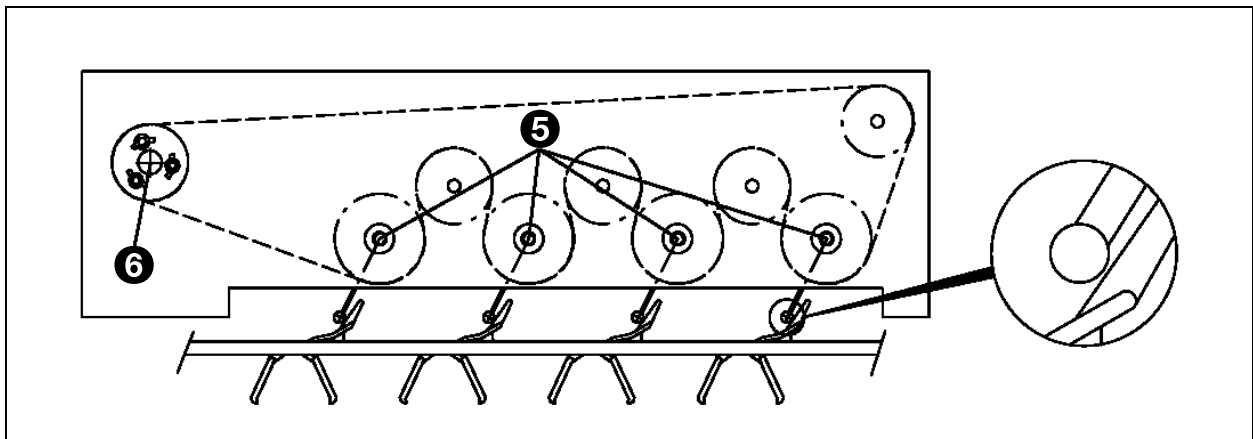


Fig. 133 - Adjustment of transfer unit



## 10.3 Adjustments of the release unit

This section discusses the adjustments of the release unit. The frame is equipped with a linear actuator that positions the release cams on the track carriers.

### 10.3.1 Mounting position of the release sets

**NOTE**

*All release sets are mounted in a fixed position in accordance with fig. 134.*

**NOTE**

*The overall height of 16.5 mm is an indicative dimension. Check the height between the reset arm on the track carrier and the release arms. The setting for this dimension (1) is shown in fig. 134.*

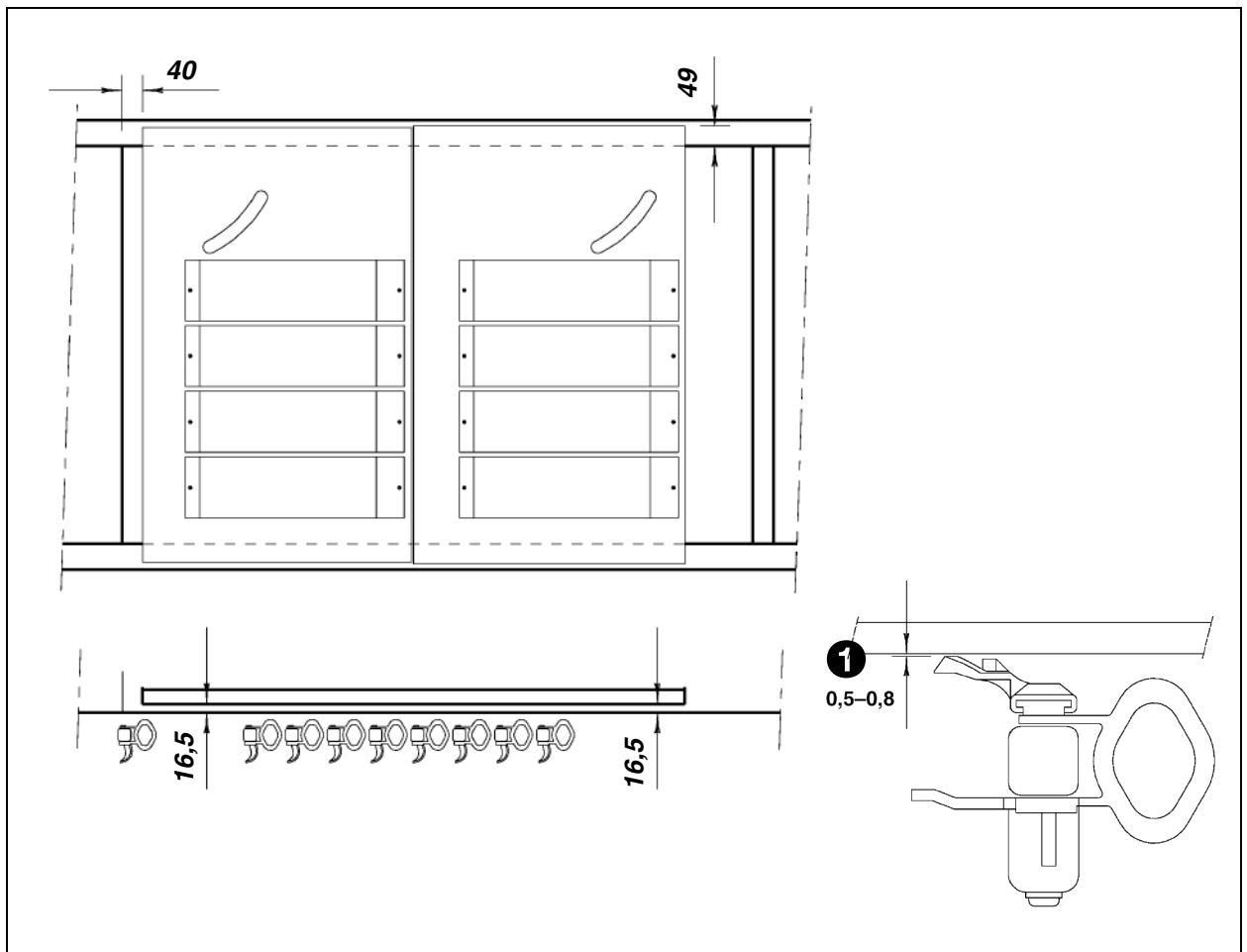


Fig. 134 - Mounting position of the release sets



### 10.3.2 0-Position and actuator stroke

1. Check this dimension between the rail and the inner side of the frame, as close to the actuator as possible. See fig. 135 and table 5.

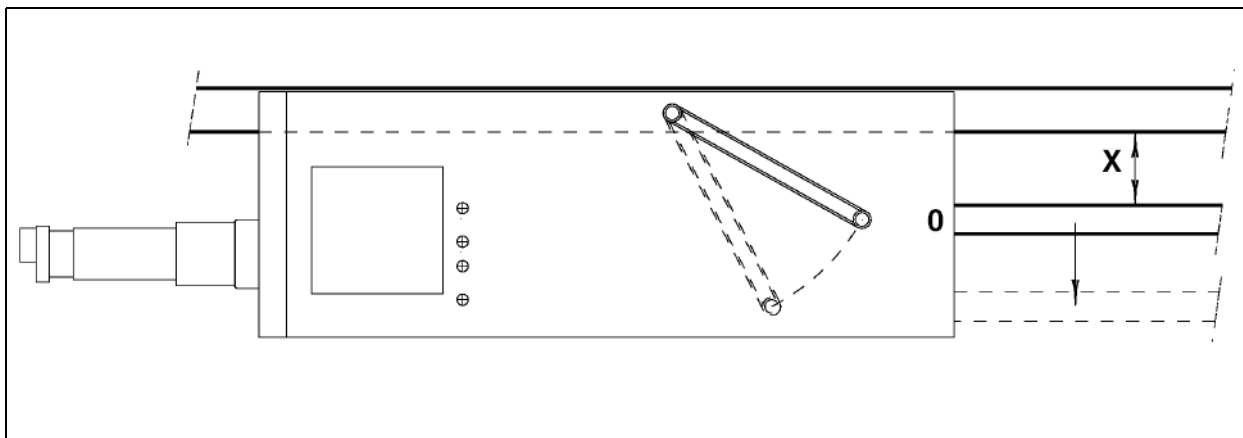


Fig. 135 - 0-position

Table 5 - Capacity adjustment

| Capacity | X = (mm) |           |           |           |           |           |
|----------|----------|-----------|-----------|-----------|-----------|-----------|
|          | Omnia 85 | Omnia 125 | Omnia 170 | Omnia 250 | Omnia 330 | Omnia 500 |
| 0        | 81       | 81        | 81        | 81        | 81        | 81        |
| 22.500   | 136      |           |           |           |           |           |
| 30.000   | 154      |           |           |           |           |           |
| 45.000   |          | 136       | 136       |           |           |           |
| 60.000   |          |           | 154       |           |           |           |
| 90.000   |          |           |           | 136       | 136       |           |
| 120.000  |          |           |           |           | 154       |           |
| 135.000  |          |           |           |           |           | 136       |
| 180.000  |          |           |           |           |           | 154       |



### 10.3.3 Adjustment of the plate under the 4 sensors

1. Stop the machine so that the actuator is in the 0-position
2. Mount the indication plate so that the first sensor '0' is activated once the actuator has shifted the switching plate 4-5 mm, see fig. 136

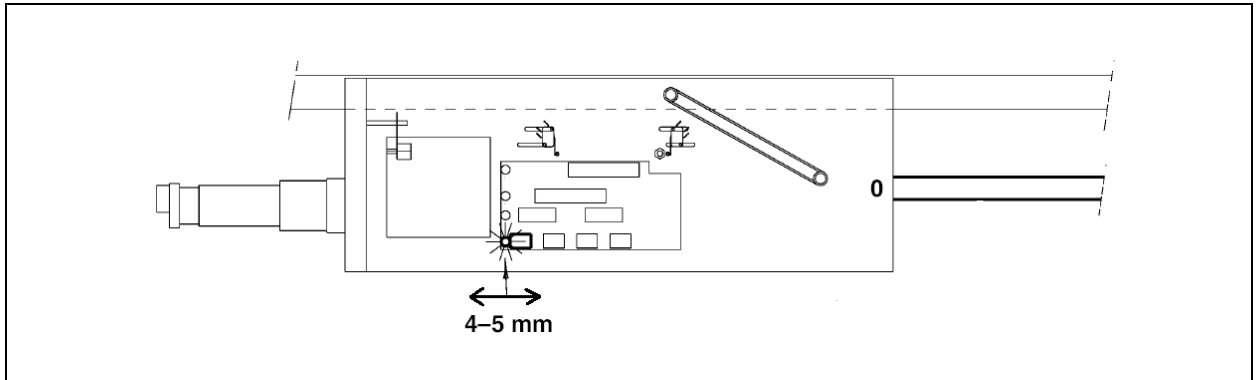


Fig. 136 - Adjustment of the plate under the 4 sensors

### 10.3.4 Mounting the potentiometer

1. Stop the machine so that the actuator is in the 0-position, see fig. 137. **! TAKE NOTE OF THE DIRECTION OF ROTATION.**
2. Turn the potentiometer's shaft 1x turn from 0-position and mount the potentiometer (2) (The potentiometer can make 2 more turns than the actuator; a potentiometer mounted in this way then has 1 turn left over at the beginning and end.)



### 10.3.5 Position of the safety switches

1. Stop the machine so that the actuator is in the 0-position.
2. Mount switch (3) in accordance with fig. 137.
3. Mount switch (4) in the beginning of the slot, see fig. 137.

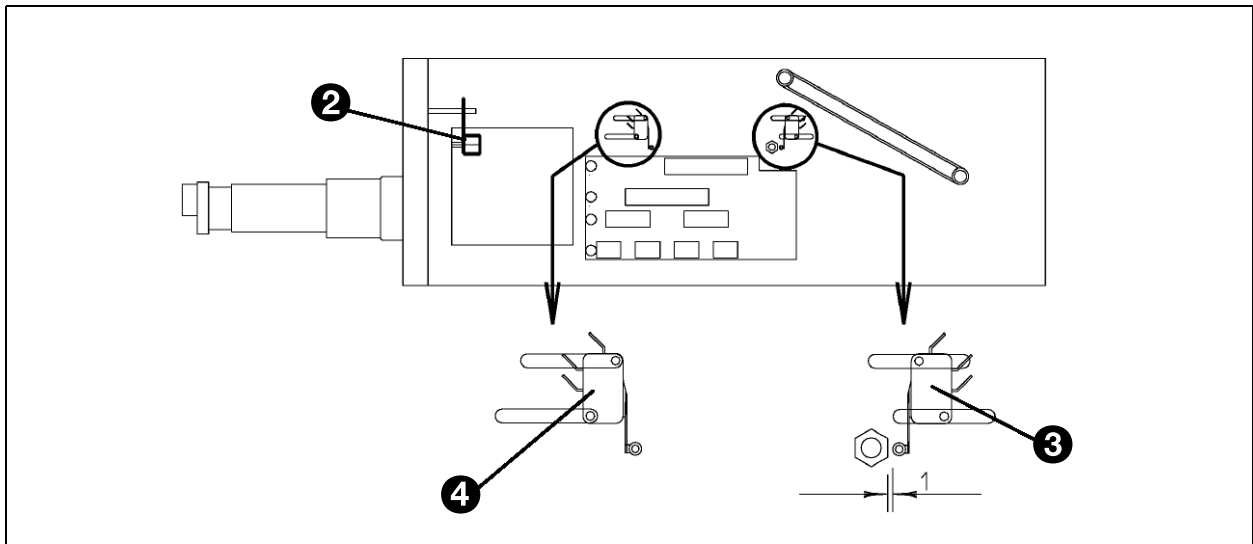


Fig. 137 - Position of the safety switches



## 10.4 Adjustment of the machine timing

Sensors are mounted at a variety of locations on the frame. A timing disk is mounted on the infeed unit's drive shaft at the transfer. This disk activates 2 sensors at various times. The signals generated by these sensors are related to the movement of various parts of the machine. fig. 138 shows which sensor signals which movements.

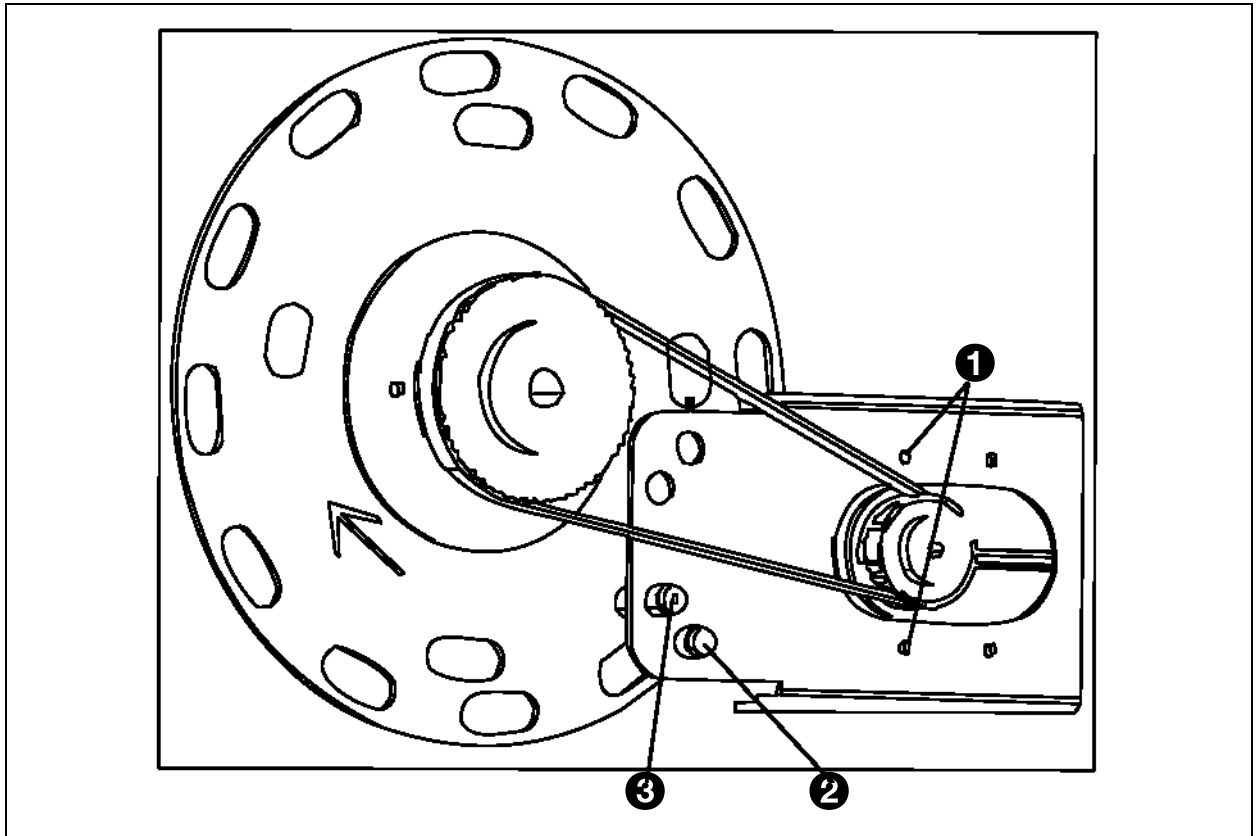


Fig. 138 - Adjustment of the machine timing

### 10.4.1 Carrier sensor (2)

This sensor signals each incremental movement of the chain with the carriers. (6x per turn of the transfer unit) See fig. 138.

### 10.4.2 Roller sensor (3)

This sensor signals each incremental movement of the infeed roller (2x per turn of the transfer unit). See fig. 138.



**NOTE**

*A different Main timing disk is used for the Omnia 85: with this disk the sensor is actuated 1 x per turn of the transfer unit.*



## 10.4.3 Tensioning the toothed belt

1. Loosen the encoder's bolts (1) and tension the toothed belt. See fig. 138.



### **NOTE**

***Pay attention to the following when fitting or tensioning a toothed belt***

- The toothed belt may not be tensioned.
- An over-tensioned toothed belt will damage the bearings.
- Determine the correct tension as follows:  
Check that the teeth of the toothed belt just do not jump over the chain-wheel while the toothed belt is turning. The toothed belt may not be tensioned any further.
- Alignment of the toothed-belt pulleys.  
The toothed belt must remain centred in the middle of the pulleys. The drive will be damaged if the toothed belt pushes against the flange of a pulley with too much force.

## 10.4.4 Adjustment of the Main timing disk

1. Place the loadcell carrier "0" in a position where the first attachment point (2) of the white tape is above the loadcell (3). See fig. 139.
2. Loosen the bolts (5) and turn the Main timing disk until the roller sensor (4) is just activated.  
The roller sensor monitors the inner 2 holes.



### **CAREFUL**

***TAKE NOTE OF THE DIRECTION OF ROTATION (see the arrow in fig. 139.)***

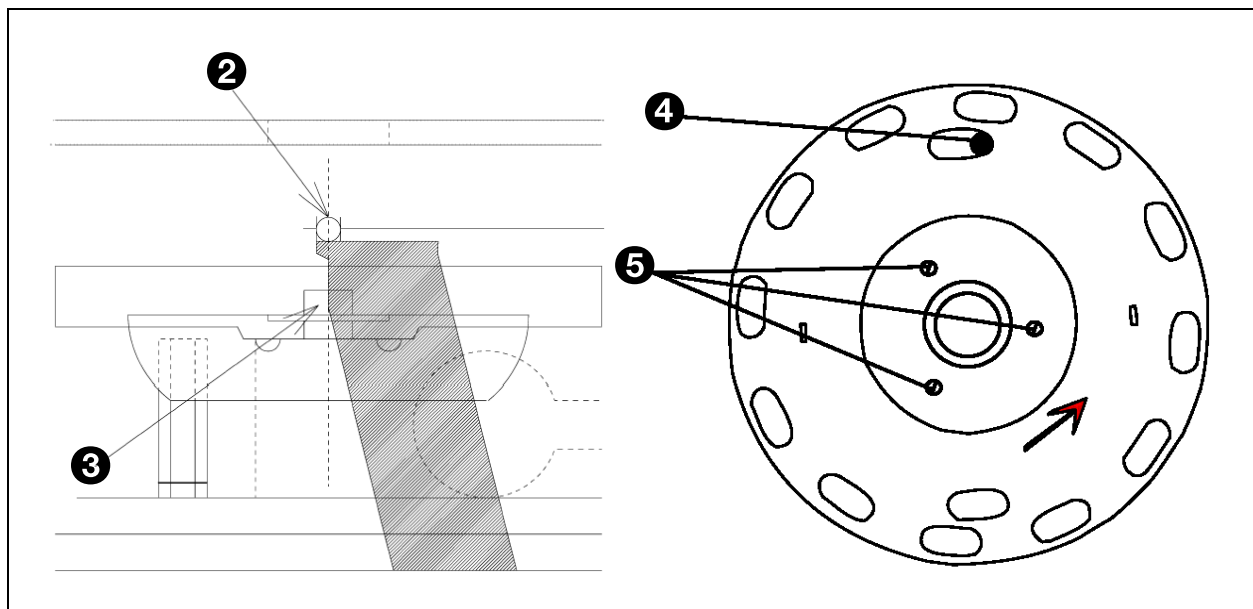


Fig. 139 - Adjustment of the Main timing disk



### 10.4.5 Adjustment of the sensor for the "0-carrier"

This sensor is mounted in the frame just after the transfer of the eggs. The sensor signals just 1 carrier, the carrier to which a magnet is fitted. This carrier with the magnet is called the "0-carrier".

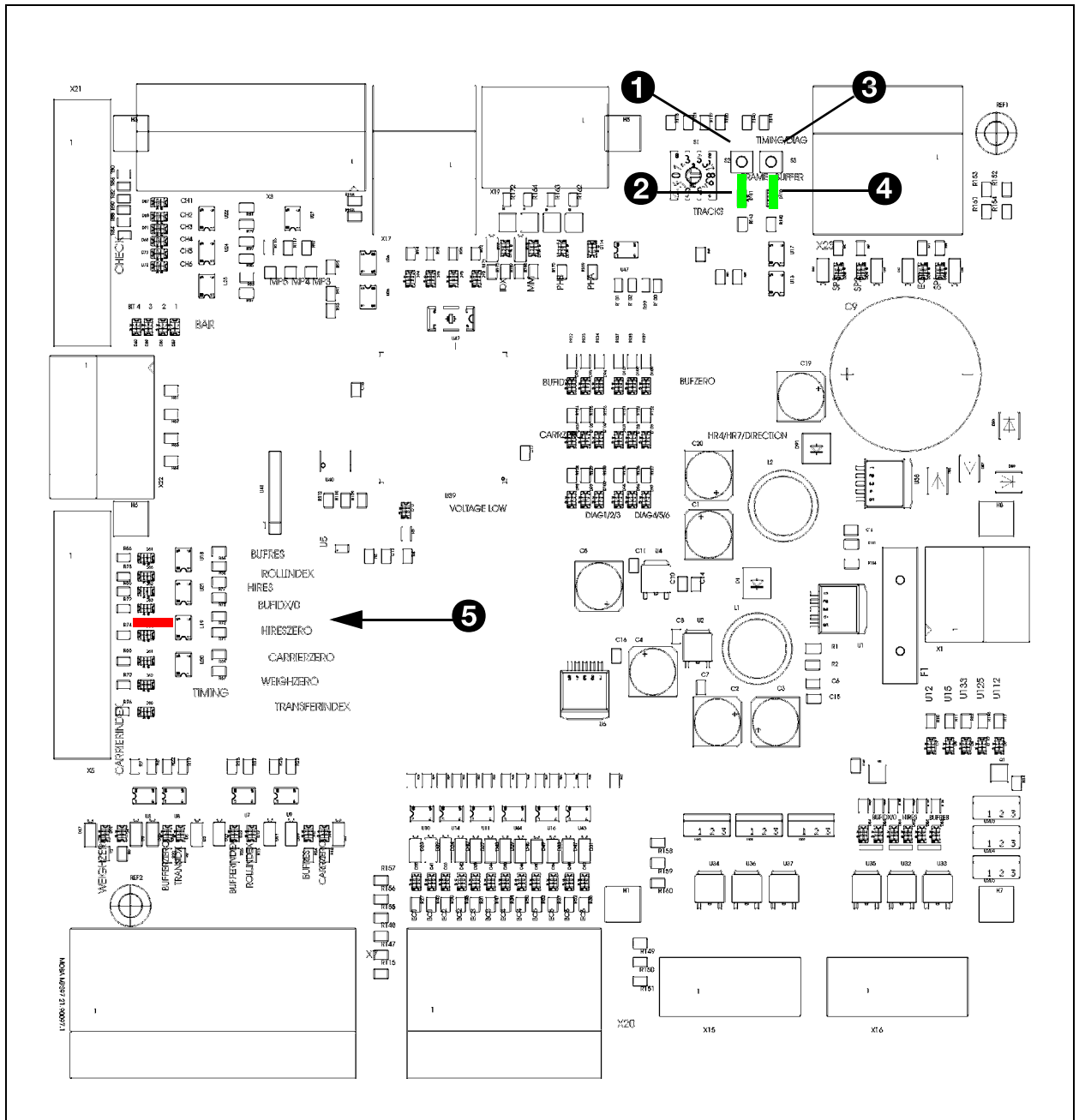


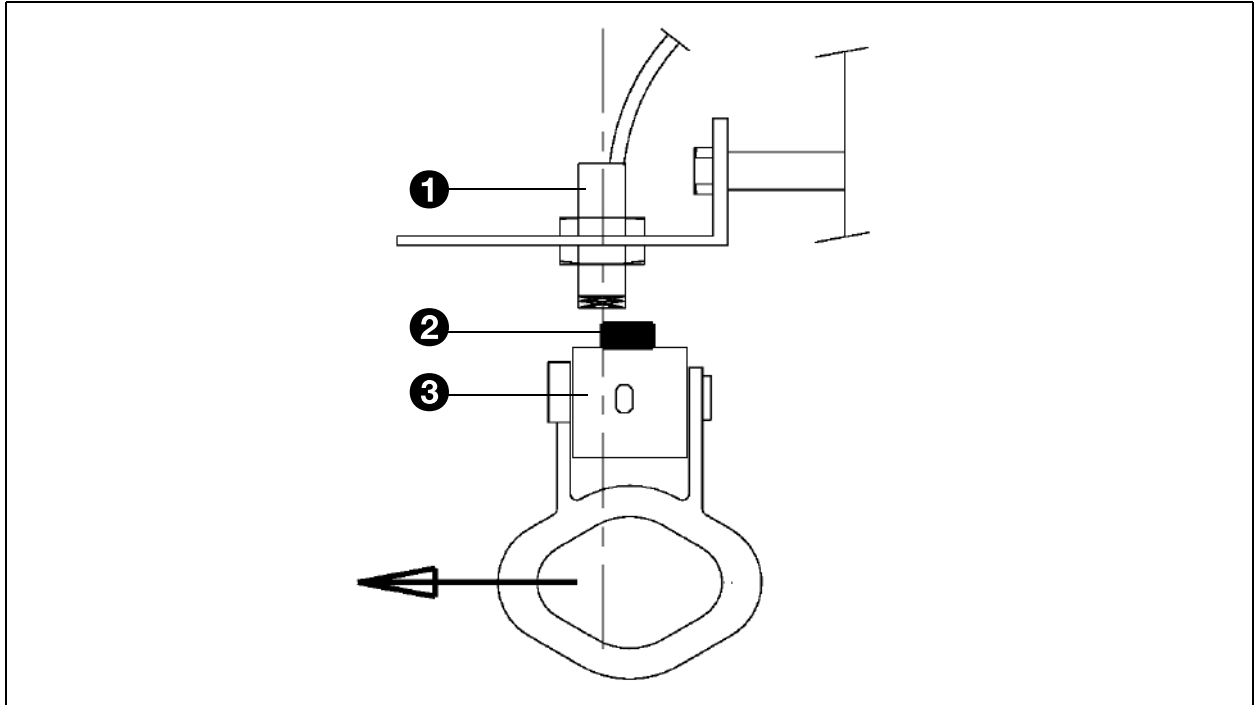
Fig. 140 - Main timing circuit board MPS97

- 1 - Switch S2 - Frame
- 2 - Status LED D111- Frame (Flashing = active)
- 3 - Switch S3 - Buffer
- 4 - Status LED D111- Frame (On = active)
- 5 - LED D64 - HIRESZERO

1. Turn the machine until the "0-carrier" is a few centimetres ahead of the sensor.
2. Turn the machine further by hand until the LED "HIRESZERO" (5) on the main timing circuit board lights. This signal is generated on each incremental movement of the chain with the carriers.



3. Turn the chain with the carriers a further 41 mm (half an increment).
4. Fit the 0-carrier sensor (1) 2 mm above the “0-carrier” as shown in fig. 141. The sensor is now switched correctly.



*Fig. 141 - Sensor adjustment on 0-carrier*

- 1 - 0-carrier sensor
- 2 - Magnet
- 3 - 0-carrier



### 10.4.6 Adjustment of the "end check"

The "end check" checks whether all carriers are open or closed correctly at the end of the frame.

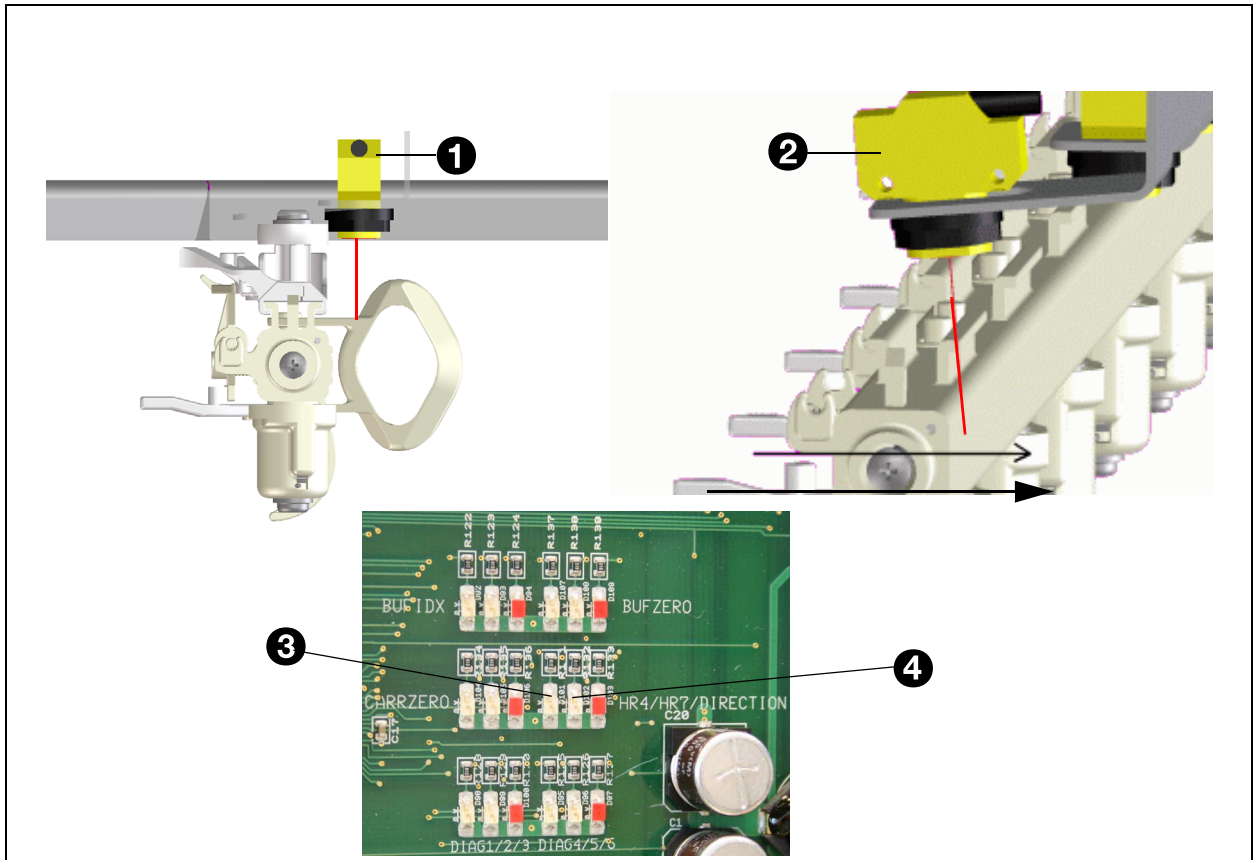


Fig. 142 - Adjustment of the end check

- 1 - End check sensor
- 2 - "Extra" sensor for end check
- 3 - Diagnosis LED D101 - HiRes 4
- 4 - Diagnosis LED D102 - HiRes 7

1. Press switch S2 (1) on the Main timing circuit board MPS97 until LED D111 (2) flashes. See fig. 140.
2. Turn the machine by hand until the D102 (4) HiRes 7 green LED lights. See fig. 142.
3. Release the carriers under the beam on which the sensors are mounted.
4. Position the beam with the sensors (1) so that the beam falls on the transition of the arm to the holder of an open carrier. See fig. 142.
5. Adjust the "Extra" sensor (2) so that it just switches on the underlying carrier bridge. See fig. 142.
6. Press switch S2 (1) on the Main timing circuit board MPS97 until LED D111 (2) goes out.



## 10.4.7 Adjustment of the "begin check" (when there is no blood detection)

The "begin check" checks whether each egg that has passed over the loadcell is still in the carrier.

**NOTE**

*This adjustment is applicable solely when the machine is not equipped with blood detection.*

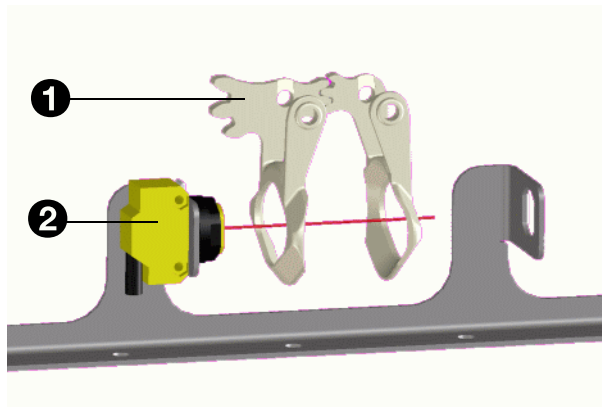


Fig. 143 - Adjustment of "begin check"

- 1 - Carrier
- 2 - Begin check sensor

1. Press switch S2 (1) on the Main timing circuit board MPS97 until LED D111 (2) flashes. See fig. 140
2. Turn the machine by hand until the D101 (3) HiRes 4 green LED lights. See fig. 142.
3. Mount the Begin check sensor (2) so that the beam of the sensor shines on the middle of a carrier (1). See fig. 143
4. Press switch S2 (1) on the Main timing circuit board MPS97 until LED D111 (2) goes out



## 10.5 PTB procedure (only applicable to Germany)

This chapter contains information about the PTB procedure.

We have designed a test in collaboration and consultation with the German PTB agency.

You will need the 12 calibrated 62-gram eggs supplied to you for this test.

### 10.5.1 The function of the test

Germany stipulates the calibration of all scales. Moba complies with the various agreements between the PTB and Moba relating to type approvals. One element of this type approval stipulates the on-site testing of the Moba weighing system. This test is prescribed by Doctor Wünsche.

### 10.5.2 What is tested?

The Moba Omnia has 6 or 12 or 18 loadcells. Each loadcell is supplied with a calibration-egg that weighs precisely 62 grams. These calibration-eggs must be sorted into the 60-65 gram category. Furthermore, it must be possible to inspect the weighing-data of these 12 calibration-eggs.

### 10.5.3 Test-procedure

1. Start the machine in the standard manner.
2. Run the machine until it is completely empty.
3. Define a "normal" product in the MMI with only class 3 (60-65 grams) and activate this product on conveyor 1.
4. Set the other belts, inclusive of the reserve belt, to pass through.
5. Place the 6 or 12 or 18 calibration eggs on the infeed rollers with the point pointing in the correct direction, about 1500 mm in front of the loadcell unit. See fig. 144.
6. Select page F7 in the mainpc.
7. Type E" (set mMessage display flags), see screen 1 (fig. 145), NOT FOLLOWED BY ENTER! Screen 2 (fig. 146) will now be displayed.)
8. Type 100 after msg\_wt ( 0). See screen 2, followed by 3x Enter". (The weight of the first 100 rollers will now be logged.)
9. Go to Diagnosis\Messages in the MMI and press the "Delete error messages" button.
10. Go to Operation>Select in the MMI and set the machine speed to the maximum.
11. Start the machine from the control panel and allow it to turn until the 12 calibrated eggs are on the set belt.
12. Stop the machine, go to Diagnosis\Messages in the MMI and press the Refresh button once. Find the 'Row weight of row x:' lines in the messages.  
WT= ... .. (the dotted section will list the weights of the calibrated eggs). Since we must comply with the 60-65 gram category all these weights will need to be within this range. See fig. 148.
13. Contact our service department if the weights fall outside this class.

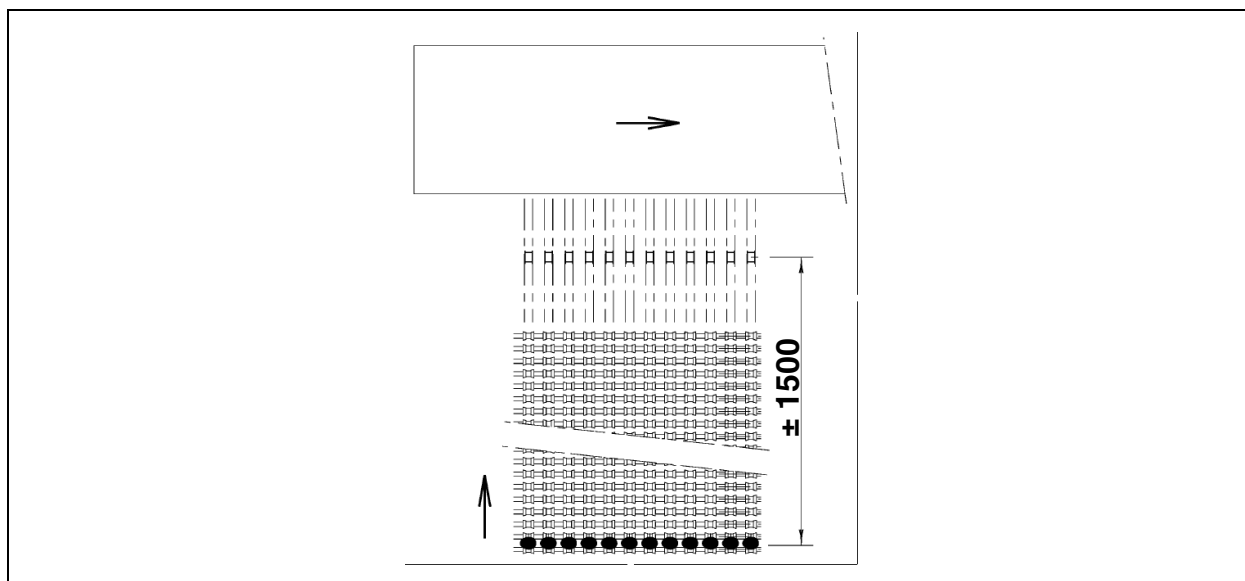


Fig. 144 - PTB test procedure

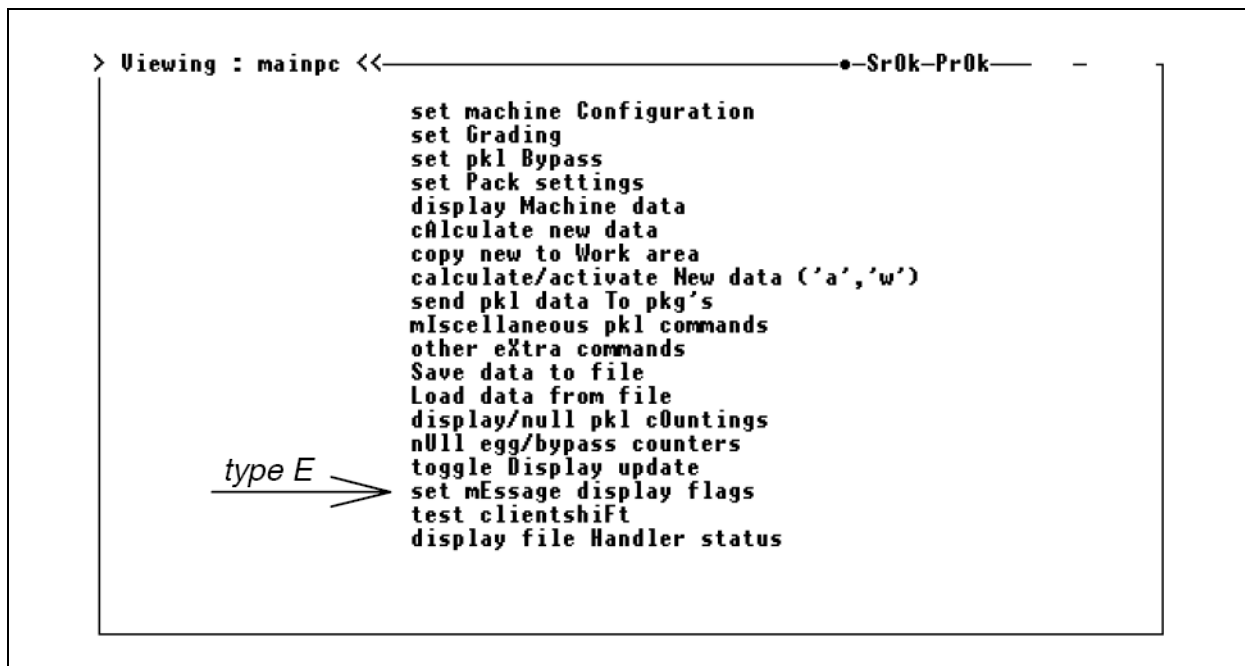


Fig. 145 - PTB test procedure screen 1



```

> Viewing : mainpc <<-----●-Sr0k-Pr0k-----

0 = not active, 1..254 = active and countdown, -1 = continu

msg_wt ( 0) 100  ← type 100
msg_candler ( 0)
msg_crack ( 0) _

```

Fig. 146 - PTB test procedure screen 2

```

Benutzer: [REDACTED]

A Infeed
B Sortierung
C Linie
D Maschine
E Mitteilung
F [REDACTED]
  alle Mitteilungen
  B Dropsetmagnetfehler (chron.)
  C Dropmagnet errors (chron.) + Start/Stop
  D Dropmagnet errors (chron.) + Magnets & Carriers
  E Start/Stop
  F HiRes errors
  G BufRes errors
  H Lane errors
  I Low weight
  J Tare errors
  K Loader errors
  L Clear message file
  M Old message files

```

Fig. 147 - PTB test procedure screen 3



**MOBA** © Moba BV 2005 - 2008, The Netherlands

**Diagnostics** Efficiency Stop data Messages Active reports Urgent Messages

Previous Month 20-1-2009 Next

Remove error messages Refresh Diagnostics Filter

| Date and time           | Level | Message                                             |
|-------------------------|-------|-----------------------------------------------------|
| 2009-02-24 09:07:33.071 | Info  | Rij gewicht van rij 7:0, 0, 612, 0, 0, 0.           |
| 2009-02-24 09:07:33.070 | Info  | Rij gewicht van rij 1:499, 0, 0, 452, 593, 0.       |
| 2009-02-24 09:07:30.051 | Info  | Rij gewicht van rij 7:793, 563, 429, 0, 0, 0.       |
| 2009-02-24 09:07:30.050 | Info  | Rij gewicht van rij 1:533, 496, 421, 789, 561, 478. |
| 2009-02-24 09:07:27.091 | Info  | Rij gewicht van rij 7:584, 550, 0, 0, 0, 0.         |
| 2009-02-24 09:07:27.090 | Info  | Rij gewicht van rij 1:0, 0, 714, 0, 468, 0.         |
| 2009-02-24 09:07:24.071 | Info  | Rij gewicht van rij 7:596, 0, 747, 0, 0, 0.         |
| 2009-02-24 09:07:24.070 | Info  | Rij gewicht van rij 1:0, 562, 668, 471, 677, 615.   |
| 2009-02-24 09:07:21.101 | Info  | Rij gewicht van rij 7:0, 0, 641, 0, 0, 0.           |
| 2009-02-24 09:07:21.100 | Info  | Rij gewicht van rij 1:570, 0, 0, 0, 0, 0.           |
| 2009-02-24 09:07:18.081 | Info  | Rij gewicht van rij 7:466, 0, 0, 0, 0, 0.           |
| 2009-02-24 09:07:18.080 | Info  | Rij gewicht van rij 1:577, 679, 0, 0, 0, 0.         |
| 2009-02-24 09:07:15.061 | Info  | Rij gewicht van rij 7:401, 413, 486, 0, 0, 0.       |
| 2009-02-24 09:07:15.060 | Info  | Rij gewicht van rij 1:0, 0, 0, 0, 680, 0.           |
| 2009-02-24 09:06:16.000 | Fout  | err_OmniaLinkStillNoConnection                      |

**MOBA - Omnia / 33999 / Server**

Operation Supplier Product Configuration History Diagnostics Exit Help Information

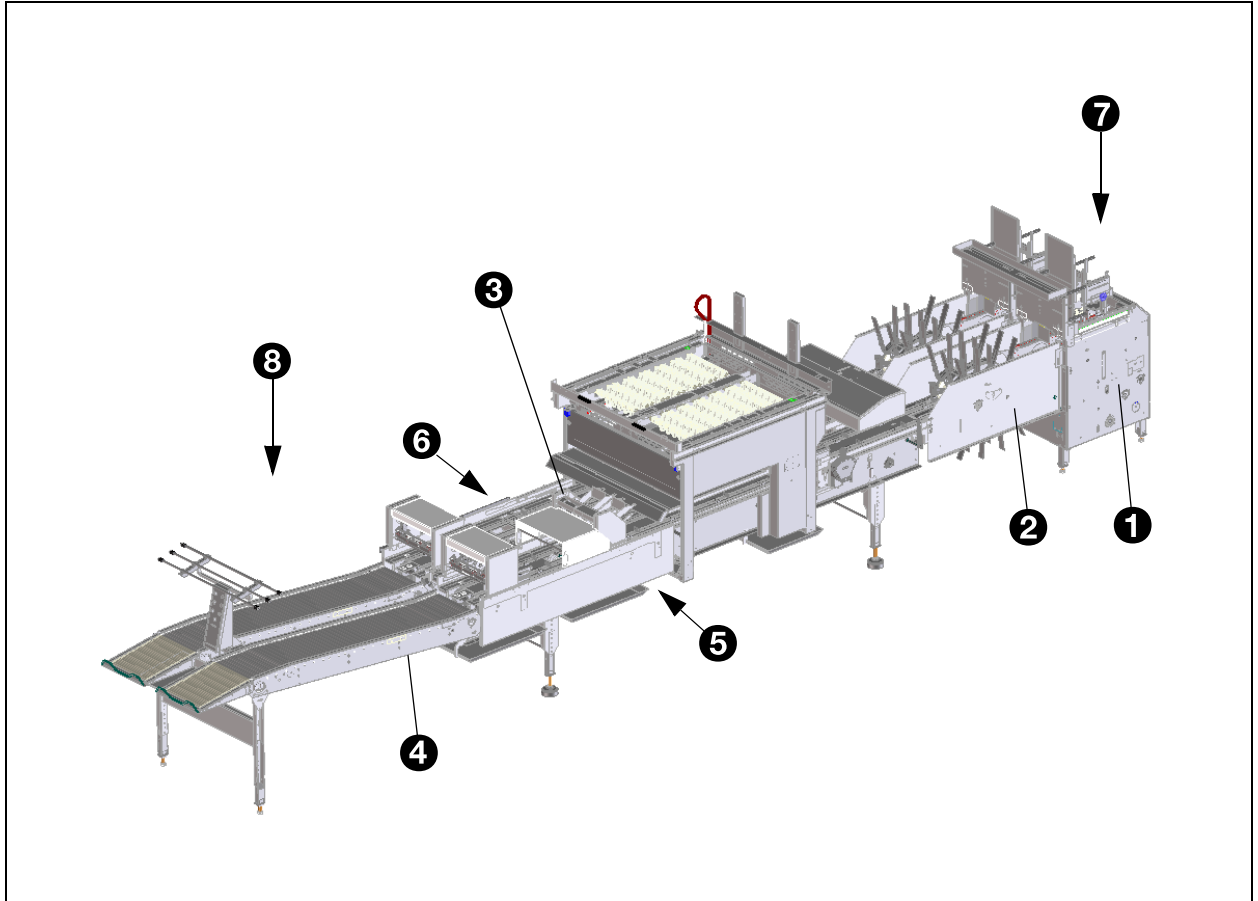
Status OK...<NoSimulate mode> Buffer stopped EG-VO activated Developer 20-3-2009 9:25:21

Fig. 148 - PTB test procedure for the MMI diagnosis screen



## 11 Packing lane

On the packing lane, sorted eggs are packed into packaging material.



*Fig. 149 - Overview of the packing lane*

- 1 - Denester
- 2 - Pin conveyor
- 3 - Packing conveyor
- 4 - Take-away conveyor
- 5 - Left-hand packing conveyor
- 6 - Right-hand packing conveyor
- 7 - Infeed-side
- 8 - Take-away side



## 11.1 Denester

The denester feeds a carton or a tray to the pin conveyor, every time the correct command is given. The denester unit will generate an alarm-signal if a jam in the supply of packaging material occurs.

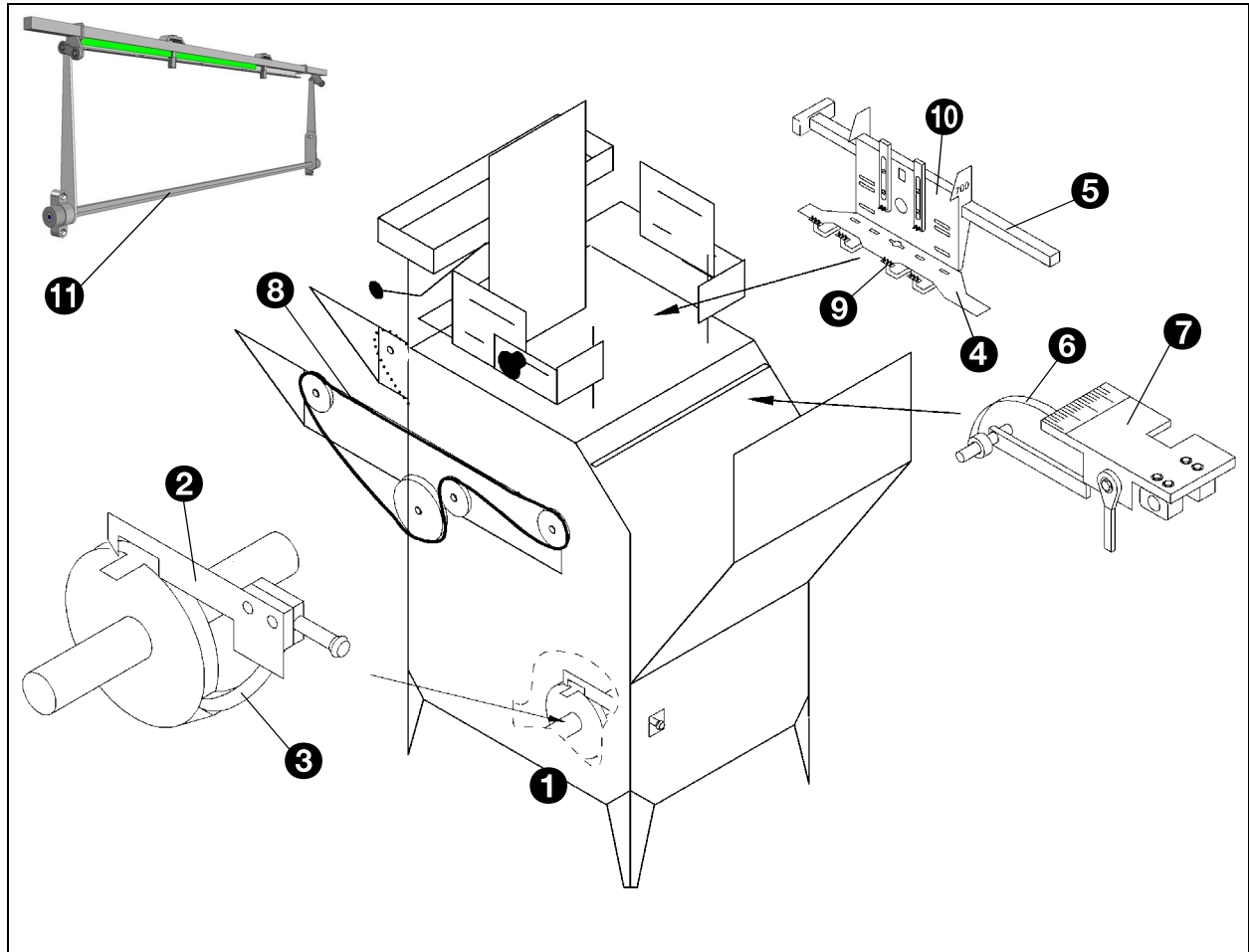


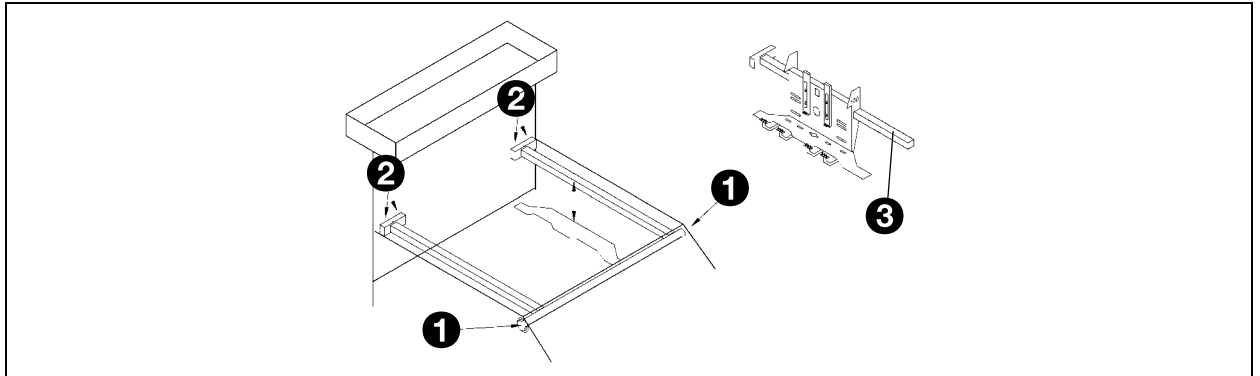
Fig. 150 - Overview of the denester

- 1 - Clutch
- 2 - Safety pawl
- 3 - Command pawl
- 4 - Plateaus
- 5 - Package-holders
- 6 - Closer-arm needle-units
- 7 - Closing-flap
- 8 - Belt guides
- 9 - Needle units
- 10 - Package
- 11 - Divider (Big Pack)



### 11.1.1 Plateau-height

The plateau-height should be adjusted using the procedure indicated below.



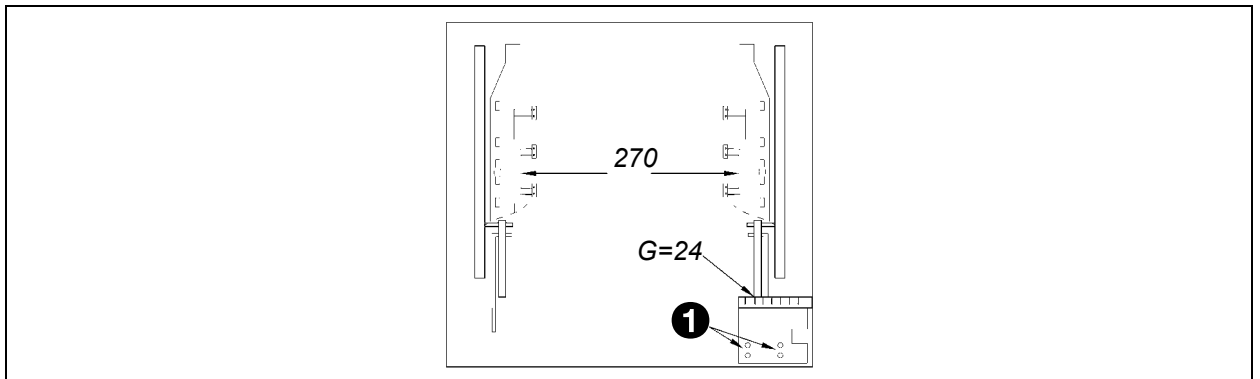
*Fig. 151 - Adjustment of the plateau-height*

Set the handle to the "0" position.

1. Adjust the height of the package-holders (3) so that the upper plate of the plateau is 60 mm below the upper surface of the package-holders.
2. To do this, loosen the bolts at the indicated points (1) and (2) in fig. 151.
3. Now adjust the height.

### 11.1.2 Plateau – adjustment-code G

For correct operation of the denester, the plateau will have to be adjusted to a standard value.



*Fig. 152 - Adjustment of the rotary-plateau*

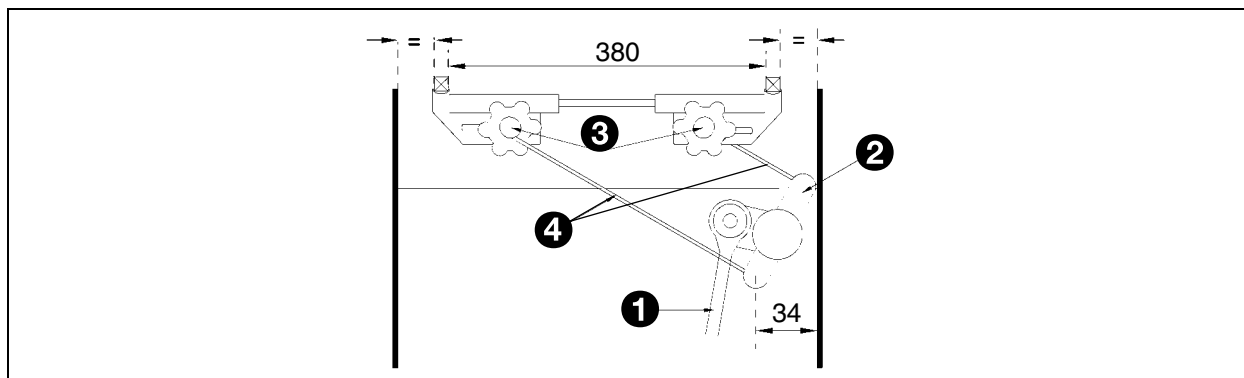
When "G" is set to 24, then the distance between the plateau-supports must be 270 mm.

1. Set the supports to an interval of 270 mm from each other.
2. Loosen the clamping blocks (1) on the closing-flap.
3. Move the closing-flap to adjustment-code '24'.
4. Refasten the clamping blocks and check the adjustment.



### 11.1.3 Plateau supports

Adjust the plateau carriers using the procedure that follows.



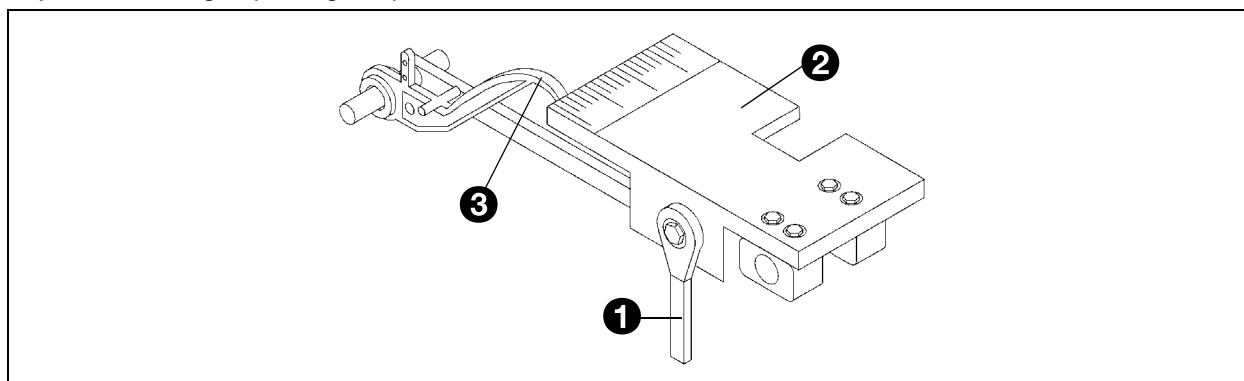
*Fig. 153 - Adjustment of plateau carriers*

Set the handle to the "0" position.

1. Adjust the pull-rod (1) to position the tilting-unit (2) as shown in fig. 153.
2. Loosen the 2 knobs (3) and slide the plateau carriers to their outermost positions.
3. Retighten the knobs (3).
4. Adjust the 2 pull-rods (4) so that the plateau carriers are set to the position describe hereafter:
  - a. The distance between plateau carriers is 380 mm.
  - b. The plateau carriers are positioned equal in comparison to the side-strips.

### 11.1.4 Closing-flap and closer-arm needle-units

Adjust the closing-flap using the procedure described below.



*Fig. 154 - Adjustment of the closing-flap / closer arm*

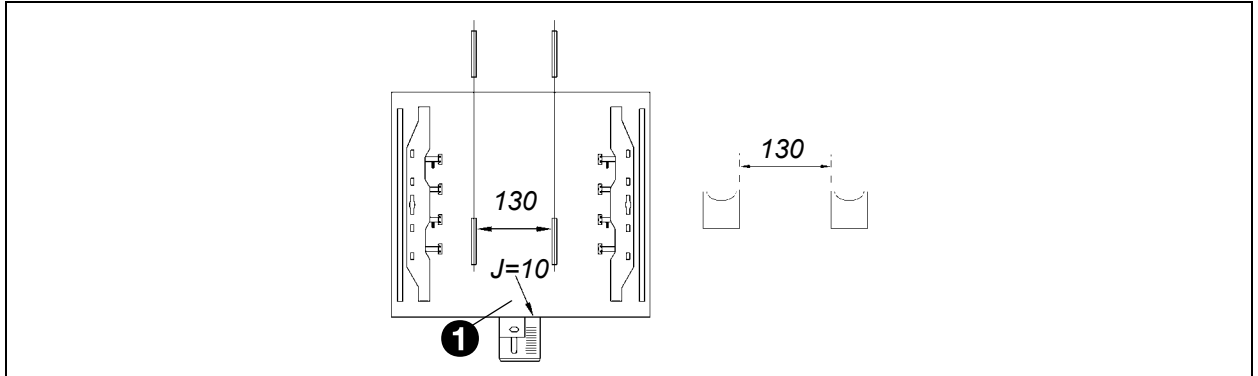
Check the adjustment of the plateau-height, see paragraph 11.1.1.

1. Adjust the pull-rod (1) so that when the closing-flap (2) switches on the needle-unit, the closing-arm (3) of the needle-units is locked.



### 11.1.5 Belt guides

The belt guides will need to be adjusted to a standard value to ensure the correct operation of the denester.



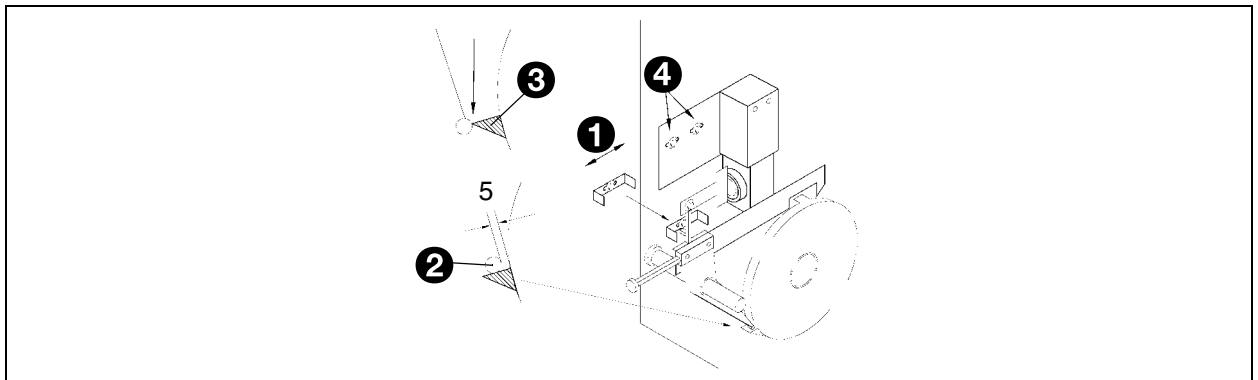
*Fig. 155 - Adjustment of the belt guides*

When "J" is set to 10, the distance between the belt guides must be 130 mm, see fig. 155.

1. Set the guides at a distance of 130 mm from each other.
2. Loosen the bolt that secures the adjustment-plate.
3. Move the adjustment-plate until the "J" adjustment code stands at 10.
4. Fasten the adjustment-plate in this position and check the adjustment.

### 11.1.6 Command pawl

The Command pawl is operated by a lifting-magnet.



*Fig. 156 - Adjustment of the command pawl*

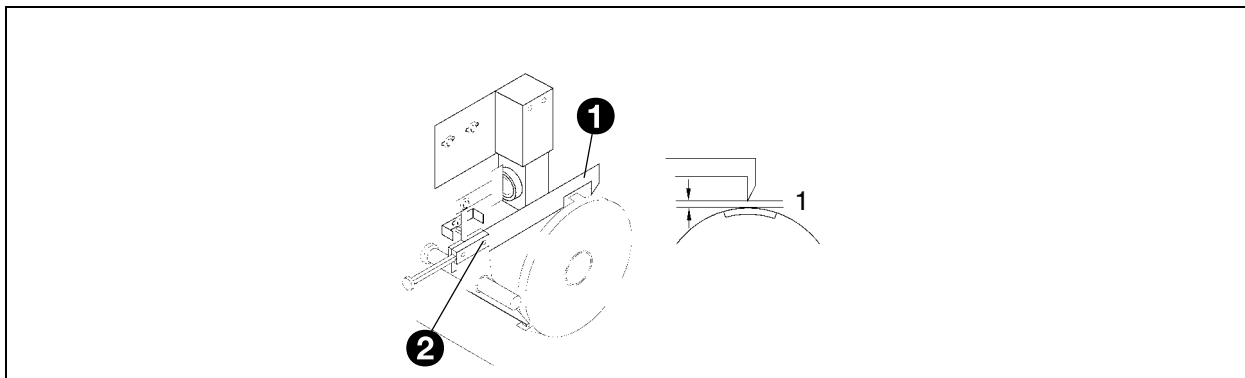
Set the handle to the "0" position.

1. Adjust the spring-retainer (1) to set the command pawl (2) to 5 mm.
2. Activate the command-clutch once.
3. Turn the denester by hand until the cam (3) passes the command pawl. The pawl may not make contact with the cam.
4. Loosen the bolts (4) in the lifting-magnet holder and adjust it.



### 11.1.7 Command-safeguard

Adjust the safeguard with the procedure described hereafter.



*Fig. 157 - Adjustment of the command-safeguard*

Activate the command-clutch once.

1. Turn the denester by hand until the safeguard-cam is centred below the safeguard pawl (1).
2. Adjust the position of the pawl [slot-attachment] (2) so that the pawl is 1 mm from the safeguard-cam.



### 11.1.8 Adjustment of the Dividers (Big Pack Denester)

Because the divider of the Big Pack Denester is longer, these are reinforced. Which means that several adjustments are required.

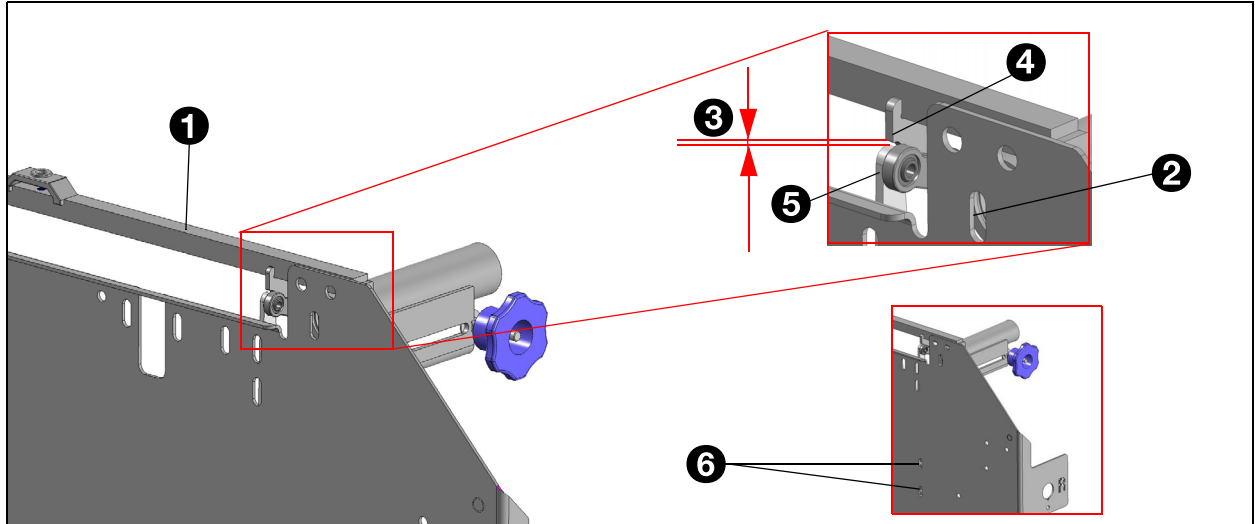


Fig. 158 - Overview extended divider adjustments

- 1 - Extended divider
- 2 - Slot for adjustment
- 3 - Tolerance (1mm)
- 4 - Cam
- 5 - Arm
- 6 - Slots of bearing blocks

#### Adjustment procedure

- 1 Adjust the height of the extended dividers so that the upper plate of the plateau is 60 mm below the upper surface of the extended dividers. See also paragraph 11.1.1.
- 2 Adjust the bearing blocks on the correct level.
- 3 Adjust the arm so, there is 1mm tolerance between the cam and the arm.

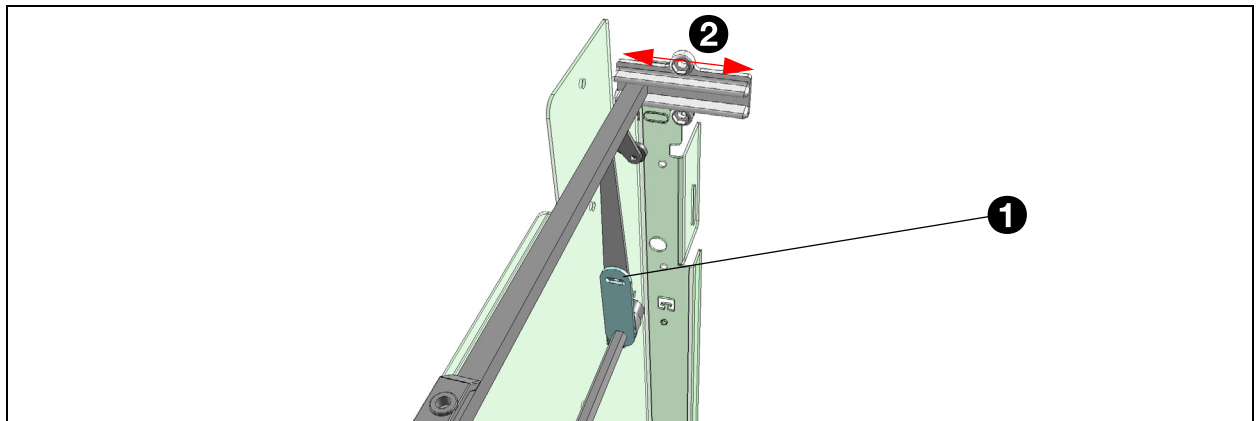


Fig. 159 - Parallel adjustment of the extended divider

- 1 - adjust-arm
- 2 - valve-facility

#### Adjustment procedure

- 1 Adjust the adjust-arm exactly parallel with the other side.



## 11.2 Pin conveyor

The pin conveyor transfers the packaging material from the denester to the packing conveyor.

### 11.2.1 Adjustment of the transfer-point from the pin conveyor to the packing conveyor.

Problems with the transfer of the packaging material can be solved by the adjustment of the transfer-point between the pin conveyor and the packing conveyor. See fig. 160.

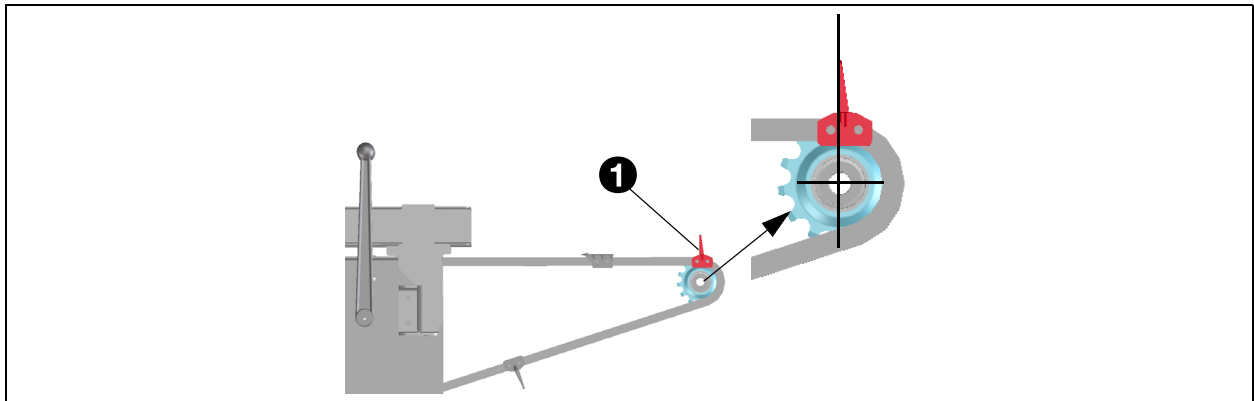


Fig. 160 - Transfer-point from pin conveyor to packing conveyor

Turn the packing conveyor by hand until a pusher (1) is centred vertically above the rear gear wheel. See fig. 160.

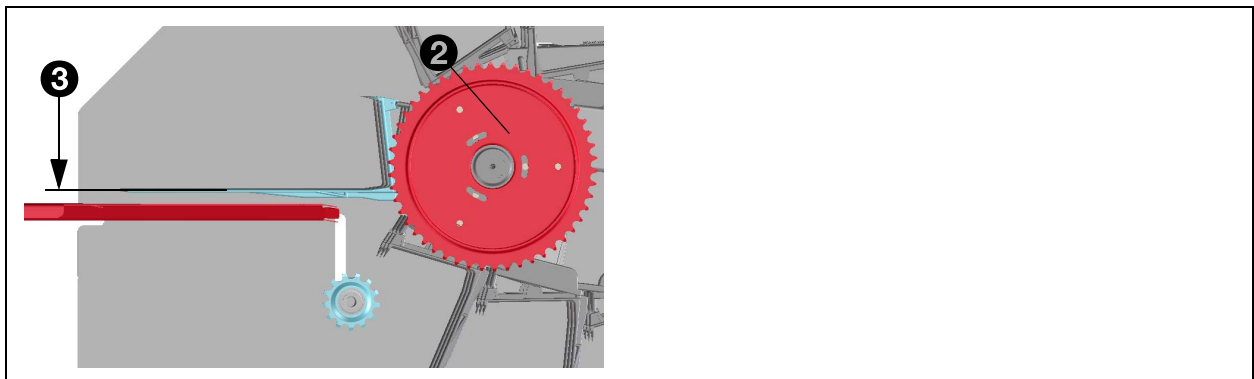


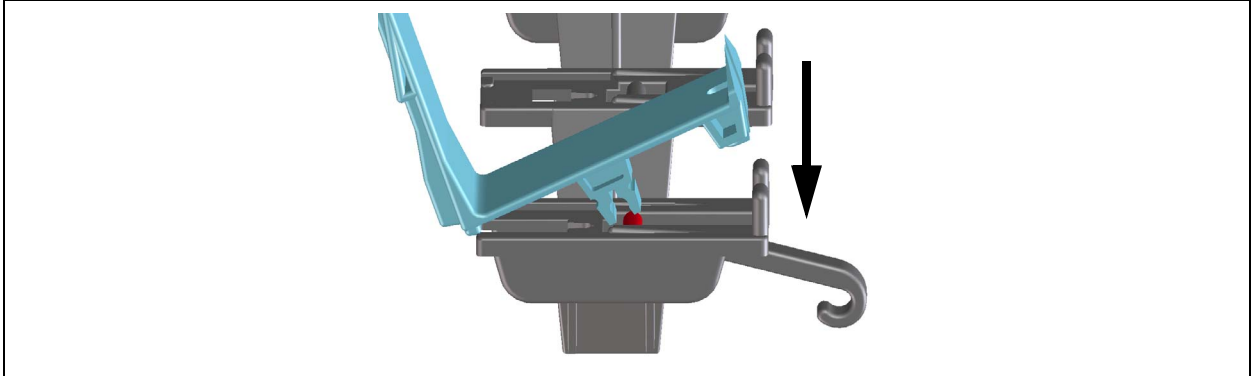
Fig. 161 - Adjustment of the pin conveyor

1. Adjust the chain-wheel (2) of the pin conveyor, so that the gap between the pins and the upper plate of the packing conveyor is 15 mm. See fig. 161.



### 11.2.2 Installation of plastic pins

The plastic pins of the pin conveyor can be displaced from the chain by a piece of packaging material that is incorrectly transported. The pins have a click-fastener that makes it easy to reinsert them. See fig. 162.



*Fig. 162 - Installation of plastic pins*

### 11.2.3 Stop-sensor Pin conveyor

The pin conveyor has a stop-sensor they stop the pins in the correct position. The sensor stops the conveyor when the pins are rotated 40 degrees and the sensor is passing the hole in the disk.

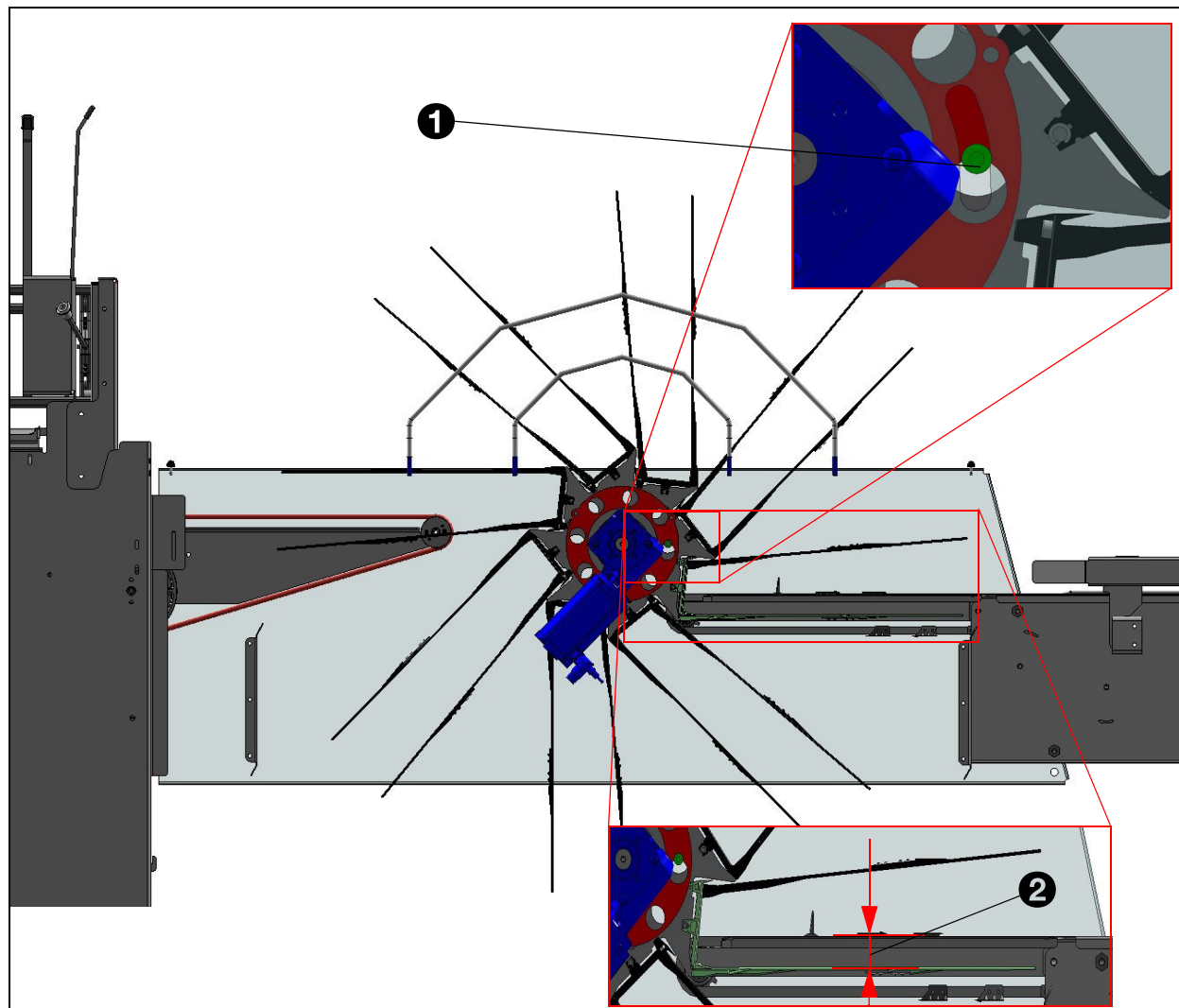


Fig. 163 - Overview of the pin conveyor

- 1 - Stop-sensor
- 2 - Height gauge of 35mm, between blade and pins

#### Adjustment procedure

- 1 Adjust the stop-sensor so, that the height of the pin by the blade is 35mm lower than the blade.

**NOTE**

**Adjust this procedure only, when the conveyor is rotating.**



### 11.3 Packing conveyor

On the packing conveyor, packaging material supplied by the pin conveyor is pushed under the Dropset™ for insertion of eggs. Once the packaging material has been filled, a number of other optional operations are possible. The optional features listed below can be used, if the correct machine configuration is available:

- Cartons closer
- Egg-stamping
- Carton-stamping
- Attachment of labels
- Packjet™
- Coupon-dispenser

The adjustments of these optional features are discussed below.

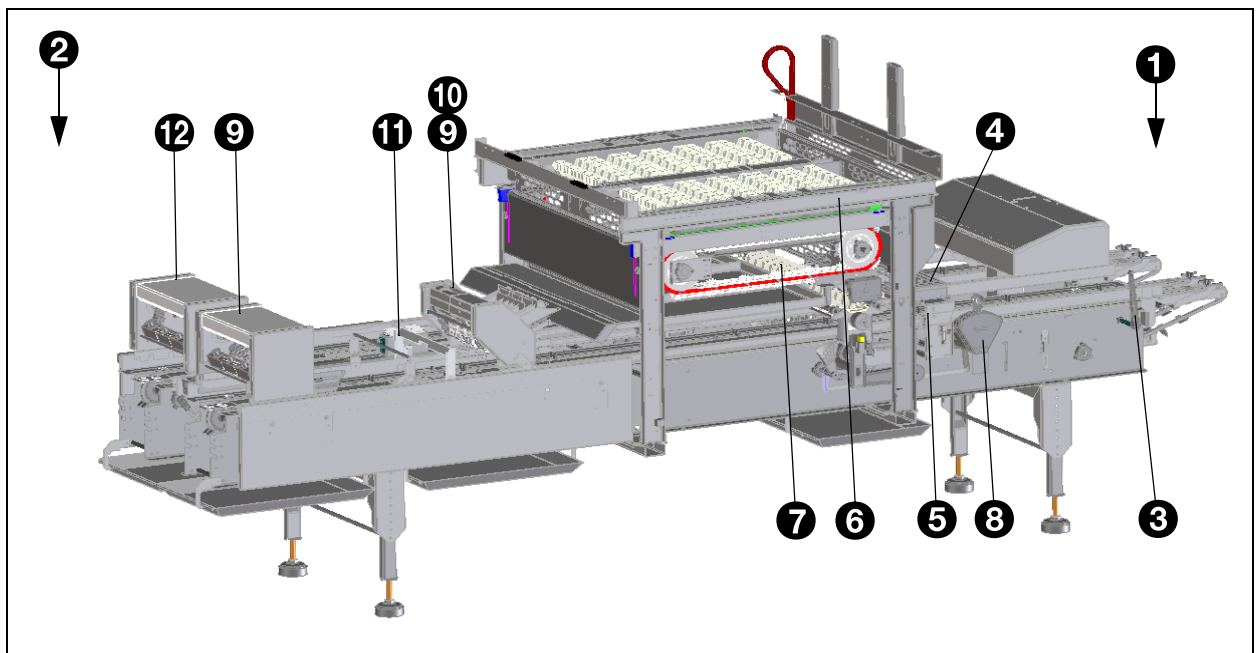


Fig. 164 - Overview of the packing lane

- 1 - Infeed-side of the packing conveyor
- 2 - Take-away side of the packing conveyor
- 3 - Box lift<sup>(\*)</sup>
- 4 - Missing Pack Unit™<sup>(\*)</sup>
- 5 - Upper guide
- 6 - Buffer container
- 7 - Dropset™<sup>(\*)</sup>
- 8 - Side-guide, drop-depth and pinching<sup>(\*)</sup>
- 9 - Eggstamp unit
- 10 - Label unit
- 11 - Lid-lifter
- 12 - Boxstamp unit

<sup>(\*)</sup> Glossary



### 11.3.1 Box-lifter

The box-lifter<sup>(\*)</sup> ensures that the cartons with a lid are carried along correctly by the pusher of the packing conveyor, see fig. 165.

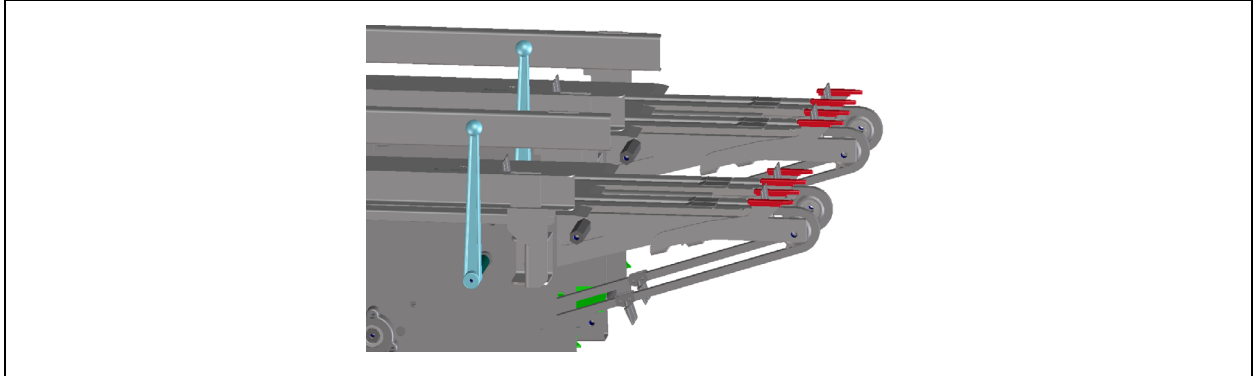


Fig. 165 - Overview of box-lifter

The box-lifter can be fitted in only one way.

### 11.3.2 Missing-Pack-Unit™

The Missing Pack Unit™ checks whether any cartons are missing.

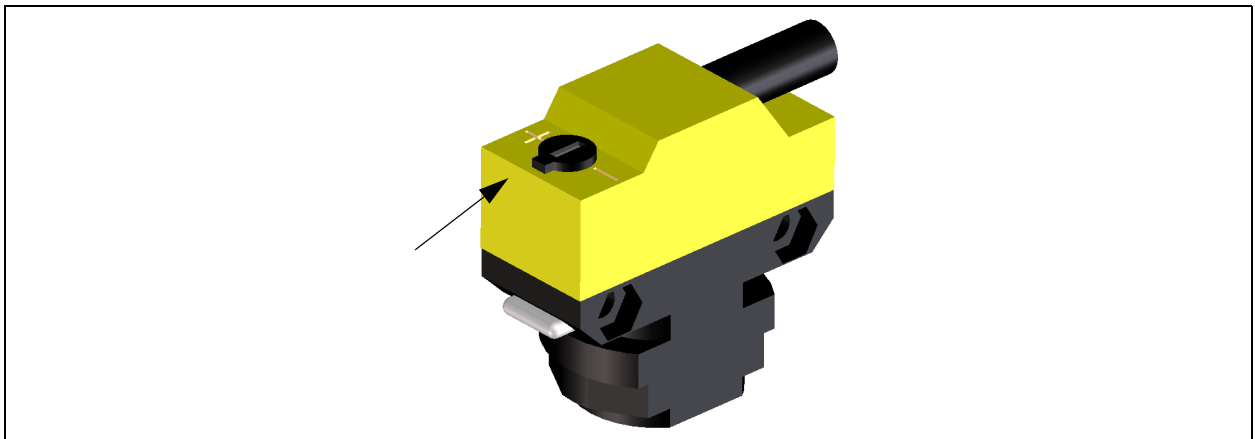


Fig. 166 - Adjusting the Missing Pack Unit™ sensor

1. The Missing Pack Unit™ is adjusted by positioning the setting dial mid-way between the plus and minus, see fig. 166.

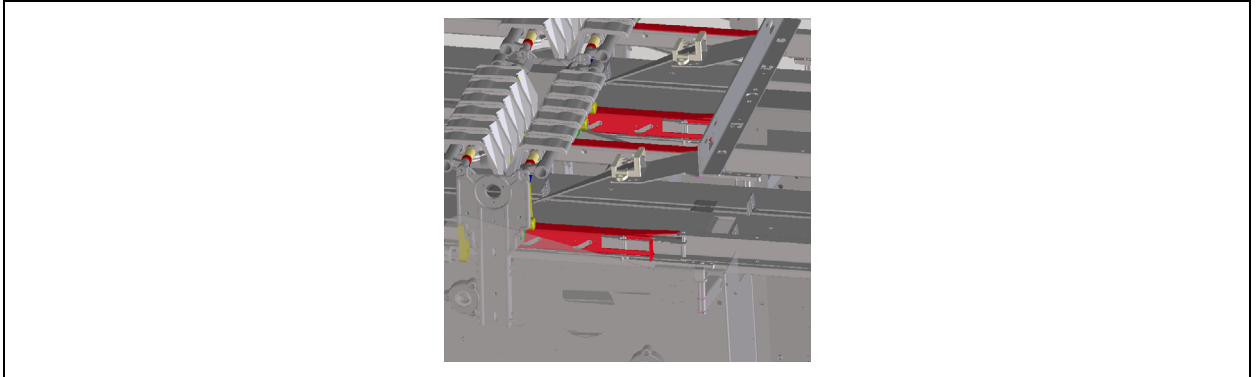
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(\*) Glossary



### 11.3.3 Upper guide

The upper guide ensures that the carton does not slide over the pusher, see fig. 167.



*Fig. 167 - Overview of upper guide*

The upper guide can be fitted in only one way.



## 11.4 Adjustment of the Receiver unit and Buffer conveyor

This chapter discusses all the adjustments of the reception of the eggs from the carriers to the reception in the dropset. This also includes the adjustment of the buffer-index sensor. This equipment consists of a number of loop wheels in the frame, with a number of sensors that determine the positions of the buffer conveyors.

### 11.4.1 Adjustment of the receiver unit time delay

This delay ensures that the eggs are dropped in the middle between the buffer conveyors' legs. Wear of the receiver legs will result in a slight change in the drop position, and the time delay will need to be adjusted in the software.

1. First, check the buffer conveyor's mechanical time setting.
2. Put the small adjustment egg in the receiver unit.

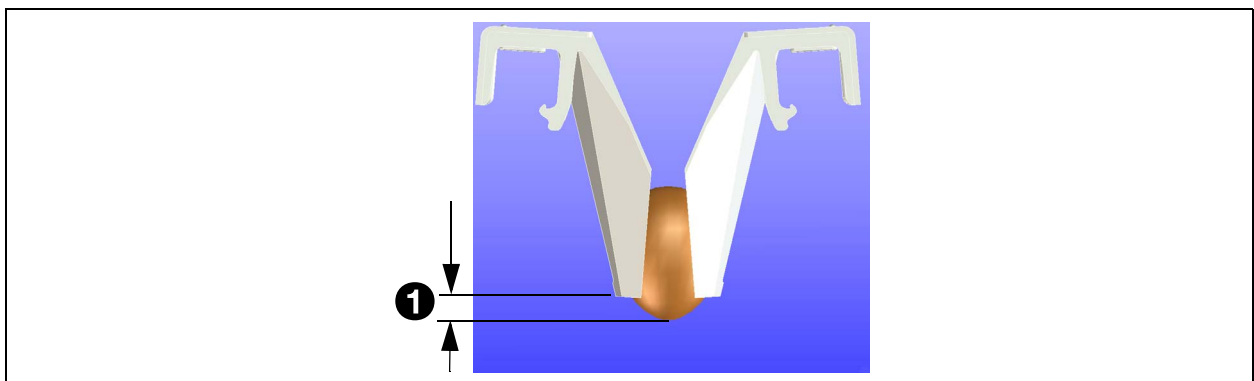


Fig. 168 - Adjustment of the egg in the receiver unit

3. Determine dimension (1) with the accompanying value in the table, see fig. 168 and table 6.
4.
  - a. DOS version: In the UserPC, go to: ->Service & Diagnosis ->Machine ->Service-machine ->Machine configuration. Go to page 14 and enter the value from table 6.

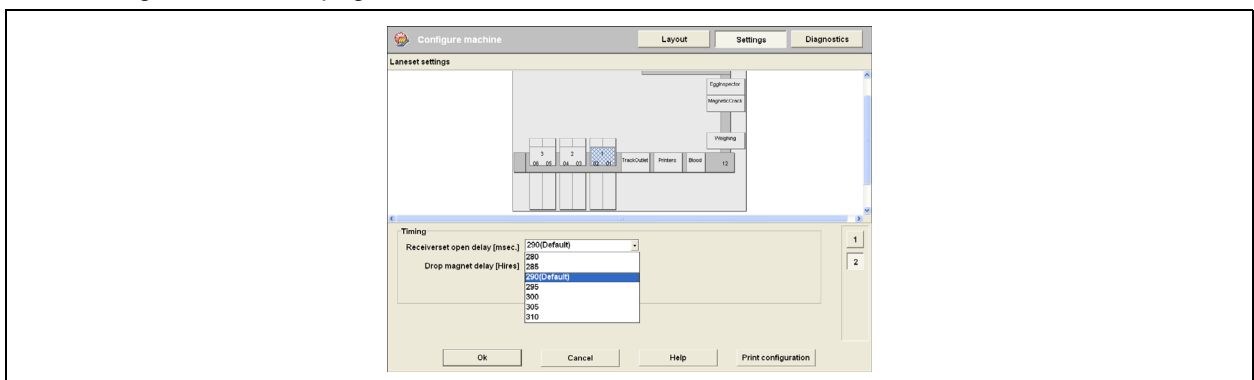


Fig. 169 - MMI Lane set settings

- b. Windows version: In the MMI, go to: -> Settings -> Login as Service -> Machine Settings -> Laneset -> Tab 2. The values can be changed on this page; see fig. 169.



Table 6 - Time delay table, receiver unit

| (1) | msec |
|-----|------|
| 0   | 290  |
| 5   | 295  |
| 10  | 300  |



## 11.4.2 Adjustment of the sensors for the buffer times and the buffer index 0

The sensors for these provisions are mounted on the frame's central shaft. This central shaft drives the buffer conveyors.

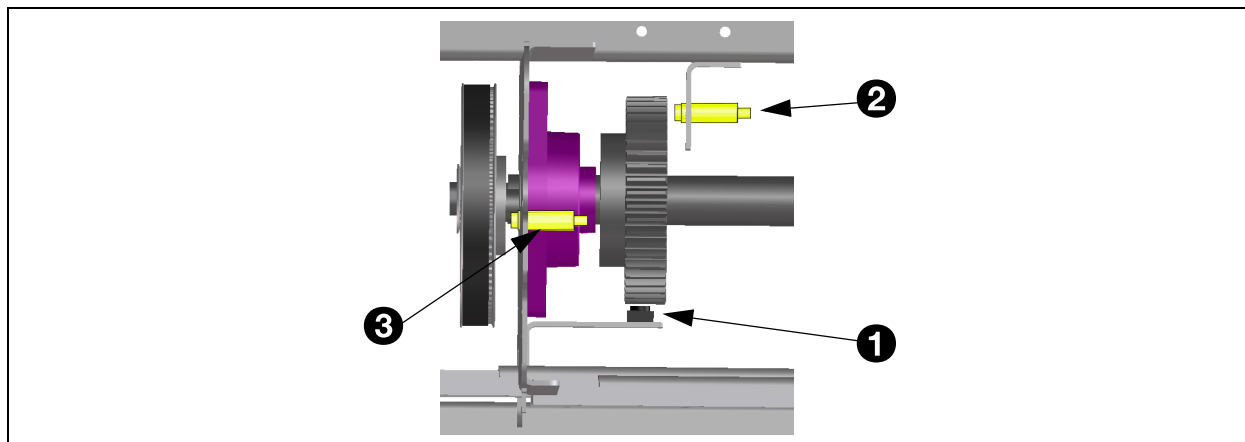


Fig. 170 - Sensor bufres and sensor buf idx

3 sensors are fitted, see fig. 170 and fig. 171.

- (1) = bufres: Gives 12 pulses per pitch movement of the buffer container.
- (2) = buf idx: Gives 1 pulse per pitch movement of the buffer container.
- (3) = buf 0: Gives 1 pulse per rotation of the buffer container.



### NOTE

*The pulses from these 3 sensors must be generated at a given time.*

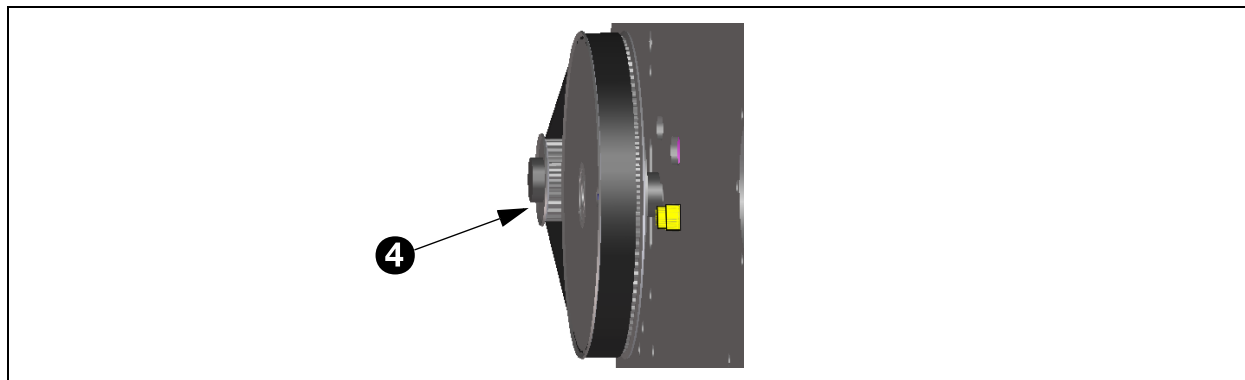
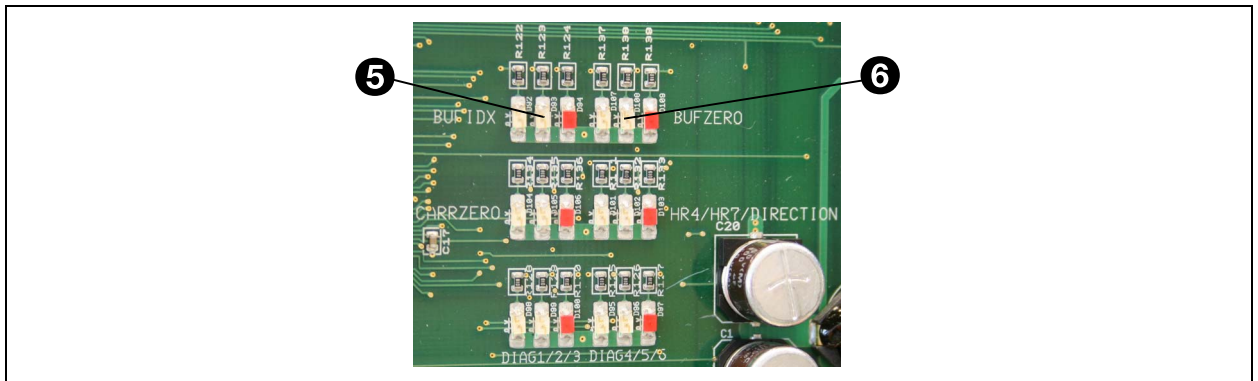


Fig. 171 - Sensor buf 0

1. Fit the sensor (1) in line with the axis of the chain-wheel as shown in fig. 170. The sensor may not touch the chain-wheel.
2. Fit sensors (2) and (3) at the standard distance of 2 mm.
3. The sensors can be checked using switch 'S3' (3) on the main timing circuit board, 'MPS97'. See fig. 140.
4. Press switch S3 until the D110 LED (5) lights. See fig. 140.



*Fig. 172 - MPS 97 Diagnosis LEDs*

5. Adjust sensor (2) so that the LED D93 (5) goes out. fig. 172
6. Adjust sensor (3) by turning chain-wheel (4) on the shaft until LED D108 (6) lights. See fig. 172
7. Press switch S3 for 4 seconds to deactivate the test function
8. .When the yellow LED is lit then the sensor is activated too early: when the red LED is lit then the sensor is activated too late
9. Check the adjustment dimensions of the buffer conveyor settings

### 11.4.3 Adjustment of the buffer conveyors timing

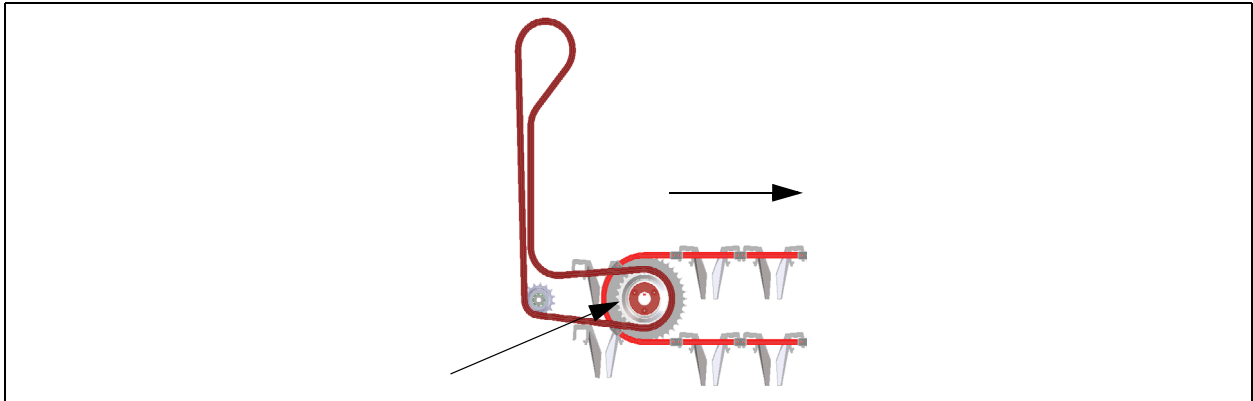


Fig. 173 - Adjusting the buffer conveyor with the chain-wheel

1. Turn the central shaft by hand until the sensor (2) of the buffer index is just activated, see paragraph 11.4.2.
2. Use the adjustable chain-wheel of the buffer conveyor, see fig. 173, to adjust the setting of block 1 and block 2 in accordance with fig. 174 and to the value from table 7.
3. The pitch between 2 rows of buffer containers is 127 mm. The time of each 2 following buffer-conveyor blocks must differ by 1/6 pitch from the previous 2, see also table 7.

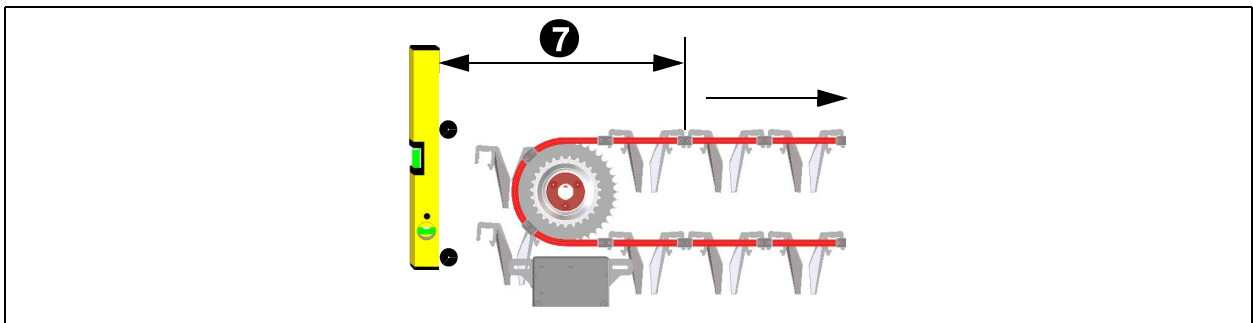


Fig. 174 - Adjustment of the buffer conveyor block timing

4. Adjust the setting of the dimension of the following blocks in accordance with fig. 174 and to the value from table 7.

Table 7 - Adjustment of buffer conveyor dimensions

| Block     | Dimension (7)<br>fig. 174 |
|-----------|---------------------------|
| 1 and 2   | 310 mm                    |
| 3 and 4   | 289 mm                    |
| 5 and 6   | 268 mm                    |
| 7 and 8   | 247 mm                    |
| 9 and 10  | 226 mm                    |
| 11 and 12 | 205 mm                    |



#### 11.4.4 Adjustment of the release unit

Calculate the dimension of the setting for the release unit, see fig. 175.

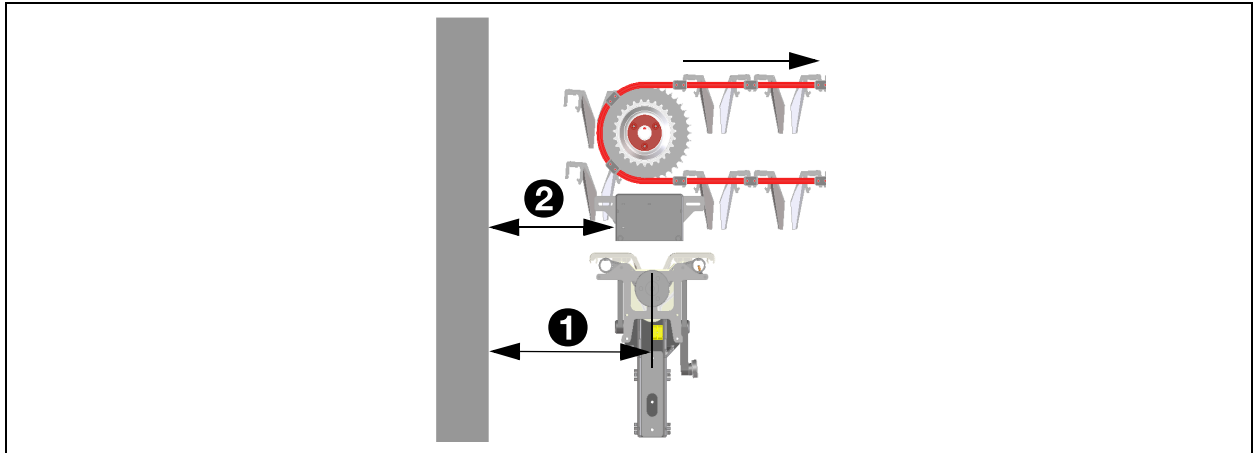


Fig. 175 - Adjustment of the release unit

1. Measure the distance (1) between the centre of the dropset and the beam of the frame.
2. Mount the release unit at a distance (2) from the beam of the frame. This distance is (1) minus 54 mm.

#### 11.4.5 Assembly of the planetary gearwheel transmission

The drive of the buffer chain wheels is called a planetary gearwheel transmission.

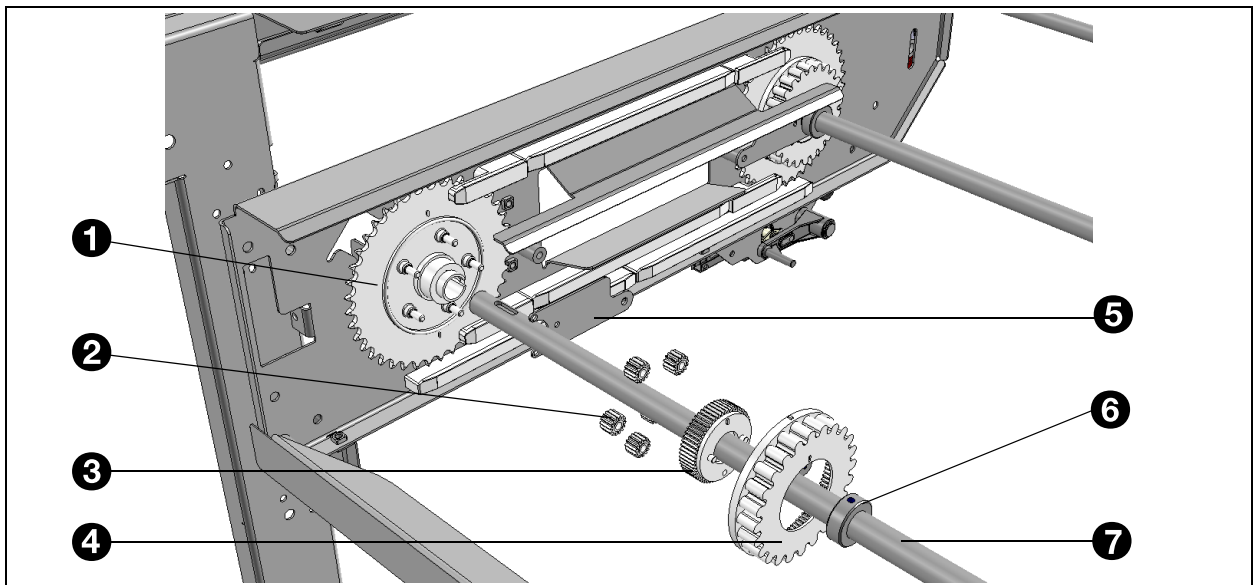


Fig. 176 - Planetary gearwheel transmission

- 1 - Sprocket wheel - 40 tooth
- 2 - Sprocket wheel - 14 tooth (5x)
- 3 - Sprocket wheel - 56 tooth
- 4 - Gear ring
- 5 - Reaction arm
- 6 - Bush
- 7 - Shaft

#### Assembly of the planetary gearwheel transmission:

- 1 Assemble the 40 tooth sprocket wheel.



- 2 Place the bush, the gear ring and the 56 tooth sprocket wheel over the shaft.
- 3 Place the shaft in the 40 tooth sprocket wheel.
- 4 Assemble the 14 tooth sprocket wheels (5 x).
- 5 Assemble the 56 tooth sprocket wheel. Take care for the markers. See paragraph 11.4.6.
- 6 Assemble the Gear ring. Take care for the markers. See paragraph 11.4.6.
- 7 Place the bush against the gear ring and lock it.
- 8 Assemble the Reaction arm and lock it.

## 11.4.6 Adjustment of the planetary gearwheel transmission

In one buffer section there are 4 gearwheels. For the 4 gearwheels there are 2 different adjustments.

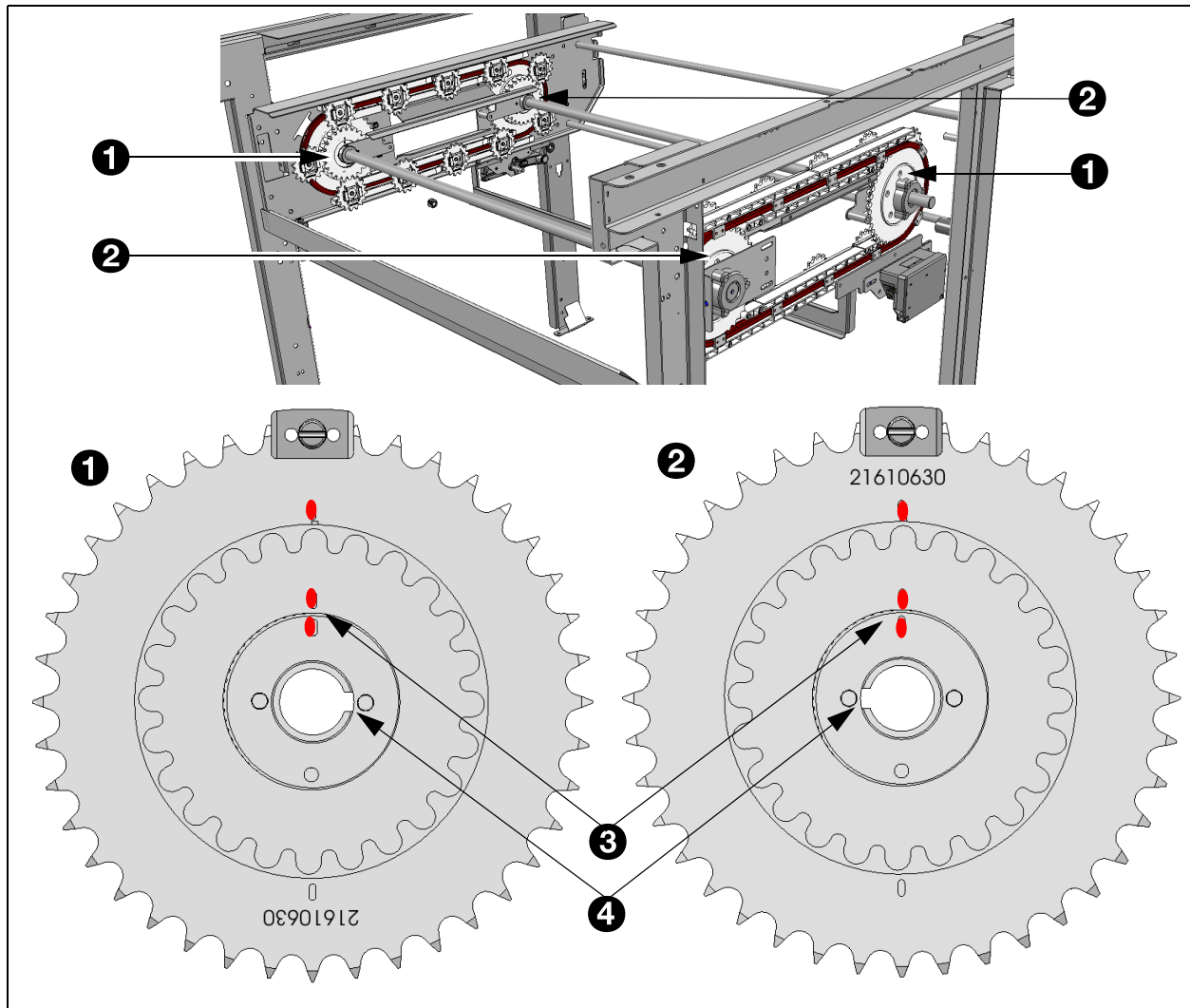


Fig. 177 - Assembly of the Gearwheels

- 1 - Assembly gearwheel #1
- 2 - Assembly gearwheel #2
- 3 - Markers
- 4 - Key-way



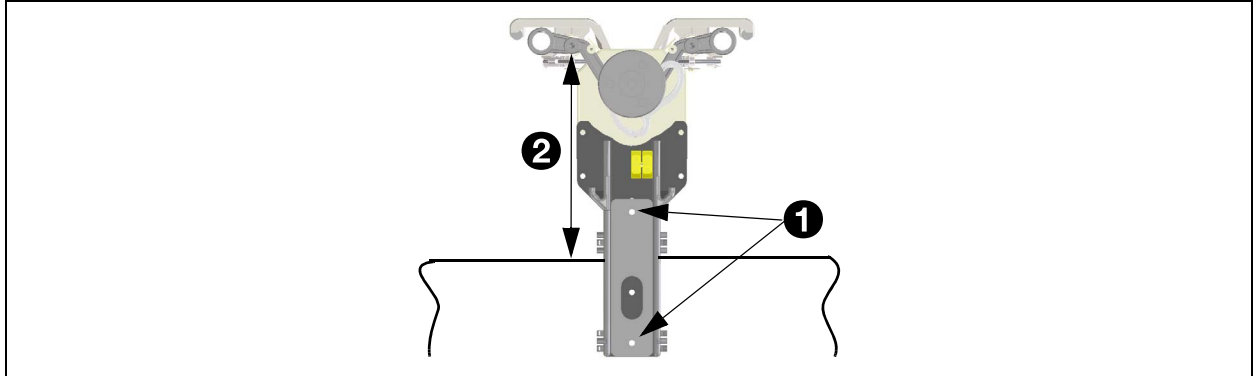
### NOTE

**Assembly the gearwheels, take care for the key-way and the markers on the several sprocket wheels.**



### 11.4.7 Adjustments of the Dropset™

This section discusses the height of the dropset. These dimensions must be checked if the dropset has been disassembled. Do not confuse this height with the drop height. The drop height is the height from which the eggs are set into the packaging material. This height is automatically adjusted when the packaging-type is selected.



*Fig. 178 - Adjustment of the dropset height*

Adjustment of 'missing egg'. These sensors check that no eggs are missing from the packaging.

1. Reset the packing lane to raise the dropset to its highest position.
2. The side shields of the dropset are located in a channel (1) attached to the arms, see fig. 178.
3. Adjust this channel fastening to set the distance (2) between the surface and the underside of the bush to 163 mm.
4. Check this dimension for all 4 corners of the dropset.
5. Next, check that the 2 dropsets in each block are aligned and positioned vertically to the packing conveyors.



## 11.5 Transport section

This section discusses the adjustments of the pusher chain. The various times for each type of packaging do not need to be adjusted, since this takes place automatically on setting the type of packaging. The pusher chain is automatically tensioned by spring tensioners.

### 11.5.1 Putting on the pusher chain

**WARNING**

*When putting on the pusher chain take note of the position of the loop wheels relative to a pusher as indicated in step 1 and 2 accompanying fig. 179.*

1. See (1) the position of the loop wheel relative to a pusher on the placement side of the packaging conveyor.
2. See (2) the position of the loop wheel relative to a pusher on the take-away side of the packaging conveyor.
3. Make sure that the pusher and auxiliary pusher are located in the notches in the tensioner chain-wheels. See (3).

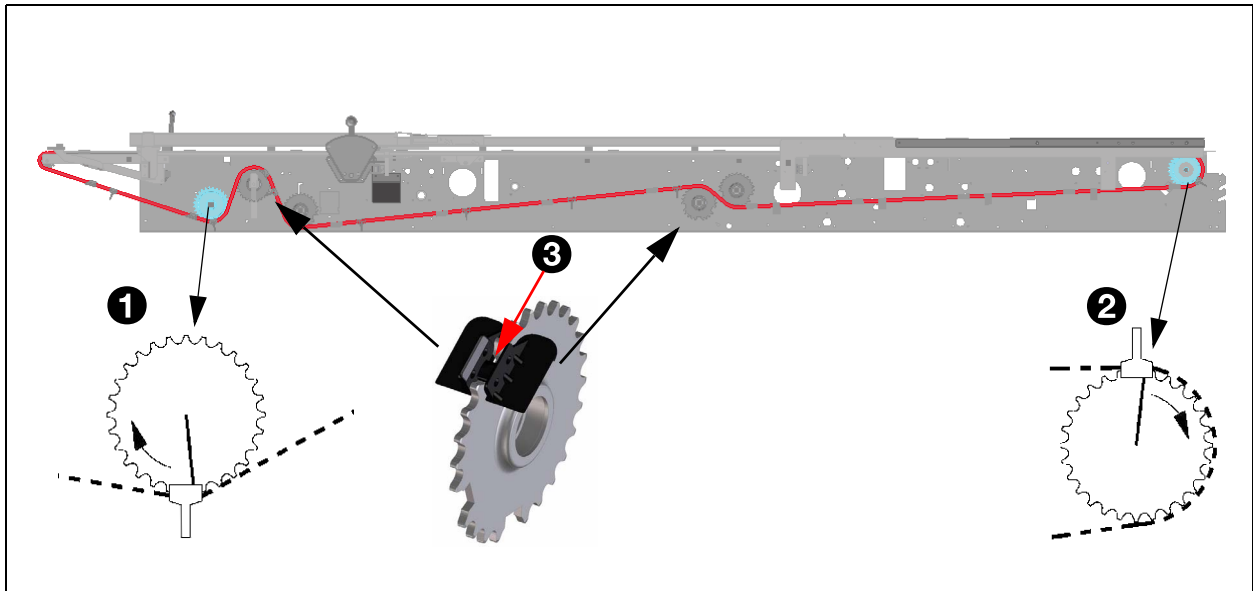


Fig. 179 - Path of transport section chain



### 11.5.2 0-position of the pusher chain (White nock and Big pack)

- 1 Turn the pusher chain by hand until the back of a pusher is in line with the indicator hole in the upper surface, see fig. 180. Check this setting using dimension (1) = 326 mm.
  - When the indicator is adjust previously (4), the distance will be bigger than 326mm.
  - When the indicator is adjust later (5), the distance will be smaller than 326mm.
- 2 Mount the indicator for the sensor (2) in this position, as shown in fig. 180. (This is on the outer side of the packing conveyor on the shaft of the front loop wheel).



**NOTE**

*For big pack lane, mount the indicator for sensor (3) in position.*

- 3 Check the adjustment while the packing conveyor is running.
- 4 Used for this the package “Index White Nock”
- 5 Make a further adjustment if distance (1) is incorrect.



**NOTE**

*Use for the Big Pack Lane, “Index Big Pack” package*

- 6 Initialize the server.
- 7 Check the “Index White Nock” or “Index Big Pack” again.



**NOTE**

*When the distance is still wrong, start the procedure again until the correct distance is reached.*

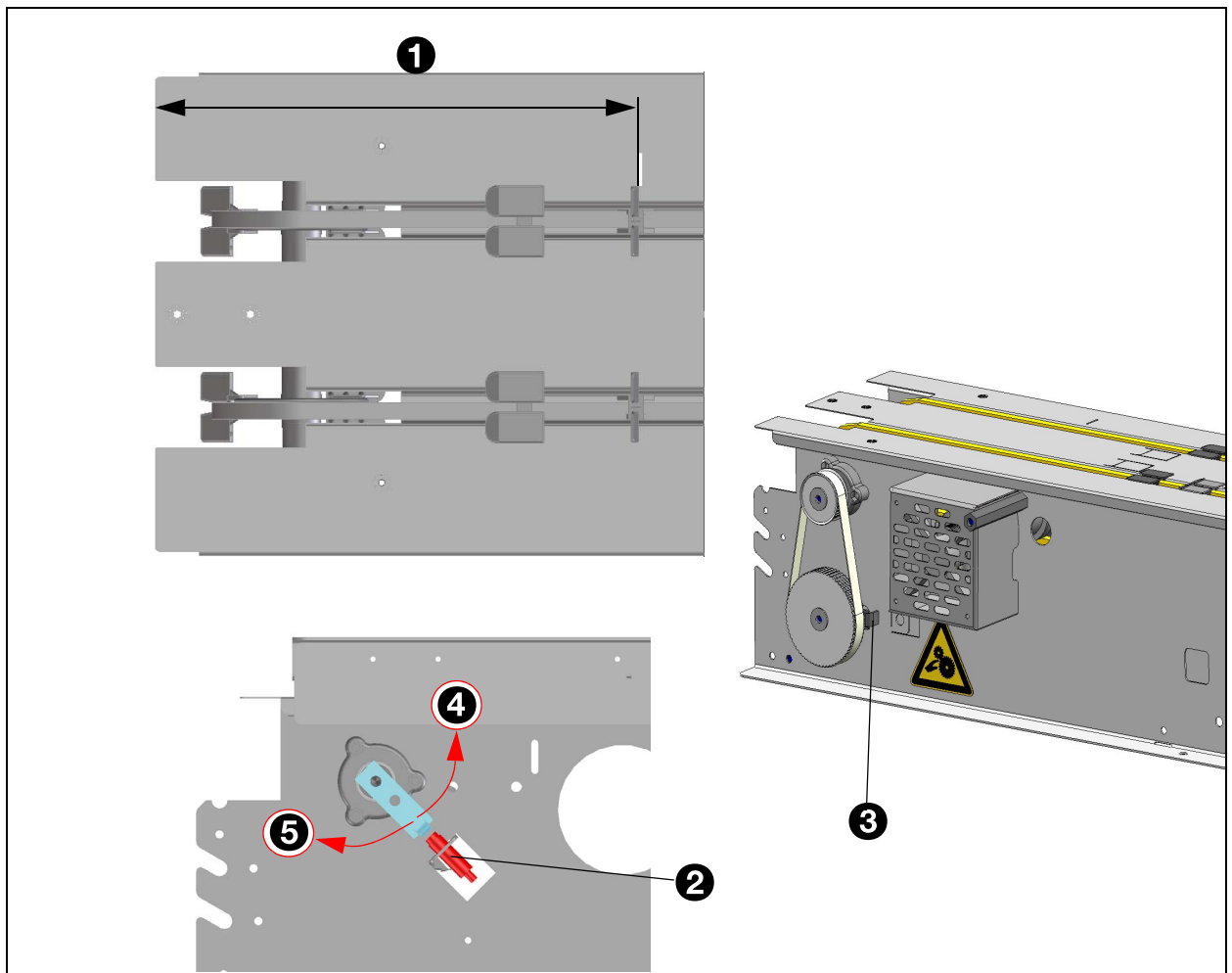


Fig. 180 - 0-position of the pusher chain



## 11.5.3 Adjustment of the spring-tensioner

The spring-tensioners (1) are installed on the infeed-side of the packing conveyor and are designed to maintain the tension on the chains, on which the pushers have been installed. These tensioners have been installed on both sides of the packing conveyor.

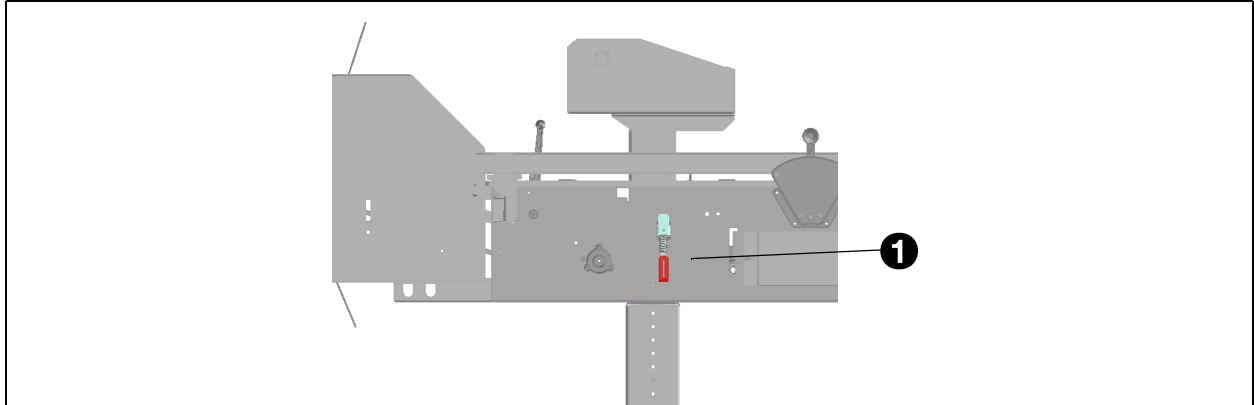


Fig. 181 - Adjustment of the spring-tensioner

Check the tension of the chains after every 400 hours of operation. The tensioner must be adjusted when it stands above the mark (6).

1. Loosen the bolts (3) of the limit-plate of the tensioner. See fig. 183.
2. Push the limit-plate (4) downwards.

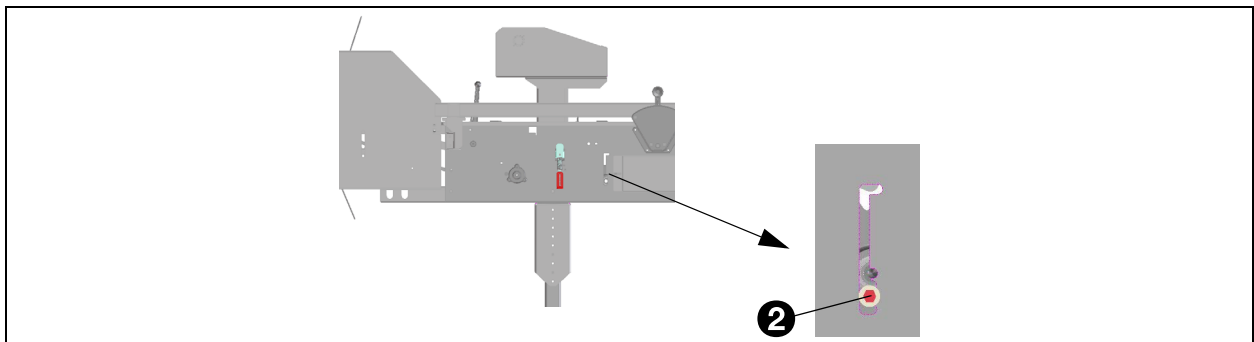


Fig. 182 - Location of the gear-wheel adjustment

3. Reposition the gear wheel (2) on the infeed-side of the packing conveyor downward by 1 hole. See fig. 182.
4. Check whether the tensioner stands under the mark and, if necessary, move the gear wheel 1 more hole downward and then retighten the bolt.
5. Mount the limit-plate (4) 1 mm under the strip (5) of the tensioner. See fig. 183.

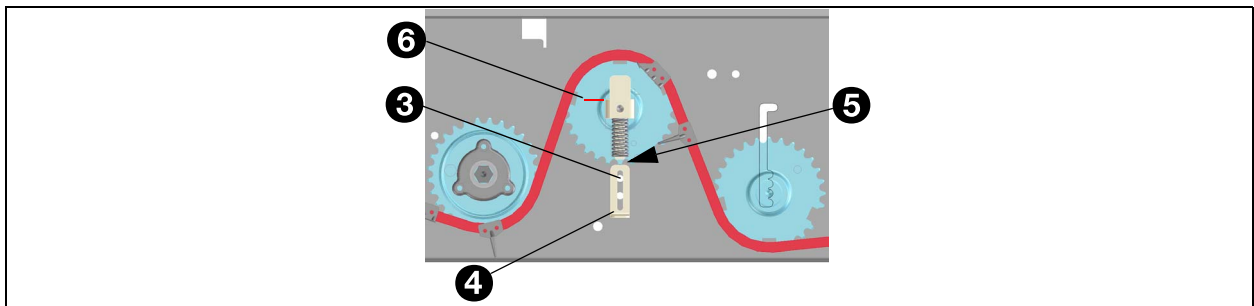
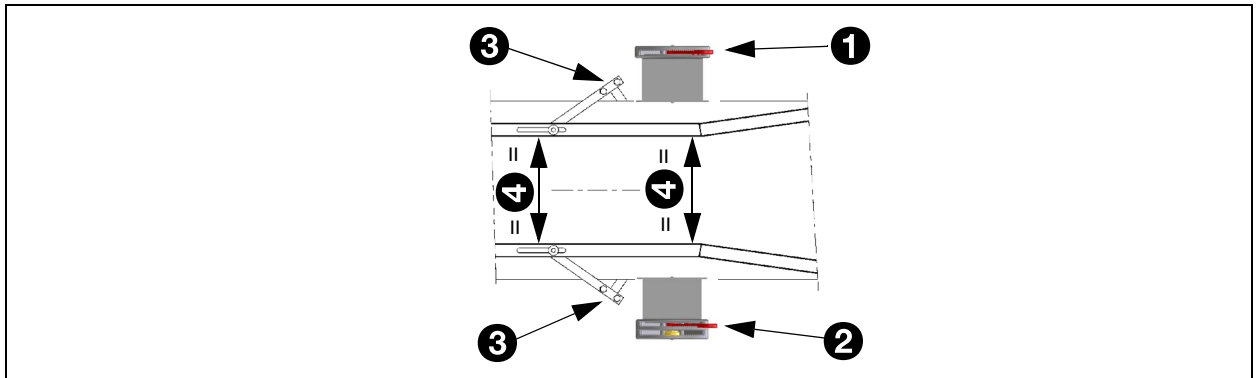


Fig. 183 - Detail of the spring-tensioner



### 11.5.4 Adjustment of the side guide

The side guide must be adjusted correctly if the eggs are to drop into the packaging properly, see fig. 184.



*Fig. 184 - Adjustment of the side guide*

1. Set lever (1) to the position 'EE'.
2. Set lever (2) to the position '3'.
3. Adjust the position of the swivelling arms (3) to set the dimension between the side guides (4) to 190 mm.



## 11.6 Closer unit

The closer unit closes consumer cartons.

### 11.6.1 Adjustment of the closer unit

The closer unit (1) is optional, and is designed to close consumer cartons. The unit can be switched on and off in the MMI software.

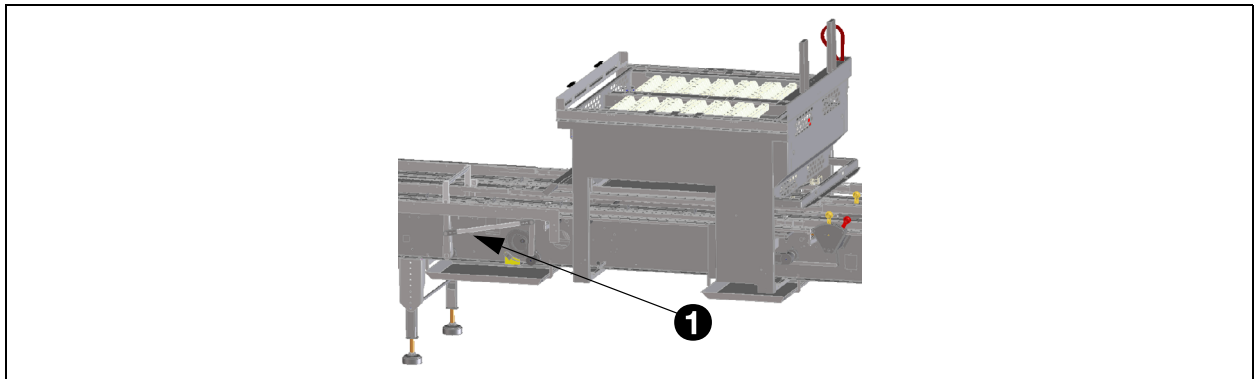


Fig. 185 - Overview of the closer unit

### 11.6.2 Adjusting the closer arm

The closer arm can be adjusted with the pull-rod (1) of the closer unit.

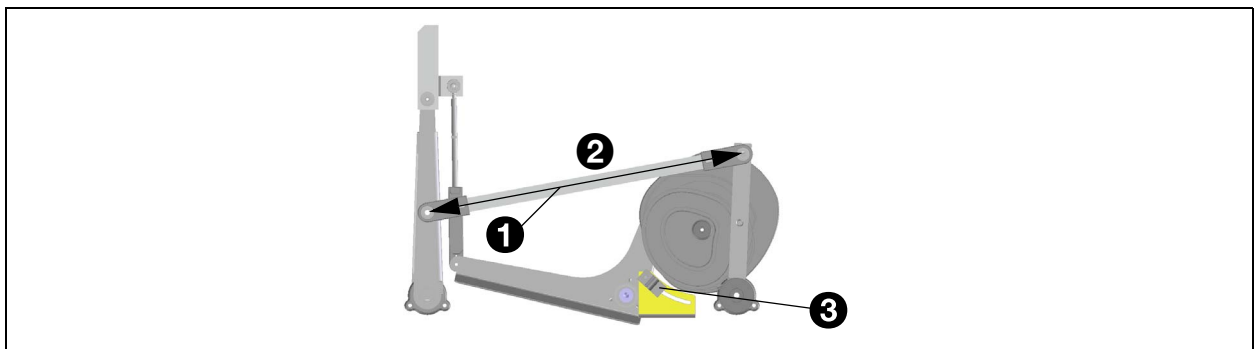


Fig. 186 - Start position of the closer arm

1. Set the closer arm (1) to the start position. The bearing must be located in the recess of the curved-disk. Turn the curved-disk to determine this point.
2. Adjust the pull-rod to 472 mm (2). See fig. 186.



Fig. 187 - Detail of sensor



3. Mount the stop-sensor (3) so that the closing unit comes to a standstill in its start-position, after 1 rotation of the curve, see fig. 187.

**NOTE**

*This adjustment must be made by trial and error.*

### 11.6.3 Adjustment of the lid-lifter (Big Pack)

The lid-lifter must be adjusted both in closed position as in raised position.

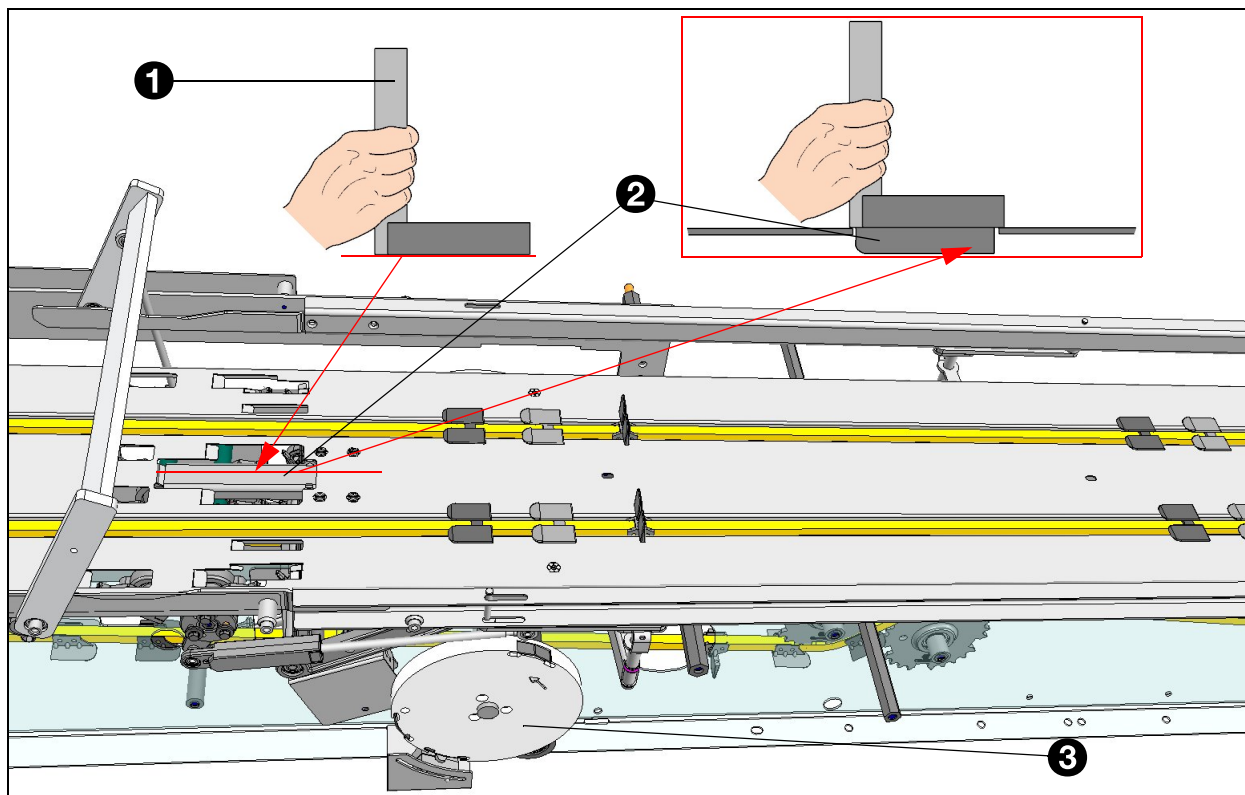


Fig. 188 - Overview of the lid-lifter

- 1 - Try-square
- 2 - Lid-lifter
- 3 - Curve-disk

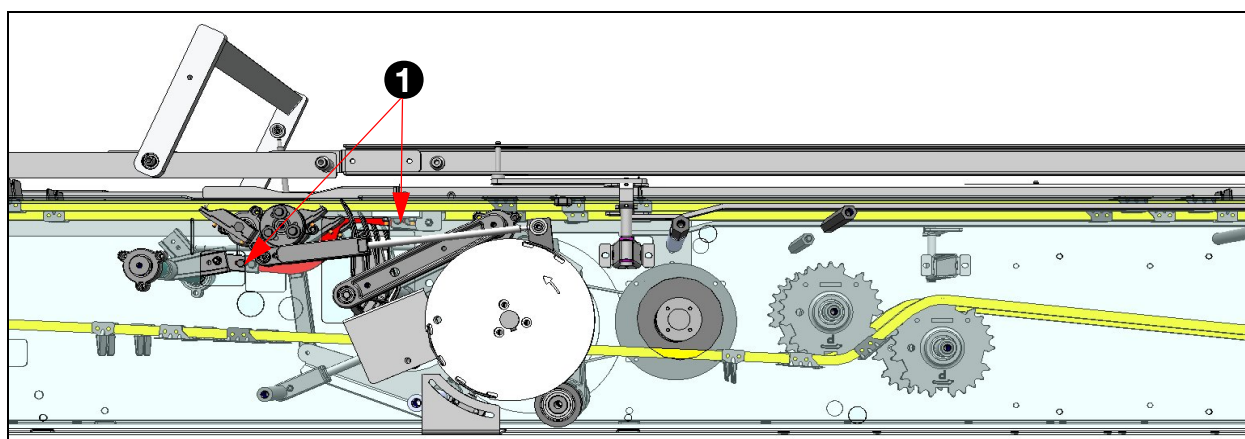


Fig. 189 - View of the pull-rod

- 1 - Pull-rod

#### Adjustment of the lid-lifter in closed position

- 1 Rotate the curve-disk in the most closed position of the lid-lifter.
- 2 Adjust the pull-rod so that the lid-lifter is equal of the upper plate.



Adjustment in raised position.

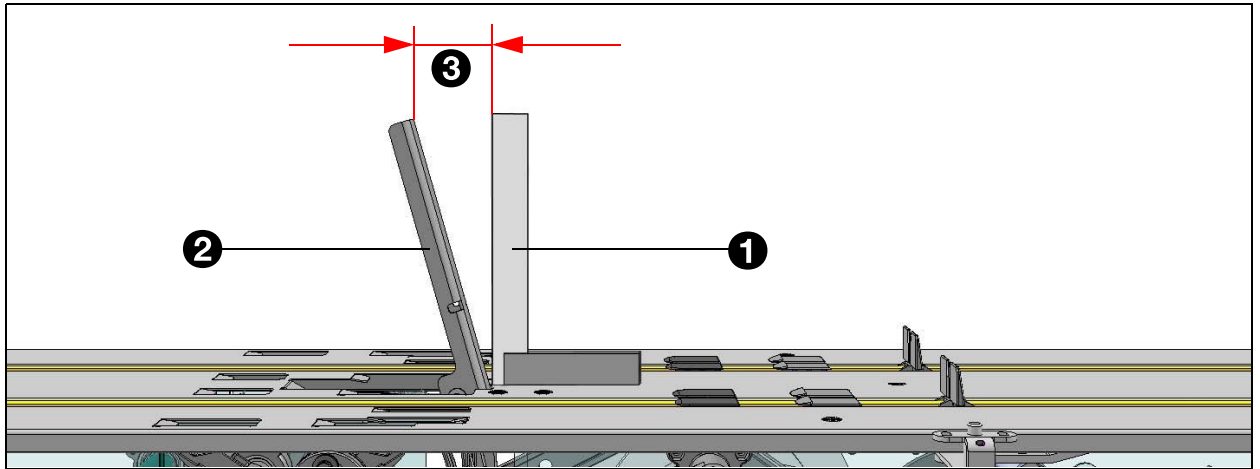


Fig. 190 - View of raised lid-lifter

- 1 - Try-square
- 2 - Lid-lifter
- 3 - Distance between 1 and 2

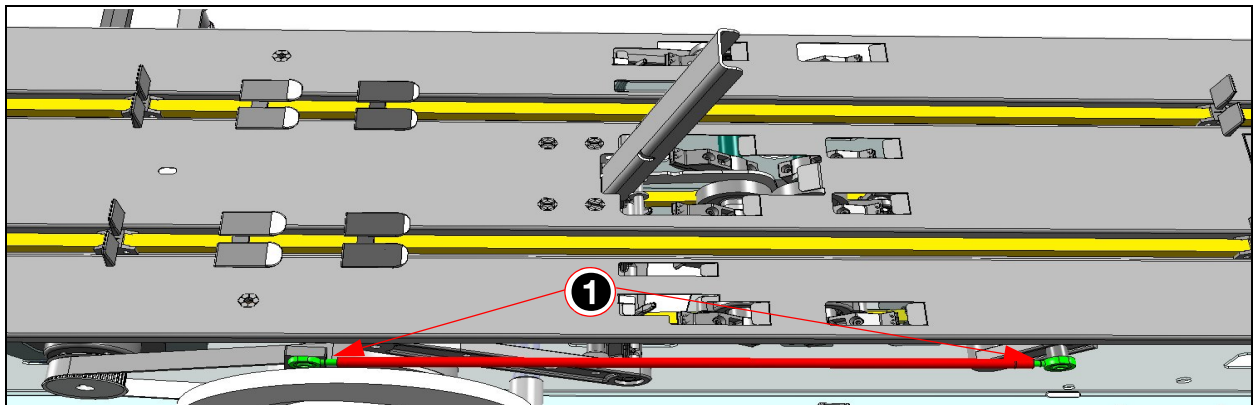


Fig. 191 - View of the pull-rod

- 1 - Pull-rod

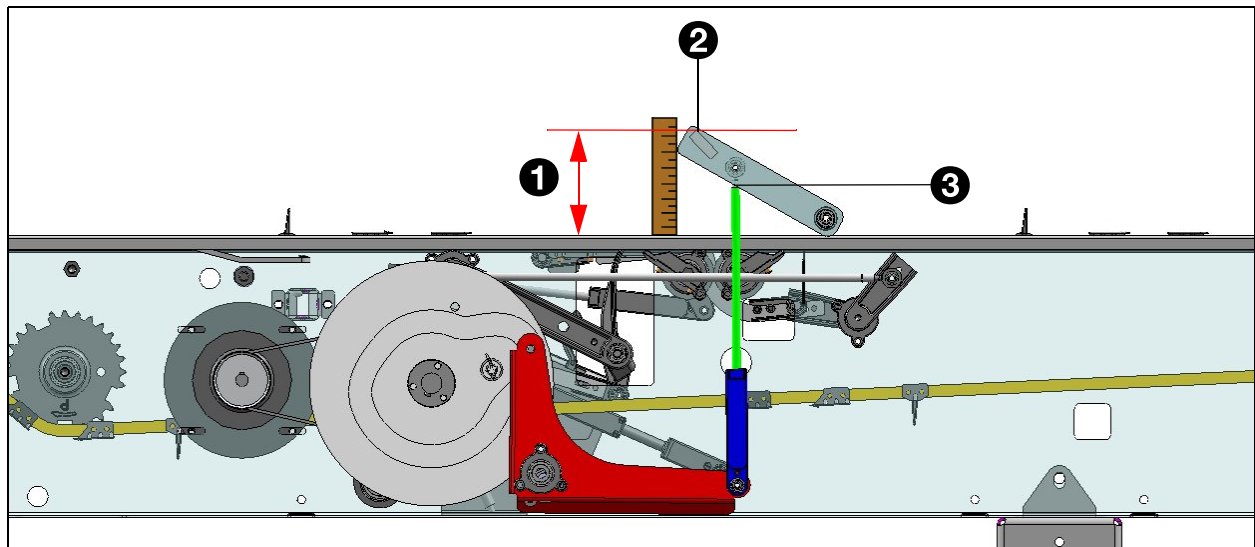
#### Adjustment of the lid-lifter in raised position

- 1 Rotate the curve-disk in the most raised position of the lid-lifter.
- 2 The distance between the try-square and the lid-lifter in the most raised position has to be 30mm.
- 3 Adjust this measure by means of the pull-rod.



#### 11.6.4 Adjustment of the lid-pusher (Big Pack)

The big pack material is closed by the lid-pusher.



*Fig. 192 - Lid-pusher of the big pack lane*

- 1 - Distance of 60mm.
- 2 - Measuring point on the highest place of the cross bar.
- 3 - adjust point.

##### **Adjustment of the lid-pusher (Big Pack)**

- 1 Rotate the curve-disk so, that the lid-pusher is standing in the deepest push position.
- 2 Adjust the pull-rod so that the distance between the upper plate and the cross bar is 60mm.



### 11.6.5 Adjustment of the lid-pusher

The packaging-material is closed by the lid-pusher.

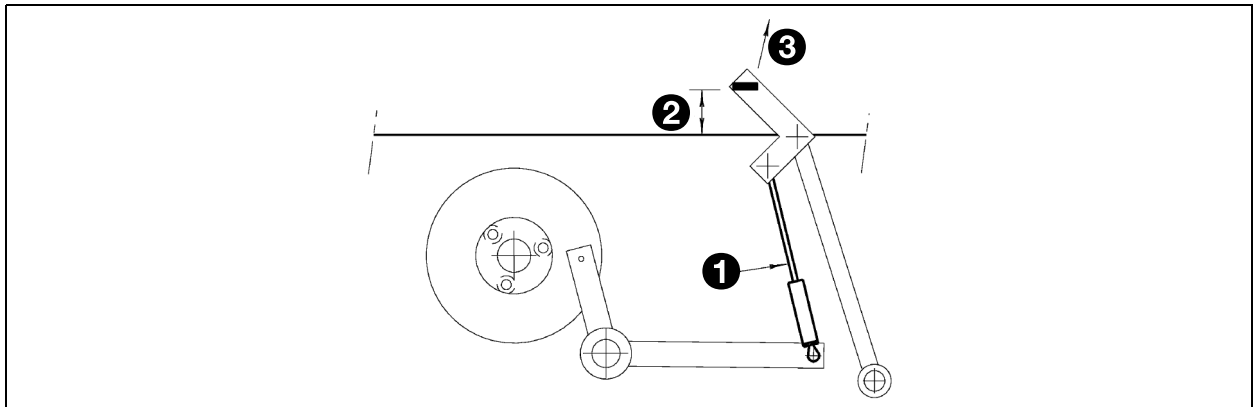


Fig. 193 - Lid-pusher

Turn the curve-disk by hand until the lid-pusher is located in the lowest position.

1. Use the pull-rod (1) to adjust the distance between the upper plate and the lid-pusher to 60 mm (2).
2. Push the play (3) upwards.



#### **WARNING**

*After they are released the curve-disks can turn by themselves due to the weight of the closer-arm. Be careful to avoid injury.*

### 11.6.6 Adjustment of the lid-lifter

The lid-lifter pushes the lid of a carton up, so that the closer-unit can push the lid onto the packaging-material.

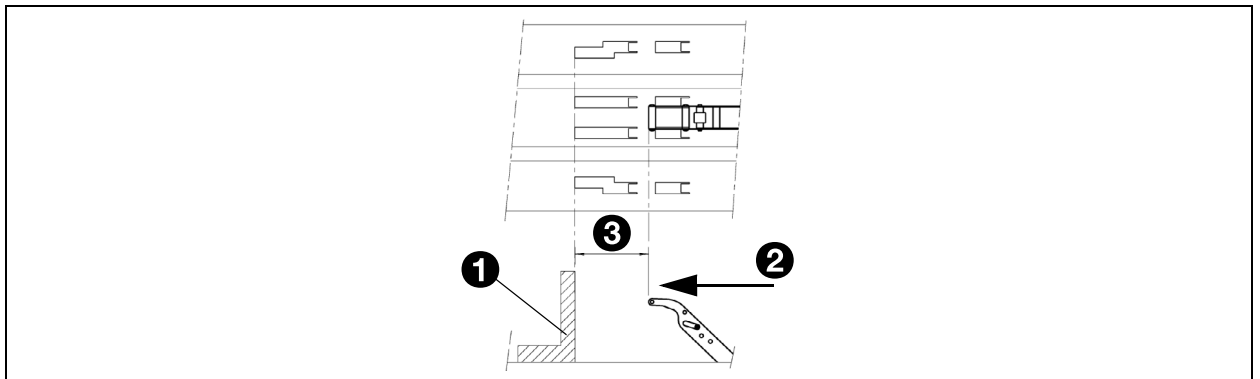
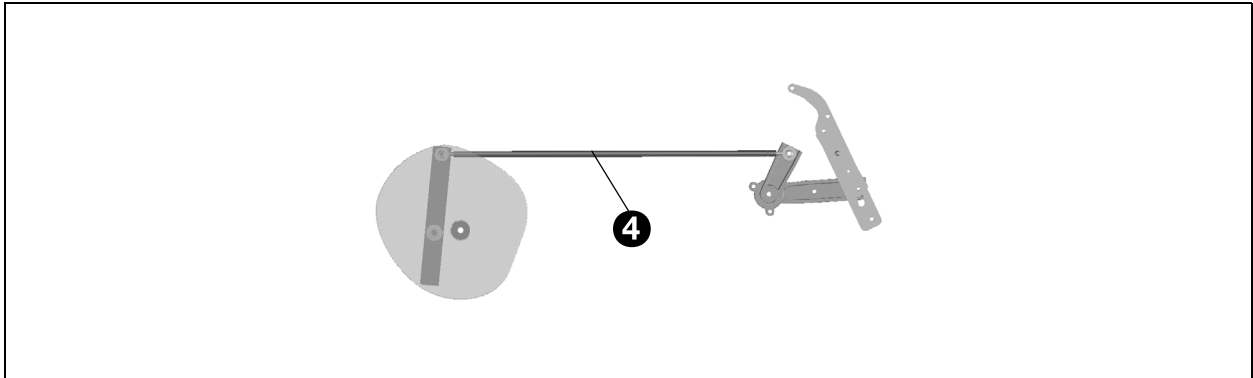


Fig. 194 - Ram

Set the lid-lifter into the position "10-12ers" (2) is the highest position. See fig. 194.

1. Turn the curve by hand, so that the lid-lifter is located in the outermost position.
2. Position a block-hook (1) at the edge of the recess.



*Fig. 195 - Pull-rod of the ram*

3. Use the length of the pull-rod (4), see fig. 195, to adjust the distance between the lid-lifter and the block-hook to 135 mm (3) (see fig. 194).



### 11.6.7 Adjustment of needle-units

The needle-units keep cartons in position, while the packaging-material is being closed.

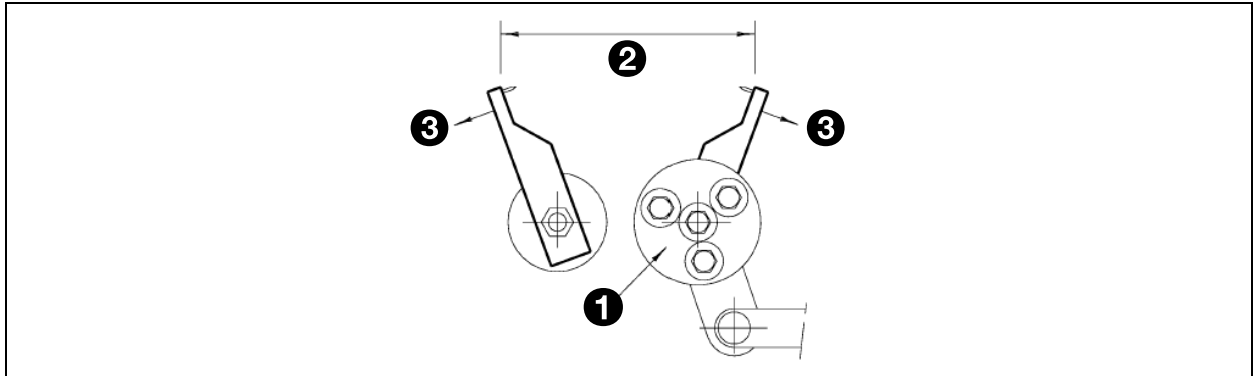


Fig. 196 - Adjustment of needle-units

- 1 - Adjustment ring
- 2 - Distance between needles = 80 [mm]
- 3 - Clearance towards the outside

#### Adjustment of the needle-units:

- 1 Turn the curve by hand, until the interval between the needle-units is the smallest.
- 2 Using the adjustment-ring (1), set the interval between the needle-units to 80 mm (2). Increase the clearance towards the outside (3) See fig. 196.

### 11.6.8 Adjustment of the closing-flap-pusher

The closing-flap-pusher pushes the flap of a carton upwards, so that the lid can fall over the flap.

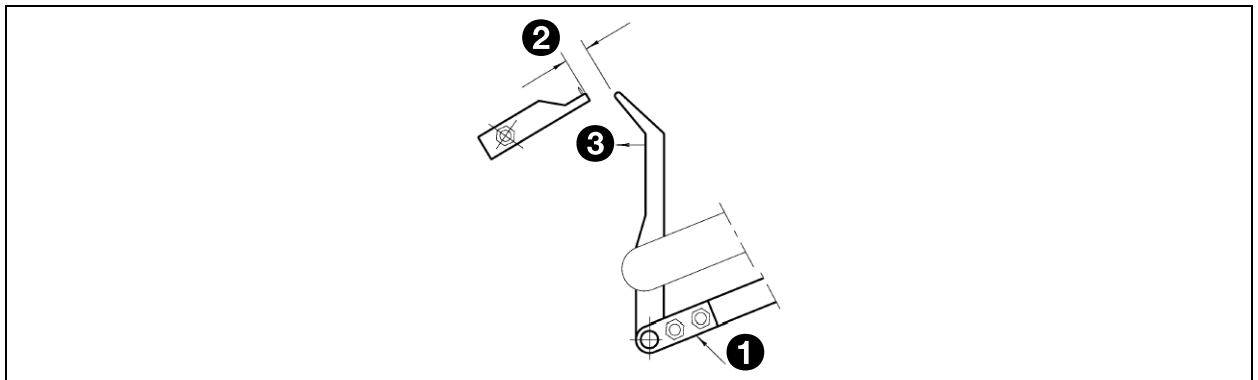


Fig. 197 - Adjustment of the forks

- 1 - Pull rod
- 2 - Distance closing-flap-pusher - needle-units = 18 [mm]
- 3 - Closing-flap-pusher

#### Adjustment of the closing-flap-pusher:

- 1 Turn the curve by hand, so that the closing unit is located in the start-position.
- 2 Use the pull-rod (1) to set the distance between the closing-flap-pusher and the needle-units to 18 mm (2). Increase the clearance (3) of the closing-flap-pusher in direction of the needle-units. See fig. 197.



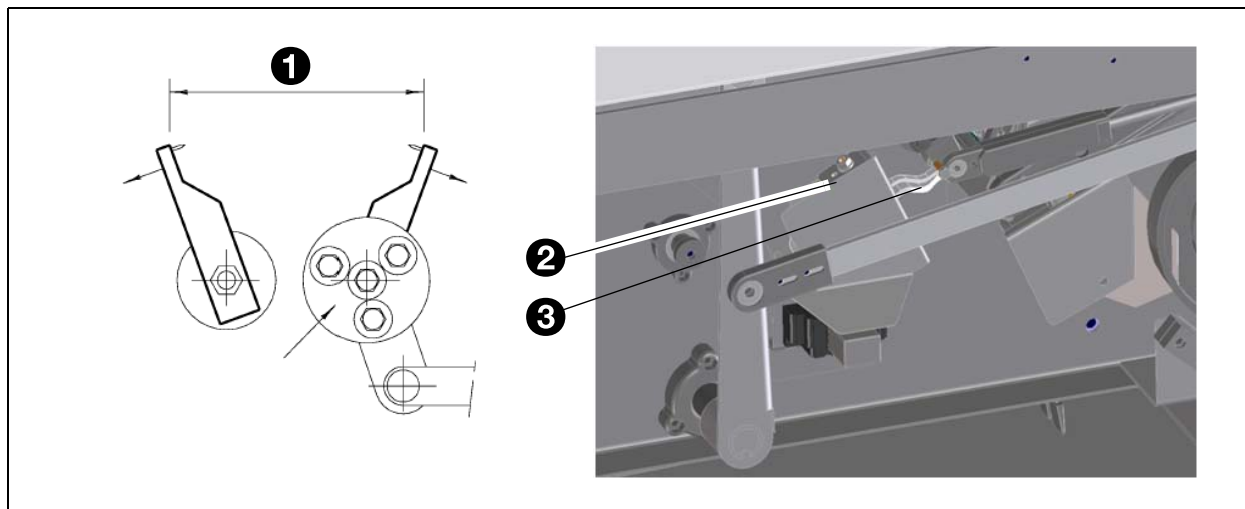
#### **WARNING**

**The closing unit contains some sharp and pointed parts. Please make sure to observe the safety instructions, while adjustments are carried out on the closing unit.**



### 11.6.9 Adjustment of Lock-system needles

In closed position the distance between the needles is 80 [mm]. For some packages this has effect on the processing of eggs. Therefore the distance between the needles is blocked at 88 [mm]. The coil is activated automatically by selecting the package where it has effect.



*Fig. 198 - Adjustment of Lock-system needles*

- 1 - Distance needles = 88 [mm]
- 2 - Coil
- 3 - Lock finger

#### **Adjustment of the Lock-system needles:**

- 1 Push the Lock finger up by hand.
- 2 Turn the curve by hand, until the interval between the needle-units is the smallest.
- 3 Check the distance between the needles is 88 [mm].



## 11.7 Lateral stamping

Adjustments of the lateral stamping, see fig. 199.

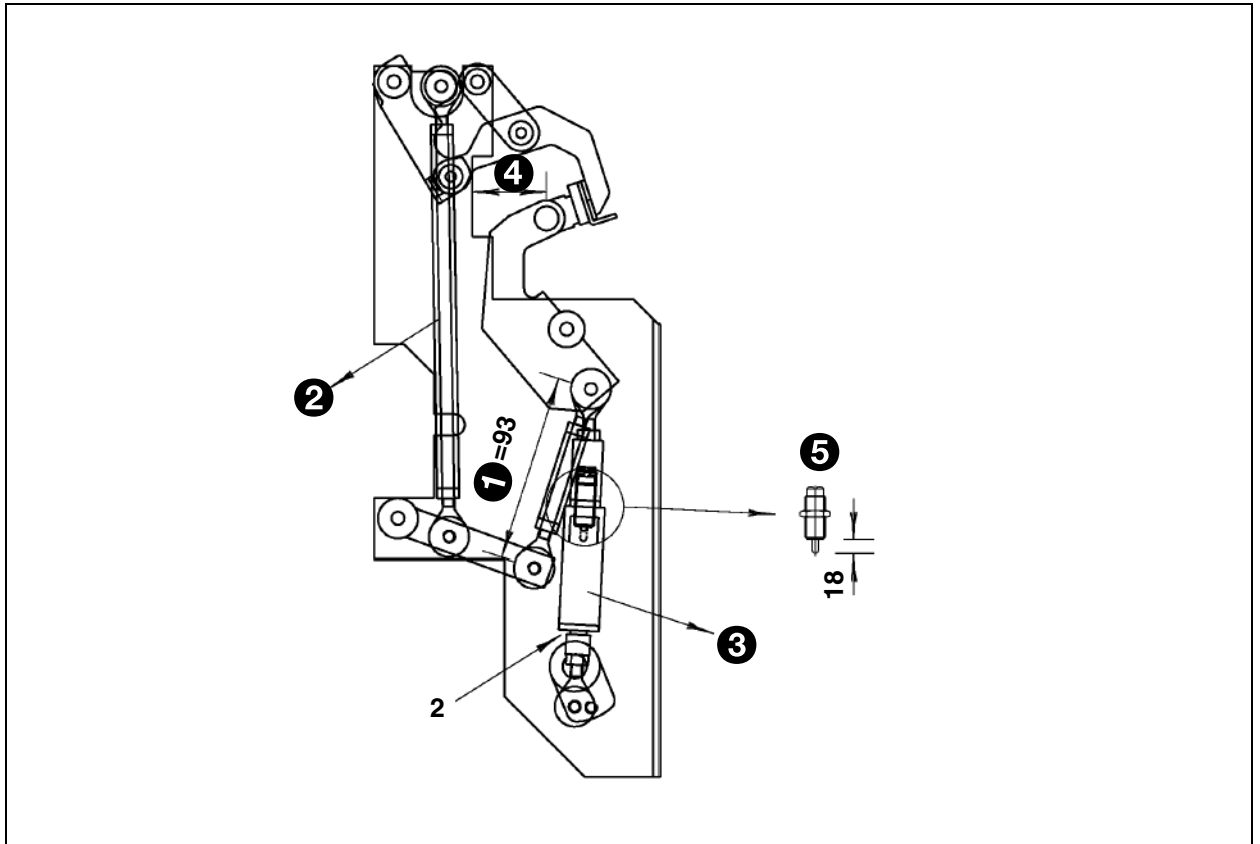


Fig. 199 - Adjustment of the lateral stamping

**Adjust the stamping in the following sequence.**

1. Check the pull-rod length (1) = (93 mm)
2. Set the stamping unit to the stamping position by hand.
3. Adjust the length of the pull-rod (2) to set dimension (4). (37 mm)
4. Adjust the length of the sprung pull-rod (3) so that it is extended by 2 mm.



**NOTE**  
*Dimension (4) can be adjusted using spacers.*

5. After replacing the impact damper (5) check the dimension in fig. 199.



## 11.8 Ink Me 59

New felt blocks must be treated with a liquid before use, see fig. 200. Use a solvent alcohol available from MOBA for this purpose.



Fig. 200 - Ink Me 59

1. Wait until the liquid has been absorbed by the felt blocks.
2. Pour the ink over the blocks.



### **NOTE**

***When the felt blocks are completely dried out then they will need to be treated with the liquid before new ink is applied.***

Order numbers:

Red ink 500 cc + 200 cc solvent: 02040021

Black ink 500 cc + 200 cc solvent: 02040022

Solvent alcohol 500 cc: 02040041



## 11.9 Take away conveyor with box shelf

When the take away conveyor is equip with a box shelf the legs need to be fixed on the floor.

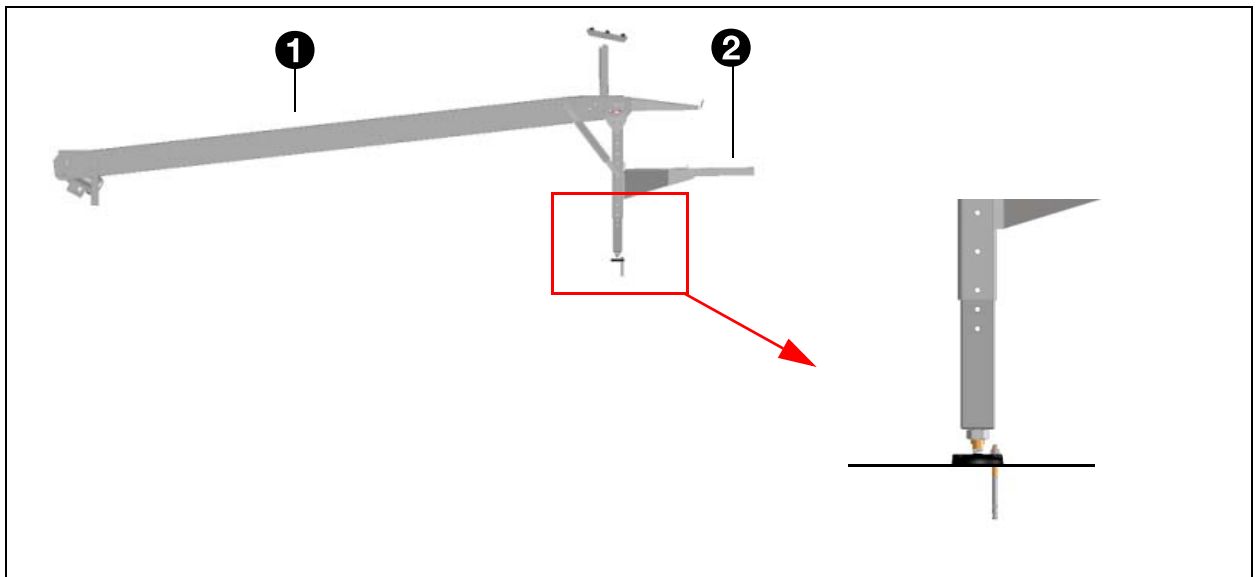


Fig. 201 - Take away conveyor fixed on the floor

- 1 - Take away conveyor
- 2 - Box shelf



**NOTE**

**Be sure all the legs of the take away conveyor are in level**



## 11.10 Carton take-away conveyor

### 11.10.1 Adjustment of carton take-away conveyor

1. Make two marks on the untensioned conveyor belt at a given distance from each other (for example, 1000 mm, see (1) in fig. 202).
2. Tension the conveyor belt to stretch it by a minimum of 0.3% and a maximum of 1%.  
! So for the 1000 mm dimension, the required stretch will be a minimum of 1003 mm and a maximum of 1010 mm, see (2) in fig. 202.
3. Allow the conveyor to run for several hours, and check the dimension of min. 1003 mm - max. 1010 mm.  
! After dismantling an already-tensioned conveyor belt then the belt must be reassembled with the same tension. You will then need to mark the position of the tensioner before removing the belt.
4. Adjust the tensioning unit such that the belt is centred on the rollers. (the guide strips indicated by (3) in fig. 202 will need to be fitted to prevent sideways movement of the belt).
5. Check that the belt is still centred on the rollers after the conveyor has run for several hours.

**NOTE**

*When making this adjustment make sure that the belt is not over-tensioned.*

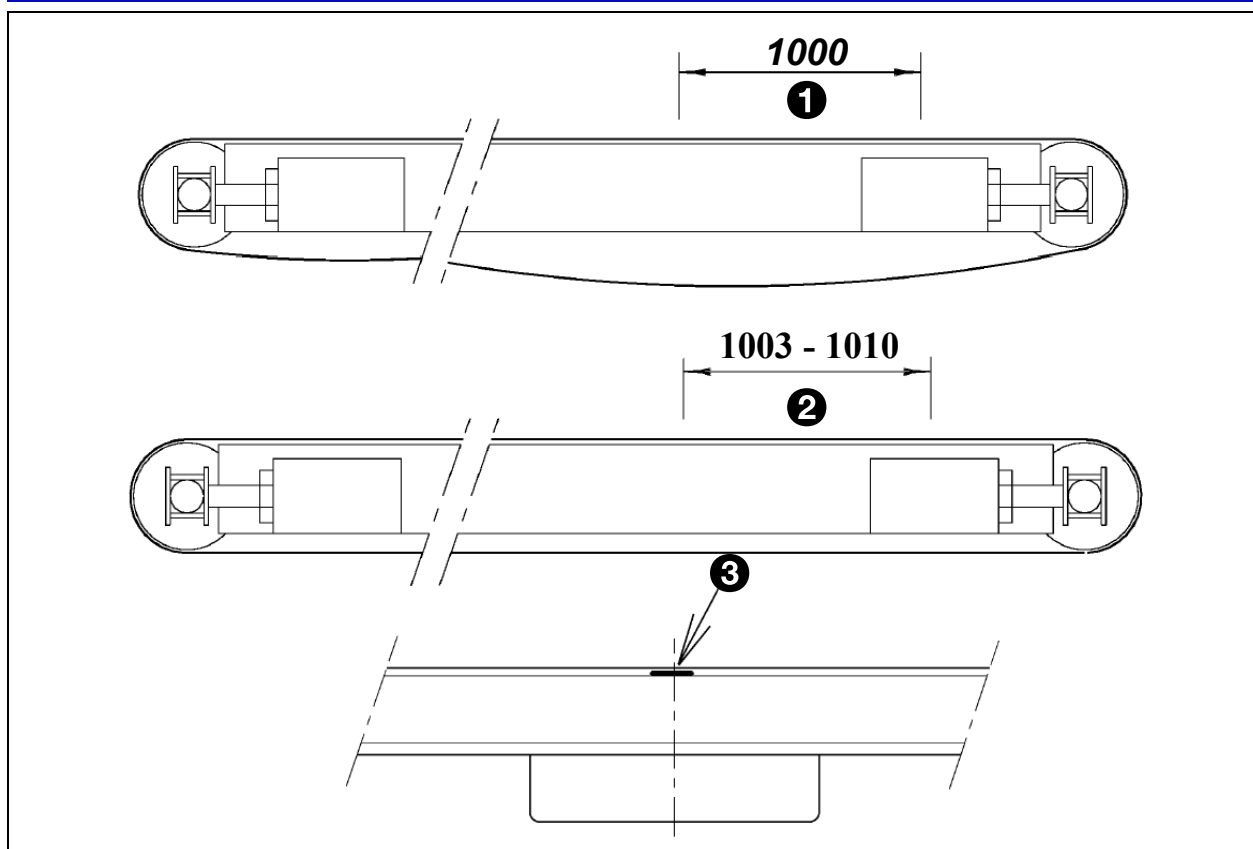


Fig. 202 - Adjustment of carton take-away conveyor



### 11.11 Manual packing lane with Dropset™ perpendicular to direction of travel

Besides the processing of eggs through an automatic packing-lane, the eggs from the Track can also be processed on a manual packing-lane. The eggs are then placed on a string conveyor. After transport by the string conveyor, the eggs are deposited on the accumulator. The eggs can be then picked up manually to place them in the cartons.

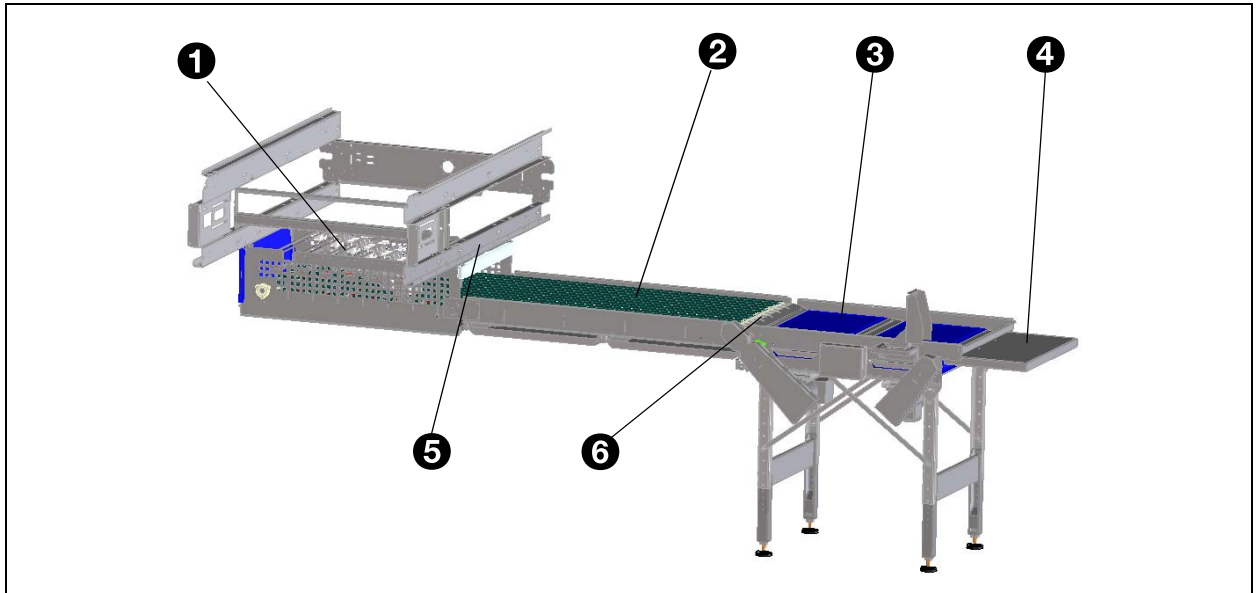


Fig. 203 - Overview of manual packing-lane

- 1 - Dropset™
- 2 - String conveyor
- 3 - Accumulator
- 4 - Pick-up side
- 5 - C-section in frame member
- 6 - Transfer point between string conveyor / accumulator



### 11.11.1 Adjustments of hand packing lane

The following height adjustments are needed to ensure that the hand packing lane and accumulator conveyor work properly.

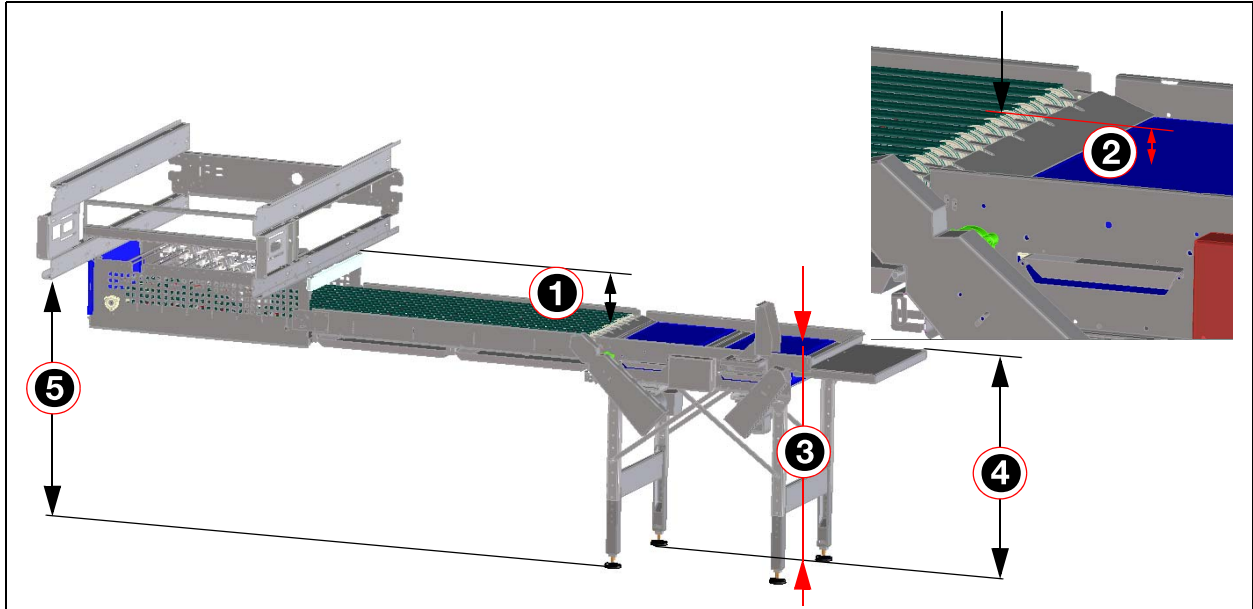


Fig. 204 - Adjustment of string conveyor / accumulator conveyor

- The height (1) is measured from the underside of the frame's C-section and the top of the belts (centre of the shaft)
- The height (2) is measured from the top of the belts (centre of the shaft) to the upper surface of the blue mat.

Table 8 - Height adjustment of belt conveyor / accumulator conveyor

| Machine model                                           | Setting (1) | Setting (2) | Take-away height (3)      |
|---------------------------------------------------------|-------------|-------------|---------------------------|
| 1-Track to breaker. Frame height (5) = 1,010 mm.        | 178 mm      | 30 mm       | 785 mm                    |
| 2 / 4 / 6-Track to breaker. Frame height (5) = 1275 mm. | 178 mm      | 30 mm       | 1,050 mm                  |
|                                                         |             |             | <b>Pick-up height (4)</b> |
| 1-Track standard frame height (5) = 1010 mm.            | 43 mm       | 30 mm       | 900 mm                    |
| 1-Track machine installed 100 mm higher, (5) = 1110 mm. | 143 mm      | 30 mm       | 1,000 mm                  |
| 2-Track. Frame height (5) = 1275 mm.                    | 308 mm      | 30 mm       | 900 mm                    |
| 4-Track. Frame height (5) = 1275 mm.                    | 308 mm      | 30 mm       | 900 mm                    |
| 6-Track. Frame height (5) = 1275 mm.                    | 308 mm      | 30 mm       | 900 mm                    |



#### **WARNING**

**If the actual dimensions differ from the above then the belt guide plates will also need to be modified.**



### 11.11.2 Fitting the mat

**WARNING**

*Fit the mat so that the chain-wheels engage in the mat's drive cams. (1) right, (2) is wrong, see example in fig. 205*

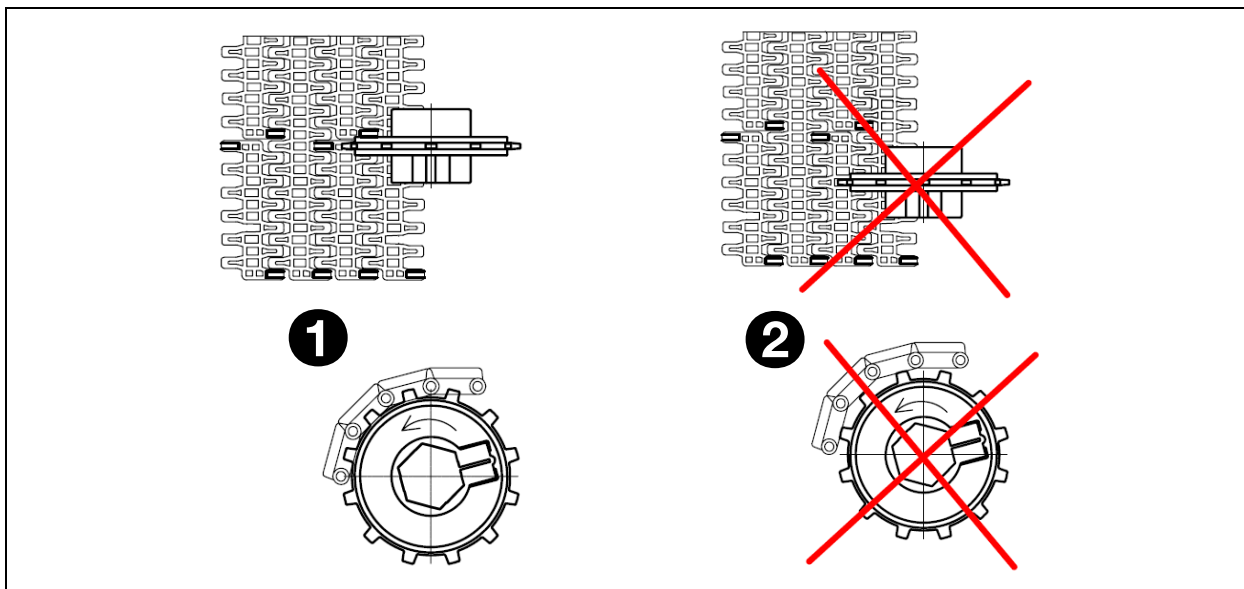
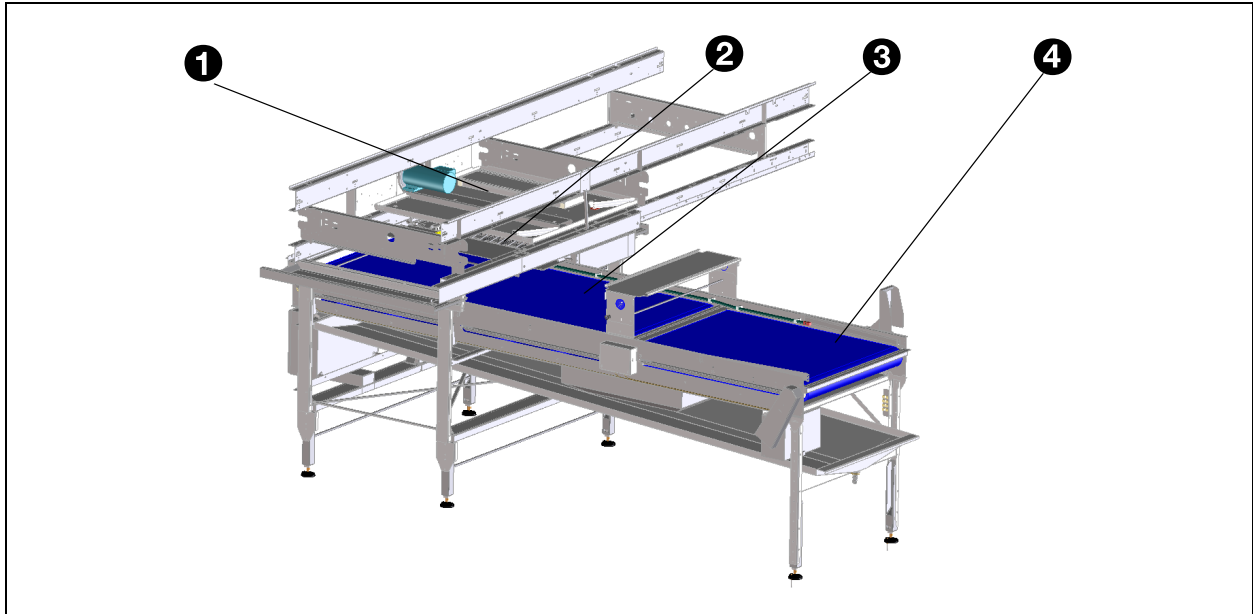


Fig. 205 - Fitting the mat



## 11.12 Egg Output Conveyor with Dropset™ in direction of travel.

Besides the processing of eggs through an automatic packing-lane, the eggs from the Track can also be processed by an E.O.C.. The eggs are then placed on a string conveyor. After transport by the string conveyor, the eggs are deposited on the accumulator. The eggs can be then picked up manually to place them in the cartons.



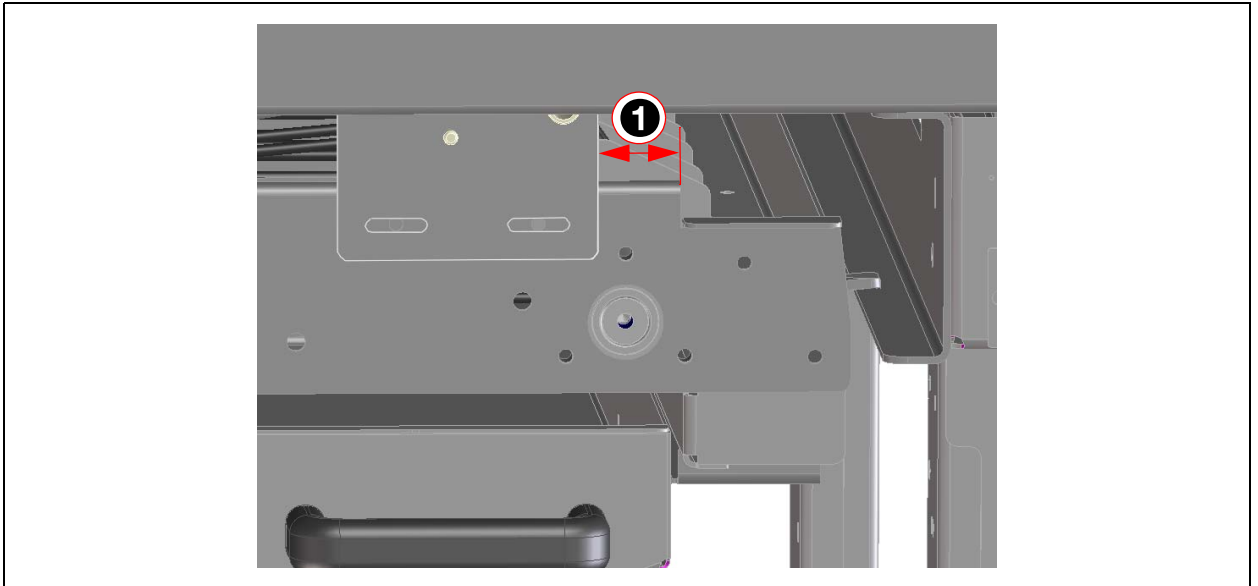
*Fig. 206 - Overview of manual packing-lane*

- 1 - Dropset™
- 2 - String conveyor
- 3 - Accumulator
- 4 - Take-away side



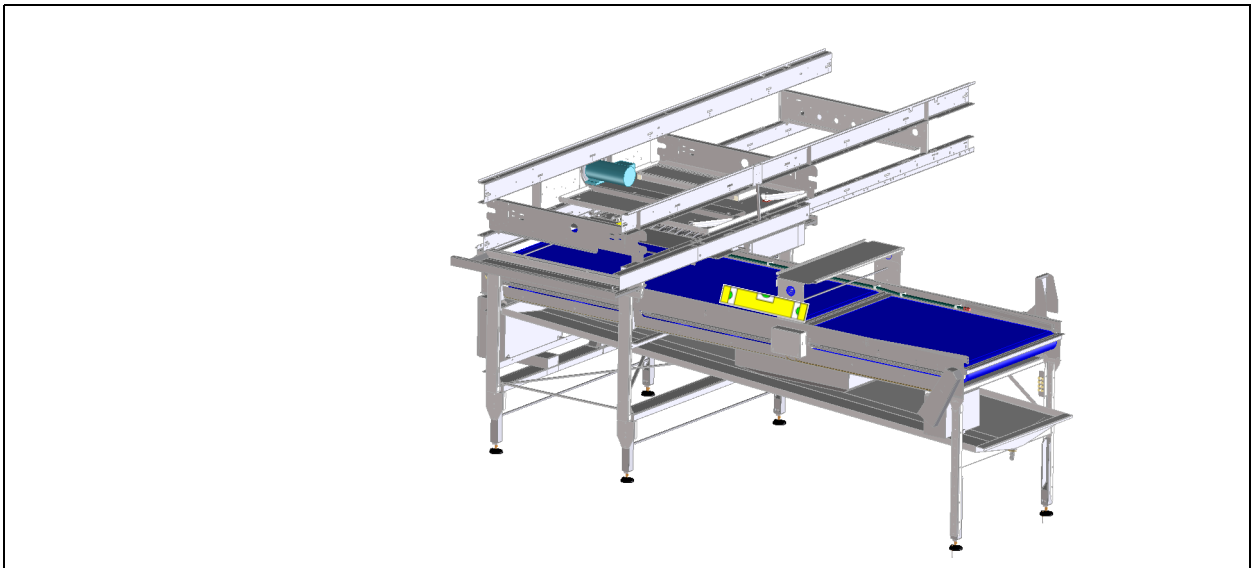
### 11.12.1 Adjustment of the E.O.C.

The following adjustments are needed to make sure that the E.O.C. works properly.



*Fig. 207 - Adjustments of the Dropset™ of the E.O.C.*

- Width (1) is 33 mm. See fig. 207
- Level the E.O.C. See fig. 208

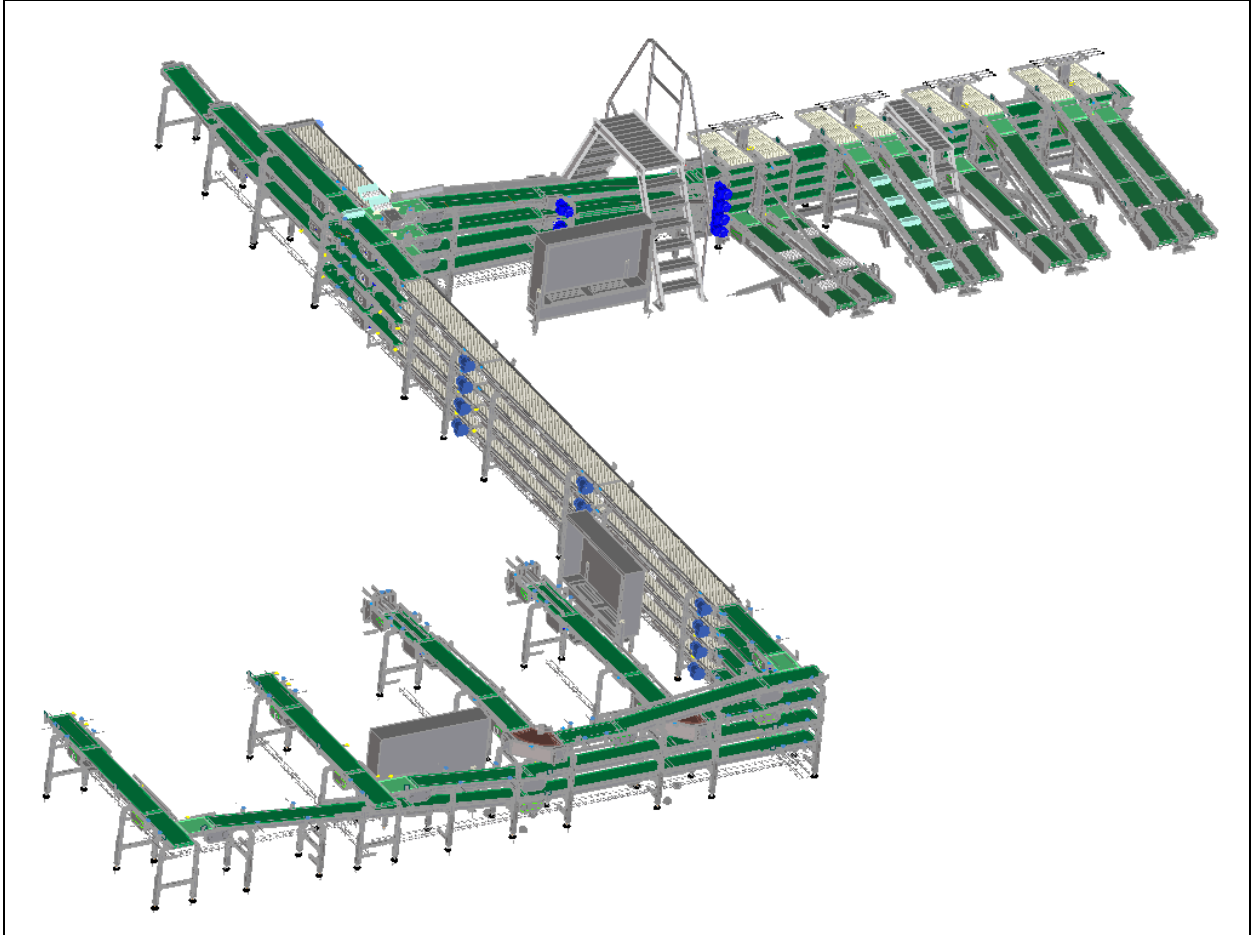


*Fig. 208 - Levelling the E.O.C*



## 12 Contiflow

This chapter discusses the adjustment and installation instructions for the contiflow installation.



*Fig. 209 - Example of the contiflow overview*



## 12.1 Installation instructions for the contiflow

These installation instructions must be followed to install the contiflow system correctly.

**WARNING**

***Check that the drop belts are aligned properly with the machine.***

1. Place the contiflow's trestles at the correct distance and to the correct height, see the drawings supplied with the machine, 14006630 (Europe) or 14005610 (Japan)
2. Mount the first and last buffer riser belts and use them to align the trestle.
3. Check the height and distance of the trestle relative to the packing conveyors, see the drawings supplied with the machine, 14006630 (Europe) or 14005610 (Japan)
4. Fasten the trestles to the floor using the wedge bolts supplied with the machine.
5. Make sure that the trestles cannot slide.
6. Mount the props (2x) of the first trestle, and mount the buffer riser belt in them.
7. Fit the locking arm into the upper notch (1) see fig. 210. **!** This is a job for 2 men.
8. Mount the gas strut in the indicated notch. See photo (2) in fig. 210. Make sure that the buffer riser belt always contacts the lowest notch (3).
9. Allow the buffer riser belt to drop into the notch (3). **!** This is a job for 2 men.
10. Once all buffer riser belts have been fitted the cross-belts can be moved. (See the indicative dimensions shown in the drawings supplied with the machine.)
11. When adjusting the buffer riser belts make sure that they are always free of the cross-belts in the vertical direction (2mm).

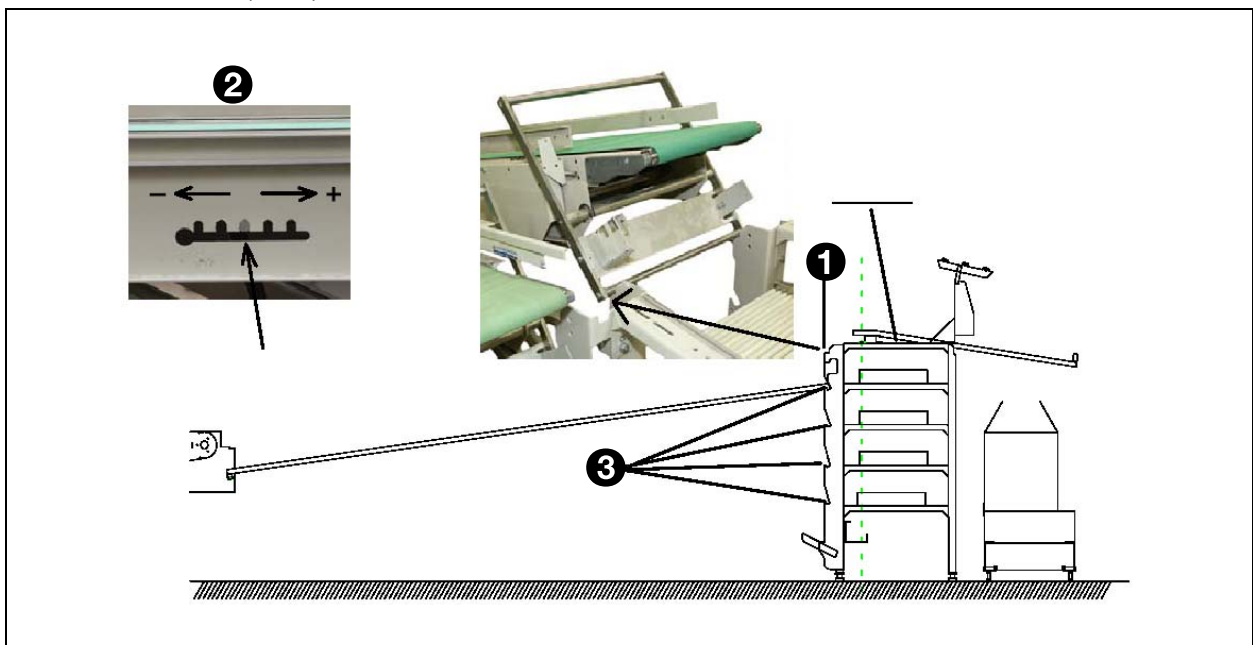


Fig. 210 - Installation of the contiflow



### 12.1.1 Tensioning and aligning the belts

The confiflow take-away system consists of a number of take-away belts. Each take-away unit is equipped with a drive unit. This unit contains the motor that drives the belt(s). The drive unit is either installed between 2 belts, see (4) fig. 211 or at the head of the belt, see (5) fig. 211 In both instances tension and align the belts in this drive.

**NOTE**

*The confiflow system can be fitted with 7 different types of belt.*

**CAREFUL**

*Check the type of belt before you tension the belt, see also page 180.*

**CAREFUL**

*Tension and align the belt with great accuracy, using the procedure explained below.*

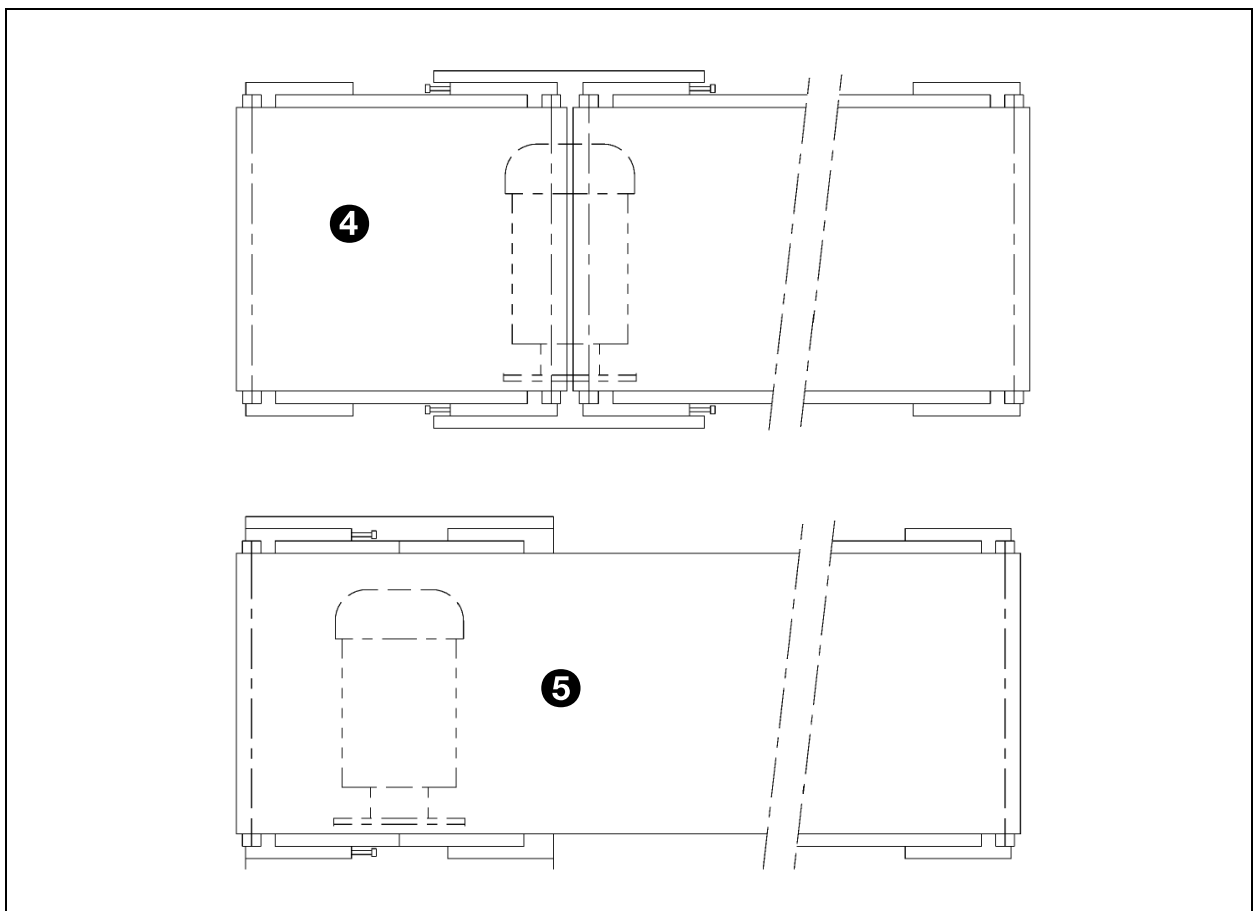


Fig. 211 - Aligning the belts



## 12.1.2 Tensioning belts of type 018741.. and 018751..

1. See the layout drawing supplied with your machine for the type of belt.
2. Make 2 marks on the untensioned belt at a given distance from each other (for example, 1000 mm (6), see fig. 212.)
3. Use the adjustment bolts (6) to tension the belt such that it stretches by 0.5%.  
With this stretch the dimension 1000 mm (7) must then be 1,005 mm (8), see fig. 213.



**CAREFUL**

**Do not tension the belt any further.**

## 12.1.3 Tensioning belts of type 018742.. , 018745.. and 018746..

1. See the layout drawing supplied with your machine for the type of belt.
2. Make 2 marks on the untensioned belt at a given distance from each other (for example, 1000 mm (7), see fig. 212.)
3. Use the adjustment bolts (6) to tension the belt such that it stretches by 0.3%.  
With this stretch the dimension 1000 mm (7) must then be 1,003 mm (9), see fig. 213.



**CAREFUL**

**Do not tension the belt any further.**

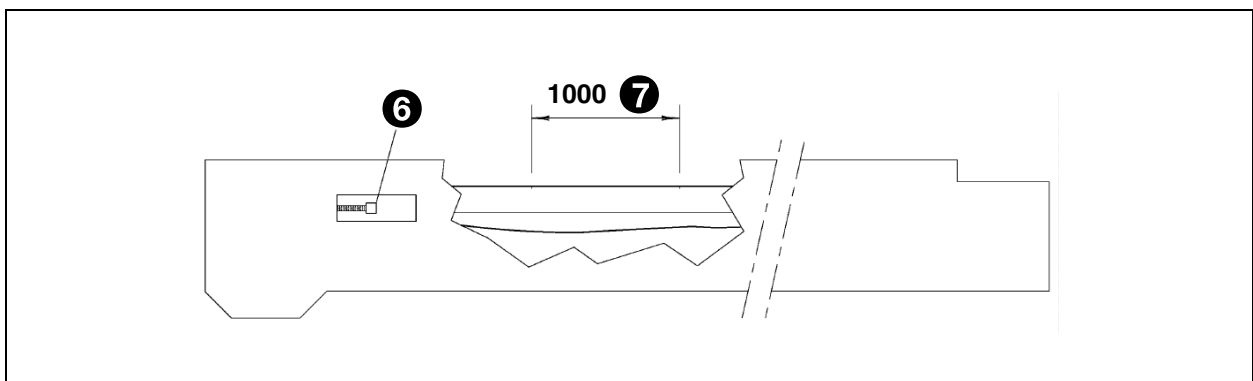


Fig. 212 - Tensioning belts of type 018741.. and 018751..

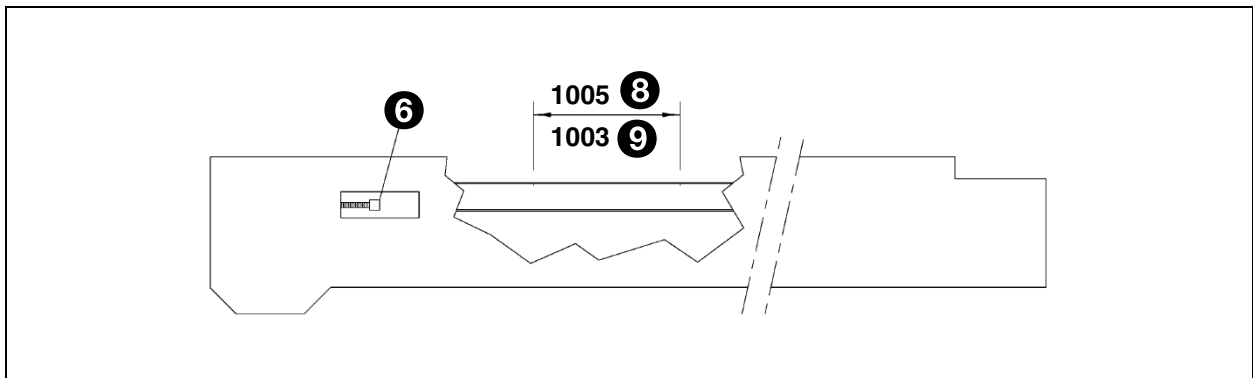


Fig. 213 - Tensioning belts of type 018742.., 018745 .. and 018746..



### 12.1.4 Tensioning belts of type 018752...

1. See the layout drawing supplied with your machine for the type of belt.
2. Make 2 marks on the untensioned belt at a given distance from each other (for example, 1000 mm (10), see fig. 214.)
3. Use the adjustment bolts (6) to tension the belt such that it stretches by 0.4%.  
With this stretch the dimension 1000 mm (10) must then be 1004 mm (12), see fig. 215.



**CAREFUL**

*Do not tension the belt any further.*

### 12.1.5 Tensioning belts of type 020984.. (with the exception of 02098417, which must not be pretensioned)

1. See the layout drawing supplied with your machine for the type of belt.
2. Make 2 marks on the untensioned belt at a given distance from each other (for example, 300 mm (11), see fig. 214.)
3. Use the adjustment bolts (6) to tension the belt such that it stretches by 1.3%.  
With this stretch the dimension 300 mm (11) must then be 304 mm (13), see fig. 215.



**CAREFUL**

*Do not tension the belt any further.*

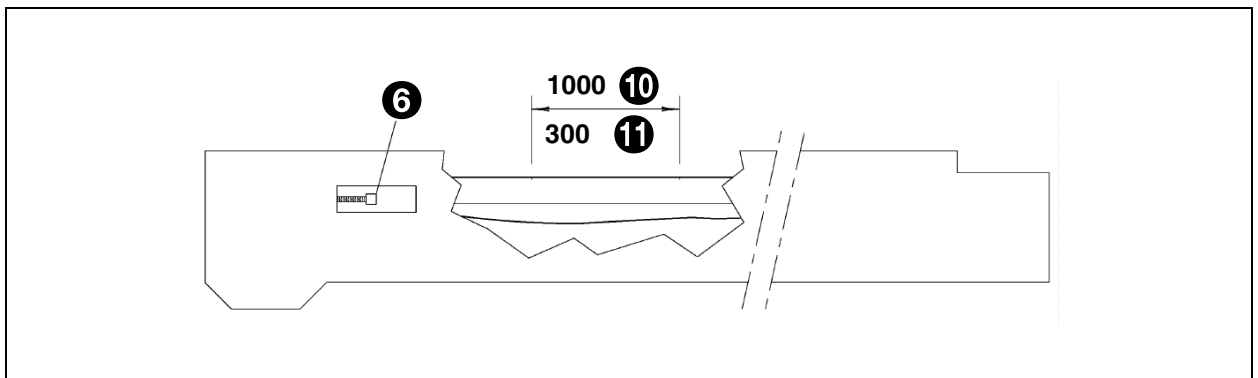


Fig. 214 - Tensioning belts of type 018752...

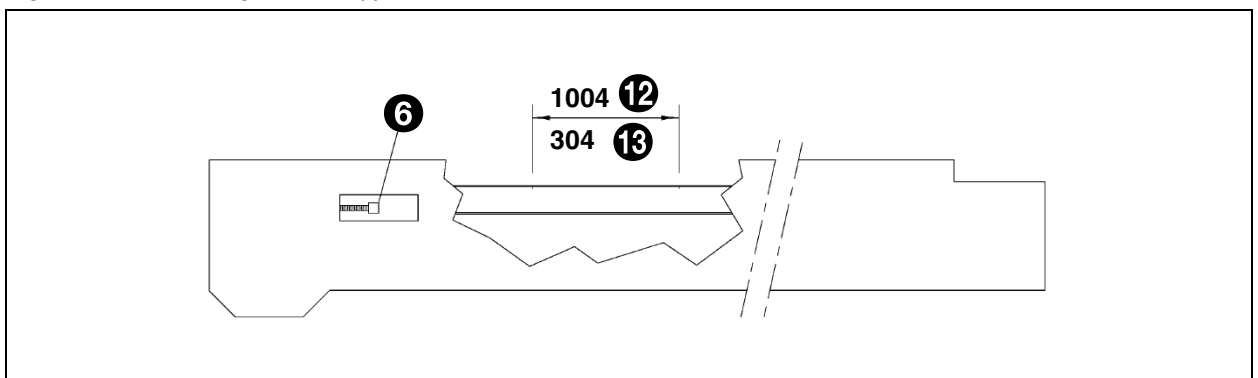


Fig. 215 - Tensioning belts of type 020984..

### 12.1.6 Alignment of the belts

1. Align the belt by on 1 side (the side with the sticker with arrows) by turning bolt (14) inwards or outwards.  
Use the following adjustment guidelines:  
Turning bolt (14) in the direction of the travel of the relevant belt will move the belt towards the sticker, see fig. 216.  
Turning bolt (14) in the opposite direction from the direction of the travel of the relevant belt will move the belt away from the sticker, see fig. 216. See also the arrows on the stickers.
2. Align the belt so that it is centred on the rollers.

**CAREFUL****CAUTION:**

*Make sure that the alignment does not increase the belt's tension.*

*Correct the alignment of the belt by turning bolt (14) inwards or outwards on only one side of the belt (on the side with the stickers!!).*

*Turning the bolt (14) by just a quarter of a turn will result in the alignment of the belt changing after operation for some time.*

*The rollers can be cylindrical and convex rollers. The alignment roller is always a cylindrical roller, and is located on the drive side. This roller can be identified by the groove in the roller, see (15) in fig. 216.*

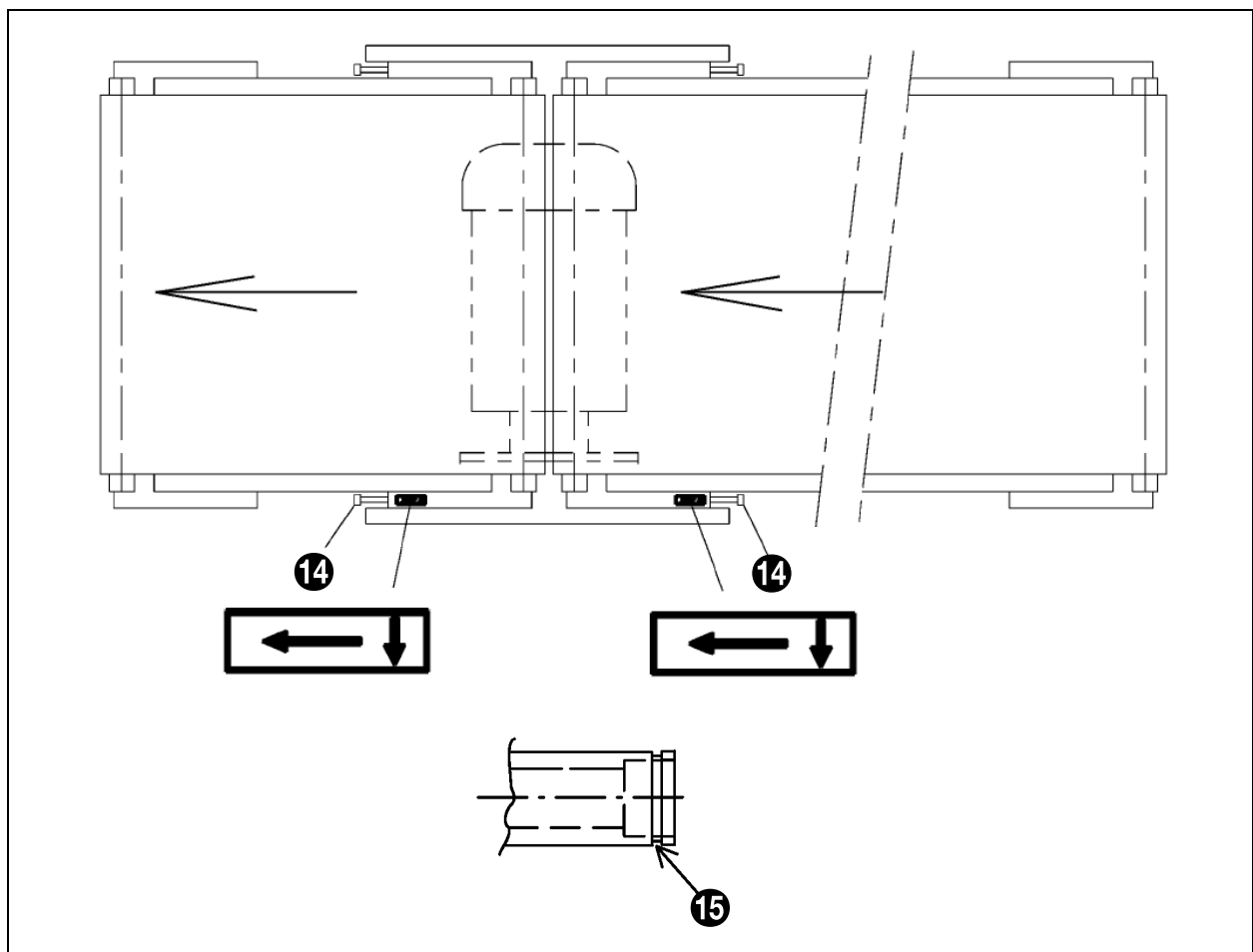


Fig. 216 - Alignment of the belts with bolts



### 12.1.7 Mounting the drive belt

1. Mount a drive belt with the canvas side facing outwards and the rubber-coated side inwards.



**CAREFUL**

*Mount the belt with the arrow in the direction of travel.*

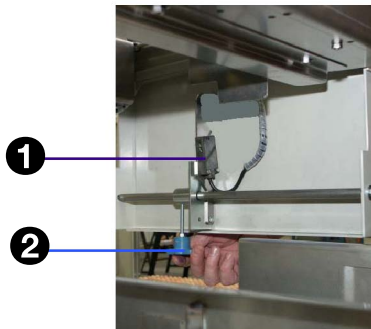


## 12.1.8 Top tray dispenser

1. Use about 10 empty trays to adjust the dispenser to the correct setting. (fig. 217)
2. Place the trays in the Top tray dispenser.
3. Position the trays symmetrically with respect to the centre.
4. Keep a 5 mm distance around the trays in the Top tray dispenser.
5. Turn the star knob (2) under the Top tray dispenser to adjust the time that the cover drops.(fig. 218)
6. 1 eye is fitted under the Top tray dispenser:  
Eye (1) ("Top tray trigger") : looks to see when a tray can be released.



*Fig. 217 - Toptray dispenser with 10 empty trays*



*Fig. 218 - Adjustment of the Toptray dispenser*

## 13 Putting out of operation

To put the machine out of operation, for a holiday period for example, the following points must be checked:

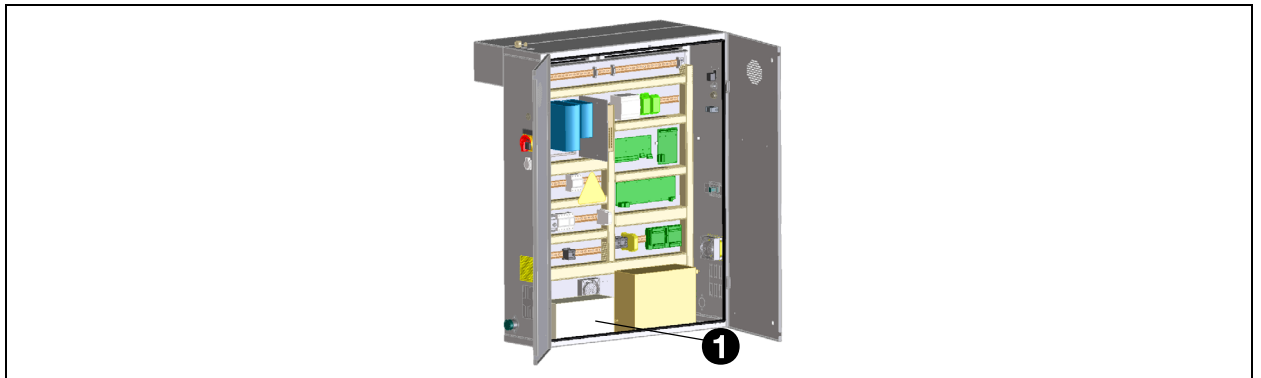
- Is the machine empty?
- Has the machine been cleaned?
- Have all maintenance points been checked?

### 13.1 UPS-system

The machine is equipped with a UPS-system (UPS = Uninterruptible Power Supply). This system supplies the power during an electricity failure. This system ensures that the loadcell-processors, the PCs and the Maintiming-prints continue to operate. To put the machine out of operation, this UPS system must be switched off.

### 13.2 Shutting down the machine

The machine must be shut down in accordance with the following procedure.



*Fig. 219 - UPS-system*

Run the machine until it is completely empty.

1. Shut down the PCs.
2. Open the main electrical cabinet.
3. Switch off the UPS-system (1).
4. Close the main electrical cabinet.

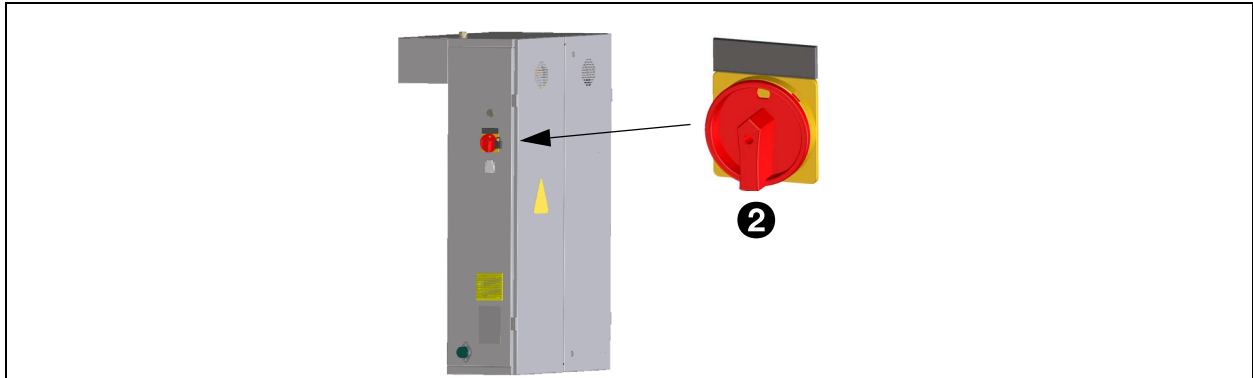


Fig. 220 - Primary power-supply-switch

5. Switch the main switch (2) on the side of the main electrical cabinet to 'OFF'.



**NOTE**

*In the case of a prolonged stoppage, read the extra preventive safety measures. See paragraph 14.2.7.*

## 14 Maintenance

This Chapter deals with the maintenance of your Omnia FT/XF. We recommend that you follow these instructions when you carry out maintenance. Follow the maintenance schedules in this book for a step to step maintenance program of your Omnia FT/XF. Keep these schedules close by the machine and draw the attention of your personnel to the proper manner of maintaining the machine. Periodical maintenance sessions, in compliance with these schedules, are absolutely necessary for the continued correct functioning of the machine and/or unit.



### NOTE

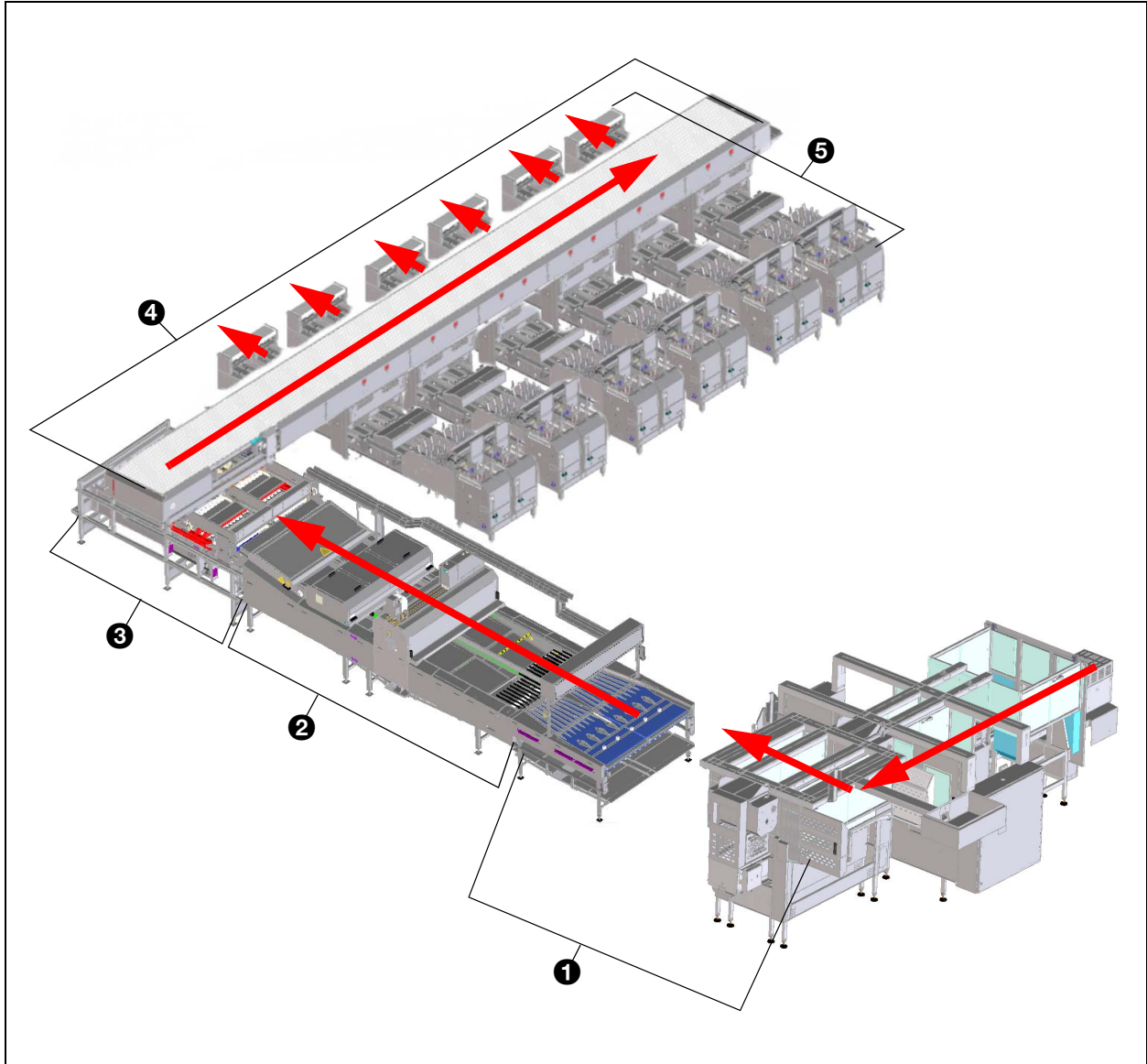
***Before carrying out any cleaning and/or maintenance work read the safety instructions in Section 2 carefully.***

Table 9 - Very important points for proper maintenance.

| KEY CONSIDERATIONS IN PERFORMING MAINTENANCE                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Your Omnia FT/XF is a precision machine. Please treat your machine accordingly.                                                                                                                                                                                                                                                    |
| Remove excess lubricants after the completion of lubrication procedures. Observe locally applicable regulations for the disposal of waste oil and grease residues.                                                                                                                                                                 |
| After your maintenance works are completed, reinstall the protective caps and covering in the correct position on the machine and/or unit.                                                                                                                                                                                         |
| Compressed air that is used for cleaning-operations must be of good quality and have a dew-point of maximum 3°C (38°F). The pressure must be at least 6 bar (87 psi).                                                                                                                                                              |
| Clean suction cups with clear water and do not use detergents. The maximum permitted water temperature is 30° C (86° F).                                                                                                                                                                                                           |
| Remove all loose drip trays from the machine and clean them separately at a washing station.                                                                                                                                                                                                                                       |
| Regularly check all toothed belts and chains for wear-and-tear. Replace worn out and/or damaged toothed belts and chains.                                                                                                                                                                                                          |
| Check all electrical connections for jamming every year.                                                                                                                                                                                                                                                                           |
| Every 40 hours of operation clean all optical sensors with the associated reflectors.                                                                                                                                                                                                                                              |
| Lubricate machine parts immediately after they have been cleaned off with water.                                                                                                                                                                                                                                                   |
| After every 40 hours of operation, remove eggshells and any feathers near the 'dropsets' in the frame.                                                                                                                                                                                                                             |
| After every 200 hours of operation clean the air filters in the machine with compressed air, with the exception of the Image House air filters of the EggInspector™. Replace the air filters after every 2400 hours of operation. We recommend that in very dusty rooms you replace the air filters after 2000 hours of operation. |
| Clean the Image House oiled air filters of the EggInspector™ with a special cleaning kit.                                                                                                                                                                                                                                          |

## 14.1 Overview of machine

The left and right side are determined on the basis of the direction of the movement of eggs. In fig. 221 the direction of movement of the eggs is indicated with the red arrows. The example below is a left-hand machine.



*Fig. 221 - Overview of machine*

- 1 - Transport conveyor and/or Accumulator unit
- 2 - Infeed unit
- 3 - Transfer
- 4 - Frame
- 5 - Packing lane

## 14.2 General maintenance instructions

Regular and good maintenance increases the operating-life of your machine. For this reason a number of guidelines are given below. For the maintenance of a specific machine model, please refer to the correct machine-module in this chapter.



### **NOTE**

*Only use environmentally-friendly and biodegradable cleaning agents.*



### **NOTE**

*Only use the lubricants recommended by Moba.*



### **NOTE**

*Consult the maintenance schedules for the location of the lubricants and parts.*

### 14.2.1 Daily or after every 8 operating hours



- Clean all parts that have been soiled with egg-drippings. Use a cloth or a soft brush.
- Never clean the black (silanized) plate under the EggInspector™ with a sharp implement
- Never clean glass plates or mirrors with a sharp object.
- Blow loadcells clean with compressed air.

### 14.2.2 Weekly or after every 40 hours of operation



- Use a dry cloth or compressed air to clean all optical sensors and their reflectors.
- Clean the sleeves of the oiled air filters of the EggInspector™ with water and soap, let them dry properly and put them back on the filters.
- Remove eggshells and any feathers present in the dropset™ of the packing lane.

### 14.2.3 Monthly or after every 200 hours of operation



- Oil all chains in accordance with the maintenance schedules.



### **NOTE**

*Use as little grease and oil as possible. Remove excess lubricants.*

- Lubricate all bearing rings in accordance with maintenance schedules.
- Remove all protective caps and clean the machine with pressurised air.
- Oil all rod-heads in accordance with maintenance schedules.
- Clean the oiled air filters of the EggInspector™ with a special cleaning kit.
- Use pressurised air to clean all other installed air-filters in the machine.

### 14.2.4 Two-monthly or after every 400 hours of operation

- Check the tension of these spring-tensioners, which are mounted on the packing conveyor, and also the drive of the tray-conveyor (optional)..

### 14.2.5 Semi-annually or after every 1200 hours of operation



- Check all chains/toothed belts and, if necessary, adjust the tension.
- If the chains/toothed belts must be tensioned, then the adjustments must be checked against the data contained in the service manual.
- Oil all lubrication points in accordance with maintenance schedules.

### 14.2.6 Annually or after every 2400 operating hours

- Replace the air-filters installed in the machine.
- Check the functions of all emergency stop switches.
- Check the functions of all interlock switches.
- Check the functions of the safety cable.
- Check the operation of all light screens with the check bar supplied with the unit.



---

**DANGER**

***Before electrical connections are tested, set the primary power-supply-switch to "0" to shut down the machine and then wait 5 minutes before checking the following.***

---



- Examine all electrical connections and make sure that they are properly and tightly connected.

### 14.2.7 Prolonged stoppage

If the Omnia FT/XF is not used for a longer period, for example, for a holiday period, the MultiDrum™ must be thoroughly cleaned. Remove all the egg-drippings from the carriers and other moving parts. Encrusted egg-drippings may cause the MultiDrum™ to no longer function properly.

## 14.3 Lubricants



The symbol  is used to indicate the lubrication-points in the lubrication charts, and a letter is used to indicate the lubricants that must be used.



The symbol  + a number that indicates the time interval in hours between cleaning-sessions.

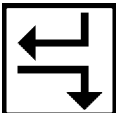
Table 10 - Lubricants

| Letter | Lubricants                                | MOBA order number | Replacement lubricant                     | MOBA order number |
|--------|-------------------------------------------|-------------------|-------------------------------------------|-------------------|
| A      | Shell T 46                                | 02045100          | None                                      | None              |
| B      | Shell Gadus S3 T220 P2                    | 02042076          | Shell Cassida Grease EPS (Food-Grade)     | 02042068          |
| C      | Shell OMALA 150                           | 80207890          | Klüber Oil 4UH1-150 (Food Grade)          | 02045249          |
| D      | Klüber Oil 4UH1-15 (Food-Grade)           | 02045249          | None                                      | None              |
| E      | Chain oil Bel Ray no tox 550 (Food-Grade) | 02045276          | None                                      | None              |
| F      | Grafloscon                                | 02042055          | None                                      | None              |
| G      | Shell Cassida Grease EPS (Food-Grade)     | 02042068          | Klübersynth UH1 14-151 (Food-Grade)       | 02042069          |
| H      | Wynn's Viscotene                          | 01635580          | Chain oil Bel Ray no tox 550 (Food-Grade) | 02045276          |

## 14.4 Explanation of symbols used in maintenance schedules

The symbols shown below are used in the maintenance plans to indicate work-activities or parts.

Table 11 - Symbols used for maintenance- and cleaning-procedures

| Symbol                                                                              | Definition                                                                                                                      |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|    | Clean and grease all chains.<br>Check the chain for jammed links.<br>Use a small brush to distribute oil evenly over the chain. |
|    | Lubricate all steel rod heads with several drops of oil.                                                                        |
|    | Lubricate all grease nipples.<br>Grease all bearings so that the new grease comes out of the bearing.<br>Remove excess grease.  |
|  | Exchange the indicated part.                                                                                                    |
|  | Blow compressed air onto the indicated part to clean it.                                                                        |
|  | Clean the indicated parts with a damp cloth or soft brush.                                                                      |
|  | Remove dirt on the indicated parts with a vacuum cleaner.                                                                       |

## 14.5 Overview of the Lubrication chart for the Infeed unit

## 14.6 Overview of units

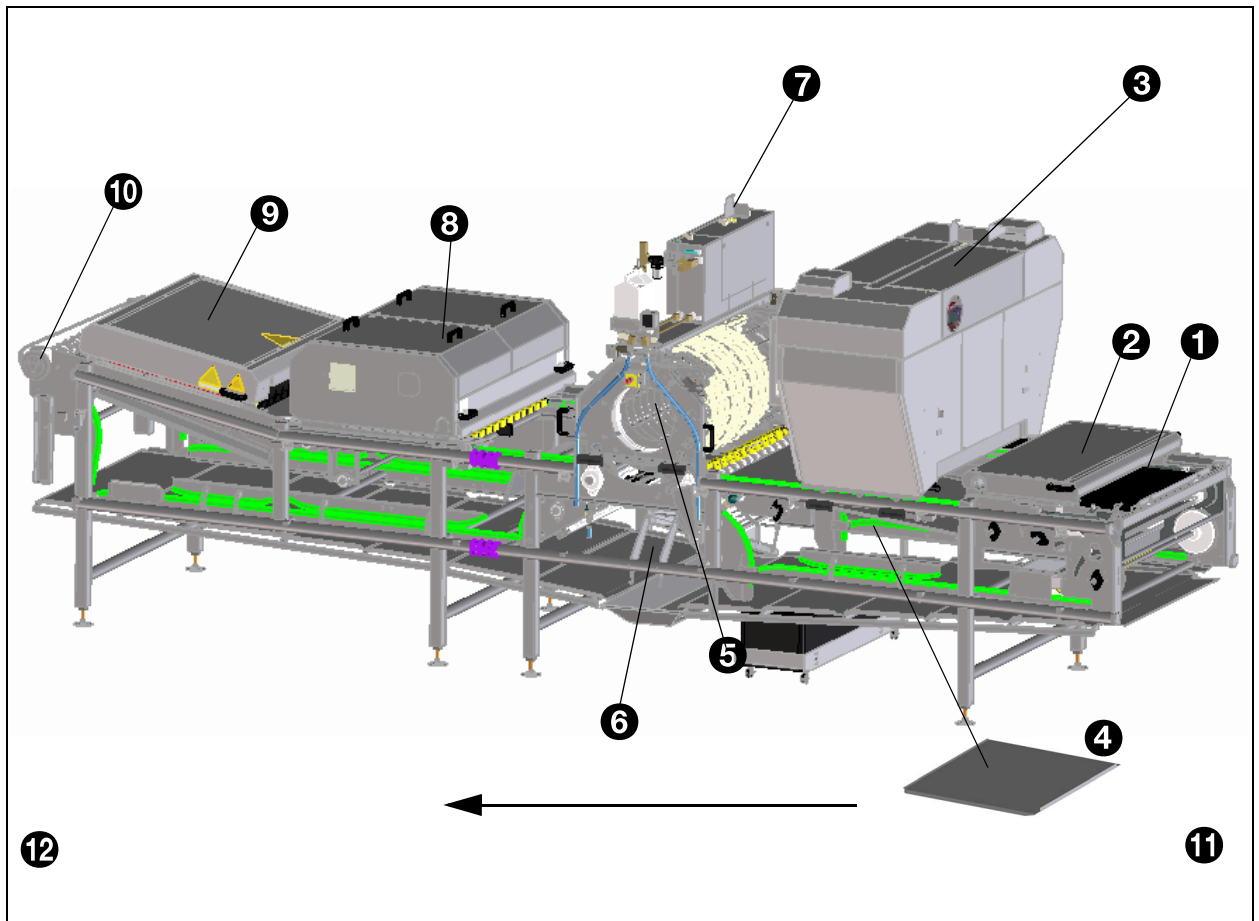


Fig. 222 - Units

- 1 - Placement unit
- 2 - Feather removal unit
- 3 - Egginspector™
- 4 - Silanized plate
- 5 - MultiDrum™
- 6 - Ejection unit
- 7 - Cleaning unit
- 8 - Crack detection unit
- 9 - UV unit
- 10 - Delivery unit
- 11 - Infeed-side
- 12 - Take-away side

The arrow in the illustration shows the direction of the movement of eggs.

## 14.6.1 Lubrication charts, Infeed unit

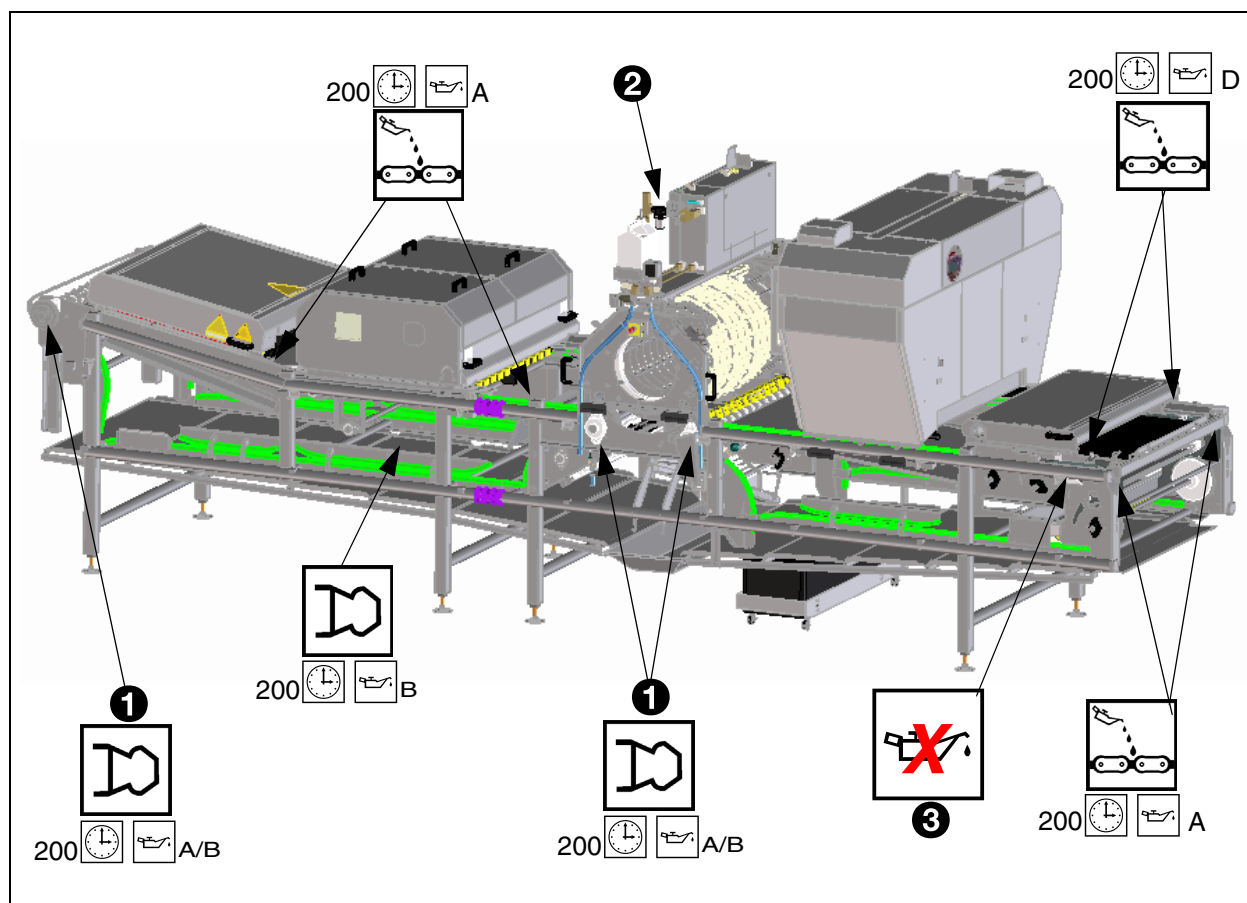


Fig. 223 - Lubrication of infeed unit

1. Lubricate bearing-rings:
  - a. when mounting blue bearing-rings with Shell Retinax EP-2
  - b. when mounting white bearing-rings with Klübersynth UH1 14-151 (Food Grade)
2. Clean the filter mounted:
  - a. to the tube placed in the jerry can with cleaning agent.
  - b. in the supply pipe.

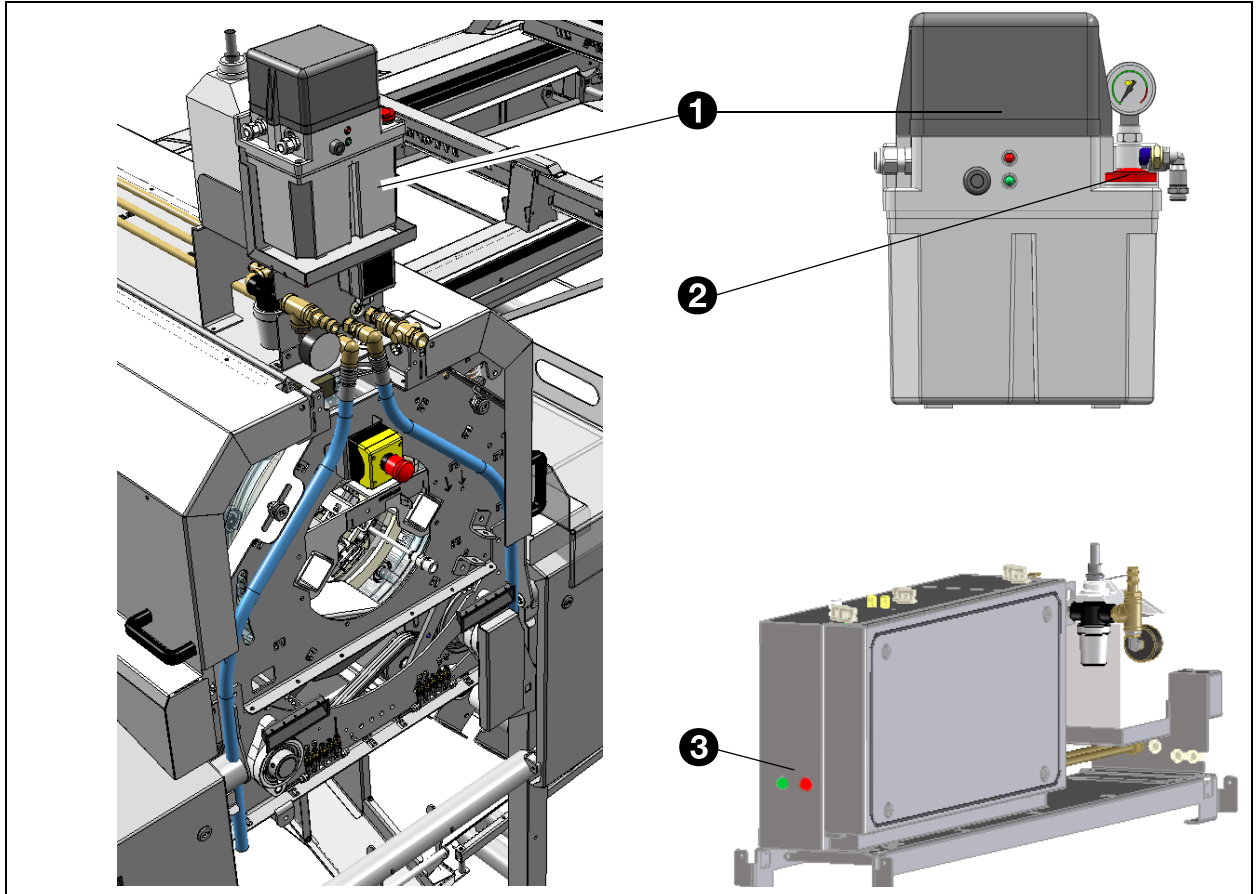


### **NOTE**

**Do not lubricate the spinner chain (3).**

### 14.6.2 Automatic lubrication system

The automatic lubrication system is available for the Omnia FT infeed unit. The most important chains in the infeed section are lubricated.



*Fig. 224 - Automatic lubrication system on the infeed*

- 1 - Oil reservoir
- 2 - Oil reservoir cover
- 3 - Cleaning unit - red lamp

#### **Automatic lubrication infeed unit Omnia FT:**

- 1 Check oil level indication , if the red lamp on the cleaning unit is flashing refill the oil reservoir.
- 2 Remove the cover of the oil reservoir.
- 3 Fill the oil reservoir with Shell T 46 or Klüber Summit HySyn FG 46 (Food-Grade).
- 4 Close the cover of the oil reservoir.

### 14.6.3 Maintenance of cleaning unit

The cleaning unit has two filters that must be cleaned once a month.

### 14.6.4 Filter in spray pipe

The nozzles mounted on the spray pipes contain a filter.

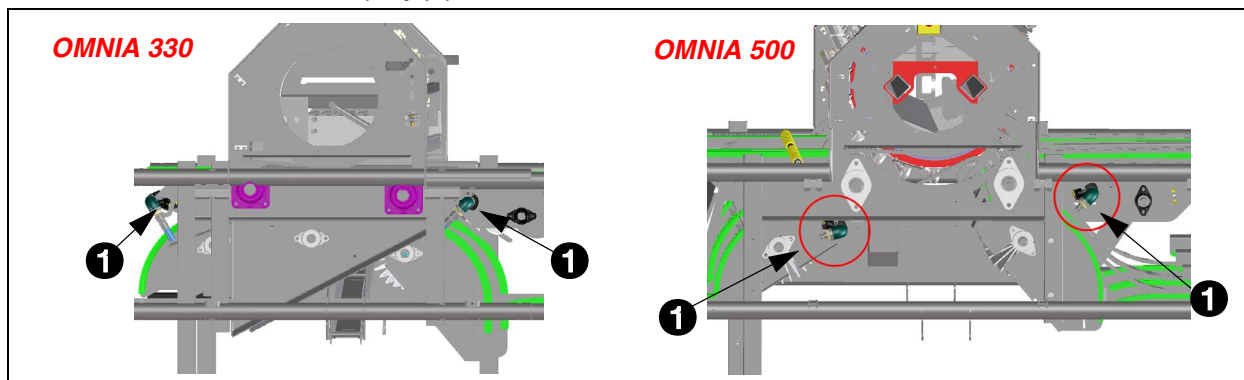


Fig. 225 - Location of spray pipes

Switch off the grader with the primary power-supply-switch. Close the water supply.

1. Remove the protection cover under the orientator.

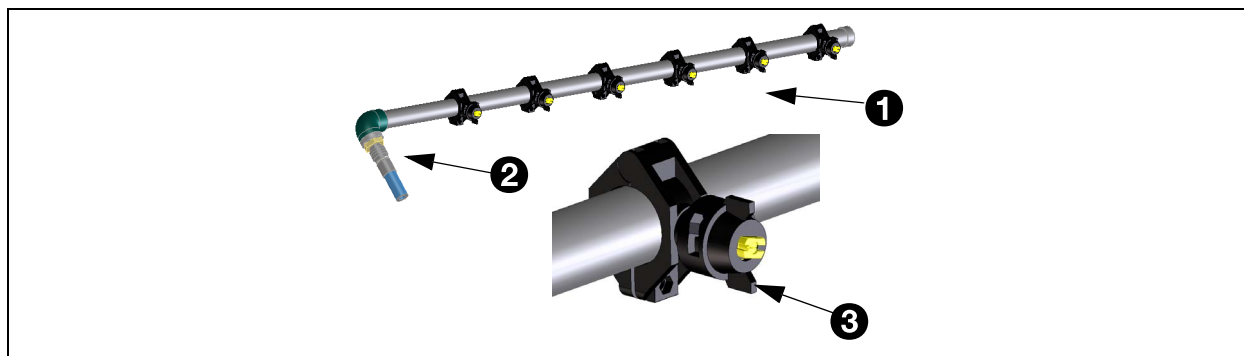


Fig. 226 - Detail of spray pipe

2. Disconnect the hoses from the spray pipes (1). The hoses are connected to the spray pipes by a quick-release connector (2). See fig. 225.
3. Take the two tubes from the frame.

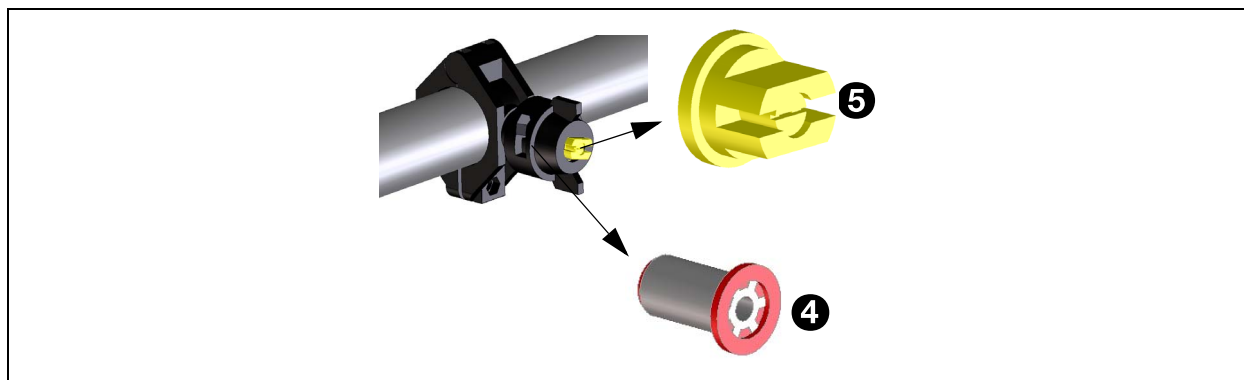


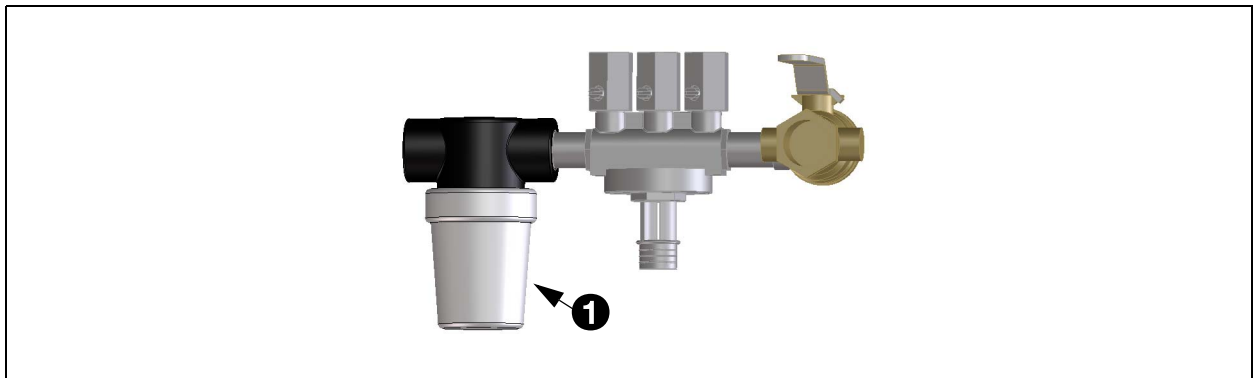
Fig. 227 - Filter



4. Remove the nozzles (3) from the spray pipe.
  - The filters (4) are now free. See fig. 227.
5. Remove the filters and clean them.
6. Clean the sprays (5).
7. Reassemble the parts.

#### 14.6.5 Liquid filter

A liquid filter (1) is installed in the water supply pipe. See fig. 228.



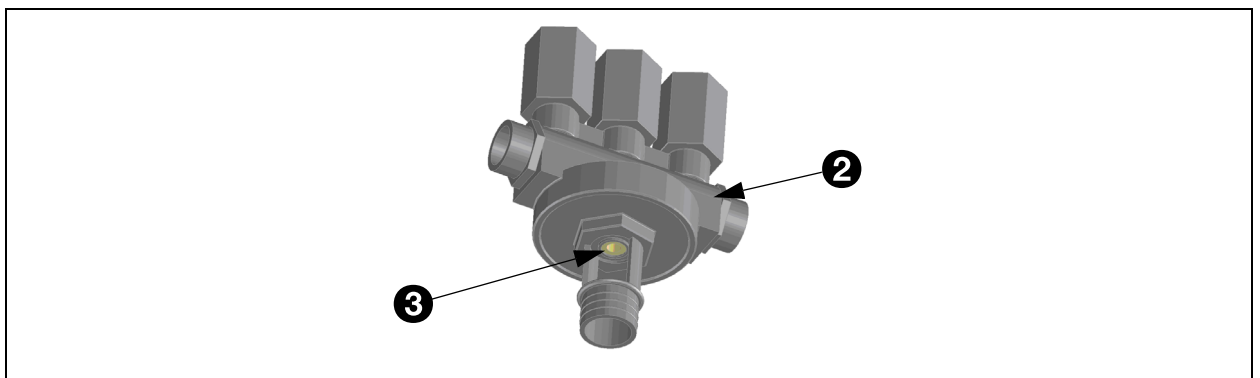
*Fig. 228 - Liquid filter*

Switch off the grader with the primary power-supply-switch. Close the water supply.

1. Loosen the lower part (1) of the filter housing.
  - The filter is now free.
2. Remove the filter and clean it.
3. Put the parts back again.

#### 14.6.6 Possible leakage of the “Triple Check Valve”

If the “Triple Check Valve” (2) leaks then this can be corrected by carrying out the following steps.



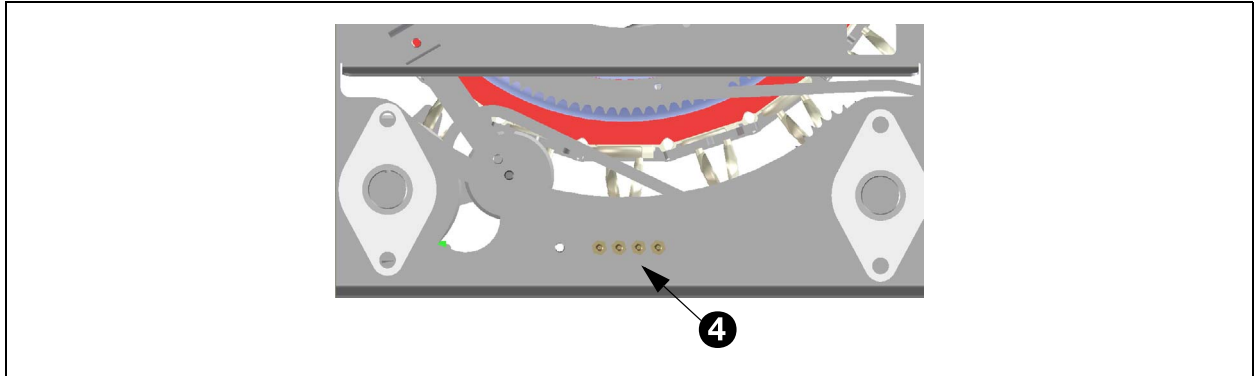
*Fig. 229 - Triple Check Valve*

1. Release the water pressure
2. Use an Allen key, for example, to push the rubber ring with bush (3) back.
3. Turn the water supply on again.



### 14.6.7 Lubrication-points MultiDrum™

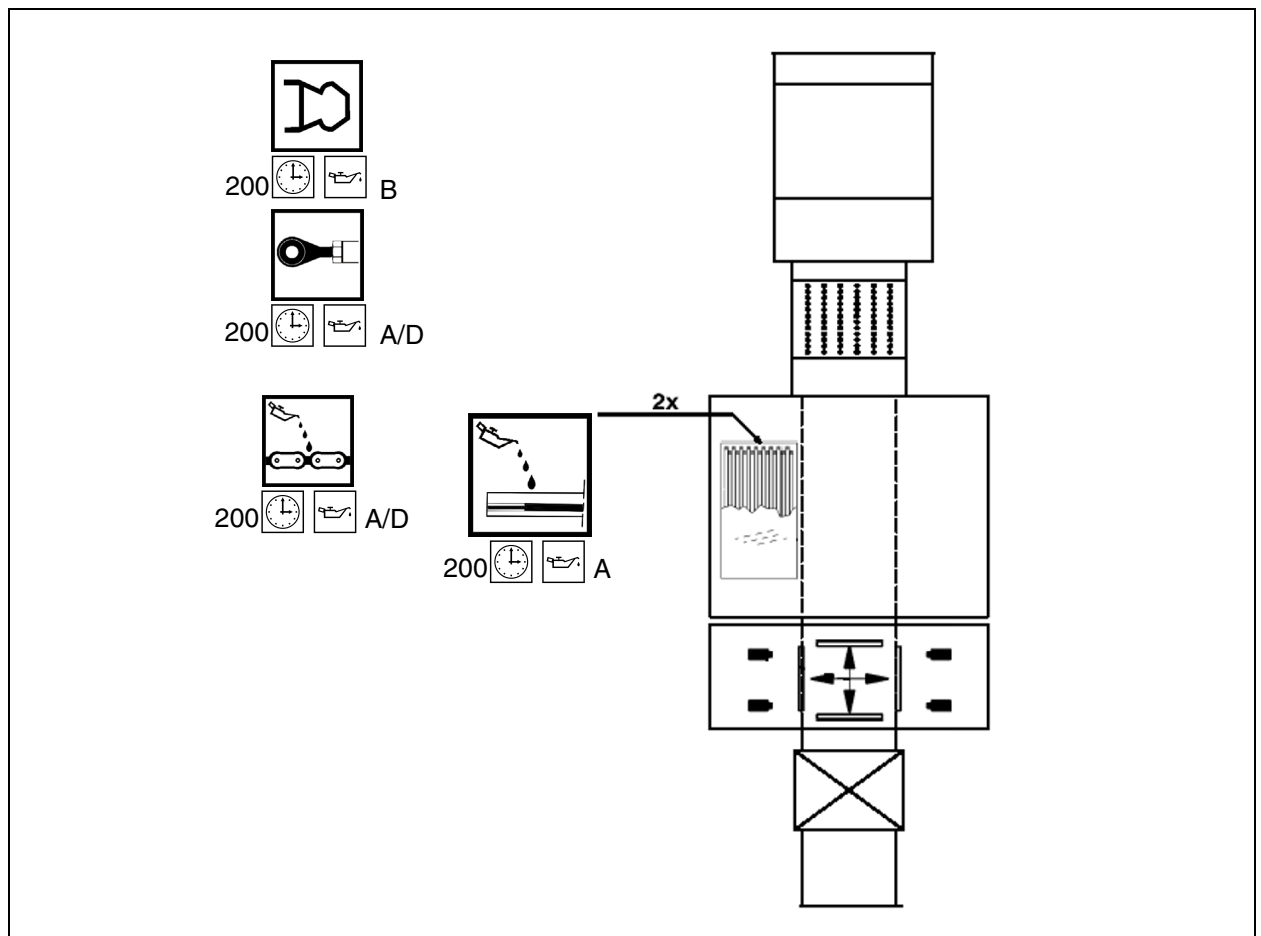
Since the bearings are difficult to reach with 18-row infeed units the lubrication points are located on the outer casing. (4). See fig. 230.



*Fig. 230 - Lubrication-points.*

Lubricate the points in accordance with the lubrication chart.

## 14.7 Lubrication chart for Omnia XF Infeed



*Fig. 231 - Lubrication of infeed unit, single roller*



## 14.7.1 Lubrication chart for Transfer

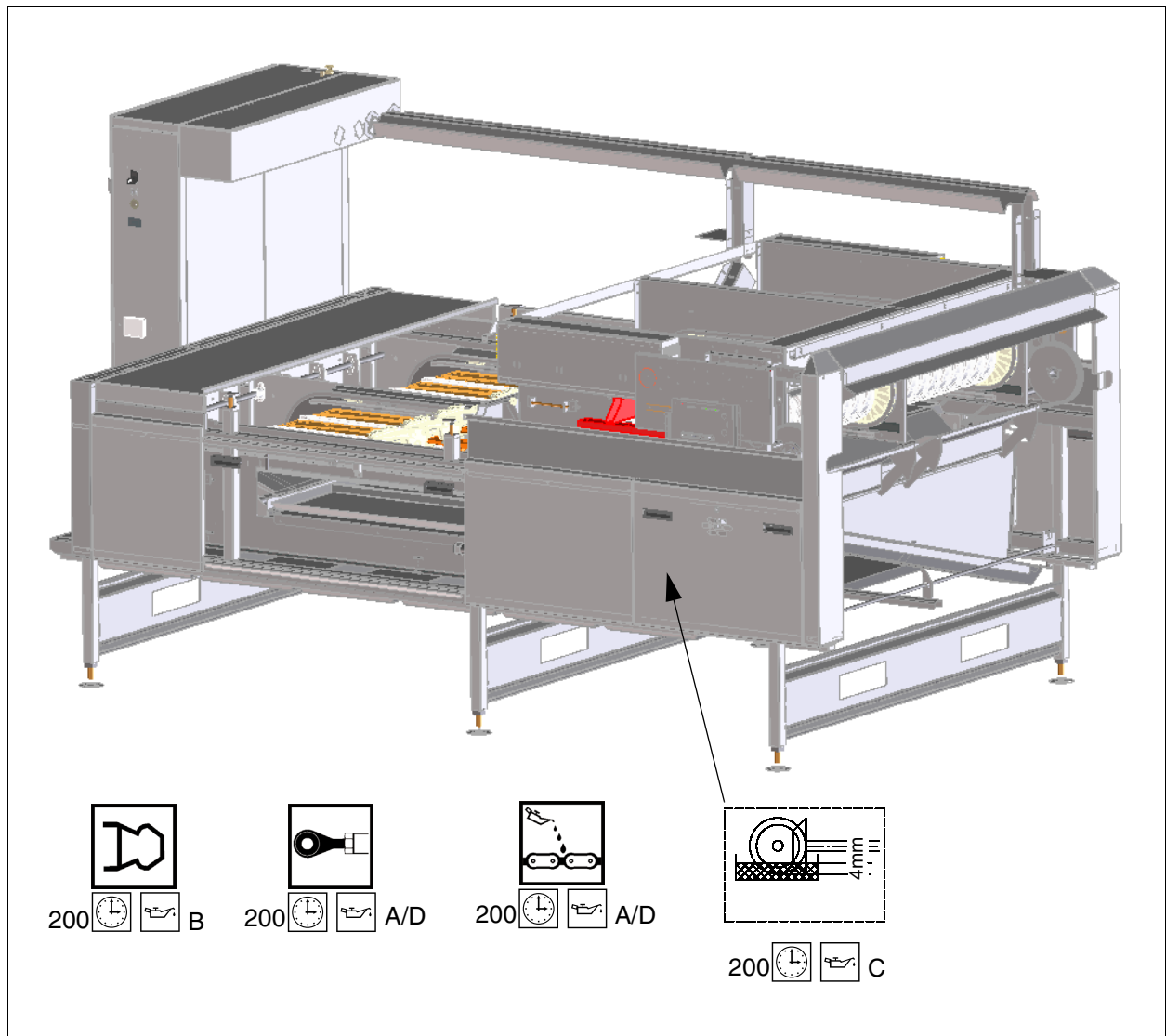


Fig. 232 - Lubrication of transfer



14.8 Lubrication chart for the frame

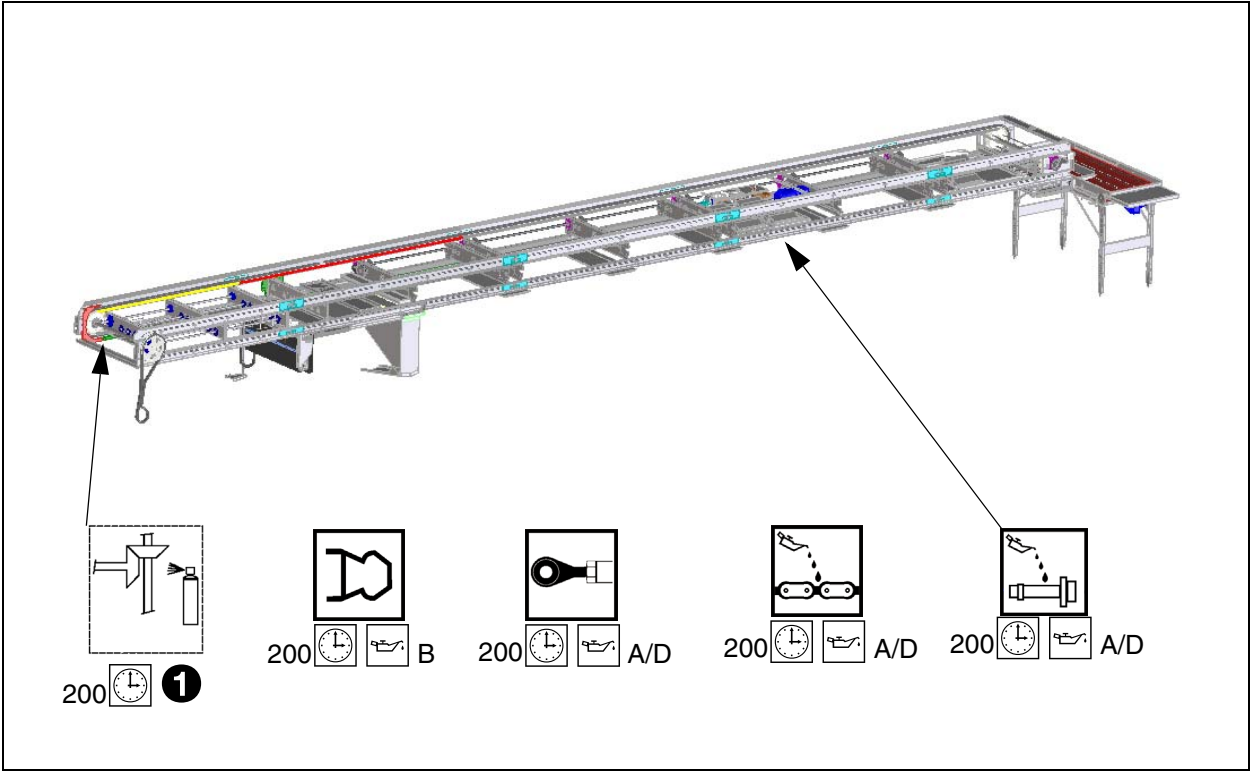


Fig. 233 - Lubrication of the frame

Table 12 - Lubricants for the Frame

| Figure | Lubricant  | MOBA order number |
|--------|------------|-------------------|
| (1)    | Grafloscon | 02042055          |
|        |            |                   |



## 14.8.1 Lubrication schedule for the receiverset™ dropset™OMNIA FT/XF 85

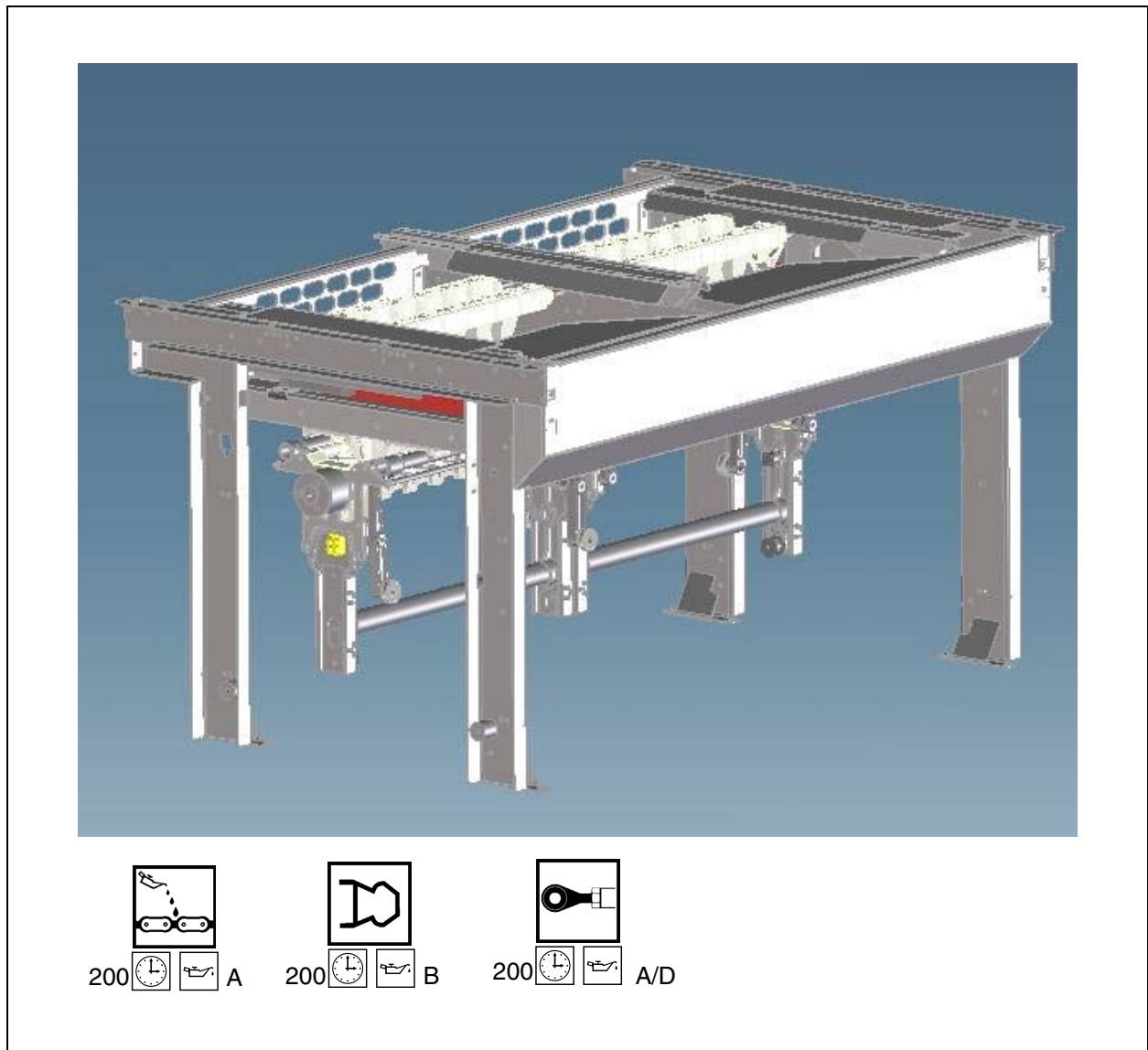


Fig. 234 - Lubrication of buffer sets



### 14.8.2 Lubrication schedule for the Buffersets OMNIA FT/XF 125/170/250/330/500

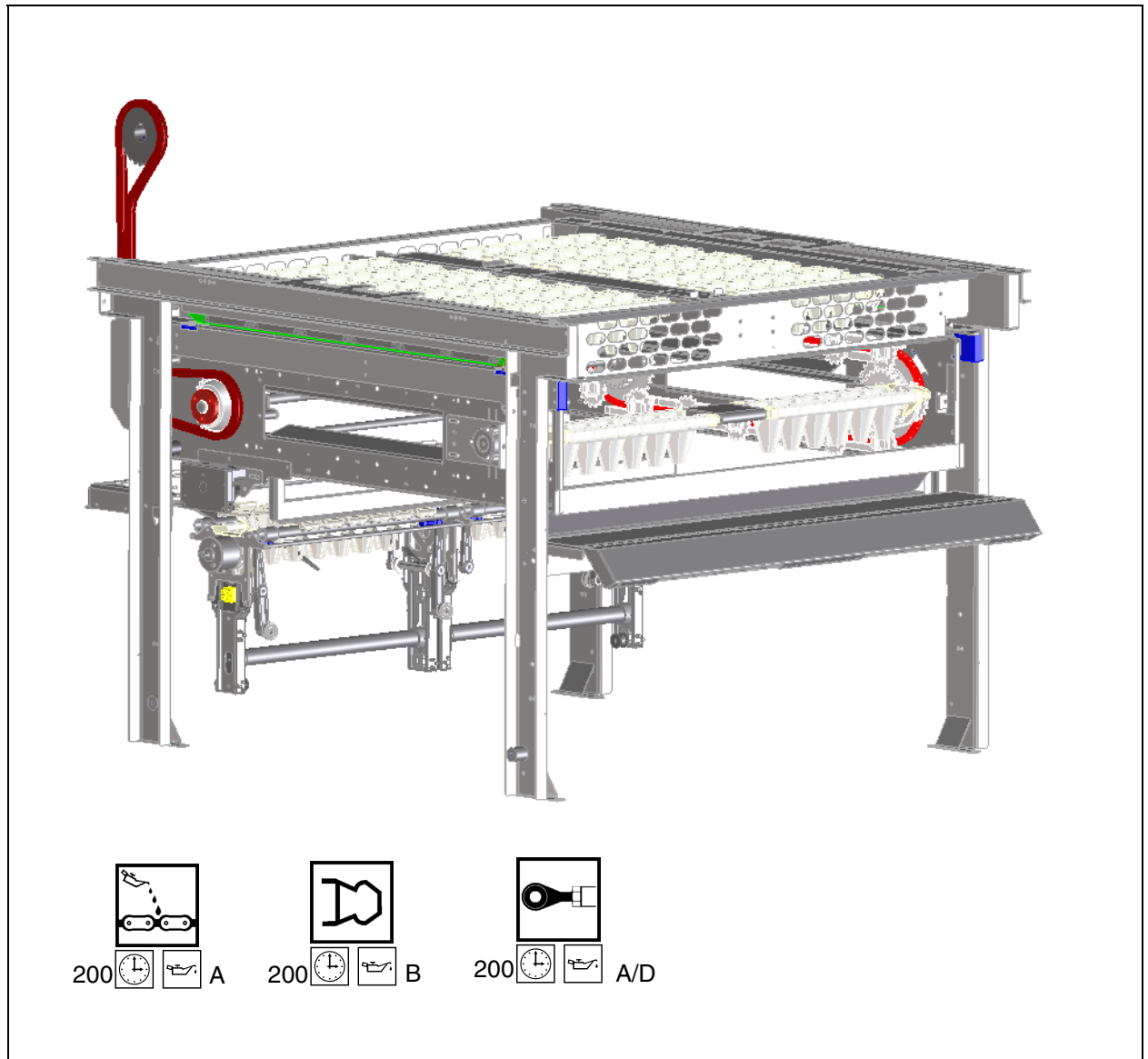


Fig. 235 - Lubrication of buffer sets



## 14.9 Lubrication chart for the Packing lane

### 14.9.1 Lubrication chart for the denester

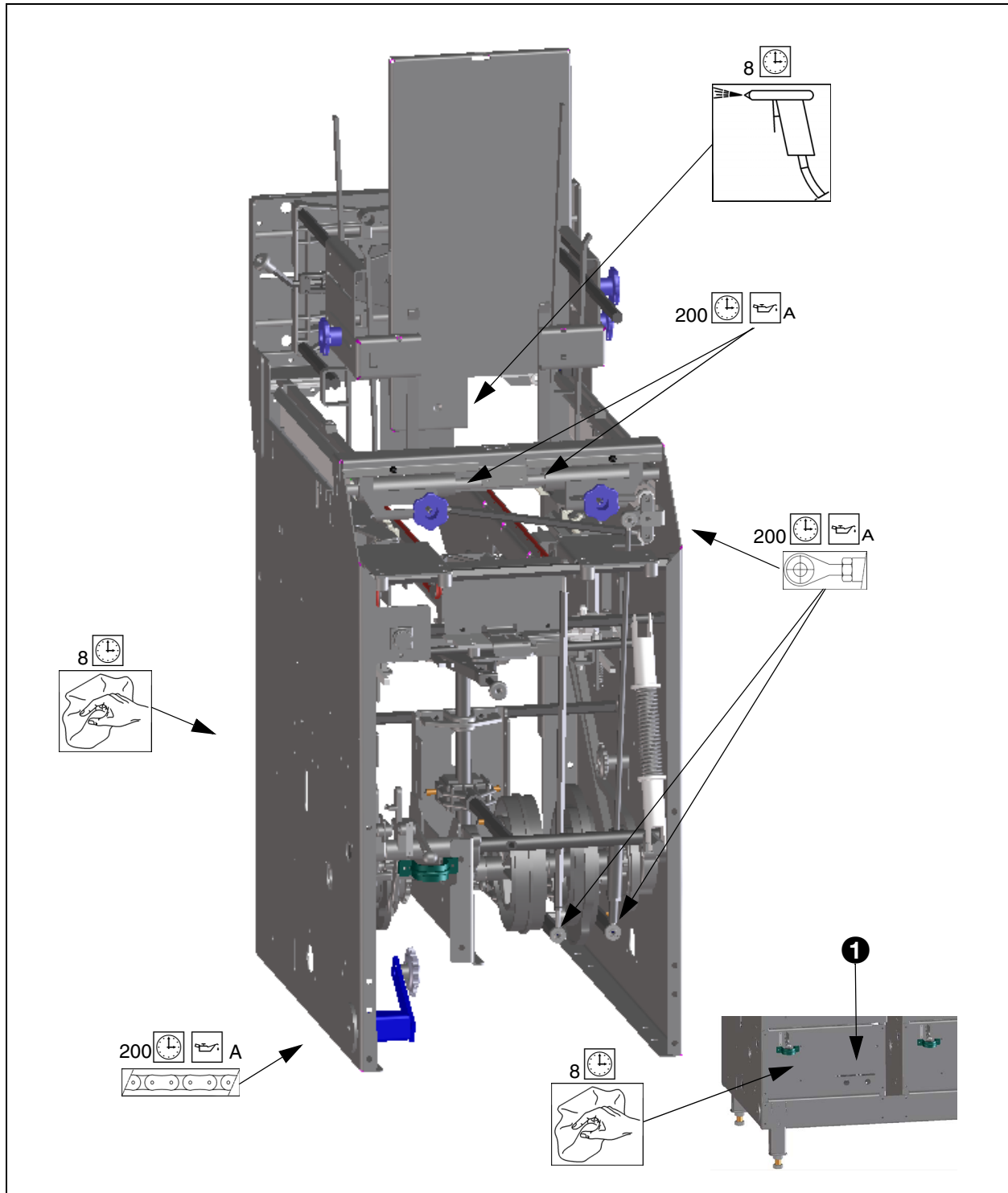


Fig. 236 - Lubrication of the denester



### 14.9.2 Lubrication schedule for the pin conveyor

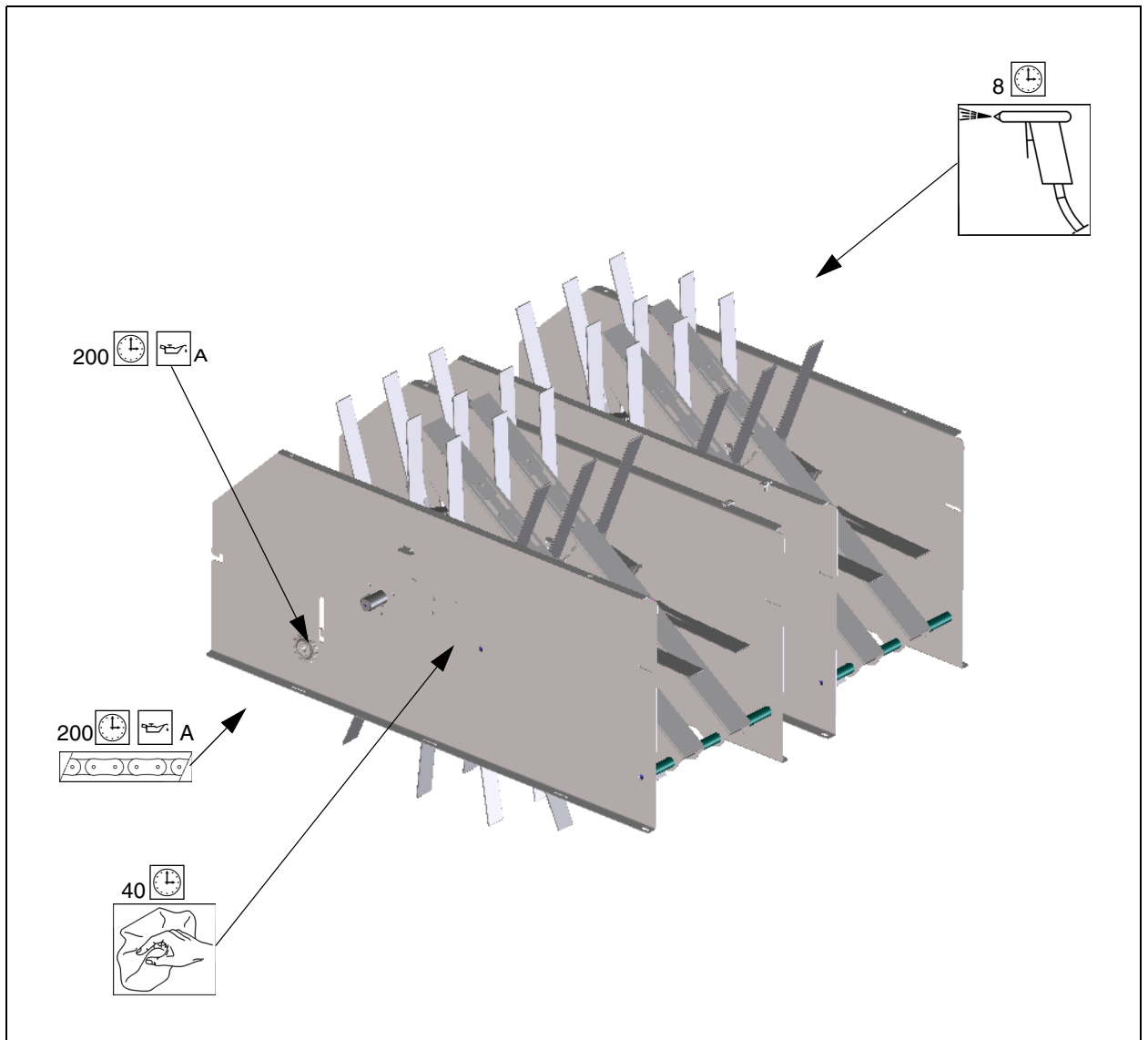


Fig. 237 - Lubrication of pin conveyor



## 14.9.3 Lubrication chart for the packing conveyor

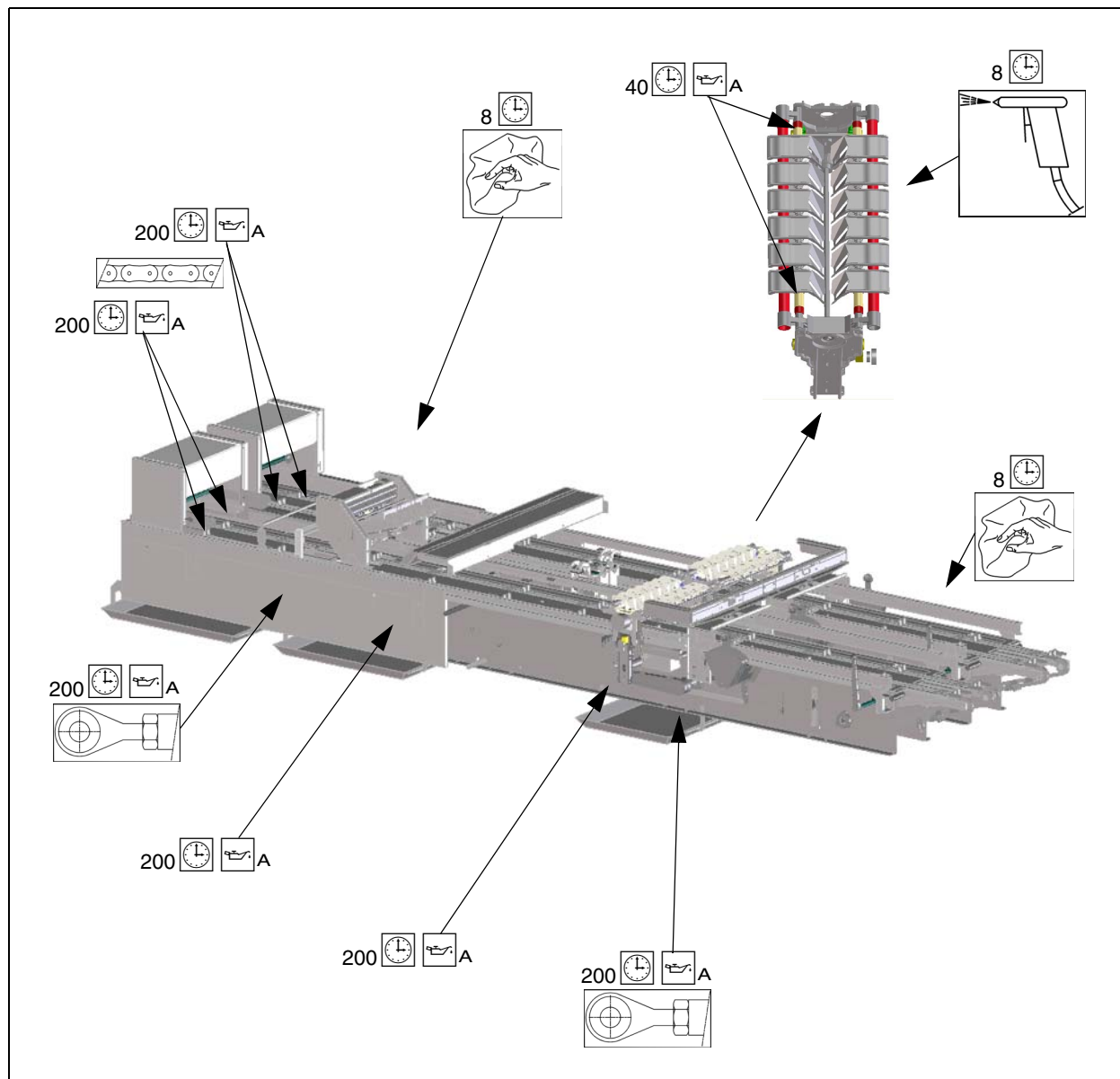


Fig. 238 - Lubrication of the packing conveyor



#### 14.9.4 Lubrication chart for take-away conveyor

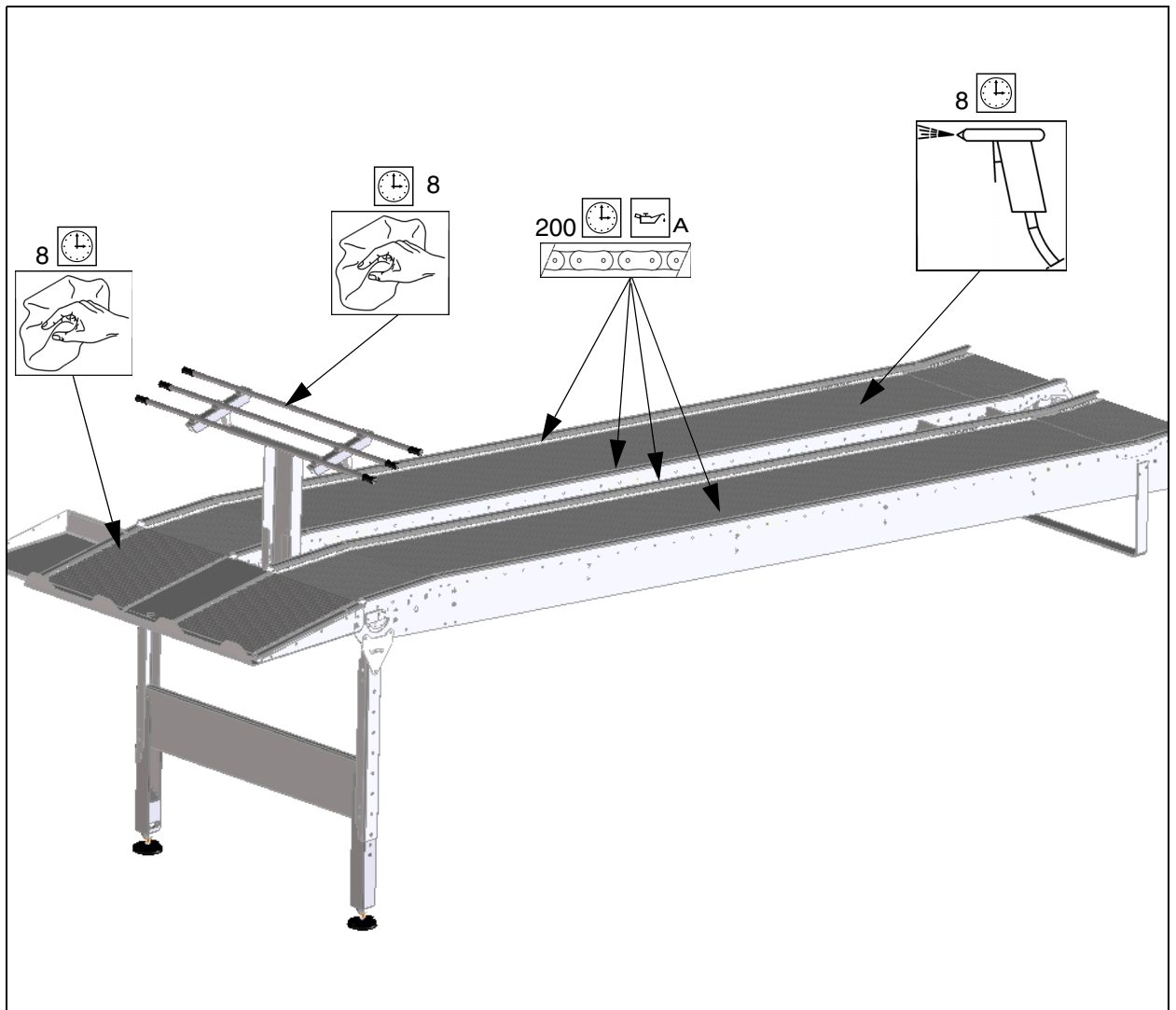


Fig. 239 - Lubrication of the infeed conveyor



## 14.9.5 Lubrication chart for the Contiflow

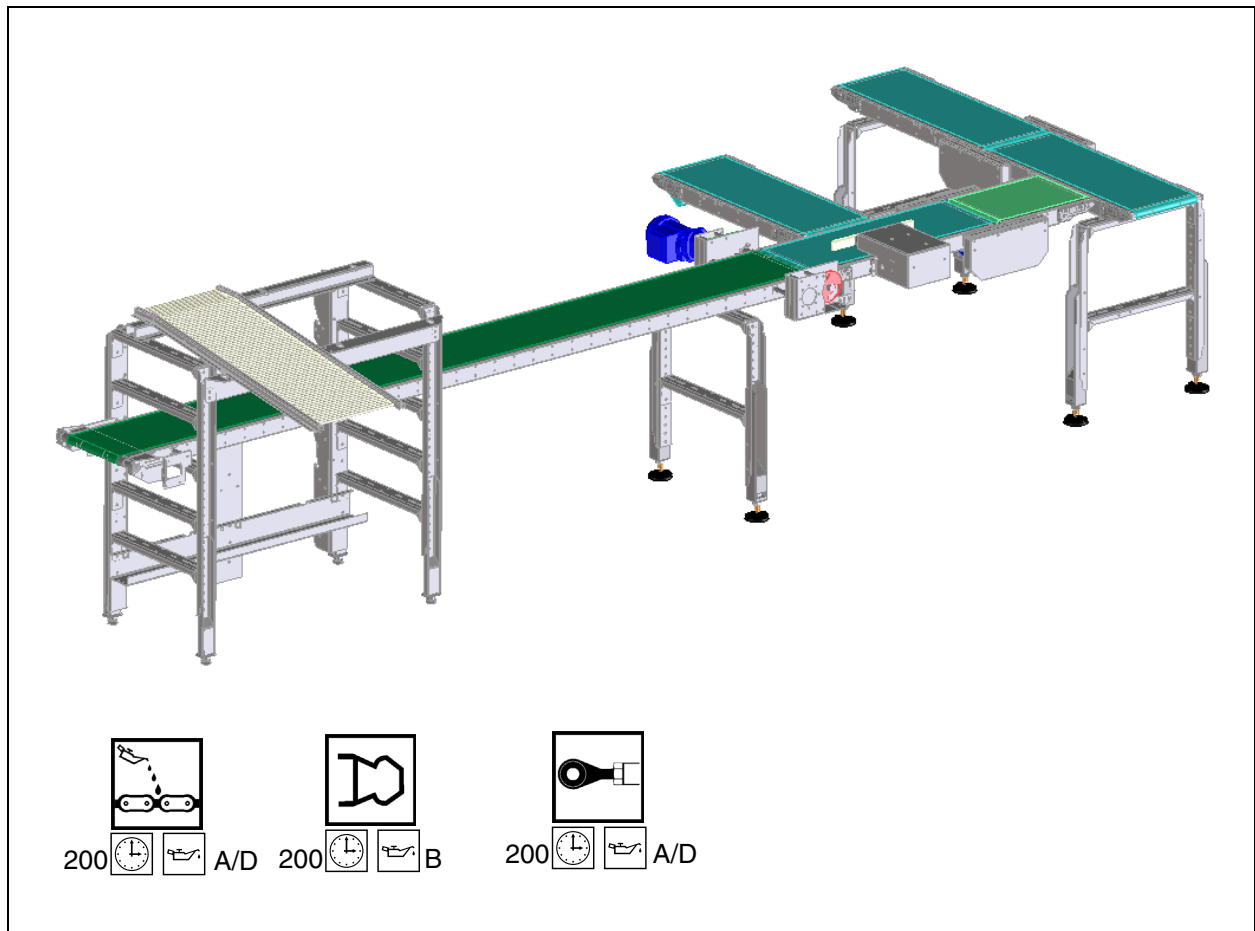


Fig. 240 - Lubrication of the contiflow

### 14.9.6 Lubrication chart for the Toplidder

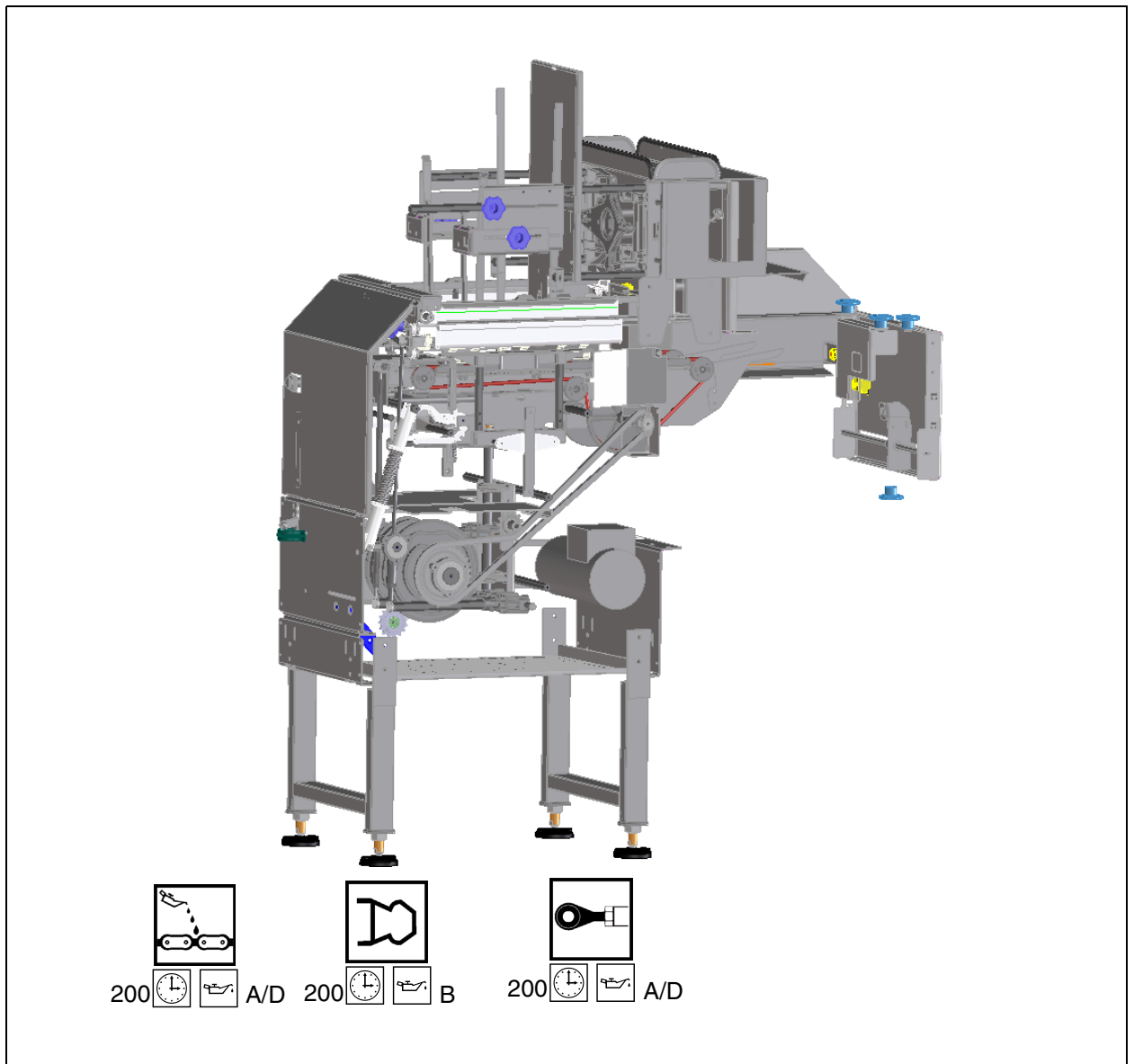


Fig. 241 - Lubrication of the Toplidder



## Appendices

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The following appendices are included with this documentation:

- Comparative table of lubricants
- TIP box
- Repair of plastic cord



## A Comparative table of lubricants

This appendix provides a list of lubricants, which are recommended by the suppliers and can be used as a replacement of lubricants in the machine. The trade-name and the type of lubricant are indicated, so that you can buy this product in your neighbourhood.



### **WARNING**

***These lubricants have not been tested by MOBA! The use of these lubricants can cause damage to your machine!***



In case of doubt about the correct lubricant to be used, please take up contact with one of our service-support-centres. See Paragraph 1.5 'How to order parts' for the address of the service-support-centre in your region.

|  |  |  |  |  | Diversen                    |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------|
| Shell T46                                                                         |                                                                                   |                                                                                   |                                                                                    | Klüber-Summit HySyn FG 46                                                           |                             |
| Shell Cassida Chain Oil 1000                                                      | Febis K 68                                                                        | Mobilgear 627                                                                     | Way lubricant                                                                      | Klüber Oil 4 UH1-15                                                                 |                             |
| Shell Retinax EP-2                                                                | Esso GP 400gr.                                                                    |                                                                                   |                                                                                    | Klübersynth UH1 14-151                                                              |                             |
| Shell Omala 220 Gear Oil                                                          | Febis K 220                                                                       | Mobilgear 630                                                                     | Meropa 220                                                                         | Klüberoil 4UH1-220                                                                  |                             |
|                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     | Finn Lube                   |
|                                                                                   |                                                                                   |                                                                                   |                                                                                    | Klüberoil 4UH1-1500 spray                                                           | Graflascon Ca901            |
|                                                                                   |                                                                                   |                                                                                   |                                                                                    | Klübersynth UH1 14-151                                                              | Kettingspray MotipUltra Gel |
| Shell Omala 150                                                                   |                                                                                   |                                                                                   |                                                                                    | Klüberoil 4UH1-150                                                                  |                             |

Fig. 242 - Comparative table of lubricants



## B TIP box OMNIA

The machine can run very slowly with the TIP box. The TIP box is supplied with the OMNIA. The box is located in the main electricity cabinet. The use of the TIP box

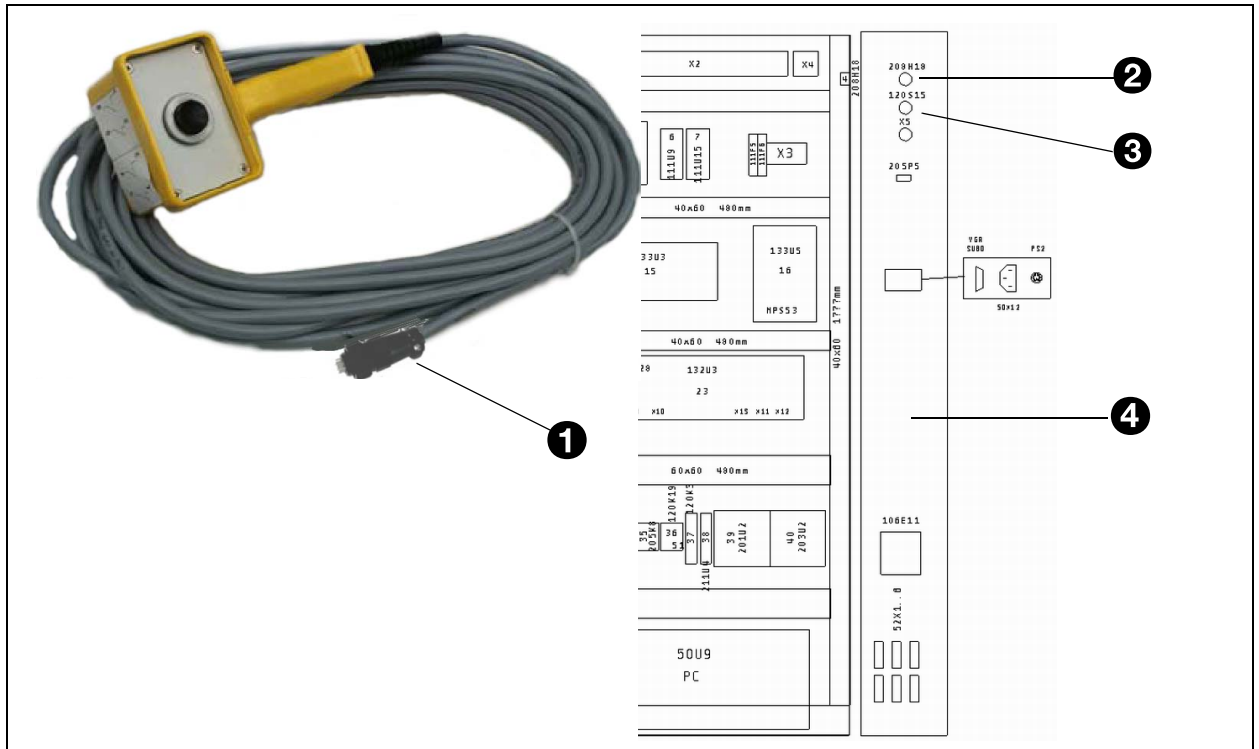


Fig. 243 - Main electricity cabinet OMNIA

- 1 - Plug
- 2 - Switch
- 3 - Socket
- 4 - Side electricity cabinet

1. Put the plug of the TIP box into the socket on the side of the main electricity cabinet of the grader.
2. Set the switch to 'Man'.
  - The machine can now be operated with the TIP box.



### **WARNING**

**Tell other employees that you plan to work with or service the machine.**

Table 13 - Operation of TIP box pushbutton

| Pushbutton       | Action                  |
|------------------|-------------------------|
| not operated     | none                    |
| half-way         | main motor in operation |
| fully pressed in | emergency stop          |

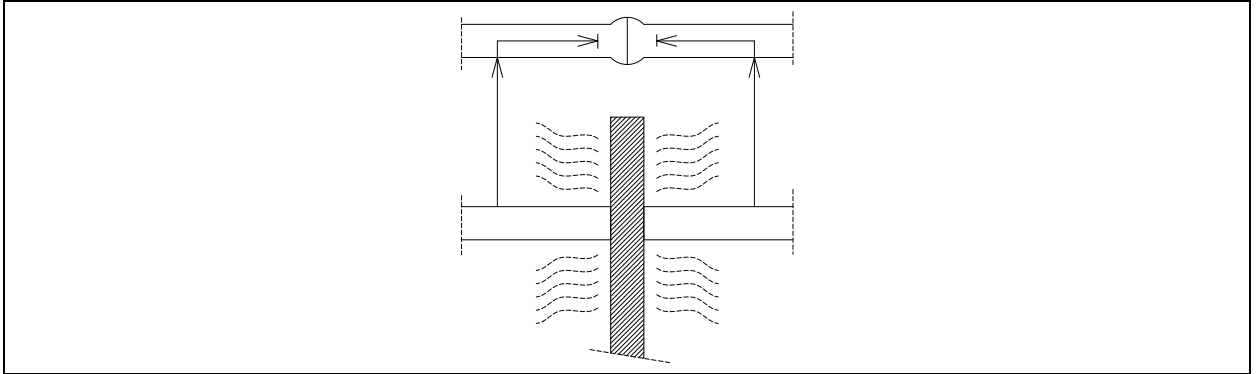
 **NOTE**  
*After using the TIP box, remove the plug and set the switch in the 'Auto' position.*

 **NOTE**  
*To use to Tip box for the Loader (except the FL loader) a special plug adapter is available.*



## C Repair of plastic cord

A manual packaging-belt and the denester are provided with plastic cords for the transport of eggs and packaging-material. A broken cord can be repaired as follows:



*Fig. 244 - Repair of plastic cord*

1. Cut off the ends of the plastic cord so they are straight.
2. Hold both ends of the plastic band simultaneously against a soldering iron with a straight element.
3. Heat the ends of the plastic band, until the material begins to melt.
4. Now press the ends of the plastic band tightly together for several minutes.
5. Remove any thickened areas from the plastic band with a sharp knife.
6. Now the cord can be used again.



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## Glossary

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### **C**

#### Carrier set

Plastic holder in which 3 carriers are placed.

### **R**

#### Return path

The return path is the path that the chain takes under the transport direction of the infeed unit.

### **T**

#### TIP box

See the user manual attachments for an explanation of the use of this operating box.

### **U**

#### U-holder

U-shaped plastic part that transports the first roller of the roller set.

#### UV-C light

Ultraviolet light (UV) is a natural part of sunlight and has a disinfecting effect on micro-organisms. UV-C light causes a photochemical reaction in DNA.



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## Afterword

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We have tried to compile this manual to be as complete and perfect as possible. However, there may be differences in your machine's model and the description of the machine in this manual. If you find any differences, please contact us.

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